Daily Briefing - November 12, 2026

Your Daily Tech & Programming Digest

Thursday, November 12, 2026

1000	123838	1177	35
ARTICLES	WORDS	MIN READ	SOURCES

📰 Today's Top Stories

RAD51 expression in pan-cancer and its prognostic and immune mechanisms in glioma

📖 1 min 📝 35 words BRAIN RESEARCH

> **Summary:** Publication date: 15 January 2027 Source: Brain Research, Volume 1895 Author(s): Liubing Hou, Yu Wang, Xiang Zhan, Jiayuan Li, Zizhou Zhang, Weijing Cai, Linlin Su, Wei Han, Xuetao Han, Yue Han, Xiaolan Lv, Xiaoying Xue, Huandi Zhou

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https://www.sciencedirect.com/science/article/pii/S0006899326004816?dgcid=rss_sd_all

Reduced hypothalamic [³H]PK11195 binding in Lewis rats after chronic high-intensity interval training

📖 1 min 📝 28 words BRAIN RESEARCH

Summary: Publication date: 15 January 2027 Source: Brain Research, Volume 1895 Author(s): Atiqur Hasany, John R. Tyler, Huy Lu, Kenneth Blum, Wei Sun, Richard J. Salvi, Igor Elman, Panayotis K. Thanos

🔗 **Read full article:**
https://www.sciencedirect.com/science/article/pii/S0006899326004622?dgcid=rss_sd_all

Alpha-beta oscillatory dynamics underlying auditory inhibitory control in children

👤 Finnilä, E., van Bijnen, S., Kujala, J., Parviainen, T. 📅 2026-11-11 📖 1 min 📝 180 words

BIORXIV NEUROSCIENCE

Summary: Response inhibition, the ability to withhold an unwanted response, is a fundamental component of cognitive control. While evoked responses in inhibitory tasks have been widely studied, induced oscillatory dynamics, particularly in the auditory modality and in children, remain largely unexplored. Her...

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https://www.biorxiv.org/content/10.64898/2026.11.10.765597v1?rss=1

How striatal neurons influence decision-making depends on broader circuit context: evidence from theory-driven cell type identification

Yu, Z., Verstynen, T., Rubin, J.

2026-11-11 1 min 249 words BIORXIV NEUROSCIENCE

Summary: Neural cell types are often conceived as having fixed computational roles, yet their behavioral contributions may depend on the broader circuit state. In the striatum, for example, direct and indirect pathway spiny projection neurons (SPNs) are typically described as facilitating or suppressing acti...

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Postpartum resource scarcity reshapes maternal care, offspring prioritization, and the steroid-sensitive medial preoptic area

Aspesi, D., Soriano, V., Rolle, S., Mahrous, M. Y., Brenhouse, H., Krishnan, K., Bangasser, D. A.

2026-11-12 1 min 249 words BIORXIV NEUROSCIENCE

Summary: Parental care requires balancing offspring needs with environmental constraints, yet how resource scarcity alters the organization of maternal responses remains poorly understood. Here, we examined maternal behavior, offspring prioritization, and steroid receptor-associated extracellular matrix orga...

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https://www.biorxiv.org/content/10.64898/2026.11.09.765420v1?rss=1

Sentence-level neural dynamics in speech motor cortex

Levi-Aharoni, H., Jude, J. J., Haro, S., Acosta, A. J., Allcroft, S. B., Ottenhoff, M. C., Card, N. S., Wairagkar, M., Iacobacci, C., Pritchard, A. L., Richards, B., Brandman, D. M., Stavisky, S. D., Au Yong, N., Pandarinath, C., Paulk, A. C., Hadar, P. N., Li, J., Williams, Z. M., Cash, S. S., Simeral, J. D., Hochberg, L. R., Rubin, D. B.

2026-11-12 1 min 155 words BIORXIV NEUROSCIENCE

Summary: Prior recordings from speech motor cortex have revealed phonemic, syllabic, articulatory and speech-sequence representations, while more recent studies indicate that activity may also reflect higher-order features of speech production. However, whether speech motor cortex is sensitive to the linguis...

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https://www.biorxiv.org/content/10.64898/2026.11.10.765804v1?rss=1

The Computational Grammar of Human Behaviour in a Biological Environment

Serratos Hernandez, U. D., Goekay, U., Kornemann, L., Hutabarat, Y., Sen, D., Bach, D. R.

2026-11-12 1 min 193 words BIORXIV NEUROSCIENCE

Summary: Human behaviour unfolds as an extended stream of spontaneous actions shaped by environmental demands and the agent's goals. The organisation of these actions, situated between setting abstract goals and concrete movement execution, remains poorly characterised. Here, we introduce DynaClade, an unsup...

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Loss of GluN2A Selectively Impairs Excitatory Synaptic Integration in Hippocampal Somatostatin Interneurons

Hunter, D. W., Kaese, H., Kind, P. C., Wyllie, D. J.
2026-11-12 1 min 224 words

BIORXIV NEUROSCIENCE

Summary: N-methyl-D-aspartate receptors (NMDARs) are widely considered key determinants of excitatory synaptic kinetics and activity-dependent plasticity, yet their specific contribution to postsynaptic potentials across distinct hippocampal CA1 cell types remains poorly defined. Null variants in GRIN2A are ...

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Same Movements, Different Skills: How Context Shapes Learning in Humans and Neural Networks

Gupta, M. W., Taylor, J. A.
2026-11-12 1 min 149 words

BIORXIV NEUROSCIENCE

Summary: What determines whether experiences are integrated into shared representations or split? Motor skill learning offers a revealing test: when two tasks require an identical sequence of movements, both intuition and theory suggest that practice in one should generalize to the other. Yet we find that ch...

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A hypothesis-generating multi-resolution post-GWAS analysis of context-dependent regulatory signals in migraine

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https://www.nature.com/articles/s41598-026-79784-5

Alterations in glymphatic system in patients with Sleep-related epilepsy: a study of PVS and DTI-ALPS index

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https://www.nature.com/articles/s41598-026-79932-x

Bridging the dementia diagnostic gap by addressing cognitive assessments globally

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https://www.nature.com/articles/s44400-026-00144-1

NeuroMemory-DiffNet: a reference-guided memory framework for brain tumor segmentation and classification

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https://www.nature.com/articles/s41598-026-79184-9

Beat perception relates to reading through phonological awareness and rapid automatized naming

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https://www.nature.com/articles/s41598-026-72872-6

When neuroscience meets machine learning: a taxonomic scoping review of machine and statistical learning approaches in sub-Saharan Africa

Deshendran Moodley
2026-11-12 1 min 146 words

FRONTIERS COMPUTATIONAL NEUROSCIENCE

Summary: In high-resource economies, computational neuroscience research frequently employs machine learning (ML) and statistical learning (SL) techniques. While these methods provide effective ways to analyse multimodal, high-dimensional, and complex data they are underexplored in low resource settings. Thi...

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https://www.frontiersin.org/articles/10.3389/fncom.2026.1975649

Validation of the Hungarian version of the Coma Recovery Scale-Revised in patients with disorders of consciousness

András Attila Horváth
2026-11-12 1 min 171 words

FRONTIERS HUMAN NEUROSCIENCE

Summary: ObjectiveOur objective was to validate the Hungarian version of the Coma Recovery Scale-Revised (CRS-R).DesignWe conducted a multicenter study across neurology, intensive care, neuro-rehabilitation, and nursing departments in Hungary. Thirty-one patients (17 males; mean age 49.1 ± 16.1 years) with d...

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Coordination without command: active inference and the route to embodied intelligence

Alexander D. Shaw 2026-11-12 1 min 181 words FRONTIERS NEUROROBOTICS

Summary: Embodied intelligence depends on bodies whose contact with the environment, distribution of load and capacity for action are only partly known and are changed by movement. We argue that this creates a practical reason to organize inference through distributed, hierarchical and plural processes, usin...

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Case Report: Clinical applications of extended-duration remote ambulatory EEG monitoring: lessons from real-world cases

Timothy Ryan 2026-11-12 1 min 191 words FRONTIERS NEUROSCIENCE

Summary: BackgroundTraditional ambulatory electroencephalography (EEG) and inpatient epilepsy monitoring unit (EMU) admissions face significant operational limitations including: short recording windows (typically 24–72 h), high costs, technician shortages, and physical restrictions on daily activities.Metho...

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https://www.frontiersin.org/articles/10.3389/fnins.2026.1976990

What shared responses reveal about audience behavior: neural prediction and the emerging potential of peripheral synchrony

Isak B. Blank 2026-11-12 1 min 214 words FRONTIERS NEUROSCIENCE

Summary: Predicting audience responses to dynamic media remains a significant challenge in neuromarketing and neuroforecasting. Intersubject correlation (ISC) has been proposed to address this need by quantifying the temporal similarity of neural, physiological, and behavioral responses among individuals exp...

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Human cortical high-gamma power scales with movement rate in healthy participants and stroke survivors

Fanny Quandt 2025-01-09 1 min 65 words OOSTENVELD ROBERT

Summary: Motor cortical high-gamma oscillations (60-90 Hz) occur at movement onset and are spatially focused over the contralateral primary motor cortex. Although high-gamma oscillations are widely recognized for their significance in human motor control, their precise function on a cortical level remains el...

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NIRS-BIDS: Brain Imaging Data Structure Extended to Near-Infrared Spectroscopy

👤 Luca Pollonini 📅 2025-01-27 📖 1 min 📝 70 words OOSTENVELD ROBERT

Summary: Functional near-infrared spectroscopy (fNIRS) is an increasingly popular neuroimaging technique that measures cortical hemodynamic activity in a non-invasive and portable fashion. Although the fNIRS community has been successful in disseminating open-source processing tools and a standard file forma...

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Pseudonymisation of neuroimages and data protection: Increasing access to data while retaining scientific utility

👤 Lyuba Zehl 📅 2025-06-26 📖 1 min 📝 67 words OOSTENVELD ROBERT

Summary: For a number of years, facial features removal techniques such as 'defacing', 'skull stripping' and 'face masking/blurring', were considered adequate privacy preserving tools to openly share brain images. Scientifically, these measures were already a compromise between data protection requirements a...

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Cycling on the Freeway: The perilous state of open-source neuroscience software

👤 Tim M Tierney 📅 2025-08-13 📖 1 min 📝 74 words OOSTENVELD ROBERT

Summary: Most scientists need software to perform their research (Barker et al., 2020;Carver et al., 2022;Hettrick, 2014;Hettrick et al., 2014;Switters & Osimo, 2019), and neuroscientists are no exception. Whether we work with reaction times, electrophysiological signals, or magnetic resonance imaging data, ...

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Optimal configuration of on-scalp OPMs with fixed channel counts

👤 Robert Oostenveld 📅 2025-08-13 📖 1 min 📝 69 words OOSTENVELD ROBERT

Summary: Recent technological developments have brought optically pumped magnetometers (OPMs) within reach of the larger neuroscientific community. The current state-of-the-art consists of whole-head systems that measure the magnetic field at >100 locations. OPM sensors can be constructed to measure the fiel...

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ARTEM-IS for ERP: Agreed Reporting Template for EEG Methodology-International Standard for Event-Related Potential Experiments

👤 Suzy J Styles 📅 2025-12-08 📖 1 min 📝 73 words OOSTENVELD ROBERT

Summary: The choices we make during the recording, preprocessing and analysis of event-related potentials (ERP) data can affect study outcomes. As such, it is critical that they are transparently reported to allow for reproducibility. Yet, systematic reviews of reporting practices in the field have shown tha...

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Eye-Tracking-BIDS: the Brain Imaging Data Structure extended to gaze position and pupil data

👤 Julia-Katharina Pfarr 📅 2026-02-12 📖 1 min 📝 68 words OOSTENVELD ROBERT

Summary: The Brain Imaging Data Structure (BIDS) is a widely adopted, community-driven standard to organize neuroimaging data and metadata. Although numerous extensions have been developed to incrementally extend coverage to new modalities and data types, an unambiguous, granular specification for eye-tracki...

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Facilitating unusual bodily experiences and out-of-body experiences across wakefulness and sleep: A high-density EEG and neurophenomenology study

👤 Robert Oostenveld 📅 2026-02-12 📖 1 min 📝 67 words OOSTENVELD ROBERT

Summary: Unusual bodily experiences (UBEs) are illusory bodily perceptions that are not coherent with typical wakeful experiences, including flying or floating sensations, body distortions, and out-of-body experiences. This study examines UBEs in a controlled sleep laboratory using meditation and light stimu...

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The Impact of Brain Tumors and Craniotomy Lesions on Scalp EEG

👤 Vitória Piai 📅 2026-02-17 📖 1 min 📝 65 words OOSTENVELD ROBERT

Summary: Electroencephalography (EEG) is widely used in both research and clinical settings, yet its accuracy can be significantly impacted by subject-specific anatomical anomalies such as brain lesions and skull defects. This study investigates the effects of glioma-related brain lesions and craniotomy-indu...

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EMG-BIDS: an extension to the Brain Imaging Data Structure for electromyography

Arnaud Delorme 2026-08-01 1 min 64 words OOSTENVELD ROBERT

Summary: Electromyography (EMG) is fundamental to clinical assessment, rehabilitation, neuromuscular research, and human-machine interfaces. Despite decades of use, no widely adopted standard exists for organizing and sharing EMG data, limiting reusability and large-scale data aggregation. We present EMG-BID...

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Ageist attitudes as a moderator of objective and subjective vision: A longitudinal study

Yoav S Bergman 2026-11-11 1 min 41 words LOW VISION

Summary: OBJECTIVES: This longitudinal study examined the relationship between objective visual acuity and subjective vision over a six-month period. Specifically, it investigated the moderating role of internalized ageist attitudes in shaping how objective vision relates to perceived sensory vision difficul...

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Registry-recorded stroke and transient ischaemic attack diagnoses in Saudi Arabia, 2016-2025: a national descriptive analysis

👤 Mohammed AlAbdulali
📅 2026-11-11 📖 1 min 📝 58 words

LOW VISION

Summary: CONCLUSION: Saudi Arabia's recorded stroke burden is concentrated among older males. The findings describe patterns captured by the PHO and should not be interpreted as prevalence, incidence, or causal associations, which indicate the need for better prevention, management of chronic diseases and a ...

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Design and parameter optimization of a bilateral staggered wheel-spoon precision seed-metering device for millet plot breeding

👤 Ankang Mu 📅 2026-11-11 📖 1 min 📝 63 words LOW VISION

Summary: Precision seeding of micro-sized millet seeds in plot breeding faces critical challenges, including high miss-seeding rates and severe seed cross-contamination during variety changeovers. This study developed a novel bilateral staggered wheel-spoon precision seed-metering device to achieve precise h...

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Representation geometry and task performance in vision-language modeling for computed tomography enterography

👤 Ryan Stidham 📅 2026-11-11 📖 1 min 📝 36 words LOW VISION

Summary: CONCLUSIONS: Slice sampling and tissue-contrast-focused encoding affect performance in CT enterography vision-language systems, and RAG improved report-label agreement over the fine-tuned baseline. These results provide initial baselines and practical guidance for multimodal learning in volumetric C...

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Incidence and Management of Ocular Hypertension Following the 0.19 mg Fluocinolone Acetonide Implant for Diabetic Macular Edema

👤 Wills Eye Retina Service Study Group, The Wills Eye Retina Service Study Group additionally includes

📅 2026-11-11 📖 1 min 📝 36 words LOW VISION

Summary: CONCLUSION: FAi was associated with low rates of OHT, with most cases managed with observation or topical therapy, with surgery rarely required. This supports the safety of FAi and highlight the importance of extended IOP monitoring.

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Passive imaging with quantum advantage

👤 A I Lvovsky

📅 2026-11-11 📖 1 min 📝 67 words LOW VISION

Summary: Far-field optical imaging inevitably involves low-pass spatial filtering, limiting the resolution. Moreover, conventional imaging suppresses high spatial frequency components close to the cutoff, making them invisible under noise-particularly the shot noise arising from discrete and random nature of...

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Memory-guided mutual information optimization in retinal imaging for early stroke detection and subtyping

Yitian Zhao 2026-11-11 1 min 64 words LOW VISION

Summary: Effective stroke prevention relies on identifying at-risk individuals before neurological symptoms arise, yet this is severely limited by the absence of scalable screening tools. To address this, we developed RetCEVD, a diagnostic framework that uses rapid, non-invasive retinal Optical Coherence Tom...

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SnowRef-Drive: A Global Instruction-Driven Traffic Video Segmentation Dataset for Adverse Weather Driving Scenarios

Yintao Wei 2026-11-11 1 min 62 words LOW VISION

Summary: Adverse weather, particularly snowfall and low visibility, severely degrades vision-based traffic perception, yet existing language-driven video segmentation datasets rarely provide pixel-level annotations for real-world driving under such conditions. We present SnowRef-Drive, a large-scale global d...

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Predictors of visual acuity change in a large cohort study of scleritis from Northeast China

Xiaoli Liu
2026-11-11 1 min 39 words LOW VISION

Summary: CONCLUSION: This Northeast Chinese scleritis cohort is predominantly anterior and female-predominant, with low rates of bilateral involvement, necrotizing type, and associated autoimmune diseases. Furthermore, this study indicates the predictors of visual acuity change, providing reference points fo...

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Comparison of outcome in patients with idiopathic intracranial hypertension (IIH) following optic nerve sheath fenestration (ONSF) using a medial transconjunctival approach with dural window or superomedial lid-crease incision approach

👤 Philip A Tenzel
📅 2026-11-12 📖 1 min 📝 37 words

LOW VISION

Summary: CONCLUSIONS: This study demonstrates that the superomedial lid-crease approach with fenestration and the medial transconjunctival approach with dural window for ONSF are both effective in preventing vision loss in patients with IIH, with similarly low side-effect profiles.

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A single session of high-definition transcranial direct current stimulation does not modulate effects of knee two-point discrimination training or sensorimotor function in healthy adults: Double-blind randomised controlled trial

👤 Paul W Hodges
📅 2026-07-30 📖 1 min 📝 53 words

TACTILE ACUITY

Summary: CONCLUSION: A single 20-minute session of 1 mA HD-tDCS, with or without brief TPD training, does not acutely modify knee somatosensory, sensorimotor, or functional performance in healthy adults. Existing evidence at the hand may not translate to the lower limb. Future work should investigate higher-...

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Variation in glabrous skin physiology across anatomical locations on the human hand

👤 Martina C Meinke 📅 2026-08-13 📖 1 min 📝 76 words TACTILE ACUITY

Summary: Skin physiology varies greatly between anatomical locations, with impact on tactile sensitivity. Research on the variations in morphology and physiology between different anatomical locations of glabrous skin is limited so far. This study investigates the morphology and physiology of glabrous skin o...

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Beyond Hamstring Shortness: Continuous Flexibility Is Associated With Balance and Stability but Not Sensory Measures-A Cross-Sectional Study

👤 Senanur Akıncı 📅 2026-08-14 📖 1 min 📝 66 words TACTILE ACUITY

Summary: CONCLUSIONS: Greater hamstring flexibility was associated with better dynamic balance performance and showed a weaker association with postural stability, whereas no robust associations were observed with sensory measures of postural control. These findings suggest that hamstring flexibility may be ...

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Resting-state EEG microstate dynamics reflect individual differences in tactile angle discriminability

👤 Liu Yang 📅 2026-08-19 📖 1 min 📝 63 words TACTILE ACUITY

Summary: Tactile spatial acuity, the ability to discriminate fine spatial details, is a fundamental sensory capacity modulated by both peripheral and central mechanisms. Here, we investigated whether resting-state EEG microstate dynamics are associated with individual differences in tactile spatial acuity. F...

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A grating orientation task with controlled dome elevation speed and contact force: stimulus conditions and day-to-day reliability

👤 Hideaki Onishi 📅 2026-09-25 📖 1 min 📝 66 words TACTILE ACUITY

Summary: The grating orientation discrimination task (GOT) is typically used to assess tactile spatial acuity. However, performance is influenced by stimulus speed, contact force between the stimulator and the skin, and the method of discrimination. This study aimed to explore suitable stimulus conditions fo...

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Epidermal microstructure and tactile sensitivity in the glabrous skin of hands and feet

Hannes P Saal 2026-09-25 1 min 69 words TACTILE ACUITY

Summary: Glabrous skin on the human palms and soles is specialized for mechanical interaction and is characterized by papillary ridges, an absence of hair follicles, and a thick stratum corneum. Although gross morphological differences, such as the increased thickness of the plantar stratum corneum, are well...

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Neurotrauma During Military Training: Results of the INVestigating traIning assoCiated blasT pAthology (INVICTA) Study

Walter S Carr 2026-10-01 1 min 67 words TACTILE ACUITY

Summary: Concern is growing that subconcussive blast exposure during military exercises such as heavy weapons training (HWT) may adversely affect brain function. HWT involves firing shoulder-mounted weapons that can expose the gunner, assistant gunner, and instructors (range safety officers [RSOs]) to blast ...

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Altered body representation in chronic non-specific low back pain: a testable framework linking body-schema-related disturbance, corticomotor alterations, and altered pain processing

👤 Zhifei Xie
📅 2026-10-24 📖 1 min 📝 66 words

TACTILE ACUITY

Summary: Low back pain is a leading contributor to disability worldwide; in chronic non-specific low back pain (CNSLBP), the processes sustaining pain and disability in the absence of identifiable structural pathology remain incompletely understood. Three partially overlapping research lines have documented ...

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Hyperexcitable primary somatosensory cortex in the MitoPark mouse model of Parkinson's disease

Alan J Emanuel 2026-10-25 1 min 67 words TACTILE ACUITY

Summary: The majority of Parkinson's disease patients suffer from decreased tactile acuity or neuropathic pain, both of which contribute to low quality of life and likely exacerbate motor dysfunction. How Parkinson's disease pathology relates to altered somatosensation remains unclear. One potential contribu...

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The Relationship Between Occlusal Tactile Acuity, Bite Force and Tooth Wear

Birgitta Häggman-Henrikson 2026-11-03 1 min 52 words TACTILE ACUITY

Summary: CONCLUSIONS: Bite force may be a clinically relevant factor associated with accumulated tooth wear, whereas OTA was not independently associated with tooth wear. Longitudinal studies are needed to determine whether bite force can predict tooth wear progression and whether more comprehensive assessme...

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Impact of Braille Booklet, Audio-Tactile Performance and Caregiver-supervised Training on Oral Hygiene Status, Knowledge, Attitude and Practices among Visually Challenged Children- A Randomized Clinical Trial

👤 Priyanga Chandrasekaran
📅 2026-09-22 📖 1 min 📝 52 words

BRAILLE

Summary: CONCLUSION: Both caregiver-supervised training and audio tactile performance using specialized model method of oral health education proved to be more effective than the braille booklet group on plaque control and improvement in the oral hygiene status, oral health knowledge, attitude and practice i...

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Design and functional verification of an automated Perkins Brailler accessory with audio feedback

👤 Ali Alhaj 📅 2026-09-23 📖 1 min 📝 67 words BRAILLE

> **Summary:** CONCLUSIONS: The study did not include learners with visual impairment, Braille teachers, rehabilitation specialists, or other end users, and it did not collect quantitative measures of accuracy across the full character set, writing speed, embossing quality, latency, durability, safety, usability, ...

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Uncertainty-Aware structured decoding for Chinese Braille document transcription

👤 Haoran Sun 📅 2026-09-25 📖 1 min 📝 64 words BRAILLE

> **Summary:** Automatic transcription of braille documents remains challenging in document digitization settings, particularly under real imaging degradations such as uneven embossing, illumination variation, paper texture, skew, blur, and ghosting, which make braille dot presence uncertain and often cause error ...

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Clinical Characteristics and Epidemiology of Isolated Dystonia: Results from a US Claims Analysis

James E Thornton 2026-10-08 1 min 54 words BRAILLE

Summary: CONCLUSIONS: This study defines an isolated dystonia cohort in the US based on real-world evidence and systematic exclusion of secondary causes of dystonia. Estimated prevalence is within range of prior epidemiological studies, and the cohort showed increased rates of comorbid psychiatric disorders....

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Effects of multisensory tactile-auditory integration teaching on systematic anatomy learning in visually impaired medical students

Fanzhen Kong 2026-10-08 1 min 50 words BRAILLE

Summary: CONCLUSION: The tactile-auditory integrated training system significantly improves objective summative knowledge acquisition and subjective learning experiences in visually impaired medical students. This approach provides a scalable, cost-constrained, and replicable pedagogical model for inclusive ...

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Physical and Environmental Accessibility of Dental Services for Persons with Disabilities in Saudi Primary Health Care: A Cross-Sectional Study

👤 Estie Kruger 📅 2026-10-16 📖 1 min 📝 66 words BRAILLE

Summary: Background/Objectives: Persons with disabilities (PWDs) may experience barriers when accessing oral health care, including those related to the physical and sensory environment of the dental-care setting. In this study we assessed the accessibility of primary health care centres (PHCs) providing den...

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Visible Light Driven Diselenide Chemistry for Simultaneous 3D Patterning and Mechanical Reinforcement

👤 Huaping Xu 📅 2026-10-16 📖 1 min 📝 63 words BRAILLE

Summary: Three-dimensional (3D) patterning on polymer surfaces is typically achieved by photolithography or related techniques, which often involve complex procedures, high-energy ultraviolet light, and risk of material damage. Here, we report a visible light-driven strategy that exploits the dual functional...

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An accessible pregnancy test for blind and low-vision users

Vanja Garaj 2026-10-16 1 min 63 words BRAILLE

Summary: CONCLUSIONS: The findings suggest that accessible pregnancy testing must prioritise not only non-visual result reading but also privacy, dignity, reproductive autonomy and independent health management. These results represent the preliminary evidence of acceptability rather than a conclusive test o...

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Genome-Informed Loop-Mediated Isothermal Amplification (LAMP) Assays for Detection and Differentiation of Ralstonia pseudosolanacearum Phylotypes I and III

Jarred Yasuhara-Bell 2026-10-27 1 min 76 words BRAILLE

Summary: The Ralstonia solanacearum species complex (RSSC) causes bacterial wilt in a broad range of economically important hosts. The RSSC has been reclassified into three species, of which R. solanacearum sequevars 1 and 2 in phylotype IIB are designated as Select Agents in the U.S. Many rapid tests for "R...

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Encoding tactile stimuli for Braille recognition with organoids

👤 Benjamin Ward-Cherrier 📅 2026-10-31 📖 1 min 📝 62 words BRAILLE

Summary: This study proposes a transferable encoding strategy that maps tactile sensor data to electrical stimulation patterns, enabling neural organoids to perform an open-loop artificial tactile Braille classification task. Human forebrain organoids cultured on a low-density microelectrode array (MEA) are ...

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More than Quaternary Ammonium Compounds: First Evidence of Tertiary Amine Compounds as Novel Contaminants Ubiquitously Present in Indoor Dust

👤 Runzeng Liu 📅 2026-11-10 📖 1 min 📝 65 words TDCS TACS TRNS

Summary: Nitrogen-containing organic compounds have attracted global concern due to their widespread production and potential environmental risks. While the environmental occurrence of quaternary ammonium compounds (QACs) has been widely studied, research on tertiary amine compounds (TACs), another important...

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Combining transcranial electrical and magnetic stimulation: a scoping review of neurophysiological and clinical studies

👤 Paul M Briley 📅 2026-11-10 📖 1 min 📝 56 words TDCS TACS TRNS

Summary: CONCLUSIONS: Studies indicate that tES could enhance or reverse expected effects of rTMS. The greatest increase in cortical excitability occurred following excitatory rTMS combined with gamma transcranial alternating current stimulation (tACS) or with cathodal tDCS, consistent with effects of metapl...

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Closed-loop transcranial electrical brain stimulation with fMRI: A proof-of-concept randomized trial of real-time fMRI-guided tACS optimization

👤 Hamed Ekhtiari 📅 2026-11-10 📖 1 min 📝 60 words TDCS TACS TRNS

Summary: Closed-loop transcranial alternating current stimulation (tACS) combined with real-time functional MRI (tACS-fMRI) enables adaptive optimization of noninvasive brain stimulation based on ongoing neural dynamics. We implemented one of the first randomized, double-arm, closed-loop tACS-fMRI protocols ...

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Effects of different electrical stimulation on depression: a systematic review and network meta-analysis

Shouping Zhao 2026-11-11 1 min 64 words TDCS TACS TRNS

Summary: CONCLUSIONS: Electrical stimulation is associated with improved clinical response rates and reduced depressive symptom severity in patients with depression, and is generally well tolerated. Among the modalities examined, taVNS demonstrated a relative advantage in improving clinical response, while t...

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Applications of non-invasive brain stimulation in the treatment of children with cerebral palsy: a narrative review

Lei Huang 2026-11-11 1 min 59 words TDCS TACS TRNS

Summary: Noninvasive brain stimulation has emerged as a promising adjunctive strategy for pediatric neurorehabilitation, particularly for children with cerebral palsy. This narrative review, informed by a structured literature search, summarizes current evidence on transcranial direct current stimulation and...

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Noninvasive brain stimulation for migraine: A bibliometric analysis of scientific research with thematic and keyword mapping

Zahra Hadi Alqallaf 2026-11-11 1 min 52 words TDCS TACS TRNS

Summary: CONCLUSIONS: This bibliometric analysis maps the evolution, collaboration patterns, thematic structure, and emerging research priorities of noninvasive brain stimulation research in migraine, highlights transcranial magnetic stimulation and transcranial direct current stimulation as established meth...

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A central hub model for remote delivery of multisite clinical trial interventions

Kanecia O Zimmerman 2026-11-11 1 min 57 words TDCS TACS TRNS

Summary: Digital health technologies enable remote delivery of clinical interventions, creating opportunities to redesign multisite trials. Traditional trials require participating sites to deliver interventions independently, potentially increasing implementation variability and operational complexity. Cent...

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Recovering Resting-State EEG During AM-tACS Using Dynamic Adaptive Stimulation Artifact Source Separation

👤 Dong Ming 📅 2026-11-11 📖 1 min 📝 58 words TDCS TACS TRNS

Summary: OBJECTIVE: Transcranial alternating current stimulation (tACS) can modulate brain oscillations in a frequency-specific manner, but strong stimulation artifacts severely contaminate concurrent electroencephalography (EEG), limiting direct assessment of ongoing neural activity. Conventional artifact s...

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Bimodal versus unimodal high-definition Transcranial random noise stimulation of auditory cortex for tinnitus: A double-blind, sham-controlled crossover trial

👤 Patrick Neff 📅 2026-11-11 📖 1 min 📝 21 words TDCS TACS TRNS

Summary: CONCLUSION: Repeated HD-tRNS produced no group-level benefit, alone or combined with acoustic stimulation, adding a well-controlled null to a mixed literature.

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Effects of Cerebellar and Primary Motor Cortex tDCS Montages on Vocal Performance in Singers: A Randomized Trial

👤 Kátia Monte-Silva 📅 2026-11-11 📖 1 min 📝 69 words TDCS TACS TRNS

Summary: Transcranial direct current stimulation (tDCS) is a non-invasive method to transiently adjust cortical excitability. Initial studies have reported changes in vocal parameters; however, it remains unclear whether tDCS can enhance the effects of structured vocal training. Therefore, this study aimed t...

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Brain-Behavior Coupling During Cognitive-Postural Dual-Tasking in Older Adults With Mild Cognitive Impairment: An fNIRS Study

Chuan Guo 2026-11-06 1 min 56 words FNIRS

Summary: CONCLUSIONS: Older adults with MCI exhibit altered brain-behavior coupling during cognitive-postural dual-task conditions involving sensory and postural constraints, characterized by altered relationships between prefrontal-sensorimotor cortical activation and behavioral performance. These findings ...

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Item-Specific Proportion Congruency Uncovers Different Forms of Attentional Control: An event-related fNIRS Study

Murat Perit Çakır 2026-11-07 1 min 70 words FNIRS

Summary: CONCLUSION: While behavioral results mostly support contingency learning, fNIRS data revealed ISPC-related activation in prefrontal channels with sensitivity to BA 46 (lateral, channel 1) and the BA 10/BA 11 overlap zone (medial, channels 10-12), with channel 11 emerging as a convergence point for b...

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Does the emotion regulation of self-directed humor engage the prefrontal cortex? An fNIRS study

Zhonglu Zhang 2026-11-07 1 min 68 words FNIRS

Summary: Previous research has revealed that self-directed humor (self-enhancing and self-defeating) can help regulate negative emotions and that the prefrontal cortex (PFC) plays a critical role in emotion regulation. However, it remains unknown whether emotion regulation of self-directed humor is associate...

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Effects of mini-maps with different reference frames on polarity-free 3D spatial navigation: a behavioural and fNIRS study

Xiaozhou Zhou 2026-11-07 1 min 62 words FNIRS

Summary: Manned spaceflight and deep-sea exploration extend human activities to extreme environments lacking gravitational and visual polarity cues, where spatial disorientation frequently occurs. Mini-map studies largely focus on terrestrial environments, while reference frame design for polarity-free 3D sp...

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Baduanjin improves motor function and cortical activation in children with developmental coordination disorder: a randomized controlled trial

👤 Wei Song 📅 2026-11-09 📖 1 min 📝 40 words FNIRS

Summary: CONCLUSIONS: Baduanjin training improved motor coordination and balance in children with DCD. The observed changes in activation within motor-related cortical regions provide preliminary support for a possible neurophysiological basis of these functional improvements, although the underlying mechani...

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Potential Applications of Brain-Computer Interface Technology in the Rehabilitation of Patients with Anxiety Disorders: A Narrative Review

Hu Lou 2026-11-10 1 min 58 words FNIRS

Summary: Anxiety disorders are among the most prevalent mental disorders worldwide and can cause persistent impairment in emotional regulation, cognition, and social functioning. Pharmacotherapy, psychotherapy, exercise, and non-invasive brain stimulation are widely used, yet their effectiveness may be const...

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Multimodal EEG-fNIRS complexity analysis can reveal the neurovascular coupling change with aging

Yicheng Zhu 2026-11-10 1 min 58 words FNIRS

Summary: CONCLUSION: Aging is associated with task-specific alterations in neural complexity, hemodynamic complexity, and the cross-modal temporal relationship between EEG and fNIRS signals. These findings suggest that age-related changes in NVC-related dynamics are better characterized as dynamic reorganiza...

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Task-state prefrontal cortex activation differences between executive dysfunction and perceptual dysfunction phenotypes of freezing of gait in Parkinson's disease during the medication ON state: a retrospective fNIRS study

Zhanhua Liang
2026-11-11 1 min 66 words FNIRS

Summary: CONCLUSION: The two proposed PD-FOG phenotypes showed distinct task-related PFC activation patterns in the medication ON state, and this task-specific double dissociation pattern was consistent with the predictions of the dual-model framework. Because fNIRS was recorded during seated cognitive tasks...

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Sensorimotor cortical responses to visuotactile synchrony during virtual foot mirror feedback: an fNIRS study

Noboru Chiba 2026-11-11 1 min 47 words FNIRS

Summary: CONCLUSIONS: The Synchronous condition was associated with changes in subjective attribution of the virtual foot without a detectable difference in medial sensorimotor HbO. These observations should be regarded as hypothesis-generating rather than as established evidence that subjective embodiment a...

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Cortical hemodynamic correlates of postural sway frequency redistribution during sensory-challenged quiet standing

Dianzhi Liu 2026-11-11 1 min 27 words FNIRS

Summary: BACKGROUND: Standing balance requires continual adjustment to the available sensory information. How changes in sway frequency composition relate to cortical hemodynamic responses during quiet standing remains unclear.

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MSTA-CNN an ultra-lightweight multi-scale temporal-spatial attention convolutional neural network for high-performance P300 spellers

👤 Mousa Shamsi
📅 2026-11-10 📖 1 min 📝 62 words

BRAIN COMPUTER INTERFACE

Summary: Efficient P300 detection in brain-computer interfaces (BCIs) remains a challenge due to low signal-to-noise ratio (SNR) and the high computational cost of existing deep learning models, which limits their deployment in real-time applications. To address these issues, we propose MSTA-CNN, an ultra-li...

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Detection of linear dependencies in EEG oscillatory sources via PARALIND and block-term decomposition: A comparative study

👤 Leonardo Jose Trejo
📅 2026-11-10 📖 1 min 📝 60 words

BRAIN COMPUTER INTERFACE

Summary: Narrowband oscillatory rhythms in electroencephalogram (EEG) signals are fundamental signatures of neural dynamics, crucial for brain-computer interfaces (BCIs) and monitoring neurological disorders. While traditionally detected and analyzed via Independent or Principal Component Analysis in the tim...

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Lower frequency ECoG features improve longitudinal stability of speech and hand-grasp detection in a patient with progressive ALS

👤 Dean J Krusienski 📅 2026-11-10 📖 1 min 📝 54 words BRAIN COMPUTER INTERFACE

Summary: Achieving long-term decoding stability is a critical challenge for chronic
intracranial brain-computer interfaces (BCIs). High-gamma (HG) activity from
intracranial recordings is the standard signal for BCI control, and lower-frequency features
can enhance HG decoding performance. This s...

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A collaborative network of bi-Mamba and CAFS for multi-task classification of unilateral lower limb

👤 Junhua Li 📅 2026-11-10 📖 1 min 📝 66 words BRAIN COMPUTER INTERFACE

Summary: Accurate decoding of motor-related electroencephalography signals remains challenging in brain-computer interface research. To date, motor imagery of the upper limbs has been extensively investigated, while research on the lower limbs remains limited. Meanwhile, multiple tasks involving the unilater...

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A theory of multilevel interactive equilibrium in neuroAI

Quanyan Zhu 2026-11-10 1 min 70 words BRAIN COMPUTER INTERFACE

Summary: We propose a class of game-theoretic framework for adaptive multi-agent intelligent systems. Unlike the classical game theory which assumes that strategies are primitive objects chosen by perfectly rational agents, the new framework provides a mathematical foundation for studying equilibrium in Neur...

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Interictal Electroencephalographic Characteristics of the Human Centromedian Thalamic Nucleus Recorded in Focal Epilepsy

Shuang Wang 2026-11-10 1 min 23 words BRAIN COMPUTER INTERFACE

Summary: CONCLUSIONS: CM SEEG demonstrates distinct, state-dependent electrophysiological dynamics across the sleep-wake cycle and preferential involvement in interictal epileptic activity compared with VL/VP nuclei.

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Case Report: Representational geometry of multi-digit coordination in the macaque grasping network for intracortical BCIs

Hansjörg Scherberger

2026-11-11 · 1 min · 65 words · BRAIN COMPUTER INTERFACE

Summary: CONCLUSION: In this animal, condition-dependent dimensionality was significantly higher in F5 than in AIP or M1, while alignment and reconstruction analyses showed descriptive area-specific patterns in how closely combined-finger activity resembled a simple combination of single-finger activity. For...

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Effects of brain-computer interface output modalities on motor recovery after stroke: a systematic review and network meta-analysis

👤 Lili Fang 📅 2026-11-11 📖 1 min 📝 54 words BRAIN COMPUTER INTERFACE

Summary: CONCLUSION: Brain-computer interface-triggered functional electrical stimulation showed a clearer signal of benefit vs. conventional rehabilitation than brain-computer interface-controlled robotic. However, the comparison between the two active modalities relies entirely on indirect evidence from a ...

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Building a safety assessment framework for brain-computer interfaces: A call for cell-free multi-omics monitoring

👤 Jilai Li 📅 2026-11-11 📖 1 min 📝 2 words BRAIN COMPUTER INTERFACE

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Spinal cord-computer interfaces for neurorehabilitation: Technical challenges and clinical perspectives

Minfei Wu 2026-11-11 1 min 35 words BRAIN COMPUTER INTERFACE

Summary: OBJECTIVE: This Perspective aims to clarify the clinical and engineering scope of spinal cord-computer interfaces (SCCIs) in neurorehabilitation, distinguishing them from adjacent brain-computer interface, neuromodulation, and assistive-technology approaches while identifying the key barriers to tra...

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Advancing neuropsychology training: Insights and advice from a qualitative analysis in 102 brain-behavior experts.

2026-07-30 1 min 266 words NEUROPSYCHOLOGY

Summary: Objective: Neuropsychologists possess knowledge that could benefit trainees as they navigate important career decisions within their unique training pathways. This wisdom is typically passed down in an individual mentorship model wherein supervisors and advisors relay advice directly. While this mod...

🔗 **Read full article:**
http://doi.org/10.1037/neu0001116

Sex-specific differences in amygdala resting-state functional connectivity with trait anxiety.

📅 2026-06-04 📖 1 min 📝 267 words NEUROPSYCHOLOGY

Summary: Objective: Anxiety disorders are the most common mental health conditions in the world, disproportionately affecting females and associated with significant morbidity. Neuroimaging can help us to understand the underpinnings of anxiety behaviors. The amygdala and its subnuclei have been strongly imp...

🔗 **Read full article:**
http://doi.org/10.1037/neu0001100

Effects of simulated hearing loss on cognitive task performance: A scoping review.

📅 2026-07-30 📖 1 min 📝 235 words NEUROPSYCHOLOGY

Summary: Objective: Hearing loss (HL) and cognitive decline often co-occur in older adults. Many cognitive tests are administered using spoken instructions or stimuli, potentially biasing test results. The current scoping review aimed to provide an overview of studies using simulated HL in cognitively health...

🔗 **Read full article:**
http://doi.org/10.1037/neu0001111

Normative cognitive test performance and typical error patterns among American Indian older adults: The Strong Heart Study.

📅 2026-06-15 📖 1 min 📝 245 words NEUROPSYCHOLOGY

Summary: Objective: The Strong Heart Study, a large cohort of American Indians from three U.S. regions, collected standardized neuropsychological assessments from participants aged 65–95 years at two visits (2010–2019), with prior documentation of lower overall score distributions and higher prevalence of co...

🔗 **Read full article:**
http://doi.org/10.1037/neu0001093

The unity and diversity of executive functions across adulthood in a diverse Brazilian sample with varying educational attainment.

📅 2026-04-16 📖 1 min 📝 261 words NEUROPSYCHOLOGY

Summary: Objective: Executive functions (EFs) are a set of cognitive abilities associated with many social and clinical outcomes that are believed to be sensitive to culture, age, and educational attainment. Yet, this was determined using EF theoretical models and tests developed for highly educated young ad...

🔗 **Read full article:**
http://doi.org/10.1037/neu0001085

Everyday executive functions in children with neurofibromatosis type 1: Data from nine institutions.

📅 2026-04-16 📖 1 min 📝 278 words NEUROPSYCHOLOGY

Summary: Objective: Executive function (EF) difficulties are prevalent in neurogenetic syndromes such as Neurofibromatosis Type 1 (NF1), the most common autosomal dominant condition associated with nervous system tumors. However, age-related differences in EF problems in children with NF1 remain poorly under...

🔗 **Read full article:**
http://doi.org/10.1037/neu0001087

Integrity in six key brain regions predicts numeric working memory performance in 30,000 middle-aged and older adults: A UK Biobank magnetic resonance imaging study.

📅 2026-06-18 📖 1 min 📝 262 words NEUROPSYCHOLOGY

Summary: Objective: This study investigated the neuroanatomical correlates of working memory using the UK Biobank Numeric Memory Test (NMT; a visual backward digit-span task) in a large sample of middle-aged and older adults. Volumes of brain regions previously implicated in digit-span tasks were hypothesize...

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http://doi.org/10.1037/neu0001096

Critical RCE Vulnerability in Apache HTTP Server

👤 soheilpro 📅 2026-11-12 📖 1 min 📝 13 words HACKER NEWS

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Lean Grimoire

👤 smilelamp 📅 2026-11-12 📖 1 min 📝 13 words HACKER NEWS

> **Summary:** Article URL: https://terrytao.wordpress.com/2026/11/11/lean-grimoire/ Comments URL: https://news.ycombinator.com/item?id=50462375 Points: 38 # Co...

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Darling Development Update – Q4 2026

👤 tech234a 📅 2026-11-12 📖 1 min 📝 13 words HACKER NEWS

> **Summary:** Article URL: https://blog.darlinghq.org/2026/11/08/development-update-q4-2026/ Comments URL: https://news.ycombinator.com/item?id=50463103 Poin...

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I'm Working on a FOSS Linux Pinbal Simulation Engine

👤 sysrpl 📅 2026-11-12 📖 1 min 📝 13 words HACKER NEWS

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Clip-to-trial: EEG-based decoding network with long short-term self-attention for DRDS recognition

📖 1 min 📝 16 words NEUROSCIENCE JOURNAL

Summary: Publication date: 9 January 2027 Source: Neuroscience, Volume 620 Author(s): Lili Shen, Yiming Wu, Zihao Yang, Xinyuan Di

🔗 **Read full article:**
https://www.sciencedirect.com/science/article/pii/S0306452226007074?dgcid=rss_sd_all

Intranasal insulin improves long-term cognitive deficits and protects hippocampal damage following neonatal hypoxia–ischemia in juvenile rats

📖 1 min 📝 41 words NEUROSCIENCE JOURNAL

Summary: Publication date: 9 January 2027 Source: Neuroscience, Volume 620 Author(s): Nilesh Dankhara, Jonathan W. Lee, Norma B. Ojeda, Michelle A. Tucci, Silu Lu, Elizabeth L. White, Lindsey E. Winborne, Sharon Xu, Carolyn K. Glendye, Lu-Tai Tien, Yi Pang, Abhay J. Bhatt, Lir-Wan Fan</...

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https://www.sciencedirect.com/science/article/pii/S0306452226007153?dgcid=rss_sd_all

Transcriptional and DNA Methylation Changes in the Hippocampus and Prefrontal Cortex after Fractionated Cranial Radiation in Mice

👤 Aksoy, T., Sumazin, P., Lu, Y., Najaf Panah, M. J., Zamorano, M., Lam, T., Zhou, W., Veselinovic, I., Duman, J. G., Gaber, M. W., Heijnen, C. J., Grosshans, D. R.

📅 2026-11-11 📖 1 min 📝 163 words BIORXIV NEUROSCIENCE

Summary: Fractionated whole-brain irradiation is essential for treating brain tumors, but many long-term survivors develop cognitive decline whose molecular basis is poorly understood. We asked which cell types and gene programs change late after irradiation, how the hippocampus and prefrontal cortex compare...

🔗 **Read full article:**
https://www.biorxiv.org/content/10.64898/2026.11.10.765716v1?rss=1

A common neural axis links image distinctiveness and memorability

👤 Utochkin, I., Chang, W., Vogel, E. K., Awh, E.

📅 2026-11-11 📖 1 min 📝 139 words

BIORXIV NEUROSCIENCE

Summary: Stimulus-specific memorability, the tendency for people to remember and forget the same stimuli, is a powerful predictor of memory performance that is independent of the study and retrieval factors that have dominated memory research. Critically, memorability can be defined in two ways: "Hit-ability...

🔗 **Read full article:**
https://www.biorxiv.org/content/10.64898/2026.11.10.765672v1?rss=1

Neuropathological assessment of cerebrovascular integrity in Chronic Traumatic Encephalopathy

👤 Martinac, A. E., DeLorenzo, A., Dik, D., Pasas, L., Bilston, L. E., Shepherd, C. E.

📅 2026-11-11

📖 1 min 📝 245 words

BIORXIV NEUROSCIENCE

Summary: Chronic traumatic encephalopathy neuropathological change (CTE-NC) is defined by perivascular deposits of phosphorylated tau, with lesions preferentially located at sulcal depths. We investigated whether CTE-NC is associated with structural changes to perivascular spaces (PVSs) or microvascular dege...

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https://www.biorxiv.org/content/10.64898/2026.11.09.765492v1?rss=1

Distinct input-output relationships between basal forebrain subnuclei underlie heterogeneous cholinergic modulation of the mouse cortex

👤 Pinto, L., Tan, J., Harter, A. M., Kolb, M., Lu, R.

📅 2026-11-11 📖 1 min 📝 150 words

BIORXIV NEUROSCIENCE

Summary: The basal forebrain (BF) supports many behavioral functions via its brain-wide projections, which can be recruited independently and task specifically. However, the anatomical underpinnings for this functional heterogeneity are poorly understood. Here, we compared two mouse BF subnuclei, the horizon...

🔗 **Read full article:**
https://www.biorxiv.org/content/10.64898/2026.11.10.765460v1?rss=1

Striatal dopamine availability peaks in adulthood and undergoes age-dependent modulation

White, S. R., Shin, J. H., Goldbach, H., Andujar, M., Swanson, E. S., Rimondini, R., Carr, A. L., Anderson, L. G., Cerveny, S. E., Harmon, P. Q., Tangen, A. C., Pelkey, K. A., Azadi, R., Cummins, A. C., McBain, C. J., Averbeck, B., Alvarez, V. A.

2026-11-11 1 min 71 words BIORXIV NEUROSCIENCE

Summary: Age is a risk factor for several psychiatric disorders. An adolescent peak in striatal dopamine availability is thought to contribute to the neuropsychiatric vulnerability. However, the lifespan trajectory of striatal dopamine release capacity is undefined in most species. Here, we measured lifespan...

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https://www.biorxiv.org/content/10.64898/2026.11.10.765699v1?rss=1

Advanced meditation stabilizes cortical dynamics through inhibitory gain control linked to absorptive depth

Panda, R., Kumar, G., Nunez, P., Subramaniam, B., Bhandare, K., Malipeddi, S., Major, G., Gosseries, O., Laureys, S., Tewarie, P.

2026-11-11 1 min 256 words BIORXIV NEUROSCIENCE

Summary: Advanced meditation can shift conscious experience from ordinary body- and world-centered awareness toward absorption, detachment and dissociation-like states, yet the neural mechanisms that stabilize these immersive states remain unclear. We studied 18 long-term meditators during eyes-open rest and...

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https://www.biorxiv.org/content/10.64898/2026.11.08.765124v1?rss=1

Learned associations constrain anticipatory visual dynamics and enhance sensory and motor readiness to facilitate perceptual decisions

Zhang, L., Yuan, M., Zhang, H., Morris Moscovitch, M., Alain, C.

2026-11-11 1 min

246 words

BIORXIV NEUROSCIENCE

Summary: Learned cue target associations can guide attention to facilitate perceptual decision making, yet how they shape the content and dynamics of neural activity, and whether this facilitation reflects anticipatory preparation, post-target sensory enhancement, or both, remains unclear. Here, we used magn...

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https://www.biorxiv.org/content/10.64898/2026.11.10.765786v1?rss=1

Tri-Modal Single-Cell Atlas of Developing Human Brain

Song, L., Wang, Z., Zhu, B., DeKuyper, F. A., Wang, C., Liao, Z., Girdhar, K., Lee, D., Tsankova, N. M., Kaji, D. A., Fullard, J. F., Dong, P., Roussos, P., Ma, S.

2026-11-11 1 min

139 words

BIORXIV NEUROSCIENCE

Summary: Epigenetic modifications, such as DNA methylation, evolve throughout development and aging, providing remarkable regulatory diversity and contributing to disease etiology, including neuropsychiatric disorders. However, how these regulatory layers interact across human brain development remains uncle...

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Hyperglycemia-responsive circRNA drives diabetic neuropathic pain by competing for PUM2 binding and promoting proNT3/CCL2 in DRG neurons

👤 Li, X., Huang, C., Govindarajalu, G., Wang, B., Zhang, H.-H., Feng, X., Sharma, D., Ma, R., Zhang, L., Zheng, B., Xiao, J., Ferreira, M. d. A., Berkman, T., Cabrera, J., Li, X., Jiang, B.-C., Zhang, X.-S., Davidson, S., TAO, Y.-X.

📅 2026-11-11 📖 1 min 📝 199 words BIORXIV NEUROSCIENCE

Summary: Diabetic neuropathic pain (DNP), one of the most common and debilitating complications of diabetes mellitus (DM), remains poorly managed by conventional analgesics, highlighting the urgent need to explore its etiology-associated mechanisms. Here, we identified a novel conserved circular RNA, cRps6kb...

🔗 **Read full article:**
https://www.biorxiv.org/content/10.64898/2026.11.10.765753v1?rss=1

A novel behavioral paradigm for characterizing anticipatory postural adjustments in mice

👤 Morina, E., Claudi, F., Murray, A. J.
📅 2026-11-11 📖 1 min 📝 274 words

BIORXIV NEUROSCIENCE

Summary: Voluntary movements perturb the body's equilibrium, and the nervous system limits this disturbance by generating anticipatory postural adjustments (APAs) that redistribute body weight before and during movement. APAs have been characterized extensively in humans and larger animal models, but the abs...

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https://www.biorxiv.org/content/10.64898/2026.11.10.765800v1?rss=1

RNA interference as a precision therapeutic platform for Parkinson’s disease: from molecular mechanisms to translational challenges

📅 2026-11-12 📖 1 min 📝 0 words

NATURE NEUROSCIENCE SUBJECTS

🔗 **Read full article:**
https://www.nature.com/articles/s44573-026-00028-8

Explainable AI-driven neural activity analysis in Parkinsonian rats under electrical stimulation

📅 2026-11-12 📖 1 min 📝 0 words NATURE NEUROSCIENCE SUBJECTS

🔗 **Read full article:**
https://www.nature.com/articles/s41598-026-79944-7

Author Correction: Social rank modulates methamphetamine-seeking in dominant and subordinate male rodents via distinct dopaminergic pathways

👤 Yingjie Zhu 📅 2026-11-03 📖 1 min 📝 24 words NATURE NEUROSCIENCE

Summary: Nature Neuroscience, Published online: 03 November 2026; doi:10.1038/s41593-026-02535-2 Author Correction: Social rank modulates methamphetamine-seeking in dominant and subordinate male rodents via distinct dopaminergic pathways

🔗 **Read full article:**
https://www.nature.com/articles/s41593-026-02535-2

Author Correction: A genome-wide analysis of the shared genetic risk architecture of complex neurological and psychiatric disorders

Ole A. Andreassen
2026-11-06 1 min 25 words NATURE NEUROSCIENCE

Summary: Nature Neuroscience, Published online: 06 November 2026; doi:10.1038/s41593-026-02536-1 Author Correction: A genome-wide analysis of the shared genetic risk architecture of complex neurological and psychiatric disorders

🔗 **Read full article:**
https://www.nature.com/articles/s41593-026-02536-1

Spatial transcriptomics reveals organizational properties of the mouse spinal cord and alterations in neuropathic pain

👤 Qimeng WangYunfei HuYuling ZhuGautham ChelliahJing PengEmmanuella Osei-AsanteMelissa SobankoBrianna SanchezDavid D. GintyXin Maizie ZhouShan Meltzerahttps://ror.org/02vm5rt34Department of Pharmacology, Vanderbilt University, Nashville, TN 37240bhttps://ror.org/02vm5rt34Department of Computer Science, Vanderbilt University, Nashville, TN 37235chttps://ror.org/02vm5rt34Department of Biomedical Engineering, Vanderbilt University, Nashville, TN 37235dDepartment of Neurobiology, HHMI, Harvard Medical School, Boston, MA 02115

📅 2026-11-10 📖 1 min 📝 48 words PNAS NEUROSCIENCE

Summary: Proceedings of the National Academy of Sciences, Volume 123, Issue 46, November 2026.
SignificanceThe spinal cord receives and integrates somatosensory inputs and executes motor and autonomic outputs, yet the spatial organization of its diverse cell types in the spinal cord remains incompletel...

🔗 **Read full article:**
https://www.pnas.org/doi/abs/10.1073/pnas.2623177123?af=R

Meditation as a bioactive intervention: evidence of translational potential via Connectivity Mapping and TMS neurophysiology

Mitali Mukerji 2026-11-09 1 min 295 words FRONTIERS HUMAN NEUROSCIENCE

Summary: BackgroundMeditation practices are often used as non-pharmacological adjunct therapy for managing stress and wellbeing benefits. Here, we investigate whether molecular signatures via meditation practices have potential overlap with those of known drug targets that have established neurophysiological...

🔗 **Read full article:**
https://www.frontiersin.org/articles/10.3389/fnhum.2026.1904068

Hearing loss as an inference problem: sensory precision, conversational repair, and partner adaptation

Sebastian Schnettler 2026-11-09 1 min 289 words FRONTIERS HUMAN NEUROSCIENCE

Summary: Conversational understanding is often approached as a problem of successful signal transmission. This paper develops a complementary account in which conversation is treated as a reciprocal process of inference and action: interlocutors must infer the latent causes of one another’s signals, estimate...

🔗 **Read full article:**
https://www.frontiersin.org/articles/10.3389/fnhum.2026.1963610

Noise resilience in brain decoding driven by spiking neural networks for responsive robot control

Alessandro De Luca
2026-11-09 1 min 291 words
FRONTIERS NEUROROBOTICS

Summary: Brain computer interface (BCI) is a promising technology, expanding on many fields ranging from medical to commercial applications. The diffusion of this technique is compromised by the complexity of the treated data and the unavoidable presence of noise in those systems. This study contributes to t...

🔗 **Read full article:**
https://www.frontiersin.org/articles/10.3389/fnbot.2026.1931325

Transformer networks are robust learners of source localization for EEG measurements

David Hart, Simon Polishchuk and Moritz Dannhauer
2026-11-06 1 min 356 words
JOURNAL NEURAL ENGINEERING

Summary: Objective. EEG source localization is an ill-posed inverse problem complicated by measurement noise, electrode dropout from sensor miscalibration, and variability in the number of active sources. Classical methods such as eLORETA and LCMV beamforming impose regularization assumptions that degrade un...

🔗 **Read full article:**
https://iopscience.iop.org/article/10.1088/1741-2552/aeb82c

ACF-Net: design of an adaptive cross-modal fusion network for cross-subject wearable EEG-EMG gesture recognition

TongLiang Yang
2026-11-06 1 min 292 words JOURNAL NEURAL ENGINEERING

Summary: Objective. Cross-subject gesture recognition in wearable brain–computer interfaces is constrained by four structural deficiencies: symmetric cross-modal attention discarding cortical precedence over muscular execution, priors confined to post hoc analysis, invariance anchored on unimodal features, a...

🔗 **Read full article:**
https://iopscience.iop.org/article/10.1088/1741-2552/aeb723

The calibration-to-accuracy mapping in P300 spellers does not transport across cohorts and was not restored by tested alignment or local recalibration

Alon Gorenshtein, Yosef Adiniaev, Tom Liba, Yiftach Barash, Eyal Klang and Oved Daniel

2026-11-06 1 min 297 words JOURNAL NEURAL ENGINEERING

Summary: Objective. A calibration-derived score has been related to online P300-speller accuracy in the same session, always within the cohort measured. We evaluated whether a mapping from that score to expected accuracy transports to withheld cohorts. Approach. Retrospective secondary analysis of BigP3BCI 1...

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Human cortical high-gamma power scales with movement rate in healthy participants and stroke survivors

👤 Fanny Quandt 📅 2025-01-09 📖 1 min 📝 65 words OOSTENVELD ROBERT

Summary: Motor cortical high-gamma oscillations (60-90 Hz) occur at movement onset and are spatially focused over the contralateral primary motor cortex. Although high-gamma oscillations are widely recognized for their significance in human motor control, their precise function on a cortical level remains el...

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NIRS-BIDS: Brain Imaging Data Structure Extended to Near-Infrared Spectroscopy

👤 Luca Pollonini 📅 2025-01-27 📖 1 min 📝 70 words OOSTENVELD ROBERT

Summary: Functional near-infrared spectroscopy (fNIRS) is an increasingly popular neuroimaging technique that measures cortical hemodynamic activity in a non-invasive and portable fashion. Although the fNIRS community has been successful in disseminating open-source processing tools and a standard file forma...

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Pseudonymisation of neuroimages and data protection: Increasing access to data while retaining scientific utility

👤 Lyuba Zehl 📅 2025-06-26 📖 1 min 📝 67 words OOSTENVELD ROBERT

Summary: For a number of years, facial features removal techniques such as 'defacing', 'skull stripping' and 'face masking/blurring', were considered adequate privacy preserving tools to openly share brain images. Scientifically, these measures were already a compromise between data protection requirements a...

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Cycling on the Freeway: The perilous state of open-source neuroscience software

👤 Tim M Tierney 📅 2025-08-13 📖 1 min 📝 74 words OOSTENVELD ROBERT

Summary: Most scientists need software to perform their research (Barker et al., 2020;Carver et al., 2022;Hettrick, 2014;Hettrick et al., 2014;Switters & Osimo, 2019), and neuroscientists are no exception. Whether we work with reaction times, electrophysiological signals, or magnetic resonance imaging data, ...

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Optimal configuration of on-scalp OPMs with fixed channel counts

👤 Robert Oostenveld 📅 2025-08-13 📖 1 min 📝 69 words OOSTENVELD ROBERT

Summary: Recent technological developments have brought optically pumped magnetometers (OPMs) within reach of the larger neuroscientific community. The current state-of-the-art consists of whole-head systems that measure the magnetic field at >100 locations. OPM sensors can be constructed to measure the fiel...

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ARTEM-IS for ERP: Agreed Reporting Template for EEG Methodology-International Standard for Event-Related Potential Experiments

👤 Suzy J Styles 📅 2025-12-08 📖 1 min 📝 73 words OOSTENVELD ROBERT

Summary: The choices we make during the recording, preprocessing and analysis of event-related potentials (ERP) data can affect study outcomes. As such, it is critical that they are transparently reported to allow for reproducibility. Yet, systematic reviews of reporting practices in the field have shown tha...

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Eye-Tracking-BIDS: the Brain Imaging Data Structure extended to gaze position and pupil data

👤 Julia-Katharina Pfarr 📅 2026-02-12 📖 1 min 📝 68 words OOSTENVELD ROBERT

Summary: The Brain Imaging Data Structure (BIDS) is a widely adopted, community-driven standard to organize neuroimaging data and metadata. Although numerous extensions have been developed to incrementally extend coverage to new modalities and data types, an unambiguous, granular specification for eye-tracki...

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Facilitating unusual bodily experiences and out-of-body experiences across wakefulness and sleep: A high-density EEG and neurophenomenology study

👤 Robert Oostenveld 📅 2026-02-12 📖 1 min 📝 67 words OOSTENVELD ROBERT

Summary: Unusual bodily experiences (UBEs) are illusory bodily perceptions that are not coherent with typical wakeful experiences, including flying or floating sensations, body distortions, and out-of-body experiences. This study examines UBEs in a controlled sleep laboratory using meditation and light stimu...

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The Impact of Brain Tumors and Craniotomy Lesions on Scalp EEG

Vitória Piai 2026-02-17 1 min 65 words OOSTENVELD ROBERT

Summary: Electroencephalography (EEG) is widely used in both research and clinical settings, yet its accuracy can be significantly impacted by subject-specific anatomical anomalies such as brain lesions and skull defects. This study investigates the effects of glioma-related brain lesions and craniotomy-indu...

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EMG-BIDS: an extension to the Brain Imaging Data Structure for electromyography

Arnaud Delorme 2026-08-01 1 min 64 words OOSTENVELD ROBERT

Summary: Electromyography (EMG) is fundamental to clinical assessment, rehabilitation, neuromuscular research, and human-machine interfaces. Despite decades of use, no widely adopted standard exists for organizing and sharing EMG data, limiting reusability and large-scale data aggregation. We present EMG-BID...

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Segmentation, Reconstruction, and Grasp-Pose Estimation Using an RGB-D Camera for Robotic Apple Harvesting

Jianqiang Lu 2026-11-10 1 min 58 words LOW VISION

Summary: Accurate segmentation, three-dimensional reconstruction, and grasp-pose estimation of fruits and surrounding structures are essential for robotic harvesting in complex orchard environments. However, occlusions caused by branches, support wires, and irregular canopy structures remain major challenges...

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Intelligent rendering optimization for animated scenes using reinforcement learning and computer vision

Li Jia 2026-11-10 1 min 63 words LOW VISION

Summary: Real-time rendering of complex animated scenes is computationally intensive and often struggles to balance visual fidelity and performance in dynamic or densely populated environments, as conventional pipelines lack adaptive resource optimization, leading to reduced frame rates and degraded visual q...

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Clinical efficacy of unilateral biportal endoscopic spinal decompression for lumbar intraspinal pneumatosis

Huiqiu Jian 2026-11-10 1 min 48 words LOW VISION

Summary: CONCLUSION: UBE for the treatment of lumbar intraspinal pneumatosis has the advantages of minimal trauma, clear field of vision, flexible operation, and rapid postoperative recovery, representing a potentially safe and effective option. Large-scale multicenter prospective trials are still needed to ...

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Photobleaching-Driven Ratiometric Fluorescent Switching of ZGO:Mn Nanorods/RhB for Iodine Vapor Monitoring with Integrated Gas-Phase Separation

Kui Zhang 2026-11-11 1 min 56 words LOW VISION

Summary: Conventional fluorescent sensors generally rely on complex, time-consuming structural engineering of recognition moieties. By contrast, leveraging photobleaching for eco-friendly, reagent-free signal transduction has long been impeded by irreversible fluorophore degradation. Additionally, direct liq...

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Ageist attitudes as a moderator of objective and subjective vision: A longitudinal study

Yoav S Bergman 2026-11-11 1 min 41 words LOW VISION

Summary: OBJECTIVES: This longitudinal study examined the relationship between objective visual acuity and subjective vision over a six-month period. Specifically, it investigated the moderating role of internalized ageist attitudes in shaping how objective vision relates to perceived sensory vision difficul...

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Registry-recorded stroke and transient ischaemic attack diagnoses in Saudi Arabia, 2016-2025: a national descriptive analysis

Mohammed AlAbdulali 2026-11-11 1 min 58 words LOW VISION

Summary: CONCLUSION: Saudi Arabia's recorded stroke burden is concentrated among older males. The findings describe patterns captured by the PHO and should not be interpreted as prevalence, incidence, or causal associations, which indicate the need for better prevention, management of chronic diseases and a ...

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Design and parameter optimization of a bilateral staggered wheel-spoon precision seed-metering device for millet plot breeding

👤 Ankang Mu 📅 2026-11-11 📖 1 min 📝 63 words LOW VISION

Summary: Precision seeding of micro-sized millet seeds in plot breeding faces critical challenges, including high miss-seeding rates and severe seed cross-contamination during variety changeovers. This study developed a novel bilateral staggered wheel-spoon precision seed-metering device to achieve precise h...

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Representation geometry and task performance in vision-language modeling for computed tomography enterography

👤 Ryan Stidham 📅 2026-11-11 📖 1 min 📝 36 words LOW VISION

Summary: CONCLUSIONS: Slice sampling and tissue-contrast-focused encoding affect performance in CT enterography vision-language systems, and RAG improved report-label agreement over the fine-tuned baseline. These results provide initial baselines and practical guidance for multimodal learning in volumetric C...

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Incidence and Management of Ocular Hypertension Following the 0.19 mg Fluocinolone Acetonide Implant for Diabetic Macular Edema

👤 Wills Eye Retina Service Study Group, The Wills Eye Retina Service Study Group additionally includes

📅 2026-11-11 📖 1 min 📝 36 words LOW VISION

Summary: CONCLUSION: FAi was associated with low rates of OHT, with most cases managed with observation or topical therapy, with surgery rarely required. This supports the safety of FAi and highlight the importance of extended IOP monitoring.

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Passive imaging with quantum advantage

👤 A I Lvovsky

📅 2026-11-11 📖 1 min 📝 67 words LOW VISION

Summary: Far-field optical imaging inevitably involves low-pass spatial filtering, limiting the resolution. Moreover, conventional imaging suppresses high spatial frequency components close to the cutoff, making them invisible under noise-particularly the shot noise arising from discrete and random nature of...

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A single session of high-definition transcranial direct current stimulation does not modulate effects of knee two-point discrimination training or sensorimotor function in healthy adults: Double-blind randomised controlled trial

👤 Paul W Hodges
📅 2026-07-30 📖 1 min 📝 53 words

TACTILE ACUITY

Summary: CONCLUSION: A single 20-minute session of 1 mA HD-tDCS, with or without brief TPD training, does not acutely modify knee somatosensory, sensorimotor, or functional performance in healthy adults. Existing evidence at the hand may not translate to the lower limb. Future work should investigate higher-...

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Variation in glabrous skin physiology across anatomical locations on the human hand

👤 Martina C Meinke 📅 2026-08-13 📖 1 min 📝 76 words TACTILE ACUITY

Summary: Skin physiology varies greatly between anatomical locations, with impact on tactile sensitivity. Research on the variations in morphology and physiology between different anatomical locations of glabrous skin is limited so far. This study investigates the morphology and physiology of glabrous skin o...

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Beyond Hamstring Shortness: Continuous Flexibility Is Associated With Balance and Stability but Not Sensory Measures-A Cross-Sectional Study

👤 Senanur Akıncı 📅 2026-08-14 📖 1 min 📝 66 words TACTILE ACUITY

Summary: CONCLUSIONS: Greater hamstring flexibility was associated with better dynamic balance performance and showed a weaker association with postural stability, whereas no robust associations were observed with sensory measures of postural control. These findings suggest that hamstring flexibility may be ...

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Resting-state EEG microstate dynamics reflect individual differences in tactile angle discriminability

Liu Yang 2026-08-19 1 min 63 words TACTILE ACUITY

Summary: Tactile spatial acuity, the ability to discriminate fine spatial details, is a fundamental sensory capacity modulated by both peripheral and central mechanisms. Here, we investigated whether resting-state EEG microstate dynamics are associated with individual differences in tactile spatial acuity. F...

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A grating orientation task with controlled dome elevation speed and contact force: stimulus conditions and day-to-day reliability

Hideaki Onishi 2026-09-25 1 min 66 words TACTILE ACUITY

Summary: The grating orientation discrimination task (GOT) is typically used to assess tactile spatial acuity. However, performance is influenced by stimulus speed, contact force between the stimulator and the skin, and the method of discrimination. This study aimed to explore suitable stimulus conditions fo...

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Epidermal microstructure and tactile sensitivity in the glabrous skin of hands and feet

Hannes P Saal 2026-09-25 1 min 69 words TACTILE ACUITY

Summary: Glabrous skin on the human palms and soles is specialized for mechanical interaction and is characterized by papillary ridges, an absence of hair follicles, and a thick stratum corneum. Although gross morphological differences, such as the increased thickness of the plantar stratum corneum, are well...

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Neurotrauma During Military Training: Results of the INVestigating traIning assoCiated blasT pAthology (INVICTA) Study

Walter S Carr 2026-10-01 1 min 67 words TACTILE ACUITY

Summary: Concern is growing that subconcussive blast exposure during military exercises such as heavy weapons training (HWT) may adversely affect brain function. HWT involves firing shoulder-mounted weapons that can expose the gunner, assistant gunner, and instructors (range safety officers [RSOs]) to blast ...

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Altered body representation in chronic non-specific low back pain: a testable framework linking body-schema-related disturbance, corticomotor alterations, and altered pain processing

👤 Zhifei Xie
📅 2026-10-24 📖 1 min 📝 66 words

TACTILE ACUITY

Summary: Low back pain is a leading contributor to disability worldwide; in chronic non-specific low back pain (CNSLBP), the processes sustaining pain and disability in the absence of identifiable structural pathology remain incompletely understood. Three partially overlapping research lines have documented ...

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Hyperexcitable primary somatosensory cortex in the MitoPark mouse model of Parkinson's disease

Alan J Emanuel 2026-10-25 1 min 67 words TACTILE ACUITY

Summary: The majority of Parkinson's disease patients suffer from decreased tactile acuity or neuropathic pain, both of which contribute to low quality of life and likely exacerbate motor dysfunction. How Parkinson's disease pathology relates to altered somatosensation remains unclear. One potential contribu...

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The Relationship Between Occlusal Tactile Acuity, Bite Force and Tooth Wear

Birgitta Häggman-Henrikson 2026-11-03 1 min 52 words TACTILE ACUITY

Summary: CONCLUSIONS: Bite force may be a clinically relevant factor associated with accumulated tooth wear, whereas OTA was not independently associated with tooth wear. Longitudinal studies are needed to determine whether bite force can predict tooth wear progression and whether more comprehensive assessme...

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Impact of Braille Booklet, Audio-Tactile Performance and Caregiver-supervised Training on Oral Hygiene Status, Knowledge, Attitude and Practices among Visually Challenged Children- A Randomized Clinical Trial

👤 Priyanga Chandrasekaran
📅 2026-09-22 📖 1 min 📝 52 words

BRAILLE

Summary: CONCLUSION: Both caregiver-supervised training and audio tactile performance using specialized model method of oral health education proved to be more effective than the braille booklet group on plaque control and improvement in the oral hygiene status, oral health knowledge, attitude and practice i...

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Design and functional verification of an automated Perkins Brailler accessory with audio feedback

👤 Ali Alhaj 📅 2026-09-23 📖 1 min 📝 67 words BRAILLE

Summary: CONCLUSIONS: The study did not include learners with visual impairment, Braille teachers, rehabilitation specialists, or other end users, and it did not collect quantitative measures of accuracy across the full character set, writing speed, embossing quality, latency, durability, safety, usability, ...

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Uncertainty-Aware structured decoding for Chinese Braille document transcription

👤 Haoran Sun 📅 2026-09-25 📖 1 min 📝 64 words BRAILLE

Summary: Automatic transcription of braille documents remains challenging in document digitization settings, particularly under real imaging degradations such as uneven embossing, illumination variation, paper texture, skew, blur, and ghosting, which make braille dot presence uncertain and often cause error ...

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Clinical Characteristics and Epidemiology of Isolated Dystonia: Results from a US Claims Analysis

James E Thornton 2026-10-08 1 min 54 words BRAILLE

Summary: CONCLUSIONS: This study defines an isolated dystonia cohort in the US based on real-world evidence and systematic exclusion of secondary causes of dystonia. Estimated prevalence is within range of prior epidemiological studies, and the cohort showed increased rates of comorbid psychiatric disorders....

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Effects of multisensory tactile-auditory integration teaching on systematic anatomy learning in visually impaired medical students

Fanzhen Kong 2026-10-08 1 min 50 words BRAILLE

Summary: CONCLUSION: The tactile-auditory integrated training system significantly improves objective summative knowledge acquisition and subjective learning experiences in visually impaired medical students. This approach provides a scalable, cost-constrained, and replicable pedagogical model for inclusive ...

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Physical and Environmental Accessibility of Dental Services for Persons with Disabilities in Saudi Primary Health Care: A Cross-Sectional Study

👤 Estie Kruger 📅 2026-10-16 📖 1 min 📝 66 words BRAILLE

Summary: Background/Objectives: Persons with disabilities (PWDs) may experience barriers when accessing oral health care, including those related to the physical and sensory environment of the dental-care setting. In this study we assessed the accessibility of primary health care centres (PHCs) providing den...

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Visible Light Driven Diselenide Chemistry for Simultaneous 3D Patterning and Mechanical Reinforcement

👤 Huaping Xu 📅 2026-10-16 📖 1 min 📝 63 words BRAILLE

Summary: Three-dimensional (3D) patterning on polymer surfaces is typically achieved by photolithography or related techniques, which often involve complex procedures, high-energy ultraviolet light, and risk of material damage. Here, we report a visible light-driven strategy that exploits the dual functional...

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An accessible pregnancy test for blind and low-vision users

Vanja Garaj 2026-10-16 1 min 63 words BRAILLE

Summary: CONCLUSIONS: The findings suggest that accessible pregnancy testing must prioritise not only non-visual result reading but also privacy, dignity, reproductive autonomy and independent health management. These results represent the preliminary evidence of acceptability rather than a conclusive test o...

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Genome-Informed Loop-Mediated Isothermal Amplification (LAMP) Assays for Detection and Differentiation of Ralstonia pseudosolanacearum Phylotypes I and III

Jarred Yasuhara-Bell 2026-10-27 1 min 76 words BRAILLE

Summary: The Ralstonia solanacearum species complex (RSSC) causes bacterial wilt in a broad range of economically important hosts. The RSSC has been reclassified into three species, of which R. solanacearum sequevars 1 and 2 in phylotype IIB are designated as Select Agents in the U.S. Many rapid tests for "R...

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Encoding tactile stimuli for Braille recognition with organoids

👤 Benjamin Ward-Cherrier 📅 2026-10-31 📖 1 min 📝 62 words BRAILLE

Summary: This study proposes a transferable encoding strategy that maps tactile sensor data to electrical stimulation patterns, enabling neural organoids to perform an open-loop artificial tactile Braille classification task. Human forebrain organoids cultured on a low-density microelectrode array (MEA) are ...

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Phase-Dependent Effects of Cerebellar Transcranial Alternating Current Stimulation on Lateral Vestibulospinal Tract Excitability

👤 Tsubasa Mitsutake 📅 2026-11-09 📖 1 min 📝 36 words TDCS TACS TRNS

Summary: CONCLUSIONS: LVST excitability differed according to the phase of the ongoing cerebellar-targeted tACS waveform. These findings provide electrophysiologic evidence of an online phase-dependent effect on vestibulospinal excitability and a physiologic basis for future phase-specific neuromodulation st...

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Effects of Transcranial Direct Current Stimulation on Core Symptoms and Sleep Disturbances in Children With Autism Spectrum Disorder: A Meta-Analysis of Randomized Controlled Trials

👤 Meimei Yang
📅 2026-11-09 📖 1 min 📝 44 words

TDCS TACS TRNS

Summary: CONCLUSION: tDCS, most commonly delivered at 1 mA over the DLPFC, was associated with improvements in some ASD-related behavioral outcomes; preliminary evidence also suggested a potential benefit for sleep disturbances, although this finding was based on only two low-certainty studies and warrants c...

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No effects of single-session concurrent tDCS on explicit motor sequence learning in healthy adults with conventional 2 mA and tangential 6 mA currents

👤 Lucas C Parra
📅 2026-11-09 📖 1 min 📝 45 words

TDCS TACS TRNS

Summary: CONCLUSION: These findings, together with a survey of the existing literature, suggest that the effects of single-session concurrent tDCS on explicit motor sequence learning are not robust across studies. Our results underscore the necessity of rigorous methodological controls to prevent false disco...

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More than Quaternary Ammonium Compounds: First Evidence of Tertiary Amine Compounds as Novel Contaminants Ubiquitously Present in Indoor Dust

👤 Runzeng Liu 📅 2026-11-10 📖 1 min 📝 65 words TDCS TACS TRNS

Summary: Nitrogen-containing organic compounds have attracted global concern due to their widespread production and potential environmental risks. While the environmental occurrence of quaternary ammonium compounds (QACs) has been widely studied, research on tertiary amine compounds (TACs), another important...

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Combining transcranial electrical and magnetic stimulation: a scoping review of neurophysiological and clinical studies

👤 Paul M Briley 📅 2026-11-10 📖 1 min 📝 56 words TDCS TACS TRNS

Summary: CONCLUSIONS: Studies indicate that tES could enhance or reverse expected effects of rTMS. The greatest increase in cortical excitability occurred following excitatory rTMS combined with gamma transcranial alternating current stimulation (tACS) or with cathodal tDCS, consistent with effects of metapl...

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Closed-loop transcranial electrical brain stimulation with fMRI: A proof-of-concept randomized trial of real-time fMRI-guided tACS optimization

Hamed Ekhtiari 2026-11-10 1 min 60 words TDCS TACS TRNS

Summary: Closed-loop transcranial alternating current stimulation (tACS) combined with real-time functional MRI (tACS-fMRI) enables adaptive optimization of noninvasive brain stimulation based on ongoing neural dynamics. We implemented one of the first randomized, double-arm, closed-loop tACS-fMRI protocols ...

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Effects of different electrical stimulation on depression: a systematic review and network meta-analysis

Shouping Zhao 2026-11-11 1 min 64 words TDCS TACS TRNS

Summary: CONCLUSIONS: Electrical stimulation is associated with improved clinical response rates and reduced depressive symptom severity in patients with depression, and is generally well tolerated. Among the modalities examined, taVNS demonstrated a relative advantage in improving clinical response, while t...

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Applications of non-invasive brain stimulation in the treatment of children with cerebral palsy: a narrative review

👤 Lei Huang 📅 2026-11-11 📖 1 min 📝 59 words TDCS TACS TRNS

Summary: Noninvasive brain stimulation has emerged as a promising adjunctive strategy for pediatric neurorehabilitation, particularly for children with cerebral palsy. This narrative review, informed by a structured literature search, summarizes current evidence on transcranial direct current stimulation and...

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Noninvasive brain stimulation for migraine: A bibliometric analysis of scientific research with thematic and keyword mapping

👤 Zahra Hadi Alqallaf 📅 2026-11-11 📖 1 min 📝 52 words TDCS TACS TRNS

Summary: CONCLUSIONS: This bibliometric analysis maps the evolution, collaboration patterns, thematic structure, and emerging research priorities of noninvasive brain stimulation research in migraine, highlights transcranial magnetic stimulation and transcranial direct current stimulation as established meth...

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A central hub model for remote delivery of multisite clinical trial interventions

👤 Kanecia O Zimmerman 📅 2026-11-11 📖 1 min 📝 57 words TDCS TACS TRNS

Summary: Digital health technologies enable remote delivery of clinical interventions, creating opportunities to redesign multisite trials. Traditional trials require participating sites to deliver interventions independently, potentially increasing implementation variability and operational complexity. Cent...

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Quality-aware mask-propagation graph attention for fNIRS-based bipolar disorder classification without channel interpolation

👤 Yajing Meng 📅 2026-11-05 📖 1 min 📝 58 words FNIRS

Summary: Bipolar disorder (BD) remains difficult to characterise objectively, motivating a neural marker to complement clinical assessment. Functional near-infrared spectroscopy (fNIRS) offers portable cortical haemodynamic measurement, but channel availability can vary across recordings because of bad chann...

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Brain-Behavior Coupling During Cognitive-Postural Dual-Tasking in Older Adults With Mild Cognitive Impairment: An fNIRS Study

👤 Chuan Guo 📅 2026-11-06 📖 1 min 📝 56 words FNIRS

Summary: CONCLUSIONS: Older adults with MCI exhibit altered brain-behavior coupling during cognitive-postural dual-task conditions involving sensory and postural constraints, characterized by altered relationships between prefrontal-sensorimotor cortical activation and behavioral performance. These findings ...

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Item-Specific Proportion Congruency Uncovers Different Forms of Attentional Control: An event-related fNIRS Study

👤 Murat Perit Çakır 📅 2026-11-07 📖 1 min 📝 70 words FNIRS

Summary: CONCLUSION: While behavioral results mostly support contingency learning, fNIRS data revealed ISPC-related activation in prefrontal channels with sensitivity to BA 46 (lateral, channel 1) and the BA 10/BA 11 overlap zone (medial, channels 10-12), with channel 11 emerging as a convergence point for b...

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Does the emotion regulation of self-directed humor engage the prefrontal cortex? An fNIRS study

Zhonglu Zhang 2026-11-07 1 min 68 words FNIRS

Summary: Previous research has revealed that self-directed humor (self-enhancing and self-defeating) can help regulate negative emotions and that the prefrontal cortex (PFC) plays a critical role in emotion regulation. However, it remains unknown whether emotion regulation of self-directed humor is associate...

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Effects of mini-maps with different reference frames on polarity-free 3D spatial navigation: a behavioural and fNIRS study

Xiaozhou Zhou 2026-11-07 1 min 62 words FNIRS

Summary: Manned spaceflight and deep-sea exploration extend human activities to extreme environments lacking gravitational and visual polarity cues, where spatial disorientation frequently occurs. Mini-map studies largely focus on terrestrial environments, while reference frame design for polarity-free 3D sp...

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Baduanjin improves motor function and cortical activation in children with developmental coordination disorder: a randomized controlled trial

👤 Wei Song 📅 2026-11-09 📖 1 min 📝 40 words FNIRS

Summary: CONCLUSIONS: Baduanjin training improved motor coordination and balance in children with DCD. The observed changes in activation within motor-related cortical regions provide preliminary support for a possible neurophysiological basis of these functional improvements, although the underlying mechani...

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Potential Applications of Brain-Computer Interface Technology in the Rehabilitation of Patients with Anxiety Disorders: A Narrative Review

Hu Lou 2026-11-10 1 min 58 words FNIRS

Summary: Anxiety disorders are among the most prevalent mental disorders worldwide and can cause persistent impairment in emotional regulation, cognition, and social functioning. Pharmacotherapy, psychotherapy, exercise, and non-invasive brain stimulation are widely used, yet their effectiveness may be const...

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Multimodal EEG-fNIRS complexity analysis can reveal the neurovascular coupling change with aging

Yicheng Zhu 2026-11-10 1 min 58 words FNIRS

Summary: CONCLUSION: Aging is associated with task-specific alterations in neural complexity, hemodynamic complexity, and the cross-modal temporal relationship between EEG and fNIRS signals. These findings suggest that age-related changes in NVC-related dynamics are better characterized as dynamic reorganiza...

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Task-state prefrontal cortex activation differences between executive dysfunction and perceptual dysfunction phenotypes of freezing of gait in Parkinson's disease during the medication ON state: a retrospective fNIRS study

Zhanhua Liang
2026-11-11 1 min 66 words FNIRS

Summary: CONCLUSION: The two proposed PD-FOG phenotypes showed distinct task-related PFC activation patterns in the medication ON state, and this task-specific double dissociation pattern was consistent with the predictions of the dual-model framework. Because fNIRS was recorded during seated cognitive tasks...

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Sensorimotor cortical responses to visuotactile synchrony during virtual foot mirror feedback: an fNIRS study

Noboru Chiba 2026-11-11 1 min 47 words FNIRS

Summary: CONCLUSIONS: The Synchronous condition was associated with changes in subjective attribution of the virtual foot without a detectable difference in medial sensorimotor HbO. These observations should be regarded as hypothesis-generating rather than as established evidence that subjective embodiment a...

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Bayesian Adaptive Stimulus Optimization in P300 Brain Computer Interfaces

Boyla O Mainsah 2026-11-10 1 min 26 words BRAIN COMPUTER INTERFACE

Summary: CONCLUSION: For computational simplicity in implementation, the optimal stimulus can generally be approximated by finding the stimulus that has a combined probability mass closest to 0.5.

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MSTA-CNN an ultra-lightweight multi-scale temporal-spatial attention convolutional neural network for high-performance P300 spellers

👤 Mousa Shamsi 📅 2026-11-10 📖 1 min 📝 62 words BRAIN COMPUTER INTERFACE

Summary: Efficient P300 detection in brain-computer interfaces (BCIs) remains a challenge due to low signal-to-noise ratio (SNR) and the high computational cost of existing deep learning models, which limits their deployment in real-time applications. To address these issues, we propose MSTA-CNN, an ultra-li...

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Detection of linear dependencies in EEG oscillatory sources via PARALIND and block-term decomposition: A comparative study

👤 Leonardo Jose Trejo
📅 2026-11-10 📖 1 min 📝 60 words

BRAIN COMPUTER INTERFACE

Summary: Narrowband oscillatory rhythms in electroencephalogram (EEG) signals are fundamental signatures of neural dynamics, crucial for brain-computer interfaces (BCIs) and monitoring neurological disorders. While traditionally detected and analyzed via Independent or Principal Component Analysis in the tim...

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Lower frequency ECoG features improve longitudinal stability of speech and hand-grasp detection in a patient with progressive ALS

👤 Dean J Krusienski 📅 2026-11-10 📖 1 min 📝 54 words BRAIN COMPUTER INTERFACE

Summary: Achieving long-term decoding stability is a critical challenge for chronic
intracranial brain-computer interfaces (BCIs). High-gamma (HG) activity from
intracranial recordings is the standard signal for BCI control, and lower-frequency features
can enhance HG decoding performance. This s...

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A collaborative network of bi-Mamba and CAFS for multi-task classification of unilateral lower limb

👤 Junhua Li 📅 2026-11-10 📖 1 min 📝 66 words BRAIN COMPUTER INTERFACE

Summary: Accurate decoding of motor-related electroencephalography signals remains challenging in brain-computer interface research. To date, motor imagery of the upper limbs has been extensively investigated, while research on the lower limbs remains limited. Meanwhile, multiple tasks involving the unilater...

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A theory of multilevel interactive equilibrium in neuroAI

Quanyan Zhu 2026-11-10 1 min 70 words BRAIN COMPUTER INTERFACE

Summary: We propose a class of game-theoretic framework for adaptive multi-agent intelligent systems. Unlike the classical game theory which assumes that strategies are primitive objects chosen by perfectly rational agents, the new framework provides a mathematical foundation for studying equilibrium in Neur...

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Interictal Electroencephalographic Characteristics of the Human Centromedian Thalamic Nucleus Recorded in Focal Epilepsy

Shuang Wang 2026-11-10 1 min 23 words BRAIN COMPUTER INTERFACE

Summary: CONCLUSIONS: CM SEEG demonstrates distinct, state-dependent electrophysiological dynamics across the sleep-wake cycle and preferential involvement in interictal epileptic activity compared with VL/VP nuclei.

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Case Report: Representational geometry of multi-digit coordination in the macaque grasping network for intracortical BCIs

Hansjörg Scherberger

2026-11-11 | 1 min | 65 words

BRAIN COMPUTER INTERFACE

Summary: CONCLUSION: In this animal, condition-dependent dimensionality was significantly higher in F5 than in AIP or M1, while alignment and reconstruction analyses showed descriptive area-specific patterns in how closely combined-finger activity resembled a simple combination of single-finger activity. For...

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Effects of brain-computer interface output modalities on motor recovery after stroke: a systematic review and network meta-analysis

👤 Lili Fang 📅 2026-11-11 📖 1 min 📝 54 words BRAIN COMPUTER INTERFACE

Summary: CONCLUSION: Brain-computer interface-triggered functional electrical stimulation showed a clearer signal of benefit vs. conventional rehabilitation than brain-computer interface-controlled robotic. However, the comparison between the two active modalities relies entirely on indirect evidence from a ...

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Building a safety assessment framework for brain-computer interfaces: A call for cell-free multi-omics monitoring

👤 Jilai Li 📅 2026-11-11 📖 1 min 📝 2 words BRAIN COMPUTER INTERFACE

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Psychomusicology: A resounding closing cadence.

📅 2024-01-22 📖 1 min 📝 256 words PSYCHOMUSICOLOGY

Summary: From 2012 to 2023, the American Psychological Association served as publisher of Psychomusicology: Music, Mind, and Brain. Annabel Cohen and Mark Schmuckler were the successive editors-in-chiefs during this time. As the journal is ceasing publication, the two editors reflect on the developm...

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http://doi.org/10.1037/pmu0000305

Suggestibility and psychedelics: From therapeutics to social context.

📅 2024-12-12 📖 1 min 📝 175 words CLINICAL NEUROSCIENCE

Summary: Suggestibility is an important aspect of the psychedelic experience that merits greater attention in the expanding field of psychedelic research. This article provides a comprehensive road map for exploring the critical questions and issues at the intersection of suggestion, suggestibility, and psyc...

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http://doi.org/10.1037/cns0000412

Evaluating dispositional emotion-related factors in the effect of acceptance on affect reactivity: A secondary analysis of an experimental study.

📅 2024-12-30 📖 1 min 📝 257 words CLINICAL NEUROSCIENCE

Summary: Acceptance as a self-regulation strategy has beneficial effects on affect reactivity in response to stressors via reducing negative affect and state anxiety as well as improving positive affect. However, dispositional emotion-related factors, namely, experiential avoidance, emotional dysregulation, ...

🔗 **Read full article:**
http://doi.org/10.1037/cns0000416

Open monitoring meditation defined as the nonreactive meta-awareness of mind wandering.

📅 2024-12-12 📖 1 min 📝 128 words CLINICAL NEUROSCIENCE

Summary: Open monitoring meditation is generally defined as the nonreactive, meta-aware observation of moment-by-moment phenomenal experiences associated with mental contents without focusing on predetermined mental or physical objects. However, in the current literature, neither general definitions nor more...

🔗 **Read full article:**
http://doi.org/10.1037/cns0000411

Multisensory interplay and theories of consciousness.

📅 2024-11-25 📖 1 min 📝 174 words CLINICAL NEUROSCIENCE

Summary: Sensory interplay typically refers to cases where the content of one sensory modality affects the content of another sensory modality. Although there is much evidence today of such interplay, it is unknown how to interpret them. Is it a one-way causal function between two separate states (e.g., a vi...

🔗 **Read full article:**
http://doi.org/10.1037/cns0000407

How ghosting and stress impact vulnerable narcissism and maladaptive daydreaming: The role of mindfulness and rumination.

📅 2024-09-12 📖 1 min 📝 204 words CLINICAL NEUROSCIENCE

Summary: The present study aims to explore the influence of ghosting and stress on maladaptive daydreaming and vulnerable narcissism within a multivariate framework, using mindfulness and rumination as mediators. We employed a cross-sectional design and conducted it among 301 university students (M<...

🔗 **Read full article:**
http://doi.org/10.1037/cns0000403

The conscious experience of thinking aloud: Are positive feelings associated with better think-aloud performance?

📅 2024-12-30 📖 1 min 📝 216 words CLINICAL NEUROSCIENCE

Summary: The think-aloud interview is a psychological procedure used to measure an individual's cognition in solving a task. However, very little is known about how an individual's conscious experience of the interview might be associated with their performance during the interview. This is consequential as ...

🔗 **Read full article:**
http://doi.org/10.1037/cns0000410

Mystical experiences in virtual reality.

📅 2024-04-04 📖 1 min 📝 176 words CLINICAL NEUROSCIENCE

Summary: Virtual reality (VR) has shown promise as a psychological means of inducing mystical experiences. Here, we compared the subjective responses of 48 undergraduate volunteers to script-guided multiuser perceptual experiences in VR (VR perception [VR-P]) with their responses to a control condit...

🔗 **Read full article:**
http://doi.org/10.1037/cns0000396

Effect of a peer-led meditation club on adolescents' well-being, empathy, and compassion.

📅 2024-09-09 📖 1 min 📝 197 words CLINICAL NEUROSCIENCE

Summary: Many studies have examined the effect of meditation sessions led by experienced teachers on personal benefits for adults. The present study investigated the effect of broad-based mindfulness practices with an adolescent peer facilitator on prosocial outcomes in addition to individual gains. The 20 s...

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http://doi.org/10.1037/cns0000400

Thursday Daily Thread: Python Careers, Courses, and Furthering Education!

👤 /u/AutoModerator 📅 2026-11-12 📖 1 min 📝 219 words REDDIT PYTHON

Summary: <!-- SC_OFF --> <h1>Weekly Thread: Professional Use, Jobs, and Education 🏢</h1> <p>Welcome to this week's discussion on Python in the professional world! This is your spot to talk about job hunting, career growth, and educational resources in Python. Please note, this thread is <stron...

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https://www.reddit.com/r/Python/comments/1xtxsyq/thursday_daily_thread_python_careers_courses_and/

United Airlines bans AI agents from booking flights on behalf of travelers

👤 herpderperator 📅 2026-11-11 📖 1 min 📝 13 words HACKER NEWS

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Jensen Huang said to have had relationship with executive who reports to him

👤 louiereederson 📅 2026-11-11 📖 1 min 📝 13 words HACKER NEWS

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Lock Your GitHub Actions

👤 gslin 📅 2026-11-11 📖 1 min 📝 13 words HACKER NEWS

> **Summary:** Article URL: https://hugovk.dev/blog/2026/lock-github-actions/ Comments URL: https://news.ycombinator.com/item?id=50459485 Points: 145 # Comments: 25</p...

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Nvidia is preparing to oust Jensen Huang

👤 jbredeche 📅 2026-11-11 📖 1 min 📝 13 words HACKER NEWS

> **Summary:** Article URL: https://www.wsj.com/business/nvidia-is-preparing-to-oust-nvidia-ceo-jensen-huang-30f421ad Comments URL: <a href="https://news.ycombinator.com/item?id=50459703">https://news....

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93% of paint splatters are valid Perl programs (2019)

👤 zenincognito 📅 2026-11-11 📖 1 min 📝 13 words HACKER NEWS

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A general purpose Apple Silicon layer for x86-64 PCs

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Multimodal MRI reveals brain structural and functional abnormalities associated with wearing-off phenomenon in Parkinson’s disease

📖 1 min 📝 26 words NEUROSCIENCE JOURNAL

> **Summary:** Publication date: 9 January 2027 Source: Neuroscience, Volume 620 Author(s): Yingyun Wang, Qing Xu, Xiaoxue Shi, Li Wang, Jinhua Zheng, Jianjun Ma, Qi Gu, Siyuan Chen, Xue Li

🔗 **Read full article:**
https://www.sciencedirect.com/science/article/pii/S0306452226007311?dgcid=rss_sd_all

Interhemispheric structural connectivity alterations and hierarchical connectome gradient reorganization following SARS-CoV-2 infection

📖 1 min 📝 15 words NEUROSCIENCE JOURNAL

> **Summary:** Publication date: 9 January 2027 Source: Neuroscience, Volume 620 Author(s): Mahya Naghipoor-Alamdari, Melina Farshbafnadi, Mohammad Hadi Aarabi

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UNC13A splice correction rescues synaptic dysfunction induced by TDP-43 loss

📅 2026-11-11 📖 1 min 📝 0 words NATURE NEUROSCIENCE SUBJECTS

🔗 **Read full article:**
https://www.nature.com/articles/s41586-026-11256-2

Correction: Selective dysregulation of serotonin dynamics in the anterior cingulate cortex and central amygdala following binge alcohol consumption

📅 2026-11-11 📖 1 min 📝 0 words NATURE NEUROSCIENCE SUBJECTS

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Defining and measuring proximity to criticality based on univariate time series

J. Samuel SooterAntonio J. FonteneleAndrea K. BarreiroCheng LyKeith B. HengenWoodrow L. ShewaDepartment of Physics, University of Arkansas, Fayetteville, AR 72701bhttps://ror.org/042tdr378Department of Mathematics, Southern Methodist University, Dallas, TX 75205chttps://ror.org/02nkdxk79Department of Mathematics and Applied Mathematics, Virginia Commonwealth University, Richmond, VA 23284dhttps://ror.org/01yc7t268Department of Biology, Washington University, St. Louis, MO 63130

2026-10-28 1 min 56 words PNAS NEUROSCIENCE

Summary: Proceedings of the National Academy of Sciences, Volume 123, Issue 44, November 2026.
SignificanceKnowledge of many complex systems comes from time series measurements of one or a few parts of the system. These dynamics depend on many detailed properties; must we keep track every detail to und...

Read full article:
https://www.pnas.org/doi/abs/10.1073/pnas.2526984123?af=R

Computational approaches to deep learning–driven EEG analysis for neurological disease diagnosis: a systematic review

Jisu Elsa Jacob 2026-10-13 1 min 229 words FRONTIERS NEUROINFORMATICS

Summary: Electroencephalography (EEG) is a widely used non-invasive neurophysiological technique for analyzing brain dynamics and diagnosing neurological diseases. However, the non-stationary nature, low signal-to-noise ratio, and inter-subject variability of EEG signals cause significant challenges to tradi...

🔗 **Read full article:**
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Editorial: Machine learning integration in computational neuroscience: enhancing neural data decoding and prediction

George Giakos 2026-11-03 1 min 0 words FRONTIERS COMPUTATIONAL NEUROSCIENCE

🔗 **Read full article:**
https://www.frontiersin.org/articles/10.3389/fncom.2026.2030505

Short excitatory projections promote burst activity and modular functional organization in neural networks

👤 Claus C. Hilgetag 📅 2026-11-03 📖 1 min 📝 165 words

FRONTIERS COMPUTATIONAL NEUROSCIENCE

Summary: Physical constraints in the brain favor short, distance-dependent connections. However, short-range projections are not merely a structural compromise; they also serve essential functional roles. Here, we explore networks with balanced excitation and inhibition, demonstrating that short excitatory p...

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https://www.frontiersin.org/articles/10.3389/fncom.2026.1965686

Exploratory evaluation of electroencephalogram-derived features for classification of mild cognitive impairment in two independent datasets

👤 Makoto Nakane 📅 2026-11-03 📖 1 min 📝 347 words

FRONTIERS COMPUTATIONAL NEUROSCIENCE

Summary: A proper diagnosis of mild cognitive impairment (MCI) is one of the important challenges in the early detection of dementia. For the classification of MCI patients and healthy controls, some electroencephalogram (EEG)-derived indices were calculated and compared. Some studies have been conducted on ...

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https://www.frontiersin.org/articles/10.3389/fncom.2026.1940507

Persistent memory through triple-loop consolidation under stochastic unit turnover

👤 Jianwei Lou 📅 2026-11-03 📖 1 min 📝 366 words FRONTIERS COMPUTATIONAL NEUROSCIENCE

Summary: Dissipative cognitive architectures maintain computation through continuous energy expenditure, where units that exhaust their energy are stochastically replaced with fresh random state. This creates a fundamental challenge: how can persistent, context-specific memory survive when all learnable stat...

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https://www.frontiersin.org/articles/10.3389/fncom.2026.1845213

Editorial: Advances in perceptual decision making and brain oscillations

👤 Francesco Di Gregorio 📅 2026-11-03 📖 1 min 📝 0 words

FRONTIERS COMPUTATIONAL NEUROSCIENCE

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https://www.frontiersin.org/articles/10.3389/fncom.2026.2023605

Editorial: The brain and social interactions: from paradigms of social emotion recognition to hyperscanning

Michela Candini
2026-11-11 1 min 0 words FRONTIERS NEUROSCIENCE

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https://www.frontiersin.org/articles/10.3389/fnins.2026.2018824

REFS: robust EEG feature selection with incomplete labels for multi-dimensional emotion recognition

Xueyuan Xu, Wenjia Dong, Fulin Wei and Li Zhuo
2026-11-06 1 min 268 words

JOURNAL NEURAL ENGINEERING

Summary: Objective. Compared to single-type features, multi-type electroencephalography (EEG) features provide a multi-level representation for analyzing multi-dimensional emotions. However, the high dimensionality of multi-type EEG features, combined with the relatively small number of high-quality EEG samp...

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https://iopscience.iop.org/article/10.1088/1741-2552/aeb97e

Human cortical high-gamma power scales with movement rate in healthy participants and stroke survivors

👤 Fanny Quandt 📅 2025-01-09 📖 1 min 📝 65 words OOSTENVELD ROBERT

Summary: Motor cortical high-gamma oscillations (60-90 Hz) occur at movement onset and are spatially focused over the contralateral primary motor cortex. Although high-gamma oscillations are widely recognized for their significance in human motor control, their precise function on a cortical level remains el...

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NIRS-BIDS: Brain Imaging Data Structure Extended to Near-Infrared Spectroscopy

👤 Luca Pollonini 📅 2025-01-27 📖 1 min 📝 70 words OOSTENVELD ROBERT

Summary: Functional near-infrared spectroscopy (fNIRS) is an increasingly popular neuroimaging technique that measures cortical hemodynamic activity in a non-invasive and portable fashion. Although the fNIRS community has been successful in disseminating open-source processing tools and a standard file forma...

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Pseudonymisation of neuroimages and data protection: Increasing access to data while retaining scientific utility

👤 Lyuba Zehl 📅 2025-06-26 📖 1 min 📝 67 words OOSTENVELD ROBERT

Summary: For a number of years, facial features removal techniques such as 'defacing', 'skull stripping' and 'face masking/blurring', were considered adequate privacy preserving tools to openly share brain images. Scientifically, these measures were already a compromise between data protection requirements a...

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Cycling on the Freeway: The perilous state of open-source neuroscience software

👤 Tim M Tierney 📅 2025-08-13 📖 1 min 📝 74 words OOSTENVELD ROBERT

Summary: Most scientists need software to perform their research (Barker et al., 2020;Carver et al., 2022;Hettrick, 2014;Hettrick et al., 2014;Switters & Osimo, 2019), and neuroscientists are no exception. Whether we work with reaction times, electrophysiological signals, or magnetic resonance imaging data, ...

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Optimal configuration of on-scalp OPMs with fixed channel counts

👤 Robert Oostenveld 📅 2025-08-13 📖 1 min 📝 69 words OOSTENVELD ROBERT

Summary: Recent technological developments have brought optically pumped magnetometers (OPMs) within reach of the larger neuroscientific community. The current state-of-the-art consists of whole-head systems that measure the magnetic field at >100 locations. OPM sensors can be constructed to measure the fiel...

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ARTEM-IS for ERP: Agreed Reporting Template for EEG Methodology-International Standard for Event-Related Potential Experiments

👤 Suzy J Styles 📅 2025-12-08 📖 1 min 📝 73 words OOSTENVELD ROBERT

Summary: The choices we make during the recording, preprocessing and analysis of event-related potentials (ERP) data can affect study outcomes. As such, it is critical that they are transparently reported to allow for reproducibility. Yet, systematic reviews of reporting practices in the field have shown tha...

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Eye-Tracking-BIDS: the Brain Imaging Data Structure extended to gaze position and pupil data

👤 Julia-Katharina Pfarr 📅 2026-02-12 📖 1 min 📝 68 words OOSTENVELD ROBERT

Summary: The Brain Imaging Data Structure (BIDS) is a widely adopted, community-driven standard to organize neuroimaging data and metadata. Although numerous extensions have been developed to incrementally extend coverage to new modalities and data types, an unambiguous, granular specification for eye-tracki...

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Facilitating unusual bodily experiences and out-of-body experiences across wakefulness and sleep: A high-density EEG and neurophenomenology study

👤 Robert Oostenveld 📅 2026-02-12 📖 1 min 📝 67 words OOSTENVELD ROBERT

Summary: Unusual bodily experiences (UBEs) are illusory bodily perceptions that are not coherent with typical wakeful experiences, including flying or floating sensations, body distortions, and out-of-body experiences. This study examines UBEs in a controlled sleep laboratory using meditation and light stimu...

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The Impact of Brain Tumors and Craniotomy Lesions on Scalp EEG

Vitória Piai 2026-02-17 1 min 65 words OOSTENVELD ROBERT

Summary: Electroencephalography (EEG) is widely used in both research and clinical settings, yet its accuracy can be significantly impacted by subject-specific anatomical anomalies such as brain lesions and skull defects. This study investigates the effects of glioma-related brain lesions and craniotomy-indu...

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EMG-BIDS: an extension to the Brain Imaging Data Structure for electromyography

Arnaud Delorme 2026-08-01 1 min 64 words OOSTENVELD ROBERT

Summary: Electromyography (EMG) is fundamental to clinical assessment, rehabilitation, neuromuscular research, and human-machine interfaces. Despite decades of use, no widely adopted standard exists for organizing and sharing EMG data, limiting reusability and large-scale data aggregation. We present EMG-BID...

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Feasibility of robotic-assisted surgery for nutcracker syndrome: a systematic review and single-arm meta-analysis

Jiang-Hua Zheng 2026-11-10 1 min 60 words LOW VISION

Summary: Nutcracker syndrome (NCS) results from compression of the left renal vein, and refractory symptoms require surgical reconstruction. Robotic platforms provide magnified three-dimensional vision and articulated instruments that are theoretically advantageous for vascular dissection and anastomosis wit...

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Segmentation, Reconstruction, and Grasp-Pose Estimation Using an RGB-D Camera for Robotic Apple Harvesting

Jianqiang Lu 2026-11-10 1 min 58 words LOW VISION

Summary: Accurate segmentation, three-dimensional reconstruction, and grasp-pose estimation of fruits and surrounding structures are essential for robotic harvesting in complex orchard environments. However, occlusions caused by branches, support wires, and irregular canopy structures remain major challenges...

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Intelligent rendering optimization for animated scenes using reinforcement learning and computer vision

Li Jia 2026-11-10 1 min 63 words LOW VISION

Summary: Real-time rendering of complex animated scenes is computationally intensive and often struggles to balance visual fidelity and performance in dynamic or densely populated environments, as conventional pipelines lack adaptive resource optimization, leading to reduced frame rates and degraded visual q...

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Clinical efficacy of unilateral biportal endoscopic spinal decompression for lumbar intraspinal pneumatosis

Huiqiu Jian 2026-11-10 1 min 48 words LOW VISION

Summary: CONCLUSION: UBE for the treatment of lumbar intraspinal pneumatosis has the advantages of minimal trauma, clear field of vision, flexible operation, and rapid postoperative recovery, representing a potentially safe and effective option. Large-scale multicenter prospective trials are still needed to ...

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Photobleaching-Driven Ratiometric Fluorescent Switching of ZGO:Mn Nanorods/RhB for Iodine Vapor Monitoring with Integrated Gas-Phase Separation

👤 Kui Zhang 📅 2026-11-11 📖 1 min 📝 56 words LOW VISION

Summary: Conventional fluorescent sensors generally rely on complex, time-consuming structural engineering of recognition moieties. By contrast, leveraging photobleaching for eco-friendly, reagent-free signal transduction has long been impeded by irreversible fluorophore degradation. Additionally, direct liq...

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Ageist attitudes as a moderator of objective and subjective vision: A longitudinal study

👤 Yoav S Bergman 📅 2026-11-11 📖 1 min 📝 41 words LOW VISION

Summary: OBJECTIVES: This longitudinal study examined the relationship between objective visual acuity and subjective vision over a six-month period. Specifically, it investigated the moderating role of internalized ageist attitudes in shaping how objective vision relates to perceived sensory vision difficul...

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Registry-recorded stroke and transient ischaemic attack diagnoses in Saudi Arabia, 2016-2025: a national descriptive analysis

👤 Mohammed AlAbdulali
📅 2026-11-11 📖 1 min 📝 58 words

LOW VISION

Summary: CONCLUSION: Saudi Arabia's recorded stroke burden is concentrated among older males. The findings describe patterns captured by the PHO and should not be interpreted as prevalence, incidence, or causal associations, which indicate the need for better prevention, management of chronic diseases and a ...

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Design and parameter optimization of a bilateral staggered wheel-spoon precision seed-metering device for millet plot breeding

👤 Ankang Mu 📅 2026-11-11 📖 1 min 📝 63 words LOW VISION

Summary: Precision seeding of micro-sized millet seeds in plot breeding faces critical challenges, including high miss-seeding rates and severe seed cross-contamination during variety changeovers. This study developed a novel bilateral staggered wheel-spoon precision seed-metering device to achieve precise h...

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Representation geometry and task performance in vision-language modeling for computed tomography enterography

👤 Ryan Stidham 📅 2026-11-11 📖 1 min 📝 36 words LOW VISION

Summary: CONCLUSIONS: Slice sampling and tissue-contrast-focused encoding affect performance in CT enterography vision-language systems, and RAG improved report-label agreement over the fine-tuned baseline. These results provide initial baselines and practical guidance for multimodal learning in volumetric C...

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Incidence and Management of Ocular Hypertension Following the 0.19 mg Fluocinolone Acetonide Implant for Diabetic Macular Edema

👤 Wills Eye Retina Service Study Group, The Wills Eye Retina Service Study Group additionally includes

📅 2026-11-11 📖 1 min 📝 36 words LOW VISION

Summary: CONCLUSION: FAi was associated with low rates of OHT, with most cases managed with observation or topical therapy, with surgery rarely required. This supports the safety of FAi and highlight the importance of extended IOP monitoring.

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A single session of high-definition transcranial direct current stimulation does not modulate effects of knee two-point discrimination training or sensorimotor function in healthy adults: Double-blind randomised controlled trial

👤 Paul W Hodges
📅 2026-07-30 📖 1 min 📝 53 words

TACTILE ACUITY

Summary: CONCLUSION: A single 20-minute session of 1 mA HD-tDCS, with or without brief TPD training, does not acutely modify knee somatosensory, sensorimotor, or functional performance in healthy adults. Existing evidence at the hand may not translate to the lower limb. Future work should investigate higher-...

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Variation in glabrous skin physiology across anatomical locations on the human hand

Martina C Meinke 2026-08-13 1 min 76 words TACTILE ACUITY

Summary: Skin physiology varies greatly between anatomical locations, with impact on tactile sensitivity. Research on the variations in morphology and physiology between different anatomical locations of glabrous skin is limited so far. This study investigates the morphology and physiology of glabrous skin o...

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Beyond Hamstring Shortness: Continuous Flexibility Is Associated With Balance and Stability but Not Sensory Measures-A Cross-Sectional Study

Senanur Akıncı 2026-08-14 1 min 66 words TACTILE ACUITY

Summary: CONCLUSIONS: Greater hamstring flexibility was associated with better dynamic balance performance and showed a weaker association with postural stability, whereas no robust associations were observed with sensory measures of postural control. These findings suggest that hamstring flexibility may be ...

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Resting-state EEG microstate dynamics reflect individual differences in tactile angle discriminability

Liu Yang 2026-08-19 1 min 63 words TACTILE ACUITY

Summary: Tactile spatial acuity, the ability to discriminate fine spatial details, is a fundamental sensory capacity modulated by both peripheral and central mechanisms. Here, we investigated whether resting-state EEG microstate dynamics are associated with individual differences in tactile spatial acuity. F...

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A grating orientation task with controlled dome elevation speed and contact force: stimulus conditions and day-to-day reliability

Hideaki Onishi 2026-09-25 1 min 66 words TACTILE ACUITY

Summary: The grating orientation discrimination task (GOT) is typically used to assess tactile spatial acuity. However, performance is influenced by stimulus speed, contact force between the stimulator and the skin, and the method of discrimination. This study aimed to explore suitable stimulus conditions fo...

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Epidermal microstructure and tactile sensitivity in the glabrous skin of hands and feet

👤 Hannes P Saal 📅 2026-09-25 📖 1 min 📝 69 words TACTILE ACUITY

Summary: Glabrous skin on the human palms and soles is specialized for mechanical interaction and is characterized by papillary ridges, an absence of hair follicles, and a thick stratum corneum. Although gross morphological differences, such as the increased thickness of the plantar stratum corneum, are well...

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Neurotrauma During Military Training: Results of the INVestigating traIning assoCiated blasT pAthology (INVICTA) Study

👤 Walter S Carr 📅 2026-10-01 📖 1 min 📝 67 words TACTILE ACUITY

Summary: Concern is growing that subconcussive blast exposure during military exercises such as heavy weapons training (HWT) may adversely affect brain function. HWT involves firing shoulder-mounted weapons that can expose the gunner, assistant gunner, and instructors (range safety officers [RSOs]) to blast ...

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Altered body representation in chronic non-specific low back pain: a testable framework linking body-schema-related disturbance, corticomotor alterations, and altered pain processing

👤 Zhifei Xie
📅 2026-10-24 📖 1 min 📝 66 words

TACTILE ACUITY

Summary: Low back pain is a leading contributor to disability worldwide; in chronic non-specific low back pain (CNSLBP), the processes sustaining pain and disability in the absence of identifiable structural pathology remain incompletely understood. Three partially overlapping research lines have documented ...

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Hyperexcitable primary somatosensory cortex in the MitoPark mouse model of Parkinson's disease

Alan J Emanuel 2026-10-25 1 min 67 words TACTILE ACUITY

Summary: The majority of Parkinson's disease patients suffer from decreased tactile acuity or neuropathic pain, both of which contribute to low quality of life and likely exacerbate motor dysfunction. How Parkinson's disease pathology relates to altered somatosensation remains unclear. One potential contribu...

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The Relationship Between Occlusal Tactile Acuity, Bite Force and Tooth Wear

Birgitta Häggman-Henrikson 2026-11-03 1 min 52 words TACTILE ACUITY

Summary: CONCLUSIONS: Bite force may be a clinically relevant factor associated with accumulated tooth wear, whereas OTA was not independently associated with tooth wear. Longitudinal studies are needed to determine whether bite force can predict tooth wear progression and whether more comprehensive assessme...

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Impact of Braille Booklet, Audio-Tactile Performance and Caregiver-supervised Training on Oral Hygiene Status, Knowledge, Attitude and Practices among Visually Challenged Children- A Randomized Clinical Trial

👤 Priyanga Chandrasekaran
📅 2026-09-22 📖 1 min 📝 52 words

BRAILLE

Summary: CONCLUSION: Both caregiver-supervised training and audio tactile performance using specialized model method of oral health education proved to be more effective than the braille booklet group on plaque control and improvement in the oral hygiene status, oral health knowledge, attitude and practice i...

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Design and functional verification of an automated Perkins Brailler accessory with audio feedback

Ali Alhaj 2026-09-23 1 min 67 words BRAILLE

Summary: CONCLUSIONS: The study did not include learners with visual impairment, Braille teachers, rehabilitation specialists, or other end users, and it did not collect quantitative measures of accuracy across the full character set, writing speed, embossing quality, latency, durability, safety, usability, ...

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Uncertainty-Aware structured decoding for Chinese Braille document transcription

Haoran Sun 2026-09-25 1 min 64 words BRAILLE

Summary: Automatic transcription of braille documents remains challenging in document digitization settings, particularly under real imaging degradations such as uneven embossing, illumination variation, paper texture, skew, blur, and ghosting, which make braille dot presence uncertain and often cause error ...

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Clinical Characteristics and Epidemiology of Isolated Dystonia: Results from a US Claims Analysis

James E Thornton 2026-10-08 1 min 54 words BRAILLE

Summary: CONCLUSIONS: This study defines an isolated dystonia cohort in the US based on real-world evidence and systematic exclusion of secondary causes of dystonia. Estimated prevalence is within range of prior epidemiological studies, and the cohort showed increased rates of comorbid psychiatric disorders....

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Effects of multisensory tactile-auditory integration teaching on systematic anatomy learning in visually impaired medical students

Fanzhen Kong 2026-10-08 1 min 50 words BRAILLE

Summary: CONCLUSION: The tactile-auditory integrated training system significantly improves objective summative knowledge acquisition and subjective learning experiences in visually impaired medical students. This approach provides a scalable, cost-constrained, and replicable pedagogical model for inclusive ...

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Physical and Environmental Accessibility of Dental Services for Persons with Disabilities in Saudi Primary Health Care: A Cross-Sectional Study

👤 Estie Kruger 📅 2026-10-16 📖 1 min 📝 66 words BRAILLE

Summary: Background/Objectives: Persons with disabilities (PWDs) may experience barriers when accessing oral health care, including those related to the physical and sensory environment of the dental-care setting. In this study we assessed the accessibility of primary health care centres (PHCs) providing den...

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Visible Light Driven Diselenide Chemistry for Simultaneous 3D Patterning and Mechanical Reinforcement

👤 Huaping Xu 📅 2026-10-16 📖 1 min 📝 63 words BRAILLE

Summary: Three-dimensional (3D) patterning on polymer surfaces is typically achieved by photolithography or related techniques, which often involve complex procedures, high-energy ultraviolet light, and risk of material damage. Here, we report a visible light-driven strategy that exploits the dual functional...

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An accessible pregnancy test for blind and low-vision users

Vanja Garaj 2026-10-16 1 min 63 words BRAILLE

Summary: CONCLUSIONS: The findings suggest that accessible pregnancy testing must prioritise not only non-visual result reading but also privacy, dignity, reproductive autonomy and independent health management. These results represent the preliminary evidence of acceptability rather than a conclusive test o...

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Genome-Informed Loop-Mediated Isothermal Amplification (LAMP) Assays for Detection and Differentiation of Ralstonia pseudosolanacearum Phylotypes I and III

Jarred Yasuhara-Bell 2026-10-27 1 min 76 words BRAILLE

Summary: The Ralstonia solanacearum species complex (RSSC) causes bacterial wilt in a broad range of economically important hosts. The RSSC has been reclassified into three species, of which R. solanacearum sequevars 1 and 2 in phylotype IIB are designated as Select Agents in the U.S. Many rapid tests for "R...

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Encoding tactile stimuli for Braille recognition with organoids

👤 Benjamin Ward-Cherrier 📅 2026-10-31 📖 1 min 📝 62 words BRAILLE

Summary: This study proposes a transferable encoding strategy that maps tactile sensor data to electrical stimulation patterns, enabling neural organoids to perform an open-loop artificial tactile Braille classification task. Human forebrain organoids cultured on a low-density microelectrode array (MEA) are ...

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From Single Cells to the Circulating Tumor Microenvironment: Label-Free Size-Selection and a Morphological Taxonomy for Liquid Biopsy in Cytopathology

👤 Satoru Kobayashi 📅 2026-11-09 📖 1 min 📝 25 words TDCS TACS TRNS

Summary: CONCLUSION: Size-selection converts a liquid biopsy into a cytology slide, placing interpretation of the circulating tumor ecosystem within the existing competence of the diagnostic laboratory.

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Phase-Dependent Effects of Cerebellar Transcranial Alternating Current Stimulation on Lateral Vestibulospinal Tract Excitability

👤 Tsubasa Mitsutake
📅 2026-11-09 📖 1 min 📝 36 words

TDCS TACS TRNS

Summary: CONCLUSIONS: LVST excitability differed according to the phase of the ongoing cerebellar-targeted tACS waveform. These findings provide electrophysiologic evidence of an online phase-dependent effect on vestibulospinal excitability and a physiologic basis for future phase-specific neuromodulation st...

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Effects of Transcranial Direct Current Stimulation on Core Symptoms and Sleep Disturbances in Children With Autism Spectrum Disorder: A Meta-Analysis of Randomized Controlled Trials

👤 Meimei Yang
📅 2026-11-09 📖 1 min 📝 44 words

TDCS TACS TRNS

Summary: CONCLUSION: tDCS, most commonly delivered at 1 mA over the DLPFC, was associated with improvements in some ASD-related behavioral outcomes; preliminary evidence also suggested a potential benefit for sleep disturbances, although this finding was based on only two low-certainty studies and warrants c...

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No effects of single-session concurrent tDCS on explicit motor sequence learning in healthy adults with conventional 2 mA and tangential 6 mA currents

👤 Lucas C Parra
📅 2026-11-09 📖 1 min 📝 45 words

TDCS TACS TRNS

Summary: CONCLUSION: These findings, together with a survey of the existing literature, suggest that the effects of single-session concurrent tDCS on explicit motor sequence learning are not robust across studies. Our results underscore the necessity of rigorous methodological controls to prevent false disco...

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More than Quaternary Ammonium Compounds: First Evidence of Tertiary Amine Compounds as Novel Contaminants Ubiquitously Present in Indoor Dust

👤 Runzeng Liu 📅 2026-11-10 📖 1 min 📝 65 words TDCS TACS TRNS

Summary: Nitrogen-containing organic compounds have attracted global concern due to their widespread production and potential environmental risks. While the environmental occurrence of quaternary ammonium compounds (QACs) has been widely studied, research on tertiary amine compounds (TACs), another important...

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Combining transcranial electrical and magnetic stimulation: a scoping review of neurophysiological and clinical studies

👤 Paul M Briley 📅 2026-11-10 📖 1 min 📝 56 words TDCS TACS TRNS

Summary: CONCLUSIONS: Studies indicate that tES could enhance or reverse expected effects of rTMS. The greatest increase in cortical excitability occurred following excitatory rTMS combined with gamma transcranial alternating current stimulation (tACS) or with cathodal tDCS, consistent with effects of metapl...

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Closed-loop transcranial electrical brain stimulation with fMRI: A proof-of-concept randomized trial of real-time fMRI-guided tACS optimization

Hamed Ekhtiari 2026-11-10 1 min 60 words TDCS TACS TRNS

Summary: Closed-loop transcranial alternating current stimulation (tACS) combined with real-time functional MRI (tACS-fMRI) enables adaptive optimization of noninvasive brain stimulation based on ongoing neural dynamics. We implemented one of the first randomized, double-arm, closed-loop tACS-fMRI protocols ...

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Effects of different electrical stimulation on depression: a systematic review and network meta-analysis

Shouping Zhao 2026-11-11 1 min 64 words TDCS TACS TRNS

Summary: CONCLUSIONS: Electrical stimulation is associated with improved clinical response rates and reduced depressive symptom severity in patients with depression, and is generally well tolerated. Among the modalities examined, taVNS demonstrated a relative advantage in improving clinical response, while t...

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Applications of non-invasive brain stimulation in the treatment of children with cerebral palsy: a narrative review

👤 Lei Huang 📅 2026-11-11 📖 1 min 📝 59 words TDCS TACS TRNS

Summary: Noninvasive brain stimulation has emerged as a promising adjunctive strategy for pediatric neurorehabilitation, particularly for children with cerebral palsy. This narrative review, informed by a structured literature search, summarizes current evidence on transcranial direct current stimulation and...

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Noninvasive brain stimulation for migraine: A bibliometric analysis of scientific research with thematic and keyword mapping

👤 Zahra Hadi Alqallaf 📅 2026-11-11 📖 1 min 📝 52 words TDCS TACS TRNS

Summary: CONCLUSIONS: This bibliometric analysis maps the evolution, collaboration patterns, thematic structure, and emerging research priorities of noninvasive brain stimulation research in migraine, highlights transcranial magnetic stimulation and transcranial direct current stimulation as established meth...

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Prefrontal Activation and Cognitive Heterogeneity in Pediatric Sleep-Disordered Breathing

Amal Isaiah 2026-11-05 1 min 39 words FNIRS

> **Summary:** CONCLUSIONS AND RELEVANCE: In this cross-sectional study, task-related PFC activation was associated with SDB severity and moderated the severity-cognition association. fNIRS may help identify children whose cognitive profiles deviate from conventional severity metrics, thereby supporting individual...

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Mirror Neural System in Hands Motor Imagery with Pictures and Videos Guidance: An EEG-fNIRS Study

Mohamad Sawan 2026-11-05 1 min 64 words FNIRS

> **Summary:** Brain-computer interfaces (BCIs) achieved by motor imagery (MI) have been widely accepted as an effective tool in rehabilitative applications. Functional near-infrared spectroscopy (fNIRS) and electroencephalography (EEG) are two common implementation forms of BCIs, capable of monitoring neuronal po...

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Quality-aware mask-propagation graph attention for fNIRS-based bipolar disorder classification without channel interpolation

👤 Yajing Meng
📅 2026-11-05 📖 1 min 📝 58 words FNIRS

Summary: Bipolar disorder (BD) remains difficult to characterise objectively, motivating a neural marker to complement clinical assessment. Functional near-infrared spectroscopy (fNIRS) offers portable cortical haemodynamic measurement, but channel availability can vary across recordings because of bad chann...

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Brain-Behavior Coupling During Cognitive-Postural Dual-Tasking in Older Adults With Mild Cognitive Impairment: An fNIRS Study

👤 Chuan Guo 📅 2026-11-06 📖 1 min 📝 56 words FNIRS

Summary: CONCLUSIONS: Older adults with MCI exhibit altered brain-behavior coupling during cognitive-postural dual-task conditions involving sensory and postural constraints, characterized by altered relationships between prefrontal-sensorimotor cortical activation and behavioral performance. These findings ...

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Item-Specific Proportion Congruency Uncovers Different Forms of Attentional Control: An event-related fNIRS Study

👤 Murat Perit Çakır 📅 2026-11-07 📖 1 min 📝 70 words FNIRS

Summary: CONCLUSION: While behavioral results mostly support contingency learning, fNIRS data revealed ISPC-related activation in prefrontal channels with sensitivity to BA 46 (lateral, channel 1) and the BA 10/BA 11 overlap zone (medial, channels 10-12), with channel 11 emerging as a convergence point for b...

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Does the emotion regulation of self-directed humor engage the prefrontal cortex? An fNIRS study

Zhonglu Zhang 2026-11-07 1 min 68 words FNIRS

Summary: Previous research has revealed that self-directed humor (self-enhancing and self-defeating) can help regulate negative emotions and that the prefrontal cortex (PFC) plays a critical role in emotion regulation. However, it remains unknown whether emotion regulation of self-directed humor is associate...

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Effects of mini-maps with different reference frames on polarity-free 3D spatial navigation: a behavioural and fNIRS study

Xiaozhou Zhou 2026-11-07 1 min 62 words FNIRS

Summary: Manned spaceflight and deep-sea exploration extend human activities to extreme environments lacking gravitational and visual polarity cues, where spatial disorientation frequently occurs. Mini-map studies largely focus on terrestrial environments, while reference frame design for polarity-free 3D sp...

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Baduanjin improves motor function and cortical activation in children with developmental coordination disorder: a randomized controlled trial

👤 Wei Song 📅 2026-11-09 📖 1 min 📝 40 words FNIRS

Summary: CONCLUSIONS: Baduanjin training improved motor coordination and balance in children with DCD. The observed changes in activation within motor-related cortical regions provide preliminary support for a possible neurophysiological basis of these functional improvements, although the underlying mechani...

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Potential Applications of Brain-Computer Interface Technology in the Rehabilitation of Patients with Anxiety Disorders: A Narrative Review

Hu Lou 2026-11-10 1 min 58 words FNIRS

Summary: Anxiety disorders are among the most prevalent mental disorders worldwide and can cause persistent impairment in emotional regulation, cognition, and social functioning. Pharmacotherapy, psychotherapy, exercise, and non-invasive brain stimulation are widely used, yet their effectiveness may be const...

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Multimodal EEG-fNIRS complexity analysis can reveal the neurovascular coupling change with aging

Yicheng Zhu 2026-11-10 1 min 58 words FNIRS

Summary: CONCLUSION: Aging is associated with task-specific alterations in neural complexity, hemodynamic complexity, and the cross-modal temporal relationship between EEG and fNIRS signals. These findings suggest that age-related changes in NVC-related dynamics are better characterized as dynamic reorganiza...

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Bayesian Adaptive Stimulus Optimization in P300 Brain Computer Interfaces

Boyla O Mainsah 2026-11-10 1 min 26 words BRAIN COMPUTER INTERFACE

Summary: CONCLUSION: For computational simplicity in implementation, the optimal stimulus can generally be approximated by finding the stimulus that has a combined probability mass closest to 0.5.

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MSTA-CNN an ultra-lightweight multi-scale temporal-spatial attention convolutional neural network for high-performance P300 spellers

Mousa Shamsi 2026-11-10 1 min 62 words BRAIN COMPUTER INTERFACE

Summary: Efficient P300 detection in brain-computer interfaces (BCIs) remains a challenge due to low signal-to-noise ratio (SNR) and the high computational cost of existing deep learning models, which limits their deployment in real-time applications. To address these issues, we propose MSTA-CNN, an ultra-li...

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Detection of linear dependencies in EEG oscillatory sources via PARALIND and block-term decomposition: A comparative study

👤 Leonardo Jose Trejo
📅 2026-11-10 📖 1 min 📝 60 words

BRAIN COMPUTER INTERFACE

Summary: Narrowband oscillatory rhythms in electroencephalogram (EEG) signals are fundamental signatures of neural dynamics, crucial for brain-computer interfaces (BCIs) and monitoring neurological disorders. While traditionally detected and analyzed via Independent or Principal Component Analysis in the tim...

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Lower frequency ECoG features improve longitudinal stability of speech and hand-grasp detection in a patient with progressive ALS

👤 Dean J Krusienski 📅 2026-11-10 📖 1 min 📝 54 words BRAIN COMPUTER INTERFACE

Summary: Achieving long-term decoding stability is a critical challenge for chronic
intracranial brain-computer interfaces (BCIs). High-gamma (HG) activity from
intracranial recordings is the standard signal for BCI control, and lower-frequency features
can enhance HG decoding performance. This s...

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A collaborative network of bi-Mamba and CAFS for multi-task classification of unilateral lower limb

👤 Junhua Li 📅 2026-11-10 📖 1 min 📝 66 words BRAIN COMPUTER INTERFACE

Summary: Accurate decoding of motor-related electroencephalography signals remains challenging in brain-computer interface research. To date, motor imagery of the upper limbs has been extensively investigated, while research on the lower limbs remains limited. Meanwhile, multiple tasks involving the unilater...

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A theory of multilevel interactive equilibrium in neuroAI

Quanyan Zhu 2026-11-10 1 min 70 words BRAIN COMPUTER INTERFACE

Summary: We propose a class of game-theoretic framework for adaptive multi-agent intelligent systems. Unlike the classical game theory which assumes that strategies are primitive objects chosen by perfectly rational agents, the new framework provides a mathematical foundation for studying equilibrium in Neur...

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Interictal Electroencephalographic Characteristics of the Human Centromedian Thalamic Nucleus Recorded in Focal Epilepsy

Shuang Wang 2026-11-10 1 min 23 words BRAIN COMPUTER INTERFACE

Summary: CONCLUSIONS: CM SEEG demonstrates distinct, state-dependent electrophysiological dynamics across the sleep-wake cycle and preferential involvement in interictal epileptic activity compared with VL/VP nuclei.

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Case Report: Representational geometry of multi-digit coordination in the macaque grasping network for intracortical BCIs

Hansjörg Scherberger

2026-11-11 · 1 min · 65 words · BRAIN COMPUTER INTERFACE

Summary: CONCLUSION: In this animal, condition-dependent dimensionality was significantly higher in F5 than in AIP or M1, while alignment and reconstruction analyses showed descriptive area-specific patterns in how closely combined-finger activity resembled a simple combination of single-finger activity. For...

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Effects of brain-computer interface output modalities on motor recovery after stroke: a systematic review and network meta-analysis

👤 Lili Fang 📅 2026-11-11 📖 1 min 📝 54 words BRAIN COMPUTER INTERFACE

Summary: CONCLUSION: Brain-computer interface-triggered functional electrical stimulation showed a clearer signal of benefit vs. conventional rehabilitation than brain-computer interface-controlled robotic. However, the comparison between the two active modalities relies entirely on indirect evidence from a ...

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Building a safety assessment framework for brain-computer interfaces: A call for cell-free multi-omics monitoring

👤 Jilai Li 📅 2026-11-11 📖 1 min 📝 2 words BRAIN COMPUTER INTERFACE

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Prefrontal cortex activity of adolescents with problematic smartphone use in response to gains and losses: A functional near-infrared spectroscopy study.

📅 2026-09-10 📖 1 min 📝 237 words BEHAVIORAL NEUROSCIENCE

Summary: Problematic smartphone use (PSU) has become a widespread problem that is highly prevalent in adolescent population. Previous studies suggest that PSU is associated with impairment in reward processing system. Thus, increased sensitivity to rewards could make individuals more vulnerable to abusive be...

🔗 **Read full article:**
http://doi.org/10.1037/bne0000668

Topiramate and anterior cingulate cortex high-frequency stimulation mitigate aggression and promote sociability in socially isolated male mice.

📅 2026-08-10 📖 1 min 📝 230 words BEHAVIORAL NEUROSCIENCE

Summary: Excessive aggression induced by social isolation (SI) has been closely linked to dysfunction in prefrontal circuits, particularly the anterior cingulate cortex (ACC). Topiramate, an antiepileptic drug known to enhance GABAergic and inhibit glutamatergic transmission, has demonstrated antiaggressive ...

🔗 **Read full article:**
http://doi.org/10.1037/bne0000663

Is memory spared in monkeys with selective lesions of the hippocampus?

📅 2026-08-03 📖 1 min 📝 239 words BEHAVIORAL NEUROSCIENCE

Summary: Normal explicit/declarative memory requires a system of anatomically related structures in the medial temporal lobe that includes the hippocampus and the entorhinal, perirhinal, and parahippocampal cortex. Although extensive medial temporal lobe damage in primates causes robust anterograde amnesia, ...

🔗 **Read full article:**
http://doi.org/10.1037/bne0000658

Distal pup cues attenuate defensive reactivity and fear-potentiated startle magnitude in postpartum rats without altering sensorimotor gating.

📅 2026-07-09 📖 1 min 📝 241 words BEHAVIORAL NEUROSCIENCE

Summary: Using a novel behavioral paradigm that integrates fear-potentiated startle (FPS) and prepulse inhibition testing, this study examined how motherhood and infant cues impact innate defensive reactivity (as measured by the magnitude of basal startle reflex), attentional processing (as measured by prepu...

🔗 **Read full article:**
http://doi.org/10.1037/bne0000661

A novel rat spatial inference task reveals rapid deduction of transitive relationships using schemas and deliberation.

📅 2026-07-13 📖 1 min 📝 264 words BEHAVIORAL NEUROSCIENCE

Summary: Inferential reasoning is a vital cognitive ability that enables animals to navigate novel situations by leveraging existing relational knowledge of memory schema, with hypothesized roles of prefrontal cortical—hippocampal circuits. Transitive inference (TI) tasks test the ability of subjects to infe...

🔗 **Read full article:**
http://doi.org/10.1037/bne0000660

Monthly Updates [April]

📅 2026-04-01 📖 4 min 📝 819 words FMHY

Summary: INFO These update threads only contain major updates. If you're interested in seeing all minor changes you can follow our Commits Page on G...

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Monthly Updates [May]

📅 2026-05-01 📖 4 min 📝 852 words FMHY

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Monthly Updates [October]

📅 2026-10-01 📖 5 min 📝 1025 words FMHY

> **Summary:** <h1 id="wiki-updates" tabindex="-1">Wiki Updates </h1>   Re-ordered Live Sports based on the polls a...

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Monthly Updates [November]

📅 2026-11-01 📖 4 min 📝 933 words FMHY

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I built a Linux sandbox in Python, and then tried to break out of it

👤 /u/mraza007 📅 2026-11-11 📖 1 min 📝 59 words REDDIT PYTHON

> **Summary:** <!-- SC_OFF --> I use coding agents a lot and wanted to know what stops one from reading my credentials. So I built a sandbox in Python with no dependencies, then tried to break out of each layer. Post: <a href="https://muhammadraza.me/2026/building-agent-sandbox-python/">ht...

🔗 **Read full article:**
https://www.reddit.com/r/Python/comments/1xtr15j/i_built_a_linux_sandbox_in_python_and_then_tried/

Is Python a good choice for learning algorithms and data structures ?

👤 /u/Minimum-Ad7352 📅 2026-11-11 📖 1 min 📝 69 words REDDIT PYTHON

> **Summary:** <!-- SC_OFF --> Is python a good choice for learning algorithms and data structures? It has so many built-in methods and libraries that make things easy. Doesn't that make it harder to understand how things actually work under the hood? Would C or C++ be a better choice for this pu...

🔗 **Read full article:**
https://www.reddit.com/r/Python/comments/1xto03d/is_python_a_good_choice_for_learning_algorithms/

Gemini Decisions API

👤 iamdelirium 📅 2026-11-11 📖 1 min 📝 13 words HACKER NEWS

> **Summary:** Article URL: https://blog.google/innovation-and-ai/technology/developers-tools/gemini-decisions-api/ Comments URL: <a href="https://news.ycombinator.com/item?id=50456791">https://news.ycom...

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Retirement Savings Are Fueling the AI Bubble

👤 speckx 📅 2026-11-11 📖 1 min 📝 13 words HACKER NEWS

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Show HN: I recreated Papers Please but you're deciding which YC startups to fund

👤 bolau 📅 2026-11-11 📖 1 min 📝 13 words HACKER NEWS

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The Utah Teapot

👤 cobbal 📅 2026-11-11 📖 1 min 📝 13 words HACKER NEWS

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IBM Quantum Nighthawk Demonstrates New Levels of Speed and Quality

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Why are we still working this hard?

👤 Wilsoniumite 📅 2026-11-11 📖 1 min 📝 13 words HACKER NEWS

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Huawei releases MACaQ, its internal quantizer for local inference

👤 thefox96 📅 2026-11-11 📖 1 min 📝 13 words HACKER NEWS

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Emacs in the Age of Magic

👤 ggcr 📅 2026-11-11 📖 1 min 📝 13 words HACKER NEWS

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Tools for exploring the souls of large language models

👤 r00tanon 📅 2026-11-11 📖 1 min 📝 13 words HACKER NEWS

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GitHub Incident with Webhooks, Actions, etc.

👤 jbredeche 📅 2026-11-11 📖 1 min 📝 13 words HACKER NEWS

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Easily download files from the Open Science Framework with Papercheck

👤 noreply@blogger.com (Daniel Lakens) 📅 2025-07-22 📖 3 min 📝 765 words

TWENTY PERCENT STATISTICIAN

Summary: Researchers increasingly use the Open Science Framework (OSF) to share files, such as data and code underlying scientific publications, or presentations and materials for scientific workshops. The OSF is an amazing service that has contributed immensely to a changed ...

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http://daniellakens.blogspot.com/2025/07/easily-download-files-from-open-science.html

How I use AI in my work

noreply@blogger.com (Daniel Lakens)
2026-10-11 21 min 4292 words

TWENTY PERCENT STATISTICIAN

Summary: Tomorrow we will have a discussion in our lab about how to use AI as a scientist. I have increasingly started to use AI, having moved from Copilot Chat that our university provides last year, to some free credit on GitHub Copilot under their educational program, to Claude Pro integrated in Visual...

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http://daniellakens.blogspot.com/2026/10/how-i-use-ai-in-my-work.html

Astragalin ameliorates cognitive dysfunction in Alzheimer's disease model mice: Role of the iron metabolism-autophagy axis

1 min 33 words

BRAIN RESEARCH

Summary: Publication date: 1 February 2027 Source: Brain Research, Volume 1896 Author(s): Can Wang, Wanlan Huang, Weishan Zhang, Li Zhang, Chang Zhou, Yaqi Yang, Shuhan Wang, Jiahong Lin, Jing Liu, Guoying Li, Yonggang Yang, Yuxin Ma

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Aquaporin channels in cytotoxic brain edema following severe head trauma: Systematic review of molecular mechanisms and therapy

📖 1 min 📝 20 words NEUROSCIENCE JOURNAL

> **Summary:** Publication date: 9 January 2027 Source: Neuroscience, Volume 620 Author(s): Clayton Rawson, Moneebah Ashraf, Srivardhan Chirasani, Dorsa Farahnak, Michael Karsy, Mehrdad Pehlevani

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Behavioral and neurobiological effects of vicarious social defeat stress in rodents: a scoping review

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> **Summary:** Publication date: 9 January 2027 Source: Neuroscience, Volume 620 Author(s): Welington dos Santos Silva, Amanda Peixoto Silva, Luciano Lazaro Vieira Lima, Luiz Antonio Chaves Viana, Maria Goretti Silva Brito

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Bilirubin in depression: redox-inflammatory mechanisms, clinical associations, and biomarker potential

📖 1 min 📝 22 words NEUROSCIENCE JOURNAL

Summary: Publication date: 9 January 2027 Source: Neuroscience, Volume 620 Author(s): Aarti Ashok Khutale, Prerana Deepak Dalvi, Pratik Prashant Doshi, Aamod Mahesh Pathak, Swathi Suresh

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Establishment of an animal model of GPIAS-defined tinnitus-like behavior with normal auditory brainstem response threshold profiles

📖 1 min 📝 24 words NEUROSCIENCE JOURNAL

Summary: Publication date: 9 January 2027 Source: Neuroscience, Volume 620 Author(s): Rong Wang, Yangyang Yu, Qianru Yu, Rui Guo, Menghua Li, Yang Li, Shusheng Gong, Ke Liu

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Corrigendum to “Neuroprotective effects of lixisenatide and liraglutide in the MPTP mouse model of Parkinson’s disease” [Neuroscience 303 (2015) 42–50]

📖 1 min 📝 20 words NEUROSCIENCE JOURNAL

Summary: Publication date: 9 January 2027 Source: Neuroscience, Volume 620 Author(s): W. Liu, J. Jalewa, M. Sharma, G. Li, L. Li, C. Hölscher

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Concurrent prefrontal neurochemical effects of high-definition transcranial direct current stimulation: directional E-field associations with GABA⁺ and Glx

📖 1 min 📝 34 words NEUROIMAGE

Summary: Publication date: 1 December 2026 Source: NeuroImage, Volume 343 Author(s): Siyuan Fang, Yihui Cheng, Ying Tang, Yupeng Wu, Yulin Wen, Qian Wu, Meiyan Lin, Jianing Kang, Huifang He, Huanping Wang, Gaiying Li, Ying Shen, Jianqi Li

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Temporal coupling shapes the effects of prolonged muscle-tendon vibration on sensorimotor EEG activity during motor imagery

📖 1 min 📝 21 words NEUROIMAGE

> **Summary:** Publication date: 1 December 2026 Source: NeuroImage, Volume 343 Author(s): Nicolas Amiez, Aymeric Guillot, Florian Monjo, Florian Marchand, Sébastien Daligault, Franck Di Rienzo

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Same networks, different dynamics: parsimony in auditory cognitive task processing

📖 1 min 📝 18 words NEUROIMAGE

> **Summary:** Publication date: 1 December 2026 Source: NeuroImage, Volume 343 Author(s): Divesh Thaploo, Adebiyi Sobitan, Kristen Wingert, Atsuko Kurosu, Nadia Biassou

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Desync and connect: Noninvasive MEG evidence of long-range coupling between locally desynchronized regions

📖 1 min 📝 12 words NEUROIMAGE

> **Summary:** Publication date: 1 December 2026 Source: NeuroImage, Volume 343 Author(s): Daria Kleeva, Alexei Ossadtchi

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https://www.sciencedirect.com/science/article/pii/S1053811926006300?dgcid=rss_sd_all

Sex-related multivariate organization of glymphatic MRI markers in healthy adults

📖 1 min 📝 22 words NEUROIMAGE

> **Summary:** Publication date: 1 December 2026 Source: NeuroImage, Volume 343 Author(s): Rui Wang, Shulan Wu, Jiajun Xu, Xiaoqi Huang, Simin Zhang, Qiyong Gong, Huaiqiang Sun

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INSULIN MODULATES FOOD VALENCE THROUGH ASTROCYTES

Zegarra-Valdivia, J. A., Fernandez de Sevilla, E., Khan, M. Z., Pignatelli, J., Torres Aleman, I.

2026-11-11 1 min 214 words BIORXIV NEUROSCIENCE

Summary: Brain cells involved in assigning affective value (i.e.: valence) to stimuli modulating brain states are not yet fully identified. Since astrocytes modulate brain states, and insulin regulates feeding behavior and exerts its central actions in part through astrocytes, we analyzed the role of the ins...

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https://www.biorxiv.org/content/10.64898/2026.11.10.765591v1?rss=1

Synaptotagmin 7 differentially regulates synaptic vesicle trafficking at tonic and phasic synapses

Inouye, M. O., Ackbergenova, Y., Guan, Z., Crane, A., Quinones-Frias, M. C., Littleton, J. T.

2026-11-11 1 min 144 words BIORXIV NEUROSCIENCE

Summary: Synaptic vesicle (SV) release probability can differ across neuronal subtypes, but the molecular mechanisms establishing such differences remain incompletely understood. In the current study we demonstrate that Synaptotagmin 7 (SYT7), a calcium-sensor regulating SV trafficking, is enriched at Drosop...

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Aperiodic Exponent as an EEG Marker of Interoceptive Attention during Gastric Fullness

👤 Liu, S., Datta, N., Lock, J. D., Coleman, T. P.

📅 2026-11-11 📖 1 min 📝 149 words

BIORXIV NEUROSCIENCE

Summary: Gastric interoceptive attention to internal, bodily signals such as hunger, satiety, and fullness are important for regulating feeding behaviors. However, abnormal processing or sensitivity to these can lead to disorders that result in overly restrictive or excessive consumption, highlighting the ne...

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Interpreting computational mechanism from multi-area neural recordings

👤 Versteeg, C., Michaels, J. A., Kashefi, M., Vermani, A., Le, T., Pruszynski, J. A., Pandarinath, C.

📅 2026-11-11 📖 1 min 📝 149 words

BIORXIV NEUROSCIENCE

Summary: Distilling computational mechanisms from neural recordings requires disambiguating intrinsic dynamics from external inputs. Here, we introduce Dynamically Interacting Systems (DynISys), a multi-compartment dynamical-systems model that infers per-area dynamics and structured inter-area communication ...

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A Complete Three-Dimensional Spiral Ganglion Neuron Map in a Human Adult using a Deep Learning Pipeline

👤 Giardina, C. K., O'Malley, J. T., Hall, A. K., Dharmarajan, A., Eckhard, A. H., Liberman, C., Arenberg, J. G., Quesnel, A. M.

📅 2026-11-11 📖 1 min 📝 241 words BIORXIV NEUROSCIENCE

Summary: The quantity and distribution of spiral ganglion neurons (SGNs) greatly influence speech and hearing ability. Historically, approaches to quantify SGNs from archival human temporal bones (TBs) have utilized only every 10th section to estimate the total SGN count. The current study evaluates a novel ...

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A Glutamate-Neuropeptide Circuit Converts Transient Mate Contact into a Persistent Local Search State in Caenorhabditis elegans Males

👤 Chen, Y.-H., Lin, Y.-C., Chen, T.-Y., You, K.-A., Ma, W.-E., Lai, Y.-S., Wang, C., Yang, J.-B., Chen, C.-H.

📅 2026-11-11 📖 1 min 📝 186 words BIORXIV NEUROSCIENCE

Summary: Transient sensory encounters can trigger behavioral states that persist long after the initiating stimulus has disappeared, yet how neural circuits initiate and sustain these states remains poorly understood. Here, we show that brief contact with a suitable mate induces a persistent local search sta...

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AKAP-CaN-NFATc3 signaling drives persistent inhibitory long-term potentiation

Wolfe, S. E., Zent, K. H., Garcia, J. D., Gookin, S. E., Dell'Acqua, M. L., Smith, K. R.

2026-11-11

1 min

247 words

BIORXIV NEUROSCIENCE

Summary: GABAergic inhibitory synaptic plasticity has critical roles in neuronal function, synaptic plasticity, and cognition. Activity-dependent alterations in gene expression are essential for generating long-term changes in inhibitory postsynaptic strength, yet the signaling mechanisms that drive inhibito...

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Sensory context gates oxytocin effects on auditory processing in macaques

Froesel, M., Wang, H., Vanduffel, W., Zhu, Q.

2026-11-11

1 min

155 words

BIORXIV NEUROSCIENCE

Summary: Oxytocin shapes social perception, yet its neural effects vary substantially across experiments and clinical trials yield inconsistent results. A key unresolved question is whether oxytocin modulates early sensory processing in primates as a function of sensory context. Using high-resolution pharmac...

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High spatiotemporal resolution in vivo imaging reveals preferential PACAP38 action on dural vasculature and a functional therapeutic readout

👤 Sperling, L. J. W., Degel, C., Zitelli, K., Zapata, J., Ollerenshaw, D. R., Bruun, T., Janjua, T., Wichern, F., Christensen, S. L. T., Asuni, A. A., Nassi, J., Hall, B. J., Botta, P.

📅 2026-11-11 📖 1 min 📝 151 words BIORXIV NEUROSCIENCE

Summary: PACAP38-induced cephalic vasodilation is used to assess functional PACAP pathway inhibition in humans, but the vascular populations underlying this response are not fully resolved. Defining these compartments in a preclinical model could help connect clinical pharmacodynamic responses with the under...

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Differential effects of AMPA receptor inhibition on seizure-like events in murine and human brain slices

Arnold, F., Podesta, A., Xu, R., Onken, J., Thomale, U.-W., Kaindl, A. M., Sauvigny, T., Kalbhenn, T., Simon, M., Bien, C., Alle, H., Geiger, J. R. P., Holtkamp, M., Plested, A. J. R., Fidzinski, P.

2026-11-11 1 min 246 words BIORXIV NEUROSCIENCE

Summary: Epilepsy is a frequent neurological condition with a worldwide prevalence of 0.5-1%. Current therapies are limited by adverse effects as well as drug resistance. This highlights the need for novel, target-specific antiseizure medications with improved efficacy and fewer side effects. AMPA (-amino-3-...

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The encephalitis blind spot of neglected orthoflaviviruses circulating in Brazil and mimicry of autoimmune CNS disorders

2026-11-11 1 min 0 words NATURE NEUROSCIENCE SUBJECTS

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A novel fluorescent ocular tracer enables detection of TDP-43 pathology in ALS and related TDP-43 proteinopathies using retinal imaging

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An experimental study on the neurological effects of hand-arm vibration: considering vibration frequency, ambient temperature, and wrist posture

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Experience modifies the synaptome architecture in the developing and adult mouse brain

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Evacuation behaviour under smoke conditions in a real-scale road tunnel experiment in Poland

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Physical fitness is prospectively associated with Chinese language performance in primary school children over ten months

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Aging remodels the suprachiasmatic nucleus and weakens ultradian rhythms and activity–temperature coupling

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Long-term training continually fine tunes motor memories

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Frequency-tagged fMRI reveals dual mechanisms of attentional enhancement and suppression

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https://www.nature.com/articles/s41467-026-79329-4

Correction: The PREP pipeline: standardized preprocessing for large-scale EEG analysis

👤 Kay A. Robbins 📅 2026-10-12 📖 1 min 📝 0 words FRONTIERS NEUROINFORMATICS

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https://www.frontiersin.org/articles/10.3389/fninf.2026.2015651

EEG emotion-recognition datasets for neuroarchitecture: an appraisal of openness, AI-readiness and ontology maturity

Han Jong Jun 2026-10-20 1 min 341 words FRONTIERS NEUROINFORMATICS

Summary: IntroductionNeuroarchitecture studies how designed space shapes emotion, often with electroencephalography (EEG), and computational models that decode emotion from EEG must be trained on labelled recordings. Yet none of the widely shared EEG emotion datasets was recorded while people moved through d...

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Efficient neuroanalytic local–global learning with dual-branch CNN–lightweight Swin Transformer for Alzheimer's disease analysis

Arshad Ali 2026-10-29 1 min 287 words FRONTIERS COMPUTATIONAL NEUROSCIENCE

Summary: Alzheimer's disease (AD) is one of the main causes of dementia, where early detection is critical for postponing the progression of the disease and timely treatment. Recent computer-aided diagnosis using deep learning with structural MRI has shown optimal performance; however, the rising number of m...

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Virtual brain models: a roadmap towards effective brain digital twins

👤 Viktor Jirsa 📅 2026-10-29 📖 1 min 📝 205 words FRONTIERS COMPUTATIONAL NEUROSCIENCE

Summary: Virtual Brain Models (VBMs) are emerging computational tools promising to revolutionize personalised and precision medicine. VBMs reproduce brain functional dynamics through a generative model that can be constructed and optimized against single subject data, thereby generating what is called a brai...

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Dolphin-inspired adaptive coordination for online multi-robot path planning in dynamic environments

👤 Lijuan Chen 📅 2026-10-29 📖 1 min 📝 200 words FRONTIERS NEUROROBOTICS

Summary: IntroductionOnline multi-robot path planning requires repeated joint motion decisions amid static obstacles, moving obstacles, and other robots.MethodsThis study introduces the Dolphin Cooperative Foraging Optimizer (DCFO), a population-based method that adapts its search strategy within each replan...

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From hemodynamics to function: the role of fNIRS and hybrid EEG-fNIRS systems in next-generation assistive technologies for rehabilitation

Vidhya Sampath 2026-11-11 1 min 220 words FRONTIERS NEUROSCIENCE

Summary: Assistive technologies (AT) serve as fundamental tools for rehabilitation because they help people with neurological or physical disabilities to restore their abilities, independence and their overall life satisfaction. Existing AT control methods, which include manual operation, eye tracking, voice...

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Real-time quaking-induced conversion detection of pathological prion protein in skeletal muscle, nasal and neural tissues of cattle with natural L-BSE

Maria Mazza 2026-11-11 1 min 195 words FRONTIERS NEUROSCIENCE

Summary: IntroductionBovine spongiform encephalopathy (BSE) includes classical BSE (C-BSE) and two atypical forms, high-type (H-BSE) and low-type (L-BSE). Compared with C-BSE, the peripheral distribution and tissue tropism of atypical BSE remain poorly characterized. The apparent spontaneous occurrence of at...

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Multineuron whole-cell patch-clamp dataset of layer 5 pyramidal cell synaptic connectivity in juvenile rat somatosensory cortex

Henry Markram 2026-11-11 1 min 268 words FRONTIERS NEUROSCIENCE

Summary: Reconstructing the wiring rules of the neocortical microcircuit ideally requires simultaneous electrophysiological access to many neighboring neurons together with a precise record of the stimuli that probe their connections. This dataset extends the whole-cell patch-clamp recordings from Perin, Ber...

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Orexinergic descending modulation of pain and itch

Tomoyuki Kuwaki 2026-11-11 1 min 246 words FRONTIERS NEUROSCIENCE

Summary: Descending pain control dynamically regulates nociceptive transmission through brainstem and spinal circuits according to behavioral and physiological demands. Within this system, accumulating evidence identifies hypothalamic orexin neurons as an important component of endogenous pain control. Orexi...

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Human cortical high-gamma power scales with movement rate in healthy participants and stroke survivors

👤 Fanny Quandt 📅 2025-01-09 📖 1 min 📝 65 words OOSTENVELD ROBERT

Summary: Motor cortical high-gamma oscillations (60-90 Hz) occur at movement onset and are spatially focused over the contralateral primary motor cortex. Although high-gamma oscillations are widely recognized for their significance in human motor control, their precise function on a cortical level remains el...

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NIRS-BIDS: Brain Imaging Data Structure Extended to Near-Infrared Spectroscopy

👤 Luca Pollonini 📅 2025-01-27 📖 1 min 📝 70 words OOSTENVELD ROBERT

Summary: Functional near-infrared spectroscopy (fNIRS) is an increasingly popular neuroimaging technique that measures cortical hemodynamic activity in a non-invasive and portable fashion. Although the fNIRS community has been successful in disseminating open-source processing tools and a standard file forma...

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Pseudonymisation of neuroimages and data protection: Increasing access to data while retaining scientific utility

Lyuba Zehl 2025-06-26 1 min 67 words OOSTENVELD ROBERT

Summary: For a number of years, facial features removal techniques such as 'defacing', 'skull stripping' and 'face masking/blurring', were considered adequate privacy preserving tools to openly share brain images. Scientifically, these measures were already a compromise between data protection requirements a...

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Cycling on the Freeway: The perilous state of open-source neuroscience software

Tim M Tierney 2025-08-13 1 min 74 words OOSTENVELD ROBERT

Summary: Most scientists need software to perform their research (Barker et al., 2020;Carver et al., 2022;Hettrick, 2014;Hettrick et al., 2014;Switters & Osimo, 2019), and neuroscientists are no exception. Whether we work with reaction times, electrophysiological signals, or magnetic resonance imaging data, ...

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Optimal configuration of on-scalp OPMs with fixed channel counts

👤 Robert Oostenveld 📅 2025-08-13 📖 1 min 📝 69 words OOSTENVELD ROBERT

Summary: Recent technological developments have brought optically pumped magnetometers (OPMs) within reach of the larger neuroscientific community. The current state-of-the-art consists of whole-head systems that measure the magnetic field at >100 locations. OPM sensors can be constructed to measure the fiel...

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ARTEM-IS for ERP: Agreed Reporting Template for EEG Methodology-International Standard for Event-Related Potential Experiments

👤 Suzy J Styles 📅 2025-12-08 📖 1 min 📝 73 words OOSTENVELD ROBERT

Summary: The choices we make during the recording, preprocessing and analysis of event-related potentials (ERP) data can affect study outcomes. As such, it is critical that they are transparently reported to allow for reproducibility. Yet, systematic reviews of reporting practices in the field have shown tha...

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Eye-Tracking-BIDS: the Brain Imaging Data Structure extended to gaze position and pupil data

👤 Julia-Katharina Pfarr 📅 2026-02-12 📖 1 min 📝 68 words OOSTENVELD ROBERT

Summary: The Brain Imaging Data Structure (BIDS) is a widely adopted, community-driven standard to organize neuroimaging data and metadata. Although numerous extensions have been developed to incrementally extend coverage to new modalities and data types, an unambiguous, granular specification for eye-tracki...

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Facilitating unusual bodily experiences and out-of-body experiences across wakefulness and sleep: A high-density EEG and neurophenomenology study

👤 Robert Oostenveld 📅 2026-02-12 📖 1 min 📝 67 words OOSTENVELD ROBERT

Summary: Unusual bodily experiences (UBEs) are illusory bodily perceptions that are not coherent with typical wakeful experiences, including flying or floating sensations, body distortions, and out-of-body experiences. This study examines UBEs in a controlled sleep laboratory using meditation and light stimu...

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The Impact of Brain Tumors and Craniotomy Lesions on Scalp EEG

👤 Vitória Piai 📅 2026-02-17 📖 1 min 📝 65 words OOSTENVELD ROBERT

Summary: Electroencephalography (EEG) is widely used in both research and clinical settings, yet its accuracy can be significantly impacted by subject-specific anatomical anomalies such as brain lesions and skull defects. This study investigates the effects of glioma-related brain lesions and craniotomy-indu...

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EMG-BIDS: an extension to the Brain Imaging Data Structure for electromyography

👤 Arnaud Delorme 📅 2026-08-01 📖 1 min 📝 64 words OOSTENVELD ROBERT

Summary: Electromyography (EMG) is fundamental to clinical assessment, rehabilitation, neuromuscular research, and human-machine interfaces. Despite decades of use, no widely adopted standard exists for organizing and sharing EMG data, limiting reusability and large-scale data aggregation. We present EMG-BID...

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Prior-guided whole-slide survival prediction with soft histopathologic priors from vision-language foundation models

👤 Zhihui Feng
📅 2026-11-09 📖 1 min 📝 62 words

LOW VISION

Summary: Whole-slide image (WSI) survival prediction is complicated by heterogeneous tumor morphology, where prognostic evidence is distributed across tumor, stroma, immune infiltration, necrosis, and residual tissue regions and often appears as mixed patch-level tissue evidence. Existing models commonly agg...

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Reduced magnocellular-biased visual responses are associated with working memory and linguistic difficulties in Chinese developmental dyslexia: A hypothesized visual-cognitive cascade

👤 Tsz Wing Leung
📅 2026-11-09 📖 1 min 📝 66 words LOW VISION

Summary: Developmental dyslexia (DD) affects 5-10% of children worldwide. Although phonological theories predominate, they may not fully explain reading difficulties in orthographically dense scripts such as Chinese. We investigated whether Chinese children with DD exhibit differences in early visual cortica...

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During natural vision, semantic novelty modulates fixation-related processing in primate cortex

👤 Lucas C Parra 📅 2026-11-09 📖 1 min 📝 74 words LOW VISION

Summary: We sample visual scenes with short gaze fixations separated by saccades. While low-level integration is known, semantic integration of foveal vision across fixations remains unclear. We hypothesized that the brain responds to changes in semantic information from one fixation to the next, and therefo...

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Health and welfare professionals' global contribution to the International Classification of Functioning, Disability and Health Core Set Vision Loss

👤 Dominique VAN DE Velde 📅 2026-11-10 📖 1 min 📝 64 words LOW VISION

Summary: CONCLUSIONS: A visual impairment exerts a broad and multidimensional impact on functioning, emphasizing the importance of emotional well-being, daily activities, and environmental supports. Regional differences highlight the need for culturally sensitive approaches in assessment and intervention. Th...

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Federated Learning: Envisioning a Pathway to More Accessible and Equitable Cleft Care

👤 Ghulam Qadir Fayyaz 📅 2026-11-10 📖 1 min 📝 66 words LOW VISION

Summary: Cleft lip and palate represents the most common congenital craniofacial anomaly worldwide, with low- and middle-income countries (LMICs) bearing nearly 84% of the global disease burden. Despite the instrumental role of global nongovernmental organizations (NGOs) in expanding access to cleft care acr...

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Dichoptic viewing reveals distinct low- and high-level mechanisms for temporal phase discrimination

👤 Keith A Schneider 📅 2026-11-10 📖 1 min 📝 68 words LOW VISION

Summary: Conscious awareness of temporal visual structure is limited to frequencies below ~10 Hz. Between ~10 and 60 Hz, flicker is visible, but temporal phase cannot be accurately discriminated; above ~60 Hz, stimuli are perceptually fused. Accurately estimating the cognitive limit of phase discrimination r...

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Trends in Near Vision Loss and Age-Sex-SDI Disparities in Belt and Road Countries Over Thirty Years

Xiaopan Li 2026-11-10 1 min 55 words LOW VISION

Summary: CONCLUSIONS: This study represents the first comprehensive analysis of NVL burden and disparities across BRI countries, revealing persistent inequalities by age, sex, and SDI. The findings reflect the diverse socio-economic and health-care contexts of BRI countries and highlight the urgent need for ...

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Feasibility of robotic-assisted surgery for nutcracker syndrome: a systematic review and single-arm meta-analysis

Jiang-Hua Zheng 2026-11-10 1 min 60 words LOW VISION

Summary: Nutcracker syndrome (NCS) results from compression of the left renal vein, and refractory symptoms require surgical reconstruction. Robotic platforms provide magnified three-dimensional vision and articulated instruments that are theoretically advantageous for vascular dissection and anastomosis wit...

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Segmentation, Reconstruction, and Grasp-Pose Estimation Using an RGB-D Camera for Robotic Apple Harvesting

Jianqiang Lu 2026-11-10 1 min 58 words LOW VISION

Summary: Accurate segmentation, three-dimensional reconstruction, and grasp-pose estimation of fruits and surrounding structures are essential for robotic harvesting in complex orchard environments. However, occlusions caused by branches, support wires, and irregular canopy structures remain major challenges...

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Intelligent rendering optimization for animated scenes using reinforcement learning and computer vision

Li Jia 2026-11-10 1 min 63 words LOW VISION

Summary: Real-time rendering of complex animated scenes is computationally intensive and often struggles to balance visual fidelity and performance in dynamic or densely populated environments, as conventional pipelines lack adaptive resource optimization, leading to reduced frame rates and degraded visual q...

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A single session of high-definition transcranial direct current stimulation does not modulate effects of knee two-point discrimination training or sensorimotor function in healthy adults: Double-blind randomised controlled trial

👤 Paul W Hodges
📅 2026-07-30 📖 1 min 📝 53 words

TACTILE ACUITY

Summary: CONCLUSION: A single 20-minute session of 1 mA HD-tDCS, with or without brief TPD training, does not acutely modify knee somatosensory, sensorimotor, or functional performance in healthy adults. Existing evidence at the hand may not translate to the lower limb. Future work should investigate higher-...

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Variation in glabrous skin physiology across anatomical locations on the human hand

👤 Martina C Meinke 📅 2026-08-13 📖 1 min 📝 76 words TACTILE ACUITY

Summary: Skin physiology varies greatly between anatomical locations, with impact on tactile sensitivity. Research on the variations in morphology and physiology between different anatomical locations of glabrous skin is limited so far. This study investigates the morphology and physiology of glabrous skin o...

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Beyond Hamstring Shortness: Continuous Flexibility Is Associated With Balance and Stability but Not Sensory Measures-A Cross-Sectional Study

👤 Senanur Akıncı 📅 2026-08-14 📖 1 min 📝 66 words TACTILE ACUITY

Summary: CONCLUSIONS: Greater hamstring flexibility was associated with better dynamic balance performance and showed a weaker association with postural stability, whereas no robust associations were observed with sensory measures of postural control. These findings suggest that hamstring flexibility may be ...

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Resting-state EEG microstate dynamics reflect individual differences in tactile angle discriminability

Liu Yang 2026-08-19 1 min 63 words TACTILE ACUITY

Summary: Tactile spatial acuity, the ability to discriminate fine spatial details, is a fundamental sensory capacity modulated by both peripheral and central mechanisms. Here, we investigated whether resting-state EEG microstate dynamics are associated with individual differences in tactile spatial acuity. F...

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A grating orientation task with controlled dome elevation speed and contact force: stimulus conditions and day-to-day reliability

Hideaki Onishi 2026-09-25 1 min 66 words TACTILE ACUITY

Summary: The grating orientation discrimination task (GOT) is typically used to assess tactile spatial acuity. However, performance is influenced by stimulus speed, contact force between the stimulator and the skin, and the method of discrimination. This study aimed to explore suitable stimulus conditions fo...

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Epidermal microstructure and tactile sensitivity in the glabrous skin of hands and feet

👤 Hannes P Saal 📅 2026-09-25 📖 1 min 📝 69 words TACTILE ACUITY

Summary: Glabrous skin on the human palms and soles is specialized for mechanical interaction and is characterized by papillary ridges, an absence of hair follicles, and a thick stratum corneum. Although gross morphological differences, such as the increased thickness of the plantar stratum corneum, are well...

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Neurotrauma During Military Training: Results of the INVestigating traIning assoCiated blasT pAthology (INVICTA) Study

👤 Walter S Carr 📅 2026-10-01 📖 1 min 📝 67 words TACTILE ACUITY

Summary: Concern is growing that subconcussive blast exposure during military exercises such as heavy weapons training (HWT) may adversely affect brain function. HWT involves firing shoulder-mounted weapons that can expose the gunner, assistant gunner, and instructors (range safety officers [RSOs]) to blast ...

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Altered body representation in chronic non-specific low back pain: a testable framework linking body-schema-related disturbance, corticomotor alterations, and altered pain processing

👤 Zhifei Xie
📅 2026-10-24 📖 1 min 📝 66 words

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Summary: Low back pain is a leading contributor to disability worldwide; in chronic non-specific low back pain (CNSLBP), the processes sustaining pain and disability in the absence of identifiable structural pathology remain incompletely understood. Three partially overlapping research lines have documented ...

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Hyperexcitable primary somatosensory cortex in the MitoPark mouse model of Parkinson's disease

Alan J Emanuel 2026-10-25 1 min 67 words TACTILE ACUITY

Summary: The majority of Parkinson's disease patients suffer from decreased tactile acuity or neuropathic pain, both of which contribute to low quality of life and likely exacerbate motor dysfunction. How Parkinson's disease pathology relates to altered somatosensation remains unclear. One potential contribu...

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The Relationship Between Occlusal Tactile Acuity, Bite Force and Tooth Wear

Birgitta Häggman-Henrikson 2026-11-03 1 min 52 words TACTILE ACUITY

Summary: CONCLUSIONS: Bite force may be a clinically relevant factor associated with accumulated tooth wear, whereas OTA was not independently associated with tooth wear. Longitudinal studies are needed to determine whether bite force can predict tooth wear progression and whether more comprehensive assessme...

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Impact of Braille Booklet, Audio-Tactile Performance and Caregiver-supervised Training on Oral Hygiene Status, Knowledge, Attitude and Practices among Visually Challenged Children- A Randomized Clinical Trial

👤 Priyanga Chandrasekaran
📅 2026-09-22 📖 1 min 📝 52 words

BRAILLE

Summary: CONCLUSION: Both caregiver-supervised training and audio tactile performance using specialized model method of oral health education proved to be more effective than the braille booklet group on plaque control and improvement in the oral hygiene status, oral health knowledge, attitude and practice i...

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Design and functional verification of an automated Perkins Brailler accessory with audio feedback

👤 Ali Alhaj 📅 2026-09-23 📖 1 min 📝 67 words BRAILLE

> **Summary:** CONCLUSIONS: The study did not include learners with visual impairment, Braille teachers, rehabilitation specialists, or other end users, and it did not collect quantitative measures of accuracy across the full character set, writing speed, embossing quality, latency, durability, safety, usability, ...

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Uncertainty-Aware structured decoding for Chinese Braille document transcription

👤 Haoran Sun 📅 2026-09-25 📖 1 min 📝 64 words BRAILLE

> **Summary:** Automatic transcription of braille documents remains challenging in document digitization settings, particularly under real imaging degradations such as uneven embossing, illumination variation, paper texture, skew, blur, and ghosting, which make braille dot presence uncertain and often cause error ...

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Clinical Characteristics and Epidemiology of Isolated Dystonia: Results from a US Claims Analysis

James E Thornton 2026-10-08 1 min 54 words BRAILLE

Summary: CONCLUSIONS: This study defines an isolated dystonia cohort in the US based on real-world evidence and systematic exclusion of secondary causes of dystonia. Estimated prevalence is within range of prior epidemiological studies, and the cohort showed increased rates of comorbid psychiatric disorders....

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Effects of multisensory tactile-auditory integration teaching on systematic anatomy learning in visually impaired medical students

Fanzhen Kong 2026-10-08 1 min 50 words BRAILLE

Summary: CONCLUSION: The tactile-auditory integrated training system significantly improves objective summative knowledge acquisition and subjective learning experiences in visually impaired medical students. This approach provides a scalable, cost-constrained, and replicable pedagogical model for inclusive ...

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Physical and Environmental Accessibility of Dental Services for Persons with Disabilities in Saudi Primary Health Care: A Cross-Sectional Study

👤 Estie Kruger 📅 2026-10-16 📖 1 min 📝 66 words BRAILLE

Summary: Background/Objectives: Persons with disabilities (PWDs) may experience barriers when accessing oral health care, including those related to the physical and sensory environment of the dental-care setting. In this study we assessed the accessibility of primary health care centres (PHCs) providing den...

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Visible Light Driven Diselenide Chemistry for Simultaneous 3D Patterning and Mechanical Reinforcement

👤 Huaping Xu 📅 2026-10-16 📖 1 min 📝 63 words BRAILLE

Summary: Three-dimensional (3D) patterning on polymer surfaces is typically achieved by photolithography or related techniques, which often involve complex procedures, high-energy ultraviolet light, and risk of material damage. Here, we report a visible light-driven strategy that exploits the dual functional...

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An accessible pregnancy test for blind and low-vision users

👤 Vanja Garaj 📅 2026-10-16 📖 1 min 📝 63 words BRAILLE

Summary: CONCLUSIONS: The findings suggest that accessible pregnancy testing must prioritise not only non-visual result reading but also privacy, dignity, reproductive autonomy and independent health management. These results represent the preliminary evidence of acceptability rather than a conclusive test o...

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Genome-Informed Loop-Mediated Isothermal Amplification (LAMP) Assays for Detection and Differentiation of Ralstonia pseudosolanacearum Phylotypes I and III

👤 Jarred Yasuhara-Bell 📅 2026-10-27 📖 1 min 📝 76 words BRAILLE

Summary: The Ralstonia solanacearum species complex (RSSC) causes bacterial wilt in a broad range of economically important hosts. The RSSC has been reclassified into three species, of which R. solanacearum sequevars 1 and 2 in phylotype IIB are designated as Select Agents in the U.S. Many rapid tests for "R...

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Encoding tactile stimuli for Braille recognition with organoids

👤 Benjamin Ward-Cherrier 📅 2026-10-31 📖 1 min 📝 62 words BRAILLE

Summary: This study proposes a transferable encoding strategy that maps tactile sensor data to electrical stimulation patterns, enabling neural organoids to perform an open-loop artificial tactile Braille classification task. Human forebrain organoids cultured on a low-density microelectrode array (MEA) are ...

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Concurrent prefrontal neurochemical effects of high-definition transcranial direct current stimulation: directional E-field associations with GABA+ and Glx

👤 Jianqi Li 📅 2026-11-05 📖 1 min 📝 58 words TDCS TACS TRNS

Summary: Transcranial direct current stimulation (tDCS) is a promising neuromodulatory technique for treating neuropsychiatric disorders. However, its neurochemical mechanisms remain poorly characterized, and clinical efficacy is limited by substantial inter-individual variability. This study used concurrent...

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Anodal tDCS alters perceived visual contrast in humans

Sang Wook Hong 2026-11-06 1 min 62 words TDCS TACS TRNS

Summary: Cortical gain modulation is thought to alter perceptual experience, most prominently demonstrated by the effects of spatial attention on contrast appearance. Whether externally induced changes in cortical excitability produce comparable perceptual consequences remains unclear. Here, we tested whethe...

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Evaluating OPTION: Benefits and Pitfalls of a Youth Trafficking Diversion Court in Florida

Joan A Reid 2026-11-06 1 min 67 words TDCS TACS TRNS

Summary: Trafficking diversion courts (TDCs) provide an alternative approach to supporting trafficked youth by promoting recovery, preventing revictimization, and diverting youth from the justice system. The current study examines the benefits and challenges of the OPTION (Offering Potentially Trafficked Ind...

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Amelioration of diabetic autonomic neuropathy symptoms following a course of transcranial direct current stimulation

👤 Ahmad Alipour 📅 2026-11-06 📖 1 min 📝 62 words TDCS TACS TRNS

Summary: Cardiovascular autonomic neuropathy (CAN) is a severe and potentially life-threatening complication of diabetes. This randomized, sham-controlled, three-arm trial evaluated the efficacy of anodal transcranial direct current stimulation (tDCS) targeting the left dorsolateral prefrontal cortex for imp...

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From Single Cells to the Circulating Tumor Microenvironment: Label-Free Size-Selection and a Morphological Taxonomy for Liquid Biopsy in Cytopathology

👤 Satoru Kobayashi 📅 2026-11-09 📖 1 min 📝 25 words TDCS TACS TRNS

Summary: CONCLUSION: Size-selection converts a liquid biopsy into a cytology slide, placing interpretation of the circulating tumor ecosystem within the existing competence of the diagnostic laboratory.

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Phase-Dependent Effects of Cerebellar Transcranial Alternating Current Stimulation on Lateral Vestibulospinal Tract Excitability

👤 Tsubasa Mitsutake
📅 2026-11-09 📖 1 min 📝 36 words

TDCS TACS TRNS

Summary: CONCLUSIONS: LVST excitability differed according to the phase of the ongoing cerebellar-targeted tACS waveform. These findings provide electrophysiologic evidence of an online phase-dependent effect on vestibulospinal excitability and a physiologic basis for future phase-specific neuromodulation st...

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Effects of Transcranial Direct Current Stimulation on Core Symptoms and Sleep Disturbances in Children With Autism Spectrum Disorder: A Meta-Analysis of Randomized Controlled Trials

👤 Meimei Yang
📅 2026-11-09 📖 1 min 📝 44 words

TDCS TACS TRNS

Summary: CONCLUSION: tDCS, most commonly delivered at 1 mA over the DLPFC, was associated with improvements in some ASD-related behavioral outcomes; preliminary evidence also suggested a potential benefit for sleep disturbances, although this finding was based on only two low-certainty studies and warrants c...

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No effects of single-session concurrent tDCS on explicit motor sequence learning in healthy adults with conventional 2 mA and tangential 6 mA currents

👤 Lucas C Parra
📅 2026-11-09 📖 1 min 📝 45 words

TDCS TACS TRNS

Summary: CONCLUSION: These findings, together with a survey of the existing literature, suggest that the effects of single-session concurrent tDCS on explicit motor sequence learning are not robust across studies. Our results underscore the necessity of rigorous methodological controls to prevent false disco...

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More than Quaternary Ammonium Compounds: First Evidence of Tertiary Amine Compounds as Novel Contaminants Ubiquitously Present in Indoor Dust

Runzeng Liu 2026-11-10 1 min 65 words TDCS TACS TRNS

Summary: Nitrogen-containing organic compounds have attracted global concern due to their widespread production and potential environmental risks. While the environmental occurrence of quaternary ammonium compounds (QACs) has been widely studied, research on tertiary amine compounds (TACs), another important...

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Combining transcranial electrical and magnetic stimulation: a scoping review of neurophysiological and clinical studies

Paul M Briley 2026-11-10 1 min 56 words TDCS TACS TRNS

Summary: CONCLUSIONS: Studies indicate that tES could enhance or reverse expected effects of rTMS. The greatest increase in cortical excitability occurred following excitatory rTMS combined with gamma transcranial alternating current stimulation (tACS) or with cathodal tDCS, consistent with effects of metapl...

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Prefrontal Activation and Cognitive Heterogeneity in Pediatric Sleep-Disordered Breathing

Amal Isaiah 2026-11-05 1 min 39 words FNIRS

> **Summary:** CONCLUSIONS AND RELEVANCE: In this cross-sectional study, task-related PFC activation was associated with SDB severity and moderated the severity-cognition association. fNIRS may help identify children whose cognitive profiles deviate from conventional severity metrics, thereby supporting individual...

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Mirror Neural System in Hands Motor Imagery with Pictures and Videos Guidance: An EEG-fNIRS Study

Mohamad Sawan 2026-11-05 1 min 64 words FNIRS

> **Summary:** Brain-computer interfaces (BCIs) achieved by motor imagery (MI) have been widely accepted as an effective tool in rehabilitative applications. Functional near-infrared spectroscopy (fNIRS) and electroencephalography (EEG) are two common implementation forms of BCIs, capable of monitoring neuronal po...

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Quality-aware mask-propagation graph attention for fNIRS-based bipolar disorder classification without channel interpolation

👤 Yajing Meng
📅 2026-11-05 📖 1 min 📝 58 words FNIRS

Summary: Bipolar disorder (BD) remains difficult to characterise objectively, motivating a neural marker to complement clinical assessment. Functional near-infrared spectroscopy (fNIRS) offers portable cortical haemodynamic measurement, but channel availability can vary across recordings because of bad chann...

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Brain-Behavior Coupling During Cognitive-Postural Dual-Tasking in Older Adults With Mild Cognitive Impairment: An fNIRS Study

Chuan Guo 2026-11-06 1 min 56 words FNIRS

Summary: CONCLUSIONS: Older adults with MCI exhibit altered brain-behavior coupling during cognitive-postural dual-task conditions involving sensory and postural constraints, characterized by altered relationships between prefrontal-sensorimotor cortical activation and behavioral performance. These findings ...

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Item-Specific Proportion Congruency Uncovers Different Forms of Attentional Control: An event-related fNIRS Study

Murat Perit Çakır 2026-11-07 1 min 70 words FNIRS

Summary: CONCLUSION: While behavioral results mostly support contingency learning, fNIRS data revealed ISPC-related activation in prefrontal channels with sensitivity to BA 46 (lateral, channel 1) and the BA 10/BA 11 overlap zone (medial, channels 10-12), with channel 11 emerging as a convergence point for b...

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Does the emotion regulation of self-directed humor engage the prefrontal cortex? An fNIRS study

Zhonglu Zhang 2026-11-07 1 min 68 words FNIRS

Summary: Previous research has revealed that self-directed humor (self-enhancing and self-defeating) can help regulate negative emotions and that the prefrontal cortex (PFC) plays a critical role in emotion regulation. However, it remains unknown whether emotion regulation of self-directed humor is associate...

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Effects of mini-maps with different reference frames on polarity-free 3D spatial navigation: a behavioural and fNIRS study

Xiaozhou Zhou 2026-11-07 1 min 62 words FNIRS

Summary: Manned spaceflight and deep-sea exploration extend human activities to extreme environments lacking gravitational and visual polarity cues, where spatial disorientation frequently occurs. Mini-map studies largely focus on terrestrial environments, while reference frame design for polarity-free 3D sp...

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Baduanjin improves motor function and cortical activation in children with developmental coordination disorder: a randomized controlled trial

👤 Wei Song 📅 2026-11-09 📖 1 min 📝 40 words FNIRS

Summary: CONCLUSIONS: Baduanjin training improved motor coordination and balance in children with DCD. The observed changes in activation within motor-related cortical regions provide preliminary support for a possible neurophysiological basis of these functional improvements, although the underlying mechani...

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Potential Applications of Brain-Computer Interface Technology in the Rehabilitation of Patients with Anxiety Disorders: A Narrative Review

Hu Lou 2026-11-10 1 min 58 words FNIRS

Summary: Anxiety disorders are among the most prevalent mental disorders worldwide and can cause persistent impairment in emotional regulation, cognition, and social functioning. Pharmacotherapy, psychotherapy, exercise, and non-invasive brain stimulation are widely used, yet their effectiveness may be const...

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Multimodal EEG-fNIRS complexity analysis can reveal the neurovascular coupling change with aging

Yicheng Zhu 2026-11-10 1 min 58 words FNIRS

Summary: CONCLUSION: Aging is associated with task-specific alterations in neural complexity, hemodynamic complexity, and the cross-modal temporal relationship between EEG and fNIRS signals. These findings suggest that age-related changes in NVC-related dynamics are better characterized as dynamic reorganiza...

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Potential Applications of Brain-Computer Interface Technology in the Rehabilitation of Patients with Anxiety Disorders: A Narrative Review

👤 Hu Lou 📅 2026-11-10 📖 1 min 📝 58 words BRAIN COMPUTER INTERFACE

Summary: Anxiety disorders are among the most prevalent mental disorders worldwide and can cause persistent impairment in emotional regulation, cognition, and social functioning. Pharmacotherapy, psychotherapy, exercise, and non-invasive brain stimulation are widely used, yet their effectiveness may be const...

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In Situ Displacement Reaction Enables a Conductive Metallized Hydrogel Interface for Smart Underwater Flexible Electronics

👤 Qigang Wang
📅 2026-11-10 📖 1 min 📝 60 words

BRAIN COMPUTER INTERFACE

Summary: Underwater wearable sensing is severely hampered by the intrinsic drawbacks of conventional materials, including water absorption, swelling, and electrical drift during long-term immersion. Here, we develop an anti-swelling hydrogel with a surface-metallized layer, which achieves strong stability of...

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Fast-Gelling, Adhesive, and Conductive Epidermal Electrode for Accurate Physiological Signal Monitoring and Real-Time Brain-Controlled Drone Flight

👤 Zhou Li 📅 2026-11-10 📖 1 min 📝 55 words BRAIN COMPUTER INTERFACE

Summary: Conductive hydrogel electrodes are vital components for achieving human-machine interfaces. However, the inherent trade-off between high conductivity and favorable mechanical/adhesive properties remains a significant challenge for hydrogel-based epidermal electrodes. Here, we developed a multifuncti...

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Bayesian Adaptive Stimulus Optimization in P300 Brain Computer Interfaces

👤 Boyla O Mainsah 📅 2026-11-10 📖 1 min 📝 26 words BRAIN COMPUTER INTERFACE

Summary: CONCLUSION: For computational simplicity in implementation, the optimal stimulus can generally be approximated by finding the stimulus that has a combined probability mass closest to 0.5.

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MSTA-CNN an ultra-lightweight multi-scale temporal-spatial attention convolutional neural network for high-performance P300 spellers

👤 Mousa Shamsi
📅 2026-11-10 📖 1 min 📝 62 words

BRAIN COMPUTER INTERFACE

Summary: Efficient P300 detection in brain-computer interfaces (BCIs) remains a challenge due to low signal-to-noise ratio (SNR) and the high computational cost of existing deep learning models, which limits their deployment in real-time applications. To address these issues, we propose MSTA-CNN, an ultra-li...

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Detection of linear dependencies in EEG oscillatory sources via PARALIND and block-term decomposition: A comparative study

👤 Leonardo Jose Trejo
📅 2026-11-10 📖 1 min 📝 60 words

BRAIN COMPUTER INTERFACE

Summary: Narrowband oscillatory rhythms in electroencephalogram (EEG) signals are fundamental signatures of neural dynamics, crucial for brain-computer interfaces (BCIs) and monitoring neurological disorders. While traditionally detected and analyzed via Independent or Principal Component Analysis in the tim...

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Lower frequency ECoG features improve longitudinal stability of speech and hand-grasp detection in a patient with progressive ALS

👤 Dean J Krusienski 📅 2026-11-10 📖 1 min 📝 54 words BRAIN COMPUTER INTERFACE

Summary: Achieving long-term decoding stability is a critical challenge for chronic
intracranial brain-computer interfaces (BCIs). High-gamma (HG) activity from
intracranial recordings is the standard signal for BCI control, and lower-frequency features
can enhance HG decoding performance. This s...

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A collaborative network of bi-Mamba and CAFS for multi-task classification of unilateral lower limb

👤 Junhua Li 📅 2026-11-10 📖 1 min 📝 66 words BRAIN COMPUTER INTERFACE

Summary: Accurate decoding of motor-related electroencephalography signals remains challenging in brain-computer interface research. To date, motor imagery of the upper limbs has been extensively investigated, while research on the lower limbs remains limited. Meanwhile, multiple tasks involving the unilater...

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A theory of multilevel interactive equilibrium in neuroAI

Quanyan Zhu 2026-11-10 1 min 70 words BRAIN COMPUTER INTERFACE

Summary: We propose a class of game-theoretic framework for adaptive multi-agent intelligent systems. Unlike the classical game theory which assumes that strategies are primitive objects chosen by perfectly rational agents, the new framework provides a mathematical foundation for studying equilibrium in Neur...

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Interictal Electroencephalographic Characteristics of the Human Centromedian Thalamic Nucleus Recorded in Focal Epilepsy

Shuang Wang 2026-11-10 1 min 23 words BRAIN COMPUTER INTERFACE

Summary: CONCLUSIONS: CM SEEG demonstrates distinct, state-dependent electrophysiological dynamics across the sleep-wake cycle and preferential involvement in interictal epileptic activity compared with VL/VP nuclei.

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Pydantic in 100 lines

/u/com_kieffer | 2026-11-11 | 1 min | 46 words | REDDIT PYTHON

> **Summary:** <!-- SC_OFF --> The title is obviously tongue in cheek, but in 100 lines we can get a quite a lot done to type-check dataclasses. https://thibaud.space/2026/10/pydantic-in-100-lines-part-1.html</p...

🔗 **Read full article:**
https://www.reddit.com/r/Python/comments/1xtcexq/pydantic_in_100_lines/

Wiimote in Python

/u/Ill_Beginning4410 | 2026-11-10 | 1 min | 54 words | REDDIT PYTHON

> **Summary:** <!-- SC_OFF --> Im having a ton of trouble trying to get a wiimote's data/Inputs. I've tried wiiuse and cwiid didnt seem to work. Idk if it has smt to do with me being a mac. <!-- SC_ON -->   submitted by   <a href="https://www.reddit.com/user/Ill_Beginning4410"> ...

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https://www.reddit.com/r/Python/comments/1xspb09/wiimote_in_python/

Next-Level Benchmarking Overfitting – Thomas Neumann

👤 artrax 📅 2026-11-11 📖 1 min 📝 13 words HACKER NEWS

> **Summary:** Article URL: http://databasearchitects.blogspot.com/2026/11/next-level-benchmarking-overfitting.html Comments URL: <a href="https://news.ycombinator.com/item?id=50452496">https://news.ycom...

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We need more goat sacrifices

👤 iamnothere 📅 2026-11-11 📖 1 min 📝 13 words HACKER NEWS

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Can a Linear Program Do Weight-Space Programming?

👤 emschwartz 📅 2026-11-11 📖 1 min 📝 13 words HACKER NEWS

> **Summary:** Article URL: https://nathan.rs/posts/weight-space-lp/ Comments URL: https://news.ycombinator.com/item?id=50452655 Points: 88 # Comments: 14

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Open research, stronger privacy: Making Firefox users harder to fingerprint

👤 Vinnl 📅 2026-11-11 📖 1 min 📝 13 words HACKER NEWS

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EventSourcingDB 1.3.0 Is Available

👤 goloroden 📅 2026-11-11 📖 1 min 📝 13 words HACKER NEWS

> **Summary:** Article URL: https://www.eventfoundation.io/resources/blog/eventsourcingdb-1-3-0-is-available Comments URL: <a href="https://news.ycombinator.com/item?id=50453080">https://news.ycombinator.com/it...

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https://www.eventfoundation.io/resources/blog/eventsourcingdb-1-3-0-is-available

Show HN: Godbolt-in-a-floating-window for Neovim (C, C++, Rust)

👤 0xdeadf1sh 📅 2026-11-11 📖 1 min 📝 22 words HACKER NEWS

> **Summary:** No internet connection needed. Uses a locally-installed compiler for assembly generation. <hr /> Comments URL: https://news.ycombinator.com/item?id=50453229 Points: 6 # Comments: 1

🔗 **Read full article:**
https://github.com/0xdeadf1sh/AssemblyForNeovim

I asked every TV manufacturer 17 questions about how they spy on you

👤 Brajeshwar 📅 2026-11-11 📖 1 min 📝 13 words HACKER NEWS

> **Summary:** Article URL: https://www.theverge.com/tech/1012372/tv-privacy-acr-questions Comments URL: https://news.ycombinator.com/item?id=50453474 Points: 40...

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https://www.theverge.com/tech/1012372/tv-privacy-acr-questions

Apple is killing its Support Community forum next week

👤 saikatsg 📅 2026-11-11 📖 1 min 📝 13 words HACKER NEWS

> **Summary:** Article URL: https://www.theverge.com/tech/1031507/apple-community-support-forum-shutting-down Comments URL: <a href="https://news.ycombinator.com/item?id=50453555">https://news.ycombinator.com/...

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https://www.theverge.com/tech/1031507/apple-community-support-forum-shutting-down

Atuin Threads: All your agent sessions on every machine

👤 linsomniac 📅 2026-11-11 📖 1 min 📝 13 words HACKER NEWS

> **Summary:** Article URL: https://blog.atuin.sh/threads-agent-sessions/ Comments URL: https://news.ycombinator.com/item?id=50453637 Points: 9 # Comments: 2

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Pharo in the Browser with WebAssembly

👤 xkriva11 📅 2026-11-11 📖 1 min 📝 13 words HACKER NEWS

> **Summary:** Article URL: https://massimo-nocentini.github.io/pharo-vm/world.html Comments URL: https://news.ycombinator.com/item?id=50453763 Points: 15 # Comm...

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Dogmatic Bayesianism Disorder

noreply@blogger.com (Daniel Lakens)
2025-12-06 4 min 910 words

TWENTY PERCENT STATISTICIAN

Summary: <p>Note: This humorously intended sarcastic blog post directly mimics classifications of psychological disorders in the Diagnostic and Statistical Manual of the American Psychological Association, but Dogmatic Bayesianism is not in the DSM-5 as an actual psychologic...

🔗 **Read full article:**
http://daniellakens.blogspot.com/2025/12/dogmatic-bayesianism-disorder.html

On the reliability and reproducibility of qualitative research

noreply@blogger.com (Daniel Lakens)
2026-02-08 14 min 2928 words

TWENTY PERCENT STATISTICIAN

Summary: <p class="MsoNormal">With my collaborators, I am increasingly performing qualitative research. I find qualitative research projects a useful way to improve my understanding of behaviors that I want to explore with other methods in the future. For example, some years ago I performe...

🔗 **Read full article:**
http://daniellakens.blogspot.com/2026/02/on-reliability-and-reproducibility-of.html

New Papers: Optimal Filter Settings for ERP Research

👤 Steve Luck 📅 2024-02-04 📖 2 min 📝 569 words ERP BOOT CAMP

> **Summary:** <p class="">Zhang, G., Garrett, D. R., & Luck, S. J. (in press). Optimal filters for ERP research I: A general approach for selecting filter settings. Psychophysiology. https://doi.org/10.1111/psyp.14531 [<a href="https://www...

🔗 **Read full article:**
https://erpinfo.org/blog/2024/2/4/optimal-filters

Q&A with Dr. Richard Carson, Professor of Biomedical Engineering and Radiology & Biomedical Imaging, Yale University and Yale School of Medicine

👤 Adriel Carridice 📅 2025-11-04 📖 1 min 📝 0 words BRAIN

🔗 **Read full article:**
https://brain.ieee.org/podcasts/qa-with-dr-richard-carson-professor-of-biomedical-engineering-and-radiology-biomedical-imaging-yale-university-and-yale-school-of-medicine/

Seeing Without Seeing: A Methodological Review of Visual Mental Imagery Reconstruction from Brain Activity

Shiyi Zhang, Dong Liang, Yihang Zhou

2026-11-11 1 min 269 words ARXIV CS HC

Summary: arXiv:2611.13691v1 Announce Type: new Abstract: Reconstructing visual mental imagery from brain activity differs from reconstructing perceived images. Imagery is generated top-down, its signal is weak, and the imagined content is never observed, so decoders are trained and scored against a proxy su...

🔗 **Read full article:**
https://arxiv.org/abs/2611.13691

How Well Do Benchmarks Evaluate LLMs' Use for Personal Health Informatics?

Fatemeh Mirhosseini, Yasaman S. Sefidgar

2026-11-11 1 min 149 words ARXIV CS HC

Summary: arXiv:2611.13652v1 Announce Type: new Abstract: Benchmarking is a valuable technique for advancing the capabilities of Large Language Models (LLMs), including in personal health informatics. There are growing efforts on benchmark creation for personal health data activities. However, it is unclear ...

🔗 **Read full article:**
https://arxiv.org/abs/2611.13652

EyeQ: Exploring Corneal Reflection for Mobile Context Sensing

Jaewook Lee, Judith Amores, Donghoon Shin, Bo-Eun Choi, Jason Kim, Davin Win Kyi, Seok-Young Kim, Jon E. Froehlich, Andrew D Wilson

2026-11-11 1 min 155 words ARXIV CS HC

Summary: arXiv:2611.13640v1 Announce Type: new Abstract: What if a single face-facing camera could sense your surroundings---through your eyes? We introduce EyeQ, which explores corneal reflection as a modality for context-aware mobile interaction. Traditional vision-based context sensing relies on world-fa...

🔗 **Read full article:**
https://arxiv.org/abs/2611.13640

EJAI: Judgment-Preserving Civic AI for Data Center Environmental Justice

Evan Pa\v{c}es-Wiles, Bryen Alvarez, Sara Gulabani, Akshara Kolipaka, Snehalkumar 'Neil' S. Gaikwad

2026-11-11 1 min 220 words ARXIV CS HC

Summary: arXiv:2611.13525v1 Announce Type: new Abstract: Communities in the United States and around the world are contesting data centers, the physical infrastructure of AI. These disputes pose environmental justice issues: residents must weigh unevenly distributed burdens using evidence scattered across m...

🔗 **Read full article:**
https://arxiv.org/abs/2611.13525

Towards Accessible, Autonomous Photosynthesis Monitoring on Living Leaves

Qitong Li, Xiaoyu Zheng, Xuanang Chen, Julia Wang, Raj Nileshbhai Dave, Dayanara Chavez, Nirvik Prakash Iyer, Nivedita Arora

2026-11-11 1 min 243 words ARXIV CS HC

Summary: arXiv:2611.13387v1 Announce Type: new Abstract: Photosynthesis is a direct biomarker of plant health and functioning. However, measuring net photosynthesis on individual leaves requires expensive (USD 42,000 - USD 89,000) instruments and manual effort. This limits access to photosynthesis sensing a...

Read full article:
https://arxiv.org/abs/2611.13387

Designing Accessible and Rigorous ML Experiments: No-Code Visual Workflows and Structured Guidance for Beginners

👤 Henrik H. J. Detjen, David Pfrang, Max Westphal, Melissa Finster, D\"orte Corr, Markus Wenzel

📅 2026-11-11 📖 1 min 📝 161 words ARXIV CS HC

Summary: arXiv:2611.13386v1 Announce Type: new Abstract: Designing machine learning (ML) experiments requires beginners to plan a methodologically sound workflow and implement it in a concrete environment. We evaluated two forms of support in two within-participant studies with 18 predominantly technically ...

🔗 **Read full article:**
https://arxiv.org/abs/2611.13386

Ender: Visual Scaffolding for Geometric Proof Comprehension

👤 Hwei-Shin Harriman, Yuchen Jin, Dominik Moritz, Joshua Sunshine

📅 2026-11-11 📖 1 min

📝 202 words

ARXIV CS HC

Summary: arXiv:2611.13368v1 Announce Type: new Abstract: Geometric proof is a foundational but notoriously difficult topic in secondary mathematics, primarily due to the high cognitive load required to integrate visual diagrams with logical text. As proofs evolve, tracking the initial givens and an expandin...

🔗 **Read full article:**
https://arxiv.org/abs/2611.13368

Project Progress Desk: Know Where All Your Claude Code and Codex Projects Stand at a Glance

Wanghao Ye, Yiting Wang, Shwai He, Guoheng Sun, Ziyao Wang, Yexiao He, Zheyu Shen, Zhaoyi Liu, Bowei Tian, Daeun Jung, Aditya Sarkar, Parastoo Pilevar, Xiaochuan Gong, Yancheng Zhu, Yingying Fan, Qiong Wang, Yuning Zhang, Meng Feng, Siyuan Peng, Zhenle Duan, Sihan Chen, Ziyi Wang, Yifei Dong, Yilong Dai, Ang Li

2026-11-11 1 min 132 words ARXIV CS HC

Summary: arXiv:2611.13287v1 Announce Type: new Abstract: Codex and Claude Code let developers work on many projects in parallel, but keeping track of them still means switching between chats, terminal sessions and providers. We present Project Progress Desk, a local desktop interface that brings overall pro...

🔗 **Read full article:**
https://arxiv.org/abs/2611.13287

Designing UIs with GenAI: Understanding the Use, Benefits, and Limitations of Prompt-to-Design Generators

Jingzhou Ye, Yao Li, Xueqiang Wang

2026-11-11 1 min 156 words ARXIV CS HC

Summary: arXiv:2611.13211v1 Announce Type: new Abstract: The integration of generative AI (GenAI) into user interface (UI) design has led to prompt-to-design generators (P2dGens), tools that enable designers to create UI prototypes from textual prompts efficiently. Despite their growing adoption, little is ...

🔗 **Read full article:**
https://arxiv.org/abs/2611.13211

TypedInk: Knowing When Not To Draw

Weiwei Yang 2026-11-11 1 min 295 words ARXIV CS HC

Summary: arXiv:2611.13027v1 Announce Type: new Abstract: Can a scientific figure generator know when it does not have enough evidence to draw? Complete-source illustrators map a finished methodology section to a schematic, or a table to plotting code, and are never asked this question because their input is...

🔗 **Read full article:**
https://arxiv.org/abs/2611.13027

Orthographic depth at two scales, with English also matched for lexicon size

Adam Y. Shavit 2026-11-11 1 min 301 words ARXIV QBIO NC

Summary: arXiv:2611.13400v1 Announce Type: cross Abstract: A reader can struggle to read in only one of their languages, and one proposed reason is how predictably spelling maps to sound. We measure that predictability as conditional entropy, the pronunciation uncertainty left once the spelling is seen, at ...

🔗 **Read full article:**
https://arxiv.org/abs/2611.13400

IC-GNN: Modeling Individualized Brain Community Organization for Disease Prediction

Yang Zhang, Xiao Zhou, Jonathan Warrell, Avram Holmes, Xuan Zhang, Mark Gerstein, for the Alzheimer's Disease Neuroimaging Initiative

2026-11-11 1 min 213 words ARXIV QBIO NC

Summary: arXiv:2611.13070v1 Announce Type: cross Abstract: Brain networks exhibit modular organization, where community structures vary substantially across individuals and diseases. Understanding these individual differences is crucial for advancing both basic neuroscience research and precision medicine. ...

Read full article:
https://arxiv.org/abs/2611.13070

Predicting Time-to-conversion for Dementia of Alzheimer's Type using Multi-modal Deep Survival Analysis

Ghazal Mirabnahrazam, Da Ma, C\'edric Beaulac, Sieun Lee, Karteek Popuri, Hyunwoo Lee, Jiguo Cao, James E Galvin, Lei Wang, Mirza Faisal Beg, the Alzheimer's Disease Neuroimaging Initiative

2026-11-11 1 min 173 words ARXIV QBIO NC

Summary: arXiv:2205.01188v1 Announce Type: cross Abstract: Dementia of Alzheimer's Type (DAT) is a complex disorder influenced by numerous factors, but it is unclear how each factor contributes to disease progression. An in-depth examination of these factors may yield an accurate estimate of time-to-convers...

Read full article:
https://arxiv.org/abs/2205.01188

Differential Diagnosis of Frontotemporal Dementia and Alzheimer's Disease using Generative Adversarial Network

Da Ma, Donghuan Lu, Karteek Popuri, Mirza Faisal Beg

2026-11-11 1 min 169 words

ARXIV QBIO NC

Summary: arXiv:2109.05627v2 Announce Type: cross Abstract: Frontotemporal dementia and Alzheimer's disease are two common forms of dementia and are easily misdiagnosed as each other due to their similar pattern of clinical symptoms. Differentiating between the two dementia types is crucial for determining d...

🔗 **Read full article:**
https://arxiv.org/abs/2109.05627

Grey matter sublayer thickness estimation in themouse cerebellum

Da Ma, Manuel J. Cardoso, Maria A. Zuluaga, Marc Modat, Nick. Powell, Frances Wiseman, Victor Tybulewicz, Elizabeth Fisher, Mark. F. Lythgoe, Sebastien Ourselin

2026-11-11 1 min 139 words

ARXIV QBIO NC

Summary: arXiv:1901.02499v1 Announce Type: cross Abstract: The cerebellar grey matter morphology is an important feature to study neurodegenerative diseases such as Alzheimer's disease or Down's syndrome. Its volume or thickness is commonly used as a surrogate imaging biomarker for such diseases. Most studi...

🔗 **Read full article:**
https://arxiv.org/abs/1901.02499

Optimization principles, revisited

William Bialek 2026-11-11 1 min 219 words ARXIV QBIO NC

Summary: arXiv:2611.14381v1 Announce Type: new Abstract: The brain solves a wide range of problems, and some of these solutions are very good indeed. It is an old idea that, in some cases, these solutions may be as good as possible given the relevant constraints. If this is true we can promote this notion o...

🔗 **Read full article:**
https://arxiv.org/abs/2611.14381

Neuromodulation Through Intrinsic Noise

Sunghyun Koo, Jin Gyu Lee, Rodolphe Sepulchre 2026-11-11 1 min 118 words

ARXIV QBIO NC

Summary: arXiv:2611.13780v1 Announce Type: new Abstract: Neuromodulation, the modulation of excitability through the regulation of neuronal parameters, is a key tool in neuromorphic control design. This paper investigates how spiking excitability can be modulated by intrinsic noise. Focusing on a simplified...

🔗 **Read full article:**
https://arxiv.org/abs/2611.13780

Similar visual tuning conceals different recovery dynamics in a model of macaque V1

Xialei Huang, Qingyu Yan, Zheng Bian, Zhuo-Cheng Xiao

2026-11-11 1 min 206 words

ARXIV QBIO NC

Summary: arXiv:2611.13692v1 Announce Type: new Abstract: Biologically detailed neuronal networks can model how brain circuits perform observed functions, but their high computational cost limits the comparisons needed to test biological hypotheses. To address this limitation, we extend a multiscale approach...

🔗 **Read full article:**
https://arxiv.org/abs/2611.13692

Human and Machine Annotations Predict Brain Responses to Naturalistic Movies Across Development

Yiran Huang, Zeyong Zhang, Rounak Rahman, Bharat Biswal, Xin Di, Mengjia Xu

2026-11-11 1 min 227 words

ARXIV QBIO NC

Summary: arXiv:2611.13637v1 Announce Type: new Abstract: Naturalistic movie watching provides a powerful framework for studying dynamic brain function, but linking movie content to neural responses typically relies on labor-intensive human annotation. We investigated whether multimodal large language models...

🔗 **Read full article:**
https://arxiv.org/abs/2611.13637

A probabilistic model of visual segmentation explains early visual cortical dynamics

Tridib K. Biswas, Ruben Coen-Cagli
2026-11-11
1 min
255 words
ARXIV QBIO NC

Summary: arXiv:2611.13316v1 Announce Type: new Abstract: Neural dynamics are a central feature of biological neural systems. In the visual cortex, the functional role of neural dynamics is poorly understood, particularly for static stimuli. The statistics of still natural images have long been used by model...

🔗 **Read full article:**
https://arxiv.org/abs/2611.13316

Disrupted imagery, disrupted scenes: mental scene construction in behavioural-variant frontotemporal dementia

1 min
18 words
BRAIN RESEARCH

Summary: Publication date: 15 January 2027 Source: Brain Research, Volume 1895 Author(s): Mohamad El Haj, Guillaume Chapelet, Dimitrios Kapogiannis, Claire Boutoleau-Bretonnière

🔗 **Read full article:**
https://www.sciencedirect.com/science/article/pii/S000689932600483X?dgcid=rss_sd_all

Paclitaxel induces region-specific microglial remodeling associated with chronic behavioral deficits and modulated by electroacupuncture in rats

📖 1 min 📝 19 words BRAIN RESEARCH

> **Summary:** Publication date: 15 January 2027 Source: Brain Research, Volume 1895 Author(s): Shinya Hiraiwa, Taiga Muramoto, Masahiro Umeda, Satomi Ebara, Fumihiko Fukuda

🔗 **Read full article:**
https://www.sciencedirect.com/science/article/pii/S0006899326004828?dgcid=rss_sd_all

Gastrin-releasing peptide in the anterior medial nucleus accumbens or paraventricular nucleus suppresses chow intake under food restriction but not palatable diet intake in nonrestricted rats

📖 1 min 📝 28 words BRAIN RESEARCH

> **Summary:** Publication date: 15 January 2027 Source: Brain Research, Volume 1895 Author(s): Regan Cochran, Ella Irvin, Yiwei Tang, Stacey Chuang, Adele Nadeau, Smruti Patel, Lindsey Wilson, Weijia Zhang, Wayne E. Pratt

🔗 **Read full article:**
https://www.sciencedirect.com/science/article/pii/S0006899326004877?dgcid=rss_sd_all

SREBP1 inhibition downregulates NR2B-PSD95 interaction in glutamate excitotoxicity following cerebral ischemic reperfusion injury

📖 1 min 📝 16 words BRAIN RESEARCH

> **Summary:** Publication date: 15 January 2027 Source: Brain Research, Volume 1895 Author(s): Lopmudra P. Sarode, Bhavesh Verma, Rajesh Ugale

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https://www.sciencedirect.com/science/article/pii/S0006899326004841?dgcid=rss_sd_all

Cell death: from basic principles to translational developments

📅 2026-11-11 📖 1 min 📝 0 words NATURE NEUROSCIENCE SUBJECTS

🔗 **Read full article:**
https://www.nature.com/articles/s41392-026-03073-w

Self-paced silent speech brain-computer interface for device control

📅 2026-11-11 📖 1 min 📝 0 words NATURE NEUROSCIENCE SUBJECTS

🔗 **Read full article:**
https://www.nature.com/articles/s42003-026-10477-5

Blood based multi biomarker panel discriminates Alzheimer disease from healthy controls in a single center nested cross validation

📅 2026-11-11 📖 1 min 📝 0 words NATURE NEUROSCIENCE SUBJECTS

🔗 **Read full article:**
https://www.nature.com/articles/s41598-026-79195-6

Misfolded tau propagates via templated seeding

Qingzhong Ren 2026-11-02 1 min 14 words NATURE NEUROSCIENCE

Summary: Nature Neuroscience, Published online: 02 November 2026; doi:10.1038/s41593-026-02512-9 Misfolded tau propagates via templated seeding

🔗 **Read full article:**
https://www.nature.com/articles/s41593-026-02512-9

Publisher Correction: Network geometry shapes the transformation of task representations across the human cortex

Michael W. Cole 2026-11-02 1 min 22 words NATURE NEUROSCIENCE

Summary: Nature Neuroscience, Published online: 02 November 2026; doi:10.1038/s41593-026-02530-7 Publisher Correction: Network geometry shapes the transformation of task representations across the human cortex

🔗 **Read full article:**
https://www.nature.com/articles/s41593-026-02530-7

Editorial Expression of Concern: NgR1 and NgR3 are receptors for chondroitin sulfate proteoglycans

👤 Roman J. Giger
📅 2026-11-05 📖 1 min 📝 21 words NATURE NEUROSCIENCE

Summary: Nature Neuroscience, Published online: 05 November 2026; doi:10.1038/s41593-026-02534-3 Editorial Expression of Concern: NgR1 and NgR3 are receptors for chondroitin sulfate proteoglycans

🔗 **Read full article:**
https://www.nature.com/articles/s41593-026-02534-3

Ddx3x deficiency in mice impairs corticofugal motor circuits and causes motor deficits that are attenuated by early-life exercise

👤 Lauren DierdorffIkrame El KhaldiMarta Garcia-FornWei WangJeronimo LukinCorinne M. SmithAdele MossaZhuhao WuSilvia De Rubeisahttps://ror.org/04a9tmd77Seaver Autism Center for Research and Treatment, Icahn School of Medicine at Mount Sinai, New York, NY 10029bhttps://ror.org/04a9tmd77Department of Psychiatry, Icahn School of Medicine at Mount Sinai, New York, NY 10029chttps://ror.org/04a9tmd77Department of Neuroscience, Icahn School of Medicine at Mount Sinai, New York, NY 10029dhttps://ror.org/04a9tmd77The Mindich Child Health and Development Institute, Icahn School of Medicine at Mount Sinai, New York, NY 10029ehttps://ror.org/04a9tmd77The Friedman Brain Institute, Icahn School of Medicine at Mount Sinai, New York, NY 10029fhttps://ror.org/04a9tmd77The Alper Center for Neural Development and Regeneration Friedman Brain Institute, Icahn School of Medicine at Mount Sinai, New York, NY 10029ghttps://ror.org/02r109517Helen & Robert Appel Alzheimer's Disease Research Institute, Feil Family Brain and Mind Research Institute, Weill Cornell Medicine, New York, NY 10021hhttps://ror.org/04a9tmd77Department of Pharmacological Sciences, Icahn School of Medicine at Mount Sinai, New York, NY 10029ihttps://ror.org/04a9tmd77Department of Graduate Education, Icahn School of Medicine at Mount Sinai, New York, NY 10029

📅 2026-10-29 📖 1 min 📝 50 words PNAS NEUROSCIENCE

Summary: Proceedings of the National Academy of Sciences, Volume 123, Issue 44, November 2026.
SignificanceAcquisition of motor skills during infancy is critical for a child's ability to interact with the physical and social environment and motor impairments are frequent in individuals with neurodevelo...

🔗 **Read full article:**
https://www.pnas.org/doi/abs/10.1073/pnas.2535732123?af=R

Respiratory phase orchestrates human olfactory cortical dynamics

👤 Alisa Y. HathawayTodd P. Colemanahttps://ror.org/00f54p054Department of Electrical Engineering, Stanford University, Stanford, CA 94305bhttps://ror.org/00f54p054Department of Bioengineering, Stanford University, Stanford, CA 94305

📅 2026-11-06 📖 1 min 📝 47 words PNAS NEUROSCIENCE

Summary: Proceedings of the National Academy of Sciences, Volume 123, Issue 45, November 2026.
SignificanceDiagnosing smell loss in patients with cognitive disorders remains challenging because conventional tests demand complex responses that measure memory and literacy instead of basic sensory functio...

🔗 **Read full article:**
https://www.pnas.org/doi/abs/10.1073/pnas.2619920123?af=R

The purinergic receptor P2Y13 controls the activation and mode of division of adult ventricular–subventricular zone neural stem cells

👤 Lucía Paniagua-HerranzJulia Serrano-LópezCelia Llorente-SáezMarina Leonor Pérez-SanzDavid de Agustín-DuránPere Duart-AbadiaAna Domingo-MuelasÁfrica Vincelle-NietoPaloma BragadoDiana Manzano-FrancoMarina Arribas-BlázquezLucía GallegoAraceli Garcia-MoraLuis Alcides Olivos-OréSergio GascónÁlvaro Gutierrez-UzquizaBenedikt BerningerRaquel Pérez-SenEsmerilda G. DelicadoRosa Gómez-VillafuertesArmando Reyes-PalomaresAntonio R. ArtalejoIsabel FariñasFelipe Ortegaahttps://ror.org/02p0gd045Departamento de Bioquímica y Biología Molecular, Universidad Complutense de Madrid, Madrid 28040, SpainbInstituto Universitario de Investigación en Neuroquímica, Departamento de Bioquímica y Biología Molecular, Madrid 28040, Spainchttps://ror.org/014v12a39Instituto de Investigación Sanitaria del Hospital Clínico San Carlos de Madrid, Madrid 28040, Spaindhttps://ror.org/02p0gd045Departamento de Farmacología y Toxicología, Facultad de Veterinaria, Universidad Complutense de Madrid, Madrid 28040, Spainehttps://ror.org/043nxc105Departamento de Biología Celular y Biología Funcional, Universidad de Valencia, Valencia 46100, Spainfhttps://ror.org/043nxc105Instituto de Biotecnología y Biomedicina, Departamento de Genética, Facultad de Biología, Universidad de Valencia, Valencia 46100, Spainghttps://ror.org/02g87qh62Centro de Investigación Biomédica en Red sobre Enfermedades Neurodegenerativas, Departamento de Biología Celular y Biología Funcional, Valencia 46100, Spainhhttps://ror.org/02gfc7t72Cajal Neuroscience Center, Laboratory of direct reprogramming, Spanish National Research Council, Madrid 28805, Spainihttps://ror.org/0220mzb33Centre for Developmental Neurobiology, Institute of Psychiatry, Psychology and Neuroscience, King’s College London, London SE1 1UL, United Kingdomjhttps://ror.org/0220mzb33Medical Research Council Centre for Neurodevelopmental Disorders, Institute of Psychiatry, Psychology and Neuroscience, King’s College London, London SE1 1UL, United Kingdomkhttps://ror.org/023b0x485Institute of Physiological Chemistry, University Medical Center Johannes Gutenberg University, Mainz 55128, GermanylFocus Program Translational Neuroscience, Johannes Gutenberg University, 55128, Mainz, Germanymhttps://ror.org/036b2ww28Department of Molecular Biology and Biochemistry, Faculty of Science, University of Malaga, Malaga SE1 1UL, Spain

📅 2026-11-04 📖 1 min 📝 48 words PNAS NEUROSCIENCE

Summary: Proceedings of the National Academy of Sciences, Volume 123, Issue 45, November 2026.
SignificanceAdult neural stem cells (NSCs) sustain lifelong neurogenesis in the mammalian brain, yet the mechanisms controlling their activation and self-renewal remain poorly understood. We identify the puri...

🔗 **Read full article:**
https://www.pnas.org/doi/abs/10.1073/pnas.2618940123?af=R

Dissociation of gravity and limb dynamics in motor control

👤 Valeriya GritsenkoTjaša KunavarMarko JamšekSergiy YakovenkoOlivier WhiteJan Babičahttps://ror.org/011vxgd24Departments of Human Performance/Neuroscience, West Virginia University, Rockefeller Neuroscience Institute, Morgantown, WV 26506bLaboratory for Neuromechanics and Biorobotics, Department of Automatics, Biocybernetics, and Robotics, Jožef Stefan Institute, Ljubljana 1000, Sloveniachttps://ror.org/02vjkv261Université Bourgogne Europe, Institut national de la santé et de la recherche médicale, Cognition, Action and Sensorimotor Plasticity, Unité Mixte de Recherche 1093, Dijon 21000, France

📅 2026-11-03 📖 1 min 📝 49 words PNAS NEUROSCIENCE

Summary: Proceedings of the National Academy of Sciences, Volume 123, Issue 45, November 2026.
SignificanceThe nervous system must control movement in presence of variable intrinsic and extrinsic forces, yet it has remained unclear whether mechanically equivalent forces arising from gravity and limb dy...

🔗 **Read full article:**
https://www.pnas.org/doi/abs/10.1073/pnas.2613283123?af=R

Lead impairs long-term memory by disrupting mitochondrial redox homeostasis in Drosophila

Cheng-Tzu HsuTe-Wei LuYa-Ting YangChun-Chao ChenYen-Hua ChiuJung-Hsuan YangZheng-Wen WuWei-Kai ChengJheng-Wei WuChen-Hao ShiuSih-Kai ChuangKuan-Lin FengHsuan-Wen LinJia-Ling YangAnn-Shyn Chiangahttps://ror.org/00zdnkx70Institute of Biotechnology, College of Life Sciences and Medicine, National Tsing-Hua University, Hsinchu 30013, Taiwanbhttps://ror.org/00mjawt10Institute of Precision Medicine, College of Medicine, National Sun Yat-sen University, Kaohsiung 804201, Taiwanchttps://ror.org/00mjawt10Department of Biological Sciences, National Sun Yat-sen University, Kaohsiung 804201, Taiwandhttps://ror.org/00zdnkx70Brain Research Center, National Tsing Hua University, Hsinchu 30013, Taiwanehttps://ror.org/00mjawt10Department of Biomedical Science and Technology, College of Medicine, National Sun Yat-sen University, Kaohsiung 804201, Taiwanfhttps://ror.org/00zdnkx70Institute of Systems Neuroscience, College of Life Sciences and Medicine, National Tsing Hua University, Hsinchu 30013, TaiwangGraduate Institute of Biomedical Sciences, China Medical University, Taichung 40402, TaiwanhInstitute of Molecular and Genomic Medicine, National Health Research Institutes, Miaoli 35053, Taiwanihttps://ror.org/03gk81f96Department of Biomedical Science and Environmental Biology, Kaohsiung Medical University, Kaohsiung 80780, Taiwan

📅 2026-11-04 📖 1 min 📝 46 words PNAS NEUROSCIENCE

Summary: Proceedings of the National Academy of Sciences, Volume 123, Issue 45, November 2026.
SignificanceEnvironmental lead exposure is strongly associated with cognitive deficits, yet how it disrupts memory at the cellular level remains unclear. Here, we show that lead selectively impairs protein-sy...

🔗 **Read full article:**
https://www.pnas.org/doi/abs/10.1073/pnas.2612578123?af=R

Hypothalamic lncRNA Synage prevents age-related sperm quality decline by suppressing A-type potassium currents in GnRH neurons

Fei WangMengyi ZhangMengtian WangShuhan TianQian BianYaohui HeShuiqiao YuanQinghua ShiWen LiuJun XiaoLi WangJi LiuXiaoyuan Songahttps://ror.org/04c4dkn09Department of Urology, The First Affiliated Hospital of University of Science & Technology of China, Hefei National Research Center for Physical Sciences at the Microscale, Ministry of Education Key Laboratory for Cellular Dynamics, Center for Advanced Interdisciplinary Science and Biomedicine of the Institute of Health and Medicine, Division of Life Sciences and Medicine, University of Science and Technology of China, Hefei, Anhui 230026, Chinabhttps://ror.org/04c4dkn09Division of Life Sciences and Medicine, School of Biomedical Engineering, University of Science and Technology of China, Hefei, Anhui 230026, Chinachttps://ror.org/04c4dkn09School of Biomedical Engineering, Suzhou Institute for Advanced Research, University of Science and Technology of China, Suzhou, Jiangsu 215123, Chinadhttps://ror.org/03hz5th67Department of Neuroscience, Faculty of Life and Health Sciences, Shenzhen University of Advanced Technology, Shenzhen, Guangdong 518107, Chinaehttps://ror.org/00mcjh785State Key Laboratory of Cellular Stress Biology, School of Pharmaceutical Sciences, Xiamen University, Xiamen, Fujian 361102, Chinafhttps://ror.org/00p991c53Institute of Reproductive Health, Tongji Medical College, Huazhong University of Science and Technology, Wuhan, Hubei 430030, Chinaghttps://ror.org/04c4dkn09Department of Urology, Anhui Provincial Hospital, The First Affiliated Hospital of University of Science & Technology of China, Division of Life Sciences and Medicine, University of Science & Technology of China, Hefei, Anhui 230001, Chinahhttps://ror.org/0220qvk04Department of Assisted Reproduction, Shanghai Ninth People's Hospital, Shanghai JiaoTong University School of Medicine, Shanghai 200011, ChinaiInstitute of Artificial Intelligence, Hefei Comprehensive National Science Center, Hefei 230026, China

📅 2026-11-03 📖 1 min 📝 46 words PNAS NEUROSCIENCE

Summary: Proceedings of the National Academy of Sciences, Volume 123, Issue 45, November 2026.
SignificanceMen increasingly delay fatherhood, yet paternal aging lowers sperm quality and increases health risks to offspring. The mechanisms underlying these reproductive changes remain incompletely defined...

🔗 **Read full article:**
https://www.pnas.org/doi/abs/10.1073/pnas.2612408123?af=R

Transcranial magnetic stimulation to the dorsolateral prefrontal cortex modulates single-neuron activity in humans

👤 Charles W. DickeyUmair HassanHiroto KawasakiAriane E. RhoneChristopher C. ClineMatthew A. HowardNicholas T. TrappAaron D. BoesJoel I. BergerCorey J. Kellerahttps://ror.org/00f54p054Department of Psychiatry and Behavioral Sciences, Stanford University, Stanford, CA 94305bhttps://ror.org/00f54p054Wu Tsai Neurosciences Institute, Stanford University, Stanford, CA 94305chttps://ror.org/036jqmy94Department of Neurosurgery, University of Iowa, Iowa City, IA 52242dhttps://ror.org/036jqmy94Department of Psychiatry, University of Iowa, Iowa City, IA 52242ehttps://ror.org/036jqmy94Department of Neurology, University of Iowa, Iowa City, IA 52242fhttps://ror.org/036jqmy94Department of Pediatrics, University of Iowa, Iowa City, IA 52242ghttps://ror.org/036jqmy94Iowa Neuroscience Institute, Carver College of Medicine, University of Iowa, Iowa City, IA 52242hVeterans Affairs Palo Alto Healthcare System, and the Sierra Pacific Mental Illness Research, Education, and Clinic Center, Palo Alto, CA 94305

📅 2026-11-02 📖 1 min 📝 49 words PNAS NEUROSCIENCE

Summary: Proceedings of the National Academy of Sciences, Volume 123, Issue 45, November 2026.
SignificanceTranscranial magnetic stimulation (TMS) to the dorsolateral prefrontal cortex is a common treatment for depression. However, it remains unknown how TMS affects single neurons in deep brain circuit...

🔗 **Read full article:**
https://www.pnas.org/doi/abs/10.1073/pnas.2610803123?af=R

Hepatocyte-derived exosomal miR-122 mediates liver-to-brain crosstalk and neuroinflammation in alcohol-exposed Alzheimer’s disease mice

👤 Veronika BrezaniMarti Ortega-RiberaNeda RezagholizadehRadhika JoshiViliam BrezaniMrigya BabutaBruna de Araujo Cardoso dos SantosAdam ZivnyPrashanth Thevkar NageshIonita GhiranElizabeth Michelle ShreyerScott Brian MinchenbergAllyson FlavinGyongyi Szaboahttps://ror.org/04drvxt59Division of Gastroenterology, Department of Medicine, Beth Israel Deaconess Medical Center, Boston, MA 02215bHarvard Medical School, Boston, MA 02115chttps://ror.org/04drvxt59Department of Anesthesia, Critical Care and Pain Medicine, Beth Israel Deaconess Medical Center, Boston, MA 02215dhttps://ror.org/05a0ya142Broad Institute of MIT and Harvard, Cambridge, MA 02142

📅 2026-11-03 📖 1 min 📝 46 words PNAS NEUROSCIENCE

Summary: Proceedings of the National Academy of Sciences, Volume 123, Issue 45, November 2026.
SignificanceExcessive alcohol use damages both the liver and the brain, but how these organs communicate during injury remains unclear. Our study identifies a previously unrecognized liver-to-brain communicat...

🔗 **Read full article:**
https://www.pnas.org/doi/abs/10.1073/pnas.2535191123?af=R

Altered interictal alpha-band organization in childhood absence epilepsy

Justyna Paprocka 2026-10-23 1 min 318 words FRONTIERS NEUROINFORMATICS

Summary: ObjectiveInterictal background EEG in childhood absence epilepsy (CAE) may appear normal. We asked whether drug-naive children with CAE differ from controls in alpha frequency and the regional distribution of phase-lagged synchronization, and how these differences depend on participant-specific surr...

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https://www.frontiersin.org/articles/10.3389/fninf.2026.1875377

Reporting framework for event-based model diagnostics

Cassie S. Mitchell 2026-11-11 1 min 211 words FRONTIERS NEUROINFORMATICS

Summary: Event-Based Models (EBMs) are widely used to infer latent disease progression from cross-sectional biomarker data, with existing studies employing diverse approaches to characterize uncertainty, stability, and validation. However, comparatively little guidance exists for systematically integrating a...

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https://www.frontiersin.org/articles/10.3389/fninf.2026.1893133

SDCL-AD: a structure-aware dual-graph contrastive learning framework for Alzheimer's disease prediction using multi-omics data

Miaomiao Sun 2026-11-11 1 min 201 words

FRONTIERS COMPUTATIONAL NEUROSCIENCE

Summary: IntroductionEarly prediction of Alzheimer's disease (AD) is of great significance for disease intervention, personalized diagnosis, and treatment. With the rapid accumulation of multi-omics data, integrating complementary information from different molecular layers has become a key challenge in impr...

🔗 **Read full article:**
https://www.frontiersin.org/articles/10.3389/fncom.2026.1973333

An EMG-driven virtual reality platform for physiology-based motor interaction: technical and functional validation

D. Sili 2026-11-11 1 min 351 words

FRONTIERS HUMAN NEUROSCIENCE

Summary: Harnessing experience-dependent neuroplasticity has become a central goal of modern neurorehabilitation. Immersive virtual reality (VR) has emerged as a promising tool, enabling task-oriented training in ecologically valid scenarios that are difficult to reproduce in conventional clinical settings. ...

🔗 **Read full article:**
https://www.frontiersin.org/articles/10.3389/fnhum.2026.1943080

The role of the right temporal lobe in neural network mechanisms during working memory encoding

👤 Yijun Song 📅 2026-11-11 📖 1 min 📝 380 words FRONTIERS HUMAN NEUROSCIENCE

Summary: BackgroundVisual working memory (vWM) impairment is common in patients with temporal lobe epilepsy (TLE) and is more severe in those with right TLE (RTLE), suggesting the significant role of right temporal lobe in WM processes.MethodsWe recruited patients with left TLE (LTLE group, n = 20), RTLE (RT...

🔗 **Read full article:**
https://www.frontiersin.org/articles/10.3389/fnhum.2026.1987904

Rethinking cerebral visual impairment: Unitary and Umbrella conceptualizations as a potential way forward

👤 John Ravenscroft 📅 2026-11-11 📖 1 min 📝 303 words FRONTIERS HUMAN NEUROSCIENCE

Summary: Over recent decades, understanding and recognition of cerebral visual impairment (CVI) have increased considerably. CVI is now recognized as the likely leading cause of certifiable childhood visual impairment in both high and low-income countries. However, differences remain concerning its definitio...

🔗 **Read full article:**
https://www.frontiersin.org/articles/10.3389/fnhum.2026.2009032

Frequency-embedded sequence learning for multi-class EEG-to-text interface

Alistair McEwan 2026-11-11 1 min 227 words FRONTIERS HUMAN NEUROSCIENCE

Summary: A brain-computer interface (BCI) is an emerging technology for decoding users' intentions, enabling speech decoding from electroencephalography (EEG) signals. This study proposes a multi-class (i.e., 5 words or phrases) EEG-to-Text classification pipeline based on a lightweight 1−dimensional convolu...

🔗 **Read full article:**
https://www.frontiersin.org/articles/10.3389/fnhum.2026.1980894

Immediate effect of gait training using the hybrid assistive limb in patients in the acute phase of stroke: a prospective pilot study

Yasushi Takagi 2026-10-23 1 min 360 words FRONTIERS NEUROROBOTICS

Summary: IntroductionThe effectiveness of Hybrid Assistive Limb (HAL) gait training in patients in the acute phase of stroke remains unclear. We prospectively assessed the immediate effects of HAL gait training in this patient population.MethodsIn this prospective pilot study, four patients with acute-phase ...

🔗 **Read full article:**
https://www.frontiersin.org/articles/10.3389/fnbot.2026.1985500

Susceptibility alterations in the hippocampus and subcortical nuclei in hemodialysis patients with end-stage renal disease: an APART-QSM study

Haiqing Yang 2026-11-11 1 min 286 words FRONTIERS NEUROSCIENCE

Summary: ObjectiveTo investigate APART-QSM-derived paramagnetic susceptibility (χpara) alterations in hippocampal, medial temporal, and subcortical regions in maintenance hemodialysis patients with end-stage renal disease (ESRD).MethodsThis cross-sectional study included 55 maintenance hemodialysis patients ...

🔗 **Read full article:**
https://www.frontiersin.org/articles/10.3389/fnins.2026.1934124

Efficacy of mindfulness-based interventions and biofeedback for migraine management: a systematic review and meta-analysis of randomized controlled trials

Irene A. Garcia-Yu 2026-11-11 1 min 292 words FRONTIERS NEUROSCIENCE

Summary: ObjectiveThis systematic review and meta-analysis of randomized controlled trials aims to analyze the efficacy of mindfulness-based interventions and biofeedback on headache frequency, pain intensity, headache-related disability, psychological distress, and quality of life in adult patients with mig...

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https://www.frontiersin.org/articles/10.3389/fnins.2026.1965907

Influence of out-of-plane electric fields in an ex vivo model of transcranial electrical stimulation

Yuhyun Lee, Vishal Jain, Yuxin Guo, Maysamreza Chamanzar, Pulkit Grover and Mats Forssell

2026-11-11 1 min 226 words JOURNAL NEURAL ENGINEERING

Summary: Objective. Electrical stimulation is the most commonly used modality for neuromodulation, with a number of clinical and scientific applications. Although the basic mechanisms of neural activation by electrical stimulation are well understood, important questions remain regarding neural responses to ...

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https://iopscience.iop.org/article/10.1088/1741-2552/aeb04a

Human cortical high-gamma power scales with movement rate in healthy participants and stroke survivors

Fanny Quandt

2025-01-09 1 min 65 words OOSTENVELD ROBERT

Summary: Motor cortical high-gamma oscillations (60-90 Hz) occur at movement onset and are spatially focused over the contralateral primary motor cortex. Although high-gamma oscillations are widely recognized for their significance in human motor control, their precise function on a cortical level remains el...

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NIRS-BIDS: Brain Imaging Data Structure Extended to Near-Infrared Spectroscopy

👤 Luca Pollonini 📅 2025-01-27 📖 1 min 📝 70 words OOSTENVELD ROBERT

> **Summary:** Functional near-infrared spectroscopy (fNIRS) is an increasingly popular neuroimaging technique that measures cortical hemodynamic activity in a non-invasive and portable fashion. Although the fNIRS community has been successful in disseminating open-source processing tools and a standard file forma...

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Pseudonymisation of neuroimages and data protection: Increasing access to data while retaining scientific utility

👤 Lyuba Zehl 📅 2025-06-26 📖 1 min 📝 67 words OOSTENVELD ROBERT

> **Summary:** For a number of years, facial features removal techniques such as 'defacing', 'skull stripping' and 'face masking/blurring', were considered adequate privacy preserving tools to openly share brain images. Scientifically, these measures were already a compromise between data protection requirements a...

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Cycling on the Freeway: The perilous state of open-source neuroscience software

👤 Tim M Tierney 📅 2025-08-13 📖 1 min 📝 74 words OOSTENVELD ROBERT

Summary: Most scientists need software to perform their research (Barker et al., 2020;Carver et al., 2022;Hettrick, 2014;Hettrick et al., 2014;Switters & Osimo, 2019), and neuroscientists are no exception. Whether we work with reaction times, electrophysiological signals, or magnetic resonance imaging data, ...

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Optimal configuration of on-scalp OPMs with fixed channel counts

👤 Robert Oostenveld 📅 2025-08-13 📖 1 min 📝 69 words OOSTENVELD ROBERT

Summary: Recent technological developments have brought optically pumped magnetometers (OPMs) within reach of the larger neuroscientific community. The current state-of-the-art consists of whole-head systems that measure the magnetic field at >100 locations. OPM sensors can be constructed to measure the fiel...

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ARTEM-IS for ERP: Agreed Reporting Template for EEG Methodology-International Standard for Event-Related Potential Experiments

👤 Suzy J Styles 📅 2025-12-08 📖 1 min 📝 73 words OOSTENVELD ROBERT

Summary: The choices we make during the recording, preprocessing and analysis of event-related potentials (ERP) data can affect study outcomes. As such, it is critical that they are transparently reported to allow for reproducibility. Yet, systematic reviews of reporting practices in the field have shown tha...

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Eye-Tracking-BIDS: the Brain Imaging Data Structure extended to gaze position and pupil data

👤 Julia-Katharina Pfarr 📅 2026-02-12 📖 1 min 📝 68 words OOSTENVELD ROBERT

Summary: The Brain Imaging Data Structure (BIDS) is a widely adopted, community-driven standard to organize neuroimaging data and metadata. Although numerous extensions have been developed to incrementally extend coverage to new modalities and data types, an unambiguous, granular specification for eye-tracki...

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Facilitating unusual bodily experiences and out-of-body experiences across wakefulness and sleep: A high-density EEG and neurophenomenology study

👤 Robert Oostenveld 📅 2026-02-12 📖 1 min 📝 67 words OOSTENVELD ROBERT

Summary: Unusual bodily experiences (UBEs) are illusory bodily perceptions that are not coherent with typical wakeful experiences, including flying or floating sensations, body distortions, and out-of-body experiences. This study examines UBEs in a controlled sleep laboratory using meditation and light stimu...

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The Impact of Brain Tumors and Craniotomy Lesions on Scalp EEG

👤 Vitória Piai 📅 2026-02-17 📖 1 min 📝 65 words OOSTENVELD ROBERT

Summary: Electroencephalography (EEG) is widely used in both research and clinical settings, yet its accuracy can be significantly impacted by subject-specific anatomical anomalies such as brain lesions and skull defects. This study investigates the effects of glioma-related brain lesions and craniotomy-indu...

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EMG-BIDS: an extension to the Brain Imaging Data Structure for electromyography

👤 Arnaud Delorme 📅 2026-08-01 📖 1 min 📝 64 words OOSTENVELD ROBERT

Summary: Electromyography (EMG) is fundamental to clinical assessment, rehabilitation, neuromuscular research, and human-machine interfaces. Despite decades of use, no widely adopted standard exists for organizing and sharing EMG data, limiting reusability and large-scale data aggregation. We present EMG-BID...

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Prior-guided whole-slide survival prediction with soft histopathologic priors from vision-language foundation models

👤 Zhihui Feng 📅 2026-11-09 📖 1 min 📝 62 words LOW VISION

Summary: Whole-slide image (WSI) survival prediction is complicated by heterogeneous tumor morphology, where prognostic evidence is distributed across tumor, stroma, immune infiltration, necrosis, and residual tissue regions and often appears as mixed patch-level tissue evidence. Existing models commonly agg...

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Reduced magnocellular-biased visual responses are associated with working memory and linguistic difficulties in Chinese developmental dyslexia: A hypothesized visual-cognitive cascade

👤 Tsz Wing Leung
📅 2026-11-09 📖 1 min 📝 66 words

LOW VISION

Summary: Developmental dyslexia (DD) affects 5-10% of children worldwide. Although phonological theories predominate, they may not fully explain reading difficulties in orthographically dense scripts such as Chinese. We investigated whether Chinese children with DD exhibit differences in early visual cortica...

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During natural vision, semantic novelty modulates fixation-related processing in primate cortex

👤 Lucas C Parra 📅 2026-11-09 📖 1 min 📝 74 words LOW VISION

Summary: We sample visual scenes with short gaze fixations separated by saccades. While low-level integration is known, semantic integration of foveal vision across fixations remains unclear. We hypothesized that the brain responds to changes in semantic information from one fixation to the next, and therefo...

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Health and welfare professionals' global contribution to the International Classification of Functioning, Disability and Health Core Set Vision Loss

👤 Dominique VAN DE Velde 📅 2026-11-10 📖 1 min 📝 64 words LOW VISION

Summary: CONCLUSIONS: A visual impairment exerts a broad and multidimensional impact on functioning, emphasizing the importance of emotional well-being, daily activities, and environmental supports. Regional differences highlight the need for culturally sensitive approaches in assessment and intervention. Th...

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Federated Learning: Envisioning a Pathway to More Accessible and Equitable Cleft Care

👤 Ghulam Qadir Fayyaz 📅 2026-11-10 📖 1 min 📝 66 words LOW VISION

Summary: Cleft lip and palate represents the most common congenital craniofacial anomaly worldwide, with low- and middle-income countries (LMICs) bearing nearly 84% of the global disease burden. Despite the instrumental role of global nongovernmental organizations (NGOs) in expanding access to cleft care acr...

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Dichoptic viewing reveals distinct low- and high-level mechanisms for temporal phase discrimination

👤 Keith A Schneider 📅 2026-11-10 📖 1 min 📝 68 words LOW VISION

Summary: Conscious awareness of temporal visual structure is limited to frequencies below ~10 Hz. Between ~10 and 60 Hz, flicker is visible, but temporal phase cannot be accurately discriminated; above ~60 Hz, stimuli are perceptually fused. Accurately estimating the cognitive limit of phase discrimination r...

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Trends in Near Vision Loss and Age-Sex-SDI Disparities in Belt and Road Countries Over Thirty Years

Xiaopan Li 2026-11-10 1 min 55 words LOW VISION

Summary: CONCLUSIONS: This study represents the first comprehensive analysis of NVL burden and disparities across BRI countries, revealing persistent inequalities by age, sex, and SDI. The findings reflect the diverse socio-economic and health-care contexts of BRI countries and highlight the urgent need for ...

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Feasibility of robotic-assisted surgery for nutcracker syndrome: a systematic review and single-arm meta-analysis

Jiang-Hua Zheng 2026-11-10 1 min 60 words LOW VISION

Summary: Nutcracker syndrome (NCS) results from compression of the left renal vein, and refractory symptoms require surgical reconstruction. Robotic platforms provide magnified three-dimensional vision and articulated instruments that are theoretically advantageous for vascular dissection and anastomosis wit...

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Segmentation, Reconstruction, and Grasp-Pose Estimation Using an RGB-D Camera for Robotic Apple Harvesting

Jianqiang Lu 2026-11-10 1 min 58 words LOW VISION

Summary: Accurate segmentation, three-dimensional reconstruction, and grasp-pose estimation of fruits and surrounding structures are essential for robotic harvesting in complex orchard environments. However, occlusions caused by branches, support wires, and irregular canopy structures remain major challenges...

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Intelligent rendering optimization for animated scenes using reinforcement learning and computer vision

Li Jia 2026-11-10 1 min 63 words LOW VISION

Summary: Real-time rendering of complex animated scenes is computationally intensive and often struggles to balance visual fidelity and performance in dynamic or densely populated environments, as conventional pipelines lack adaptive resource optimization, leading to reduced frame rates and degraded visual q...

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A single session of high-definition transcranial direct current stimulation does not modulate effects of knee two-point discrimination training or sensorimotor function in healthy adults: Double-blind randomised controlled trial

👤 Paul W Hodges
📅 2026-07-30 📖 1 min 📝 53 words

TACTILE ACUITY

Summary: CONCLUSION: A single 20-minute session of 1 mA HD-tDCS, with or without brief TPD training, does not acutely modify knee somatosensory, sensorimotor, or functional performance in healthy adults. Existing evidence at the hand may not translate to the lower limb. Future work should investigate higher-...

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Variation in glabrous skin physiology across anatomical locations on the human hand

Martina C Meinke 2026-08-13 1 min 76 words TACTILE ACUITY

Summary: Skin physiology varies greatly between anatomical locations, with impact on tactile sensitivity. Research on the variations in morphology and physiology between different anatomical locations of glabrous skin is limited so far. This study investigates the morphology and physiology of glabrous skin o...

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Beyond Hamstring Shortness: Continuous Flexibility Is Associated With Balance and Stability but Not Sensory Measures-A Cross-Sectional Study

Senanur Akıncı 2026-08-14 1 min 66 words TACTILE ACUITY

Summary: CONCLUSIONS: Greater hamstring flexibility was associated with better dynamic balance performance and showed a weaker association with postural stability, whereas no robust associations were observed with sensory measures of postural control. These findings suggest that hamstring flexibility may be ...

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Resting-state EEG microstate dynamics reflect individual differences in tactile angle discriminability

Liu Yang 2026-08-19 1 min 63 words TACTILE ACUITY

Summary: Tactile spatial acuity, the ability to discriminate fine spatial details, is a fundamental sensory capacity modulated by both peripheral and central mechanisms. Here, we investigated whether resting-state EEG microstate dynamics are associated with individual differences in tactile spatial acuity. F...

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A grating orientation task with controlled dome elevation speed and contact force: stimulus conditions and day-to-day reliability

Hideaki Onishi 2026-09-25 1 min 66 words TACTILE ACUITY

Summary: The grating orientation discrimination task (GOT) is typically used to assess tactile spatial acuity. However, performance is influenced by stimulus speed, contact force between the stimulator and the skin, and the method of discrimination. This study aimed to explore suitable stimulus conditions fo...

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Epidermal microstructure and tactile sensitivity in the glabrous skin of hands and feet

Hannes P Saal 2026-09-25 1 min 69 words TACTILE ACUITY

Summary: Glabrous skin on the human palms and soles is specialized for mechanical interaction and is characterized by papillary ridges, an absence of hair follicles, and a thick stratum corneum. Although gross morphological differences, such as the increased thickness of the plantar stratum corneum, are well...

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Neurotrauma During Military Training: Results of the INVestigating traIning assoCiated blasT pAthology (INVICTA) Study

Walter S Carr 2026-10-01 1 min 67 words TACTILE ACUITY

Summary: Concern is growing that subconcussive blast exposure during military exercises such as heavy weapons training (HWT) may adversely affect brain function. HWT involves firing shoulder-mounted weapons that can expose the gunner, assistant gunner, and instructors (range safety officers [RSOs]) to blast ...

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Altered body representation in chronic non-specific low back pain: a testable framework linking body-schema-related disturbance, corticomotor alterations, and altered pain processing

👤 Zhifei Xie
📅 2026-10-24 📖 1 min 📝 66 words

TACTILE ACUITY

Summary: Low back pain is a leading contributor to disability worldwide; in chronic non-specific low back pain (CNSLBP), the processes sustaining pain and disability in the absence of identifiable structural pathology remain incompletely understood. Three partially overlapping research lines have documented ...

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Hyperexcitable primary somatosensory cortex in the MitoPark mouse model of Parkinson's disease

Alan J Emanuel 2026-10-25 1 min 67 words TACTILE ACUITY

Summary: The majority of Parkinson's disease patients suffer from decreased tactile acuity or neuropathic pain, both of which contribute to low quality of life and likely exacerbate motor dysfunction. How Parkinson's disease pathology relates to altered somatosensation remains unclear. One potential contribu...

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The Relationship Between Occlusal Tactile Acuity, Bite Force and Tooth Wear

Birgitta Häggman-Henrikson 2026-11-03 1 min 52 words TACTILE ACUITY

Summary: CONCLUSIONS: Bite force may be a clinically relevant factor associated with accumulated tooth wear, whereas OTA was not independently associated with tooth wear. Longitudinal studies are needed to determine whether bite force can predict tooth wear progression and whether more comprehensive assessme...

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Impact of Braille Booklet, Audio-Tactile Performance and Caregiver-supervised Training on Oral Hygiene Status, Knowledge, Attitude and Practices among Visually Challenged Children- A Randomized Clinical Trial

👤 Priyanga Chandrasekaran
📅 2026-09-22 📖 1 min 📝 52 words

BRAILLE

Summary: CONCLUSION: Both caregiver-supervised training and audio tactile performance using specialized model method of oral health education proved to be more effective than the braille booklet group on plaque control and improvement in the oral hygiene status, oral health knowledge, attitude and practice i...

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Design and functional verification of an automated Perkins Brailler accessory with audio feedback

👤 Ali Alhaj 📅 2026-09-23 📖 1 min 📝 67 words BRAILLE

Summary: CONCLUSIONS: The study did not include learners with visual impairment, Braille teachers, rehabilitation specialists, or other end users, and it did not collect quantitative measures of accuracy across the full character set, writing speed, embossing quality, latency, durability, safety, usability, ...

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Uncertainty-Aware structured decoding for Chinese Braille document transcription

👤 Haoran Sun 📅 2026-09-25 📖 1 min 📝 64 words BRAILLE

Summary: Automatic transcription of braille documents remains challenging in document digitization settings, particularly under real imaging degradations such as uneven embossing, illumination variation, paper texture, skew, blur, and ghosting, which make braille dot presence uncertain and often cause error ...

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Clinical Characteristics and Epidemiology of Isolated Dystonia: Results from a US Claims Analysis

James E Thornton 2026-10-08 1 min 54 words BRAILLE

Summary: CONCLUSIONS: This study defines an isolated dystonia cohort in the US based on real-world evidence and systematic exclusion of secondary causes of dystonia. Estimated prevalence is within range of prior epidemiological studies, and the cohort showed increased rates of comorbid psychiatric disorders....

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Effects of multisensory tactile-auditory integration teaching on systematic anatomy learning in visually impaired medical students

Fanzhen Kong 2026-10-08 1 min 50 words BRAILLE

Summary: CONCLUSION: The tactile-auditory integrated training system significantly improves objective summative knowledge acquisition and subjective learning experiences in visually impaired medical students. This approach provides a scalable, cost-constrained, and replicable pedagogical model for inclusive ...

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Physical and Environmental Accessibility of Dental Services for Persons with Disabilities in Saudi Primary Health Care: A Cross-Sectional Study

👤 Estie Kruger 📅 2026-10-16 📖 1 min 📝 66 words BRAILLE

Summary: Background/Objectives: Persons with disabilities (PWDs) may experience barriers when accessing oral health care, including those related to the physical and sensory environment of the dental-care setting. In this study we assessed the accessibility of primary health care centres (PHCs) providing den...

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Visible Light Driven Diselenide Chemistry for Simultaneous 3D Patterning and Mechanical Reinforcement

👤 Huaping Xu 📅 2026-10-16 📖 1 min 📝 63 words BRAILLE

Summary: Three-dimensional (3D) patterning on polymer surfaces is typically achieved by photolithography or related techniques, which often involve complex procedures, high-energy ultraviolet light, and risk of material damage. Here, we report a visible light-driven strategy that exploits the dual functional...

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An accessible pregnancy test for blind and low-vision users

👤 Vanja Garaj 📅 2026-10-16 📖 1 min 📝 63 words BRAILLE

Summary: CONCLUSIONS: The findings suggest that accessible pregnancy testing must prioritise not only non-visual result reading but also privacy, dignity, reproductive autonomy and independent health management. These results represent the preliminary evidence of acceptability rather than a conclusive test o...

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Genome-Informed Loop-Mediated Isothermal Amplification (LAMP) Assays for Detection and Differentiation of Ralstonia pseudosolanacearum Phylotypes I and III

👤 Jarred Yasuhara-Bell 📅 2026-10-27 📖 1 min 📝 76 words BRAILLE

Summary: The Ralstonia solanacearum species complex (RSSC) causes bacterial wilt in a broad range of economically important hosts. The RSSC has been reclassified into three species, of which R. solanacearum sequevars 1 and 2 in phylotype IIB are designated as Select Agents in the U.S. Many rapid tests for "R...

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Encoding tactile stimuli for Braille recognition with organoids

👤 Benjamin Ward-Cherrier 📅 2026-10-31 📖 1 min 📝 62 words BRAILLE

Summary: This study proposes a transferable encoding strategy that maps tactile sensor data to electrical stimulation patterns, enabling neural organoids to perform an open-loop artificial tactile Braille classification task. Human forebrain organoids cultured on a low-density microelectrode array (MEA) are ...

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Concurrent prefrontal neurochemical effects of high-definition transcranial direct current stimulation: directional E-field associations with GABA+ and Glx

👤 Jianqi Li 📅 2026-11-05 📖 1 min 📝 58 words TDCS TACS TRNS

Summary: Transcranial direct current stimulation (tDCS) is a promising neuromodulatory technique for treating neuropsychiatric disorders. However, its neurochemical mechanisms remain poorly characterized, and clinical efficacy is limited by substantial inter-individual variability. This study used concurrent...

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Anodal tDCS alters perceived visual contrast in humans

Sang Wook Hong 2026-11-06 1 min 62 words TDCS TACS TRNS

Summary: Cortical gain modulation is thought to alter perceptual experience, most prominently demonstrated by the effects of spatial attention on contrast appearance. Whether externally induced changes in cortical excitability produce comparable perceptual consequences remains unclear. Here, we tested whethe...

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Evaluating OPTION: Benefits and Pitfalls of a Youth Trafficking Diversion Court in Florida

Joan A Reid 2026-11-06 1 min 67 words TDCS TACS TRNS

Summary: Trafficking diversion courts (TDCs) provide an alternative approach to supporting trafficked youth by promoting recovery, preventing revictimization, and diverting youth from the justice system. The current study examines the benefits and challenges of the OPTION (Offering Potentially Trafficked Ind...

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Amelioration of diabetic autonomic neuropathy symptoms following a course of transcranial direct current stimulation

👤 Ahmad Alipour 📅 2026-11-06 📖 1 min 📝 62 words TDCS TACS TRNS

Summary: Cardiovascular autonomic neuropathy (CAN) is a severe and potentially life-threatening complication of diabetes. This randomized, sham-controlled, three-arm trial evaluated the efficacy of anodal transcranial direct current stimulation (tDCS) targeting the left dorsolateral prefrontal cortex for imp...

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From Single Cells to the Circulating Tumor Microenvironment: Label-Free Size-Selection and a Morphological Taxonomy for Liquid Biopsy in Cytopathology

👤 Satoru Kobayashi 📅 2026-11-09 📖 1 min 📝 25 words TDCS TACS TRNS

Summary: CONCLUSION: Size-selection converts a liquid biopsy into a cytology slide, placing interpretation of the circulating tumor ecosystem within the existing competence of the diagnostic laboratory.

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Phase-Dependent Effects of Cerebellar Transcranial Alternating Current Stimulation on Lateral Vestibulospinal Tract Excitability

👤 Tsubasa Mitsutake
📅 2026-11-09 📖 1 min 📝 36 words

TDCS TACS TRNS

Summary: CONCLUSIONS: LVST excitability differed according to the phase of the ongoing cerebellar-targeted tACS waveform. These findings provide electrophysiologic evidence of an online phase-dependent effect on vestibulospinal excitability and a physiologic basis for future phase-specific neuromodulation st...

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Effects of Transcranial Direct Current Stimulation on Core Symptoms and Sleep Disturbances in Children With Autism Spectrum Disorder: A Meta-Analysis of Randomized Controlled Trials

👤 Meimei Yang
📅 2026-11-09 📖 1 min 📝 44 words

TDCS TACS TRNS

Summary: CONCLUSION: tDCS, most commonly delivered at 1 mA over the DLPFC, was associated with improvements in some ASD-related behavioral outcomes; preliminary evidence also suggested a potential benefit for sleep disturbances, although this finding was based on only two low-certainty studies and warrants c...

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No effects of single-session concurrent tDCS on explicit motor sequence learning in healthy adults with conventional 2 mA and tangential 6 mA currents

👤 Lucas C Parra
📅 2026-11-09 📖 1 min 📝 45 words

TDCS TACS TRNS

Summary: CONCLUSION: These findings, together with a survey of the existing literature, suggest that the effects of single-session concurrent tDCS on explicit motor sequence learning are not robust across studies. Our results underscore the necessity of rigorous methodological controls to prevent false disco...

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More than Quaternary Ammonium Compounds: First Evidence of Tertiary Amine Compounds as Novel Contaminants Ubiquitously Present in Indoor Dust

👤 Runzeng Liu 📅 2026-11-10 📖 1 min 📝 65 words TDCS TACS TRNS

Summary: Nitrogen-containing organic compounds have attracted global concern due to their widespread production and potential environmental risks. While the environmental occurrence of quaternary ammonium compounds (QACs) has been widely studied, research on tertiary amine compounds (TACs), another important...

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Combining transcranial electrical and magnetic stimulation: a scoping review of neurophysiological and clinical studies

👤 Paul M Briley 📅 2026-11-10 📖 1 min 📝 56 words TDCS TACS TRNS

Summary: CONCLUSIONS: Studies indicate that tES could enhance or reverse expected effects of rTMS. The greatest increase in cortical excitability occurred following excitatory rTMS combined with gamma transcranial alternating current stimulation (tACS) or with cathodal tDCS, consistent with effects of metapl...

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Prefrontal Activation and Cognitive Heterogeneity in Pediatric Sleep-Disordered Breathing

Amal Isaiah 2026-11-05 1 min 39 words FNIRS

Summary: CONCLUSIONS AND RELEVANCE: In this cross-sectional study, task-related PFC activation was associated with SDB severity and moderated the severity-cognition association. fNIRS may help identify children whose cognitive profiles deviate from conventional severity metrics, thereby supporting individual...

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Mirror Neural System in Hands Motor Imagery with Pictures and Videos Guidance: An EEG-fNIRS Study

Mohamad Sawan 2026-11-05 1 min 64 words FNIRS

Summary: Brain-computer interfaces (BCIs) achieved by motor imagery (MI) have been widely accepted as an effective tool in rehabilitative applications. Functional near-infrared spectroscopy (fNIRS) and electroencephalography (EEG) are two common implementation forms of BCIs, capable of monitoring neuronal po...

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Quality-aware mask-propagation graph attention for fNIRS-based bipolar disorder classification without channel interpolation

👤 Yajing Meng
📅 2026-11-05 📖 1 min 📝 58 words FNIRS

Summary: Bipolar disorder (BD) remains difficult to characterise objectively, motivating a neural marker to complement clinical assessment. Functional near-infrared spectroscopy (fNIRS) offers portable cortical haemodynamic measurement, but channel availability can vary across recordings because of bad chann...

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Brain-Behavior Coupling During Cognitive-Postural Dual-Tasking in Older Adults With Mild Cognitive Impairment: An fNIRS Study

Chuan Guo 2026-11-06 1 min 56 words FNIRS

Summary: CONCLUSIONS: Older adults with MCI exhibit altered brain-behavior coupling during cognitive-postural dual-task conditions involving sensory and postural constraints, characterized by altered relationships between prefrontal-sensorimotor cortical activation and behavioral performance. These findings ...

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Item-Specific Proportion Congruency Uncovers Different Forms of Attentional Control: An event-related fNIRS Study

Murat Perit Çakır 2026-11-07 1 min 70 words FNIRS

Summary: CONCLUSION: While behavioral results mostly support contingency learning, fNIRS data revealed ISPC-related activation in prefrontal channels with sensitivity to BA 46 (lateral, channel 1) and the BA 10/BA 11 overlap zone (medial, channels 10-12), with channel 11 emerging as a convergence point for b...

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Does the emotion regulation of self-directed humor engage the prefrontal cortex? An fNIRS study

Zhonglu Zhang 2026-11-07 1 min 68 words FNIRS

Summary: Previous research has revealed that self-directed humor (self-enhancing and self-defeating) can help regulate negative emotions and that the prefrontal cortex (PFC) plays a critical role in emotion regulation. However, it remains unknown whether emotion regulation of self-directed humor is associate...

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Effects of mini-maps with different reference frames on polarity-free 3D spatial navigation: a behavioural and fNIRS study

Xiaozhou Zhou 2026-11-07 1 min 62 words FNIRS

Summary: Manned spaceflight and deep-sea exploration extend human activities to extreme environments lacking gravitational and visual polarity cues, where spatial disorientation frequently occurs. Mini-map studies largely focus on terrestrial environments, while reference frame design for polarity-free 3D sp...

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Baduanjin improves motor function and cortical activation in children with developmental coordination disorder: a randomized controlled trial

👤 Wei Song 📅 2026-11-09 📖 1 min 📝 40 words FNIRS

Summary: CONCLUSIONS: Baduanjin training improved motor coordination and balance in children with DCD. The observed changes in activation within motor-related cortical regions provide preliminary support for a possible neurophysiological basis of these functional improvements, although the underlying mechani...

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Potential Applications of Brain-Computer Interface Technology in the Rehabilitation of Patients with Anxiety Disorders: A Narrative Review

Hu Lou 2026-11-10 1 min 58 words FNIRS

Summary: Anxiety disorders are among the most prevalent mental disorders worldwide and can cause persistent impairment in emotional regulation, cognition, and social functioning. Pharmacotherapy, psychotherapy, exercise, and non-invasive brain stimulation are widely used, yet their effectiveness may be const...

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Multimodal EEG-fNIRS complexity analysis can reveal the neurovascular coupling change with aging

Yicheng Zhu 2026-11-10 1 min 58 words FNIRS

Summary: CONCLUSION: Aging is associated with task-specific alterations in neural complexity, hemodynamic complexity, and the cross-modal temporal relationship between EEG and fNIRS signals. These findings suggest that age-related changes in NVC-related dynamics are better characterized as dynamic reorganiza...

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Potential Applications of Brain-Computer Interface Technology in the Rehabilitation of Patients with Anxiety Disorders: A Narrative Review

👤 Hu Lou 📅 2026-11-10 📖 1 min 📝 58 words BRAIN COMPUTER INTERFACE

Summary: Anxiety disorders are among the most prevalent mental disorders worldwide and can cause persistent impairment in emotional regulation, cognition, and social functioning. Pharmacotherapy, psychotherapy, exercise, and non-invasive brain stimulation are widely used, yet their effectiveness may be const...

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In Situ Displacement Reaction Enables a Conductive Metallized Hydrogel Interface for Smart Underwater Flexible Electronics

👤 Qigang Wang
📅 2026-11-10 📖 1 min 📝 60 words

BRAIN COMPUTER INTERFACE

Summary: Underwater wearable sensing is severely hampered by the intrinsic drawbacks of conventional materials, including water absorption, swelling, and electrical drift during long-term immersion. Here, we develop an anti-swelling hydrogel with a surface-metallized layer, which achieves strong stability of...

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Fast-Gelling, Adhesive, and Conductive Epidermal Electrode for Accurate Physiological Signal Monitoring and Real-Time Brain-Controlled Drone Flight

Zhou Li 2026-11-10 1 min 55 words BRAIN COMPUTER INTERFACE

Summary: Conductive hydrogel electrodes are vital components for achieving human-machine interfaces. However, the inherent trade-off between high conductivity and favorable mechanical/adhesive properties remains a significant challenge for hydrogel-based epidermal electrodes. Here, we developed a multifuncti...

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Bayesian Adaptive Stimulus Optimization in P300 Brain Computer Interfaces

Boyla O Mainsah 2026-11-10 1 min 26 words BRAIN COMPUTER INTERFACE

Summary: CONCLUSION: For computational simplicity in implementation, the optimal stimulus can generally be approximated by finding the stimulus that has a combined probability mass closest to 0.5.

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MSTA-CNN an ultra-lightweight multi-scale temporal-spatial attention convolutional neural network for high-performance P300 spellers

👤 Mousa Shamsi
📅 2026-11-10 📖 1 min 📝 62 words

BRAIN COMPUTER INTERFACE

Summary: Efficient P300 detection in brain-computer interfaces (BCIs) remains a challenge due to low signal-to-noise ratio (SNR) and the high computational cost of existing deep learning models, which limits their deployment in real-time applications. To address these issues, we propose MSTA-CNN, an ultra-li...

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Detection of linear dependencies in EEG oscillatory sources via PARALIND and block-term decomposition: A comparative study

👤 Leonardo Jose Trejo
📅 2026-11-10 📖 1 min 📝 60 words

BRAIN COMPUTER INTERFACE

Summary: Narrowband oscillatory rhythms in electroencephalogram (EEG) signals are fundamental signatures of neural dynamics, crucial for brain-computer interfaces (BCIs) and monitoring neurological disorders. While traditionally detected and analyzed via Independent or Principal Component Analysis in the tim...

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Lower frequency ECoG features improve longitudinal stability of speech and hand-grasp detection in a patient with progressive ALS

👤 Dean J Krusienski 📅 2026-11-10 📖 1 min 📝 54 words BRAIN COMPUTER INTERFACE

Summary: Achieving long-term decoding stability is a critical challenge for chronic
intracranial brain-computer interfaces (BCIs). High-gamma (HG) activity from
intracranial recordings is the standard signal for BCI control, and lower-frequency features
can enhance HG decoding performance. This s...

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A collaborative network of bi-Mamba and CAFS for multi-task classification of unilateral lower limb

👤 Junhua Li 📅 2026-11-10 📖 1 min 📝 66 words BRAIN COMPUTER INTERFACE

Summary: Accurate decoding of motor-related electroencephalography signals remains challenging in brain-computer interface research. To date, motor imagery of the upper limbs has been extensively investigated, while research on the lower limbs remains limited. Meanwhile, multiple tasks involving the unilater...

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A theory of multilevel interactive equilibrium in neuroAI

Quanyan Zhu 2026-11-10 1 min 70 words BRAIN COMPUTER INTERFACE

Summary: We propose a class of game-theoretic framework for adaptive multi-agent intelligent systems. Unlike the classical game theory which assumes that strategies are primitive objects chosen by perfectly rational agents, the new framework provides a mathematical foundation for studying equilibrium in Neur...

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Interictal Electroencephalographic Characteristics of the Human Centromedian Thalamic Nucleus Recorded in Focal Epilepsy

Shuang Wang 2026-11-10 1 min 23 words BRAIN COMPUTER INTERFACE

Summary: CONCLUSIONS: CM SEEG demonstrates distinct, state-dependent electrophysiological dynamics across the sleep-wake cycle and preferential involvement in interictal epileptic activity compared with VL/VP nuclei.

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Tim Peters has passed away

👤 /u/ExoticMandibles 📅 2026-11-11 📖 1 min 📝 142 words REDDIT PYTHON

> **Summary:** <!-- SC_OFF --> Tim died July 15, at his home in Ashland, Wisconsin. He was 74. We don't know any specifics yet. In case you don't know who Tim Peters is, he was an early pioneer in helping shape the Python language and its runtime. He invented a new sorting algorithm (&quot...

🔗 **Read full article:**
https://www.reddit.com/r/Python/comments/1xt65ah/tim_peters_has_passed_away/

$30M Raised to Redo the CPU

👤 shioyama 📅 2026-11-10 📖 1 min 📝 13 words HACKER NEWS

> **Summary:** Article URL: https://tendrils.co/blog/2026-11-10/30m-raised Comments URL: https://news.ycombinator.com/item?id=50447036 Points: 75 # Comments: 67

🔗 **Read full article:**
https://tendrils.co/blog/2026-11-10/30m-raised

AI is becoming an information gateway. Audiences still don’t want it making news

👤 rdmuser 📅 2026-11-11 📖 1 min 📝 13 words HACKER NEWS

> **Summary:** Article URL: https://www.niemanlab.org/2026/11/ai-is-becoming-an-information-gateway-audiences-still-dont-want-it-making-their-news/ Comments URL: <a href="...

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https://www.niemanlab.org/2026/11/ai-is-becoming-an-information-gateway-audiences-still-dont-want-it-making-their-news/

LLMs Play Zachtronics' TIS-100

👤 fairestvalue 📅 2026-11-11 📖 1 min 📝 13 words HACKER NEWS

> **Summary:** Article URL: https://tis-100.wincon.dev/ Comments URL: https://news.ycombinator.com/item?id=50448340 Points: 41 # Comments: 7

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https://tis-100.wincon.dev/

A great old-timey game-programming hack (2013)

👤 palsecam 📅 2026-11-11 📖 1 min 📝 13 words HACKER NEWS

> **Summary:** Article URL: https://blog.moertel.com/posts/2013-12-14-great-old-timey-game-programming-hack.html Comments URL: <a href="https://news.ycombinator.com/item?id=50448921">https://news.ycombinato...

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https://blog.moertel.com/posts/2013-12-14-great-old-timey-game-programming-hack.html

Hooked on Tuna

👤 dnetesn 📅 2026-11-11 📖 1 min 📝 14 words HACKER NEWS

> **Summary:** https://blue-continuum.com/hooked-on-tuna <hr /> Comments URL: https://news.ycombinator.com/item?id=50449080 Points: 25 # Comments: 2

🔗 **Read full article:**
https://theconversation.com/hooked-on-tuna-people-have-been-harvesting-these-ocean-predators-for-44-000-years-292428

To deter bioterrorism, U.S. imposes new rules on ordering synthetic DNA

👤 Metacelsus 📅 2026-11-11 📖 1 min 📝 13 words HACKER NEWS

> **Summary:** Article URL: https://www.science.org/content/article/deter-bioterrorism-u-s-imposes-new-rules-ordering-synthetic-dna Comments URL: <a href="https://news.ycombinator.com/ite...

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https://www.science.org/content/article/deter-bioterrorism-u-s-imposes-new-rules-ordering-synthetic-dna

Protecting SSH Keys with macOS Secure Enclave

👤 js2 📅 2026-11-11 📖 1 min 📝 13 words HACKER NEWS

> **Summary:** Article URL: https://gatezh.com/posts/macos-secure-enclave-ssh-keys/ Comments URL: https://news.ycombinator.com/item?id=50449748 Points: 83 # Comm...

🔗 **Read full article:**
https://gatezh.com/posts/macos-secure-enclave-ssh-keys/

Greg Kurtzer's Ginux is now a userspace kernel

👤 d3Xt3r 📅 2026-11-11 📖 1 min 📝 13 words HACKER NEWS

> **Summary:** Article URL: https://github.com/gmkurtzer/ginux Comments URL: https://news.ycombinator.com/item?id=50450007 Points: 34 # Comments: 6

🔗 **Read full article:**
https://github.com/gmkurtzer/ginux

There are 13 ways to analyse a replication study, but only one of them is coherent.

👤 noreply@blogger.com (Daniel Lakens) 📅 2026-05-03 📖 6 min 📝 1361 words

TWENTY PERCENT STATISTICIAN

> **Summary:** Paul Simon’s says there are 50 ways to leave your lover, and a similar abundance characterizes the contemporary literature on how to analyse a replication study. The recent SCORE replication project in Nature makes this explicit by reporting “13 replication success metrics along with the number o...

🔗 **Read full article:**
http://daniellakens.blogspot.com/2026/05/there-are-13-ways-to-analyse.html

Evaluating Dr. Cuddy’s Claim that the Debunking of Power Posing is a Myth

noreply@blogger.com (Daniel Lakens)
2026-05-25 30 min 6008 words

TWENTY PERCENT STATISTICIAN

Summary: In this blog post I will analyse the arguments that Dr. Amy Cuddy provided in a LinkedIn post “The "Power Posing Was Debunked" Myth: What the Research Actually Shows — and Why Scientific Discourse Matters” on February 26. You can find the LinkedIn post here: <span lang="...

🔗 **Read full article:**
http://daniellakens.blogspot.com/2026/05/evaluating-dr-cuddys-claim-that.html

Which Data Repository Should You Use?

noreply@blogger.com (Daniel Lakens)
2026-08-15 17 min 3434 words

TWENTY PERCENT STATISTICIAN

Summary: Update 28-10-2026: ResearchBox now has an API. This new information has been incorporated in the text below. The Center for Open Science has announced that from November 16, 2026, no new projects can be created on the Open Sci...

🔗 **Read full article:**
http://daniellakens.blogspot.com/2026/08/which-data-repository-should-you-use.html

Episode 29: Q&A with Dr. Anahita Kyani, Director of Neuroscience, Data Science and Digital Health at Johnson & Johnson Innovative Medicine

👤 Adriel Carridice 📅 2026-06-22 📖 1 min 📝 0 words BRAIN

🔗 **Read full article:**
https://brain.ieee.org/podcasts/episode-29-qa-with-dr-anahita-kyani-director-of-neuroscience-data-science-and-digital-health-at-johnson-johnson-innovative-medicine/

Q&A with Felix Schwock, PhD student at University of Washington and Best Poster Award Winner at the 2025 IEEE Brain Discovery and Neurotechnology Workshop

👤 ieeebrain 📅 2026-06-22 📖 1 min 📝 0 words BRAIN

🔗 **Read full article:**
https://brain.ieee.org/podcasts/qa-with-felix-schwock-phd-student-at-university-of-washington-and-best-poster-award-winner-at-the-2025-ieee-brain-discovery-and-neurotechnology-workshop/

Neuroethical Perspectives on Neurotechnology in Latin America

Adriel Carridice 2026-09-23 1 min 0 words BRAIN

🔗 **Read full article:**
https://brain.ieee.org/publications/research-and-white-papers/neuroethical-perspectives-on-neurotechnology-in-latin-america/

Transactions on Neural Systems and Rehabilitation Engineering (TNSRE) Editor in Chief Search

Deidre Artis 2026-07-17 1 min 34 words EMBS

Summary: Transactions on Neural Systems and Rehabilitation Engineering (TNSRE) Editor-in-Chief Search Submit your Application The post <a href="https://www.embs.org/featured-news/transactions-on-neural-systems-and-rehabilitation-engineering-tnsre-editor-in-chief-search/">Transactions on Neural Syst...

🔗 **Read full article:**
https://www.embs.org/featured-news/transactions-on-neural-systems-and-rehabilitation-engineering-tnsre-editor-in-chief-search/

A complete adult Drosophila melanogaster brain tracheolar network

👤 Cho, H., Ng, T., Schoenberg, K., Huntress, S., Suh, J., Manandhar, A., Lin, S., Woldetsadik, B. M., Kang, R. S., Bui, M., Zebrowski, L. M., Reynolds-Cornell, F., Agbedun, E., Cruz, Y., Lewis, C. A., Bhasiin, K., Lin, K., Mooney, C., Dacks, A. M., Schlegel, P., Bock, D. D., Colodner, K. J.

📅 2026-11-10 📖 1 min 📝 222 words BIORXIV NEUROSCIENCE

Summary: The tracheal system of the fruit fly, Drosophila melanogaster, is a complex network of epithelial tubules that supplies oxygen and removes carbon dioxide throughout the organism, with gases entering and exiting through spiracles along the body. While tracheal development and patterning in the embryo...

🔗 **Read full article:**
https://www.biorxiv.org/content/10.64898/2026.11.09.765327v1?rss=1

HMGB1/CXCL12 heterocomplex activates CXCR4 to suppress the ability of Muller glia to become proliferating progenitor-like cells in the chick retina

👤 Kelly, L., Taylor, O., El-Hodiri, H., Lee, S., Parekh, E., Hawthorn, E., Wright, J., Guest, M., Tangemen, A., Fischer, A.

📅 2026-11-10 📖 1 min 📝 229 words BIORXIV NEUROSCIENCE

Summary: We investigated how High Mobility Group Box 1 (HMGB1) signaling regulates glial reactivity and Muller glia-derived progenitor cell (MGPC) formation in the chick retina. HMGB1 released from stressed cells is known to form a heterocomplex with C-X-C chemokine ligand 12 (CXCL12) to activate C-X-C chemo...

🔗 **Read full article:**
https://www.biorxiv.org/content/10.64898/2026.11.09.765360v1?rss=1

Pharmaco-genetic evidence of presymptomatic lysosomal dysfunction coupled to detrimental DPR secretion in Drosophila Models of C9orf72-linked ALS/FTD.

Smeele, P. H., Ferrari, V., Spina, G., Dona', E., Cauchi, R., Poletti, A., Vaccari, T.

2026-11-10

1 min

205 words

BIORXIV NEUROSCIENCE

Summary: Hexanucleotide repeat expansions in C9orf72, the most common genetic cause of Amyotrophic lateral sclerosis (ALS) and frontotemporal dementia (FTD), drive neurodegeneration through both loss of gene function and toxic gain-of-function mechanisms, including transcription of repeat RNA and repeat-asso...

🔗 **Read full article:**
https://www.biorxiv.org/content/10.64898/2026.11.09.765393v1?rss=1

Capturing continuous spectrotemporal pupil--brain coupling with simultaneous wearable EEG/Eye tracking

Liu, T., Kotz, S. A., Schwartze, M., Criscuolo, A.

2026-11-10

1 min

181 words

BIORXIV NEUROSCIENCE

Summary: Spontaneous fluctuations in pupil diameter are closely linked to neural dynamics, yet most evidence for pupil--brain coupling has been obtained in conventional laboratory settings. We therefore asked whether the spectrotemporal and directional organization of such coupling can be captured with weara...

🔗 **Read full article:**
https://www.biorxiv.org/content/10.64898/2026.11.08.765166v1?rss=1

Sensory and Motor Building Blocks of an Extra Robotic Digit Representation

👤 Dowdall, L., Amoruso, E., Dominijanni, G., Molina-Sanchez, M., Diedrichsen, J., Clode, D., Makin, T. R.

📅 2026-11-10 📖 1 min 📝 195 words BIORXIV NEUROSCIENCE

Summary: Supernumerary limbs provide a unique opportunity to ask how the brain represents an entirely novel effector within an existing sensorimotor system. Here, we investigated how primary motor cortex (M1) represents a robotic extra digit (the Third Thumb; Dani Clode Design), and how this representation c...

🔗 **Read full article:**
https://www.biorxiv.org/content/10.64898/2026.11.09.765353v1?rss=1

Whole-Brain MRSI Metabolic Connectomics Reveals Network Disruptions and Altered Gradient Coherence in Psychosis Risk

👤 Lucchetti, F., Celereau, E., Jenni, R., Petrova, T., Abrahamyan, L., Conus, P., Hagmann, P., Aleman-Gomez, Y., Klauser, A., Klauser, P.

📅 2026-11-10 📖 1 min 📝 235 words BIORXIV NEUROSCIENCE

Summary: Brain network alterations preceding first-episode psychosis often escape connectomic approaches based on conventional neuroimaging, such as diffusion or functional magnetic resonance imaging (MRI). Whole-brain magnetic resonance spectroscopic imaging (MRSI) offers a potentially more sensitive altern...

🔗 **Read full article:**
https://www.biorxiv.org/content/10.64898/2026.11.09.765372v1?rss=1

Spatial proteomics on repurposed sequencers reveals early astrocytic remodeling in an ALS mouse model

Pandit, K., Fouts, C., Rodwin, S., Maniatis, S., Dhingra, K., Marrakchi, S., Domovic, D., Singh, J., Dumitrascu, B., Phatnani, H., Pleska, M., Vickovic, S.

2026-11-10 1 min 216 words BIORXIV NEUROSCIENCE

Summary: Spatial proteomics can reveal how disease reshapes cells and their local environments, but extending these measurements across replicated tissue cohorts requires validated antibodies, consistent sample processing and integrated analysis. These requirements are particularly pressing in studies of amy...

Read full article:
https://www.biorxiv.org/content/10.64898/2026.11.08.765207v1?rss=1

Evidence-based re-identification: Leveraging appearance cues to correct tracking mistakes rather than trying to prevent them

Coulombe, V., Leboulleux, Q., Gosselin, B., Testard, C., Labonte, B.

2026-11-10 1 min

300 words

BIORXIV NEUROSCIENCE

Summary: Naturalistic behavioral neuroscience experiments, in which groups of animals interact in enriched environments over prolonged periods, require each animal's identity to remain correct throughout the recording. Current multi-animal tracking tools assign identities frame by frame without revisiting pa...

Read full article:
https://www.biorxiv.org/content/10.64898/2026.11.08.765193v1?rss=1

Competing routes back to firing shape population hyperexcitability and its selective suppression

Kamaraj, A. K., Szuromi, M. P.

2026-11-10 1 min 200 words

BIORXIV NEUROSCIENCE

Summary: Whether a neuron will fire again is fundamentally a different question from whether a quiescent neuron will activate. Even within the same neuron, successive spikes can follow distinct post-spike trajectories, with after-depolarization permitting early return to firing and depolarization block prolo...

Read full article:
https://www.biorxiv.org/content/10.64898/2026.11.09.765141v1?rss=1

Object features survive motion at saccadic speed but lose their link to specific objects

Ince, M., Hübner, C., Rolfs, M. 2026-11-10 1 min 202 words BIORXIV NEUROSCIENCE

Summary: Saccadic eye movements sweep objects across the retina at high speed, and this motion is routinely omitted from conscious perception. Whether object identity survives such motion is unclear. In two preregistered experiments (N = 20 and N = 19), we asked how object motion at saccadic speeds, while th...

🔗 **Read full article:**
https://www.biorxiv.org/content/10.64898/2026.11.09.765345v1?rss=1

Human cortical high-gamma power scales with movement rate in healthy participants and stroke survivors

Fanny Quandt 2025-01-09 1 min 65 words OOSTENVELD ROBERT

Summary: Motor cortical high-gamma oscillations (60-90 Hz) occur at movement onset and are spatially focused over the contralateral primary motor cortex. Although high-gamma oscillations are widely recognized for their significance in human motor control, their precise function on a cortical level remains el...

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NIRS-BIDS: Brain Imaging Data Structure Extended to Near-Infrared Spectroscopy

👤 Luca Pollonini 📅 2025-01-27 📖 1 min 📝 70 words OOSTENVELD ROBERT

> **Summary:** Functional near-infrared spectroscopy (fNIRS) is an increasingly popular neuroimaging technique that measures cortical hemodynamic activity in a non-invasive and portable fashion. Although the fNIRS community has been successful in disseminating open-source processing tools and a standard file forma...

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Pseudonymisation of neuroimages and data protection: Increasing access to data while retaining scientific utility

👤 Lyuba Zehl 📅 2025-06-26 📖 1 min 📝 67 words OOSTENVELD ROBERT

> **Summary:** For a number of years, facial features removal techniques such as 'defacing', 'skull stripping' and 'face masking/blurring', were considered adequate privacy preserving tools to openly share brain images. Scientifically, these measures were already a compromise between data protection requirements a...

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Cycling on the Freeway: The perilous state of open-source neuroscience software

👤 Tim M Tierney 📅 2025-08-13 📖 1 min 📝 74 words OOSTENVELD ROBERT

Summary: Most scientists need software to perform their research (Barker et al., 2020;Carver et al., 2022;Hettrick, 2014;Hettrick et al., 2014;Switters & Osimo, 2019), and neuroscientists are no exception. Whether we work with reaction times, electrophysiological signals, or magnetic resonance imaging data, ...

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Optimal configuration of on-scalp OPMs with fixed channel counts

👤 Robert Oostenveld 📅 2025-08-13 📖 1 min 📝 69 words OOSTENVELD ROBERT

Summary: Recent technological developments have brought optically pumped magnetometers (OPMs) within reach of the larger neuroscientific community. The current state-of-the-art consists of whole-head systems that measure the magnetic field at >100 locations. OPM sensors can be constructed to measure the fiel...

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ARTEM-IS for ERP: Agreed Reporting Template for EEG Methodology-International Standard for Event-Related Potential Experiments

👤 Suzy J Styles 📅 2025-12-08 📖 1 min 📝 73 words OOSTENVELD ROBERT

Summary: The choices we make during the recording, preprocessing and analysis of event-related potentials (ERP) data can affect study outcomes. As such, it is critical that they are transparently reported to allow for reproducibility. Yet, systematic reviews of reporting practices in the field have shown tha...

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Eye-Tracking-BIDS: the Brain Imaging Data Structure extended to gaze position and pupil data

👤 Julia-Katharina Pfarr 📅 2026-02-12 📖 1 min 📝 68 words OOSTENVELD ROBERT

Summary: The Brain Imaging Data Structure (BIDS) is a widely adopted, community-driven standard to organize neuroimaging data and metadata. Although numerous extensions have been developed to incrementally extend coverage to new modalities and data types, an unambiguous, granular specification for eye-tracki...

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Facilitating unusual bodily experiences and out-of-body experiences across wakefulness and sleep: A high-density EEG and neurophenomenology study

👤 Robert Oostenveld 📅 2026-02-12 📖 1 min 📝 67 words OOSTENVELD ROBERT

Summary: Unusual bodily experiences (UBEs) are illusory bodily perceptions that are not coherent with typical wakeful experiences, including flying or floating sensations, body distortions, and out-of-body experiences. This study examines UBEs in a controlled sleep laboratory using meditation and light stimu...

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The Impact of Brain Tumors and Craniotomy Lesions on Scalp EEG

👤 Vitória Piai 📅 2026-02-17 📖 1 min 📝 65 words OOSTENVELD ROBERT

Summary: Electroencephalography (EEG) is widely used in both research and clinical settings, yet its accuracy can be significantly impacted by subject-specific anatomical anomalies such as brain lesions and skull defects. This study investigates the effects of glioma-related brain lesions and craniotomy-indu...

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EMG-BIDS: an extension to the Brain Imaging Data Structure for electromyography

👤 Arnaud Delorme 📅 2026-08-01 📖 1 min 📝 64 words OOSTENVELD ROBERT

Summary: Electromyography (EMG) is fundamental to clinical assessment, rehabilitation, neuromuscular research, and human-machine interfaces. Despite decades of use, no widely adopted standard exists for organizing and sharing EMG data, limiting reusability and large-scale data aggregation. We present EMG-BID...

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Prior-guided whole-slide survival prediction with soft histopathologic priors from vision-language foundation models

👤 Zhihui Feng 📅 2026-11-09 📖 1 min 📝 62 words LOW VISION

Summary: Whole-slide image (WSI) survival prediction is complicated by heterogeneous tumor morphology, where prognostic evidence is distributed across tumor, stroma, immune infiltration, necrosis, and residual tissue regions and often appears as mixed patch-level tissue evidence. Existing models commonly agg...

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Reduced magnocellular-biased visual responses are associated with working memory and linguistic difficulties in Chinese developmental dyslexia: A hypothesized visual-cognitive cascade

👤 Tsz Wing Leung
📅 2026-11-09 📖 1 min 📝 66 words

LOW VISION

Summary: Developmental dyslexia (DD) affects 5-10% of children worldwide. Although phonological theories predominate, they may not fully explain reading difficulties in orthographically dense scripts such as Chinese. We investigated whether Chinese children with DD exhibit differences in early visual cortica...

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During natural vision, semantic novelty modulates fixation-related processing in primate cortex

👤 Lucas C Parra 📅 2026-11-09 📖 1 min 📝 74 words LOW VISION

Summary: We sample visual scenes with short gaze fixations separated by saccades. While low-level integration is known, semantic integration of foveal vision across fixations remains unclear. We hypothesized that the brain responds to changes in semantic information from one fixation to the next, and therefo...

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Health and welfare professionals' global contribution to the International Classification of Functioning, Disability and Health Core Set Vision Loss

👤 Dominique VAN DE Velde 📅 2026-11-10 📖 1 min 📝 64 words LOW VISION

Summary: CONCLUSIONS: A visual impairment exerts a broad and multidimensional impact on functioning, emphasizing the importance of emotional well-being, daily activities, and environmental supports. Regional differences highlight the need for culturally sensitive approaches in assessment and intervention. Th...

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Federated Learning: Envisioning a Pathway to More Accessible and Equitable Cleft Care

👤 Ghulam Qadir Fayyaz 📅 2026-11-10 📖 1 min 📝 66 words LOW VISION

Summary: Cleft lip and palate represents the most common congenital craniofacial anomaly worldwide, with low- and middle-income countries (LMICs) bearing nearly 84% of the global disease burden. Despite the instrumental role of global nongovernmental organizations (NGOs) in expanding access to cleft care acr...

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Dichoptic viewing reveals distinct low- and high-level mechanisms for temporal phase discrimination

👤 Keith A Schneider 📅 2026-11-10 📖 1 min 📝 68 words LOW VISION

Summary: Conscious awareness of temporal visual structure is limited to frequencies below ~10 Hz. Between ~10 and 60 Hz, flicker is visible, but temporal phase cannot be accurately discriminated; above ~60 Hz, stimuli are perceptually fused. Accurately estimating the cognitive limit of phase discrimination r...

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Trends in Near Vision Loss and Age-Sex-SDI Disparities in Belt and Road Countries Over Thirty Years

Xiaopan Li 2026-11-10 1 min 55 words LOW VISION

Summary: CONCLUSIONS: This study represents the first comprehensive analysis of NVL burden and disparities across BRI countries, revealing persistent inequalities by age, sex, and SDI. The findings reflect the diverse socio-economic and health-care contexts of BRI countries and highlight the urgent need for ...

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Feasibility of robotic-assisted surgery for nutcracker syndrome: a systematic review and single-arm meta-analysis

Jiang-Hua Zheng 2026-11-10 1 min 60 words LOW VISION

Summary: Nutcracker syndrome (NCS) results from compression of the left renal vein, and refractory symptoms require surgical reconstruction. Robotic platforms provide magnified three-dimensional vision and articulated instruments that are theoretically advantageous for vascular dissection and anastomosis wit...

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Segmentation, Reconstruction, and Grasp-Pose Estimation Using an RGB-D Camera for Robotic Apple Harvesting

Jianqiang Lu 2026-11-10 1 min 58 words LOW VISION

Summary: Accurate segmentation, three-dimensional reconstruction, and grasp-pose estimation of fruits and surrounding structures are essential for robotic harvesting in complex orchard environments. However, occlusions caused by branches, support wires, and irregular canopy structures remain major challenges...

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Intelligent rendering optimization for animated scenes using reinforcement learning and computer vision

Li Jia 2026-11-10 1 min 63 words LOW VISION

Summary: Real-time rendering of complex animated scenes is computationally intensive and often struggles to balance visual fidelity and performance in dynamic or densely populated environments, as conventional pipelines lack adaptive resource optimization, leading to reduced frame rates and degraded visual q...

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A single session of high-definition transcranial direct current stimulation does not modulate effects of knee two-point discrimination training or sensorimotor function in healthy adults: Double-blind randomised controlled trial

👤 Paul W Hodges
📅 2026-07-30 📖 1 min 📝 53 words

TACTILE ACUITY

Summary: CONCLUSION: A single 20-minute session of 1 mA HD-tDCS, with or without brief TPD training, does not acutely modify knee somatosensory, sensorimotor, or functional performance in healthy adults. Existing evidence at the hand may not translate to the lower limb. Future work should investigate higher-...

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Variation in glabrous skin physiology across anatomical locations on the human hand

👤 Martina C Meinke 📅 2026-08-13 📖 1 min 📝 76 words TACTILE ACUITY

Summary: Skin physiology varies greatly between anatomical locations, with impact on tactile sensitivity. Research on the variations in morphology and physiology between different anatomical locations of glabrous skin is limited so far. This study investigates the morphology and physiology of glabrous skin o...

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Beyond Hamstring Shortness: Continuous Flexibility Is Associated With Balance and Stability but Not Sensory Measures-A Cross-Sectional Study

👤 Senanur Akıncı 📅 2026-08-14 📖 1 min 📝 66 words TACTILE ACUITY

Summary: CONCLUSIONS: Greater hamstring flexibility was associated with better dynamic balance performance and showed a weaker association with postural stability, whereas no robust associations were observed with sensory measures of postural control. These findings suggest that hamstring flexibility may be ...

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Resting-state EEG microstate dynamics reflect individual differences in tactile angle discriminability

Liu Yang 2026-08-19 1 min 63 words TACTILE ACUITY

Summary: Tactile spatial acuity, the ability to discriminate fine spatial details, is a fundamental sensory capacity modulated by both peripheral and central mechanisms. Here, we investigated whether resting-state EEG microstate dynamics are associated with individual differences in tactile spatial acuity. F...

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A grating orientation task with controlled dome elevation speed and contact force: stimulus conditions and day-to-day reliability

Hideaki Onishi 2026-09-25 1 min 66 words TACTILE ACUITY

Summary: The grating orientation discrimination task (GOT) is typically used to assess tactile spatial acuity. However, performance is influenced by stimulus speed, contact force between the stimulator and the skin, and the method of discrimination. This study aimed to explore suitable stimulus conditions fo...

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Epidermal microstructure and tactile sensitivity in the glabrous skin of hands and feet

Hannes P Saal 2026-09-25 1 min 69 words TACTILE ACUITY

Summary: Glabrous skin on the human palms and soles is specialized for mechanical interaction and is characterized by papillary ridges, an absence of hair follicles, and a thick stratum corneum. Although gross morphological differences, such as the increased thickness of the plantar stratum corneum, are well...

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Neurotrauma During Military Training: Results of the INVestigating traIning assoCiated blasT pAthology (INVICTA) Study

Walter S Carr 2026-10-01 1 min 67 words TACTILE ACUITY

Summary: Concern is growing that subconcussive blast exposure during military exercises such as heavy weapons training (HWT) may adversely affect brain function. HWT involves firing shoulder-mounted weapons that can expose the gunner, assistant gunner, and instructors (range safety officers [RSOs]) to blast ...

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Altered body representation in chronic non-specific low back pain: a testable framework linking body-schema-related disturbance, corticomotor alterations, and altered pain processing

👤 Zhifei Xie
📅 2026-10-24 📖 1 min 📝 66 words

TACTILE ACUITY

Summary: Low back pain is a leading contributor to disability worldwide; in chronic non-specific low back pain (CNSLBP), the processes sustaining pain and disability in the absence of identifiable structural pathology remain incompletely understood. Three partially overlapping research lines have documented ...

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Hyperexcitable primary somatosensory cortex in the MitoPark mouse model of Parkinson's disease

Alan J Emanuel 2026-10-25 1 min 67 words TACTILE ACUITY

Summary: The majority of Parkinson's disease patients suffer from decreased tactile acuity or neuropathic pain, both of which contribute to low quality of life and likely exacerbate motor dysfunction. How Parkinson's disease pathology relates to altered somatosensation remains unclear. One potential contribu...

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The Relationship Between Occlusal Tactile Acuity, Bite Force and Tooth Wear

Birgitta Häggman-Henrikson 2026-11-03 1 min 52 words TACTILE ACUITY

Summary: CONCLUSIONS: Bite force may be a clinically relevant factor associated with accumulated tooth wear, whereas OTA was not independently associated with tooth wear. Longitudinal studies are needed to determine whether bite force can predict tooth wear progression and whether more comprehensive assessme...

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Impact of Braille Booklet, Audio-Tactile Performance and Caregiver-supervised Training on Oral Hygiene Status, Knowledge, Attitude and Practices among Visually Challenged Children- A Randomized Clinical Trial

👤 Priyanga Chandrasekaran
📅 2026-09-22 📖 1 min 📝 52 words

BRAILLE

Summary: CONCLUSION: Both caregiver-supervised training and audio tactile performance using specialized model method of oral health education proved to be more effective than the braille booklet group on plaque control and improvement in the oral hygiene status, oral health knowledge, attitude and practice i...

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Design and functional verification of an automated Perkins Brailler accessory with audio feedback

Ali Alhaj 2026-09-23 1 min 67 words BRAILLE

Summary: CONCLUSIONS: The study did not include learners with visual impairment, Braille teachers, rehabilitation specialists, or other end users, and it did not collect quantitative measures of accuracy across the full character set, writing speed, embossing quality, latency, durability, safety, usability, ...

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Uncertainty-Aware structured decoding for Chinese Braille document transcription

Haoran Sun 2026-09-25 1 min 64 words BRAILLE

Summary: Automatic transcription of braille documents remains challenging in document digitization settings, particularly under real imaging degradations such as uneven embossing, illumination variation, paper texture, skew, blur, and ghosting, which make braille dot presence uncertain and often cause error ...

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Clinical Characteristics and Epidemiology of Isolated Dystonia: Results from a US Claims Analysis

James E Thornton 2026-10-08 1 min 54 words BRAILLE

Summary: CONCLUSIONS: This study defines an isolated dystonia cohort in the US based on real-world evidence and systematic exclusion of secondary causes of dystonia. Estimated prevalence is within range of prior epidemiological studies, and the cohort showed increased rates of comorbid psychiatric disorders....

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Effects of multisensory tactile-auditory integration teaching on systematic anatomy learning in visually impaired medical students

Fanzhen Kong 2026-10-08 1 min 50 words BRAILLE

Summary: CONCLUSION: The tactile-auditory integrated training system significantly improves objective summative knowledge acquisition and subjective learning experiences in visually impaired medical students. This approach provides a scalable, cost-constrained, and replicable pedagogical model for inclusive ...

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Physical and Environmental Accessibility of Dental Services for Persons with Disabilities in Saudi Primary Health Care: A Cross-Sectional Study

👤 Estie Kruger 📅 2026-10-16 📖 1 min 📝 66 words BRAILLE

Summary: Background/Objectives: Persons with disabilities (PWDs) may experience barriers when accessing oral health care, including those related to the physical and sensory environment of the dental-care setting. In this study we assessed the accessibility of primary health care centres (PHCs) providing den...

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Visible Light Driven Diselenide Chemistry for Simultaneous 3D Patterning and Mechanical Reinforcement

👤 Huaping Xu 📅 2026-10-16 📖 1 min 📝 63 words BRAILLE

Summary: Three-dimensional (3D) patterning on polymer surfaces is typically achieved by photolithography or related techniques, which often involve complex procedures, high-energy ultraviolet light, and risk of material damage. Here, we report a visible light-driven strategy that exploits the dual functional...

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An accessible pregnancy test for blind and low-vision users

Vanja Garaj 2026-10-16 1 min 63 words BRAILLE

Summary: CONCLUSIONS: The findings suggest that accessible pregnancy testing must prioritise not only non-visual result reading but also privacy, dignity, reproductive autonomy and independent health management. These results represent the preliminary evidence of acceptability rather than a conclusive test o...

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Genome-Informed Loop-Mediated Isothermal Amplification (LAMP) Assays for Detection and Differentiation of Ralstonia pseudosolanacearum Phylotypes I and III

Jarred Yasuhara-Bell 2026-10-27 1 min 76 words BRAILLE

Summary: The Ralstonia solanacearum species complex (RSSC) causes bacterial wilt in a broad range of economically important hosts. The RSSC has been reclassified into three species, of which R. solanacearum sequevars 1 and 2 in phylotype IIB are designated as Select Agents in the U.S. Many rapid tests for "R...

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Encoding tactile stimuli for Braille recognition with organoids

👤 Benjamin Ward-Cherrier 📅 2026-10-31 📖 1 min 📝 62 words BRAILLE

Summary: This study proposes a transferable encoding strategy that maps tactile sensor data to electrical stimulation patterns, enabling neural organoids to perform an open-loop artificial tactile Braille classification task. Human forebrain organoids cultured on a low-density microelectrode array (MEA) are ...

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The Non-Invasive Brain Stimulation Side Effect Scale (NIBS-SES): development of a brief, multilingual questionnaire for monitoring adverse effects in transcranial magnetic, electrical and ultrasound stimulation

👤 Maximilian Kiebs 📅 2026-11-05 📖 1 min 📝 59 words TDCS TACS TRNS

Summary: CONCLUSIONS: Our findings support the feasibility of repeated session-level assessment with the initial version, from which the final NIBS-SES was derived. This multilingual, open-access instrument comprises 10 within-session and 9 between-session items developed for monitoring across TMS, tES, and ...

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Concurrent prefrontal neurochemical effects of high-definition transcranial direct current stimulation: directional E-field associations with GABA+ and Glx

👤 Jianqi Li 📅 2026-11-05 📖 1 min 📝 58 words TDCS TACS TRNS

Summary: Transcranial direct current stimulation (tDCS) is a promising neuromodulatory technique for treating neuropsychiatric disorders. However, its neurochemical mechanisms remain poorly characterized, and clinical efficacy is limited by substantial inter-individual variability. This study used concurrent...

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Anodal tDCS alters perceived visual contrast in humans

👤 Sang Wook Hong 📅 2026-11-06 📖 1 min 📝 62 words TDCS TACS TRNS

Summary: Cortical gain modulation is thought to alter perceptual experience, most prominently demonstrated by the effects of spatial attention on contrast appearance. Whether externally induced changes in cortical excitability produce comparable perceptual consequences remains unclear. Here, we tested whethe...

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Evaluating OPTION: Benefits and Pitfalls of a Youth Trafficking Diversion Court in Florida

Joan A Reid 2026-11-06 1 min 67 words TDCS TACS TRNS

Summary: Trafficking diversion courts (TDCs) provide an alternative approach to supporting trafficked youth by promoting recovery, preventing revictimization, and diverting youth from the justice system. The current study examines the benefits and challenges of the OPTION (Offering Potentially Trafficked Ind...

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Amelioration of diabetic autonomic neuropathy symptoms following a course of transcranial direct current stimulation

Ahmad Alipour 2026-11-06 1 min 62 words TDCS TACS TRNS

Summary: Cardiovascular autonomic neuropathy (CAN) is a severe and potentially life-threatening complication of diabetes. This randomized, sham-controlled, three-arm trial evaluated the efficacy of anodal transcranial direct current stimulation (tDCS) targeting the left dorsolateral prefrontal cortex for imp...

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From Single Cells to the Circulating Tumor Microenvironment: Label-Free Size-Selection and a Morphological Taxonomy for Liquid Biopsy in Cytopathology

Satoru Kobayashi 2026-11-09 1 min 25 words TDCS TACS TRNS

Summary: CONCLUSION: Size-selection converts a liquid biopsy into a cytology slide, placing interpretation of the circulating tumor ecosystem within the existing competence of the diagnostic laboratory.

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Phase-Dependent Effects of Cerebellar Transcranial Alternating Current Stimulation on Lateral Vestibulospinal Tract Excitability

Tsubasa Mitsutake 2026-11-09 1 min 36 words TDCS TACS TRNS

Summary: CONCLUSIONS: LVST excitability differed according to the phase of the ongoing cerebellar-targeted tACS waveform. These findings provide electrophysiologic evidence of an online phase-dependent effect on vestibulospinal excitability and a physiologic basis for future phase-specific neuromodulation st...

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Effects of Transcranial Direct Current Stimulation on Core Symptoms and Sleep Disturbances in Children With Autism Spectrum Disorder: A Meta-Analysis of Randomized Controlled Trials

👤 Meimei Yang
📅 2026-11-09 📖 1 min 📝 44 words

TDCS TACS TRNS

Summary: CONCLUSION: tDCS, most commonly delivered at 1 mA over the DLPFC, was associated with improvements in some ASD-related behavioral outcomes; preliminary evidence also suggested a potential benefit for sleep disturbances, although this finding was based on only two low-certainty studies and warrants c...

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No effects of single-session concurrent tDCS on explicit motor sequence learning in healthy adults with conventional 2 mA and tangential 6 mA currents

👤 Lucas C Parra
📅 2026-11-09 📖 1 min 📝 45 words

TDCS TACS TRNS

Summary: CONCLUSION: These findings, together with a survey of the existing literature, suggest that the effects of single-session concurrent tDCS on explicit motor sequence learning are not robust across studies. Our results underscore the necessity of rigorous methodological controls to prevent false disco...

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More than Quaternary Ammonium Compounds: First Evidence of Tertiary Amine Compounds as Novel Contaminants Ubiquitously Present in Indoor Dust

👤 Runzeng Liu 📅 2026-11-10 📖 1 min 📝 65 words TDCS TACS TRNS

Summary: Nitrogen-containing organic compounds have attracted global concern due to their widespread production and potential environmental risks. While the environmental occurrence of quaternary ammonium compounds (QACs) has been widely studied, research on tertiary amine compounds (TACs), another important...

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Prefrontal Activation and Cognitive Heterogeneity in Pediatric Sleep-Disordered Breathing

👤 Amal Isaiah 📅 2026-11-05 📖 1 min 📝 39 words FNIRS

Summary: CONCLUSIONS AND RELEVANCE: In this cross-sectional study, task-related PFC activation was associated with SDB severity and moderated the severity-cognition association. fNIRS may help identify children whose cognitive profiles deviate from conventional severity metrics, thereby supporting individual...

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Mirror Neural System in Hands Motor Imagery with Pictures and Videos Guidance: An EEG-fNIRS Study

👤 Mohamad Sawan 📅 2026-11-05 📖 1 min 📝 64 words FNIRS

Summary: Brain-computer interfaces (BCIs) achieved by motor imagery (MI) have been widely accepted as an effective tool in rehabilitative applications. Functional near-infrared spectroscopy (fNIRS) and electroencephalography (EEG) are two common implementation forms of BCIs, capable of monitoring neuronal po...

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Quality-aware mask-propagation graph attention for fNIRS-based bipolar disorder classification without channel interpolation

👤 Yajing Meng 📅 2026-11-05 📖 1 min 📝 58 words FNIRS

Summary: Bipolar disorder (BD) remains difficult to characterise objectively, motivating a neural marker to complement clinical assessment. Functional near-infrared spectroscopy (fNIRS) offers portable cortical haemodynamic measurement, but channel availability can vary across recordings because of bad chann...

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Brain-Behavior Coupling During Cognitive-Postural Dual-Tasking in Older Adults With Mild Cognitive Impairment: An fNIRS Study

Chuan Guo 2026-11-06 1 min 56 words FNIRS

Summary: CONCLUSIONS: Older adults with MCI exhibit altered brain-behavior coupling during cognitive-postural dual-task conditions involving sensory and postural constraints, characterized by altered relationships between prefrontal-sensorimotor cortical activation and behavioral performance. These findings ...

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Item-Specific Proportion Congruency Uncovers Different Forms of Attentional Control: An event-related fNIRS Study

Murat Perit Çakır 2026-11-07 1 min 70 words FNIRS

Summary: CONCLUSION: While behavioral results mostly support contingency learning, fNIRS data revealed ISPC-related activation in prefrontal channels with sensitivity to BA 46 (lateral, channel 1) and the BA 10/BA 11 overlap zone (medial, channels 10-12), with channel 11 emerging as a convergence point for b...

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Does the emotion regulation of self-directed humor engage the prefrontal cortex? An fNIRS study

Zhonglu Zhang 2026-11-07 1 min 68 words FNIRS

Summary: Previous research has revealed that self-directed humor (self-enhancing and self-defeating) can help regulate negative emotions and that the prefrontal cortex (PFC) plays a critical role in emotion regulation. However, it remains unknown whether emotion regulation of self-directed humor is associate...

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Effects of mini-maps with different reference frames on polarity-free 3D spatial navigation: a behavioural and fNIRS study

Xiaozhou Zhou 2026-11-07 1 min 62 words FNIRS

Summary: Manned spaceflight and deep-sea exploration extend human activities to extreme environments lacking gravitational and visual polarity cues, where spatial disorientation frequently occurs. Mini-map studies largely focus on terrestrial environments, while reference frame design for polarity-free 3D sp...

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Baduanjin improves motor function and cortical activation in children with developmental coordination disorder: a randomized controlled trial

👤 Wei Song 📅 2026-11-09 📖 1 min 📝 40 words FNIRS

Summary: CONCLUSIONS: Baduanjin training improved motor coordination and balance in children with DCD. The observed changes in activation within motor-related cortical regions provide preliminary support for a possible neurophysiological basis of these functional improvements, although the underlying mechani...

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Potential Applications of Brain-Computer Interface Technology in the Rehabilitation of Patients with Anxiety Disorders: A Narrative Review

Hu Lou 2026-11-10 1 min 58 words FNIRS

Summary: Anxiety disorders are among the most prevalent mental disorders worldwide and can cause persistent impairment in emotional regulation, cognition, and social functioning. Pharmacotherapy, psychotherapy, exercise, and non-invasive brain stimulation are widely used, yet their effectiveness may be const...

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Multimodal EEG-fNIRS complexity analysis can reveal the neurovascular coupling change with aging

Yicheng Zhu 2026-11-10 1 min 58 words FNIRS

Summary: CONCLUSION: Aging is associated with task-specific alterations in neural complexity, hemodynamic complexity, and the cross-modal temporal relationship between EEG and fNIRS signals. These findings suggest that age-related changes in NVC-related dynamics are better characterized as dynamic reorganiza...

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Neural interface instructs layer 4-appropriate circuit reconstruction with engrafted human brain organoids and improves visually guided behavior

👤 Xiao-Hong Li
📅 2026-11-06 📖 1 min 📝 65 words

BRAIN COMPUTER INTERFACE

Summary: Repairing precise neural microcircuitry after cortical injury remains a pivotal challenge. Human brain organoids (hBOs) are a promising cell source, but disorganized integration limits their therapeutic potential. Here, we demonstrated that targeted neuromodulation, steered by endogenous narrowband ...

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Self-Guided Beta-Responsive Visual Neurofeedback in a Public Exhibition: Feasibility and Usability Randomized Controlled rial

Joseph Ollier
2026-11-07 1 min 57 words

BRAIN COMPUTER INTERFACE

Summary: CONCLUSIONS: This trial evaluated an autonomous consumer-EEG neurofeedback system integrating self-fitting, signal-quality checking, baseline calibration, adaptive feedback, randomization, and self-report without operator support. Many first-time users completed the autonomous workflow with acceptab...

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Auditory rehabilitation in mixed hearing loss due to chronic otitis media sequelae: passive versus active transcutaneous bone conduction implant

👤 Esther Sofía Angulo Martínez 📅 2026-11-08 📖 1 min 📝 47 words BRAIN COMPUTER INTERFACE

Summary: CONCLUSIONS: The findings of the present study suggest that AT-BCI provides a greater and clinically meaningful auditory gain for HF-PTA in patients with MHL secondary to COM. In ATBCI group a stable audiological performance was achieved as early as the 3-month follow-up and is maintained over time.

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Selective dissociation of coordination bonds enables rapid rigidity switching of hydrogel microfibre bioelectrodes

👤 Lidong Zhang 📅 2026-11-09 📖 1 min 📝 64 words BRAIN COMPUTER INTERFACE

Summary: Bioelectrodes with tunable rigidity deliver excellent mechanical compliance and can effectively alleviate tissue damage. Herein, we report hydrogel microfibre electrodes (diameter: 50-200 μm) fabricated by cross-linking sodium alginate (SA) microfibres with Ca^(2+)/Al^(3+) dual ions. The resultant S...

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Constructing Neurorights in the Era of Clinical Brain-Computer Interfaces : China's Emerging Governance Pathway

Yuan Lin 2026-11-09 1 min 67 words BRAIN COMPUTER INTERFACE

Summary: In March 2026, China approved the world's first implantable brain-computer interface (BCI) medical device for clinical use outside clinical trials, a milestone that raises urgent questions about how legal and ethical systems should protect the brain and mind. This article examines China's emerging n...

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DicomDoodle: Human-in-the-Loop Annotation of 3D MRI DICOM Images with AI Support and DICOM-SEG Compatibility

👤 Debnath Pal 📅 2026-11-09 📖 1 min 📝 62 words BRAIN COMPUTER INTERFACE

Summary: Accurate medical image annotation requires curation and remains a major bottleneck due to its time-intensive nature and inter-observer variability. To address this, we developed DicomDoodle, an open-source, Python-based tool for human-in-the-loop annotation that integrates a fine-tuned YOLO11 detect...

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Potential Applications of Brain-Computer Interface Technology in the Rehabilitation of Patients with Anxiety Disorders: A Narrative Review

👤 Hu Lou 📅 2026-11-10 📖 1 min 📝 58 words BRAIN COMPUTER INTERFACE

Summary: Anxiety disorders are among the most prevalent mental disorders worldwide and can cause persistent impairment in emotional regulation, cognition, and social functioning. Pharmacotherapy, psychotherapy, exercise, and non-invasive brain stimulation are widely used, yet their effectiveness may be const...

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In Situ Displacement Reaction Enables a Conductive Metallized Hydrogel Interface for Smart Underwater Flexible Electronics

👤 Qigang Wang
📅 2026-11-10 📖 1 min 📝 60 words

BRAIN COMPUTER INTERFACE

Summary: Underwater wearable sensing is severely hampered by the intrinsic drawbacks of conventional materials, including water absorption, swelling, and electrical drift during long-term immersion. Here, we develop an anti-swelling hydrogel with a surface-metallized layer, which achieves strong stability of...

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Fast-Gelling, Adhesive, and Conductive Epidermal Electrode for Accurate Physiological Signal Monitoring and Real-Time Brain-Controlled Drone Flight

Zhou Li 2026-11-10 1 min 55 words BRAIN COMPUTER INTERFACE

Summary: Conductive hydrogel electrodes are vital components for achieving human-machine interfaces. However, the inherent trade-off between high conductivity and favorable mechanical/adhesive properties remains a significant challenge for hydrogel-based epidermal electrodes. Here, we developed a multifuncti...

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Bayesian Adaptive Stimulus Optimization in P300 Brain Computer Interfaces

Boyla O Mainsah 2026-11-10 1 min 26 words BRAIN COMPUTER INTERFACE

Summary: CONCLUSION: For computational simplicity in implementation, the optimal stimulus can generally be approximated by finding the stimulus that has a combined probability mass closest to 0.5.

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A research on the typical images of animals in the dreams of Naxi.

📅 2026-09-10 📖 1 min 📝 157 words DREAMING

Summary: Animal images in dreams have long been studied, yet existing research is not systematic or in-depth. The Naxi, a small Chinese ethnic group (about 320,000), possess a rich cultural heritage. In their belief, animals mediate between humans, deities, and nature, while dreams are key communication chan...

🔗 **Read full article:**
http://doi.org/10.1037/drm0000353

Philosophical and medical analysis of dreams in avicenna’s perspective: Nature, mechanism, and functions.

📅 2026-06-15 📖 1 min 📝 218 words DREAMING

Summary: Avicenna was an eminent Islamic philosopher and physician. Drawing primarily on The Book of Healing and The Canon of Medicine, this article presents a systematic account of his theory of dreaming within the framework of his psychology and natural philosophy. Avicenna conceives drea...

🔗 **Read full article:**
http://doi.org/10.1037/drm0000341

Threatening events and other themes in COVID-19-related dreams: Examining pandemic dreams from the perspective of current dream simulation theories.

📅 2026-06-25 📖 1 min 📝 218 words DREAMING

Summary: In this study on pandemic dream experiences in rural Nepal, 54 interviews were collected 1 year into the COVID-19 pandemic and compared with 62 interviews collected some years earlier. As in comparable studies, the dream narratives were categorized and analyzed according to their content, employing ...

🔗 **Read full article:**
http://doi.org/10.1037/drm0000348

Correction to “Big-five personality traits and dream experiences: A systematic review and meta-analysis” by Marin et al. (2026).

📅 2026-09-10 📖 1 min 📝 399 words DREAMING

Summary: Reports an error in "Big-Five personality traits and dream experiences: A systematic review and meta-analysis" by Jai Marin, Noah Thomas, Lan Nguyen and Jared Walters (Dreaming, Advanced Online Publication, Mar 26, 2026, np; see record 2027-47158-001). The article is being made available op...

🔗 **Read full article:**
http://doi.org/10.1037/drm0000349

When the body suffers, the mind dreams: The lived experience of cancer.

📅 2026-09-10 📖 1 min 📝 194 words DREAMING

Summary: Dreams have intrigued humanity across centuries, serving as portals to the unconscious, reflections of lived experiences, and sources of spiritual guidance. The study of dreaming spans neuroscience, psychology, anthropology, sociology, and religious thoughts, with each discipline offering unique ins...

🔗 **Read full article:**
http://doi.org/10.1037/drm0000355

Big-Five personality traits and dream experiences: A systematic review and meta-analysis.

📅 2026-03-26 📖 1 min 📝 369 words DREAMING

Summary: [Correction Notice: An Erratum for this article was reported in Vol 36(3) of Dreaming (see record 2028-23196-002). The article is being made available open access under the CC-BY license following the Griffith University opt-in to the CAUL/ APA Read and Publish agreement. The correct copyrig...

🔗 **Read full article:**
http://doi.org/10.1037/drm0000336

The dream insight framework: Harnessing the power of dreams in leadership.

📅 2026-08-20 📖 1 min 📝 80 words DREAMING

Summary: This article introduces the Dream Insight Framework, an approach for integrating dreamwork into leadership development. It explores how dreams, often dismissed in existing leadership approaches, can be powerful tools for enhancing creativity, decision-making, and problem-solving. By reviewing existi...

🔗 **Read full article:**
http://doi.org/10.1037/drm0000354

Interpretability of dreams elicited by progressive-interval rapid-eye-movement and fixed-schedule awakening methods: Trajectories of dream affect, themes, and insight across the night.

📅 2026-07-30 📖 1 min 📝 201 words DREAMING

Summary: This study examined whether interpreting REM dream reports elicited using a REM-cycle-accommodating awakening protocol yielded more associations and greater insight than interpreting REM and non-REM dream reports obtained through a fixed-schedule awakening method. It also explored how interpretabili...

🔗 **Read full article:**
http://doi.org/10.1037/drm0000350

Monthly Updates [March]

📅 2026-03-01 📖 2 min 📝 539 words FMHY

> **Summary:** INFO These update threads only contain major updates. If you're interested in seeing all minor changes you can follow our Commits Page on G...

🔗 **Read full article:**
https://fmhy.net/posts/mar-2026

Monthly Updates [June]

📅 2026-06-01 📖 4 min 📝 899 words FMHY

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https://fmhy.net/posts/june-2026

Monthly Updates [August]

📅 2026-08-01 📖 3 min 📝 666 words FMHY

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https://fmhy.net/posts/aug-2026

Monthly Updates [Sept]

📅 2026-08-31 📖 5 min 📝 1013 words FMHY

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🔗 **Read full article:**
https://fmhy.net/posts/sept-2026

A comprehensive study of how fingerprintable users are

👤 tomrittervg 📅 2026-11-10 📖 1 min 📝 13 words HACKER NEWS

> **Summary:** Article URL: https://fingerprinting.study/ Comments URL: https://news.ycombinator.com/item?id=50444684 Points: 58 # Comments: 26

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What objective is reasoning solving?

👤 gmays 📅 2026-11-10 📖 1 min 📝 13 words HACKER NEWS

> **Summary:** Article URL: https://alexwang.ai/posts/reasoning-as-pathfinding/ Comments URL: https://news.ycombinator.com/item?id=50445013 Points: 85 # Comments: 40...

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Toyota brand Scion is back and it's going to sell a kei truck in America

👤 NaOH 📅 2026-11-10 📖 1 min 📝 13 words HACKER NEWS

> **Summary:** Article URL: https://www.theautopian.com/scion-is-back-and-its-going-to-sell-a-freaking-kei-truck-in-america/ Comments URL: <a href="https://news.ycombinator.com/item?id=50445401"...

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Sherman Anti-Trust Act (1890)

👤 hashmap 📅 2026-11-10 📖 1 min 📝 13 words HACKER NEWS

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Six Math Essentials by Terence Tao

👤 digital55 📅 2026-11-10 📖 1 min 📝 13 words HACKER NEWS

> **Summary:** Article URL: https://teorth.github.io/tao-web/six-math-essentials.html Comments URL: https://news.ycombinator.com/item?id=50446246 Points: 96 # ...

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https://teorth.github.io/tao-web/six-math-essentials.html

Apple has new Mac app coming to make running local AI models easy

👤 linolevan 📅 2026-11-10 📖 1 min 📝 13 words HACKER NEWS

> **Summary:** Article URL: https://9to5mac.com/2026/11/10/apple-has-new-mac-app-coming-to-make-running-local-ai-models-easy/ Comments URL: <a href="https://news.ycombinator.com/item?id=5044652...

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https://9to5mac.com/2026/11/10/apple-has-new-mac-app-coming-to-make-running-local-ai-models-easy/

rVPN – stealth VPN with end-to-end encryption

👤 michaelsbradley 📅 2026-11-10 📖 1 min 📝 13 words HACKER NEWS

> **Summary:** Article URL: https://rvpn.org/ Comments URL: https://news.ycombinator.com/item?id=50446714 Points: 50 # Comments: 38

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https://rvpn.org/

After 7 Months, Glasswing Overwelms Maintainers with Underwhelming Findings

👤 panic 📅 2026-11-10 📖 1 min 📝 13 words HACKER NEWS

> **Summary:** Article URL: https://www.vulncheck.com/blog/anthropic-glasswing-november-update Comments URL: https://news.ycombinator.com/item?id=50447074 Po...

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Type S and M errors as a “rhetorical tool”

👤 noreply@blogger.com (Daniel Lakens)
📅 2025-09-28 📖 17 min 📝 3572 words

TWENTY PERCENT STATISTICIAN

> **Summary:** Update 30/09/2025: I have added a reply by Andrew Gelman below my original blog post. We recently posted a preprint criticizing the idea of Type S and M errors (https://osf.io/2phzb_v1). From our abstract: “While these concepts have been pr...

🔗 **Read full article:**
http://daniellakens.blogspot.com/2025/09/type-s-and-m-errors-as-rhetorical-tool.html

Why we should stop using statistical techniques that have not been adequately vetted by experts in psychology

👤 noreply@blogger.com (Daniel Lakens)
📅 2025-10-29 📖 27 min 📝 5516 words

TWENTY PERCENT STATISTICIAN

> **Summary:** In a recent post on Bluesky, where Richard Morey reflects on a paper he published with Clintin Davis-Stober that points out concerns with the p-curve method (Morey & Davis-Stober, 2025), he writes: <p cla...

🔗 **Read full article:**
http://daniellakens.blogspot.com/2025/10/why-we-should-stop-using-statistical.html

ERP Decoding for Everyone: Software and Webinar

👤 Steve Luck 📅 2023-06-23 📖 2 min 📝 420 words ERP BOOT CAMP

Summary: You can access the recording h strong>.
You can access the final PDF of the slides <a href="https://ucdavis.box.com/s/f...

🔗 **Read full article:**
https://erpinfo.org/blog/2023/6/23/decoding-webinar

Important Changes to the 2024 ERP Boot Camp

👤 Steve Luck 📅 2024-03-05 📖 2 min 📝 444 words ERP BOOT CAMP

Summary: We are disappointed to announce that we will not be holding a regular 10-day ERP Boot Camp this summer. We have held Boot Camps nearly every summer since 2007, supported by a series of generous grants from NIMH that allowed us to provide scholarships for all attendees. Unf...

🔗 **Read full article:**
https://erpinfo.org/blog/2024/3/5/changes-to-the-2024-erp-boot-camp

Standardized Measurement Error now available in BrainVision Analyzer and MNE-Python

👤 Steve Luck 📅 2025-12-19 📖 1 min 📝 253 words ERP BOOT CAMP

Summary: <p class="">A few years ago, we developed a universal metric of data quality for averaged ERPs called the standardized measurement error (click here for more information or watch our <a href="https://www.youtube.com/watch?v=tEKsx0p53rs" t...

🔗 **Read full article:**
https://erpinfo.org/blog/2025/12/19/sme-now-available-in-bva-and-mne-python

Transcranial Focused Ultrasound: A Transformative Tool for Intracranial Ablation, Drug Delivery, and Neuromodulation

📅 2025-11-11 📖 1 min 📝 136 words REVIEWS BIOMEDICAL ENGINEERING

Summary: Transcranial focused ultrasound (tFUS) is an emerging neuromodulation and therapeutic technology offering noninvasive, submillimeter precision for targeting deep brain structures. Unlike transcranial magnetic stimulation (TMS) and transcranial electric stimulation (tES), which are limited by depth-f...

🔗 **Read full article:**
http://ieeexplore.ieee.org/document/11241049

Network Models of Neurodegeneration: Bridging Neuronal Dynamics and Disease Progression

📅 2026-01-16 📖 1 min 📝 196 words REVIEWS BIOMEDICAL ENGINEERING

Summary: Neurodegenerative diseases are characterized by the accumulation of misfolded proteins and widespread disruptions in brain function. Computational modeling has advanced our understanding of these processes, but efforts have traditionally focused on either neuronal dynamics or the biological processe...

🔗 **Read full article:**
http://ieeexplore.ieee.org/document/11357276

Emerging Noninvasive Approaches for the Suppression of Pathological Tremor

📅 2025-12-16 📖 1 min 📝 215 words REVIEWS BIOMEDICAL ENGINEERING

Summary: Pathological tremor affects over 40 million people worldwide, significantly impairing daily activities and quality of life. Pharmacological treatments show limited efficacy, with up to 30% discontinuation rates, while surgical interventions like deep brain stimulation achieve significant tremor redu...

🔗 **Read full article:**
http://ieeexplore.ieee.org/document/11301615

Assistive Trajectory Planning for Lower Limb Exoskeletons: Strategies From Laboratory-Optimized Gait to Environmentally-Adaptive Locomotion Through Multimodal Parameter Awareness

📅 2025-12-30 📖 1 min 📝 182 words REVIEWS BIOMEDICAL ENGINEERING

Summary: Lower limb assistive exoskeletons (LLEs) show great potential to reduce metabolic cost, improve walking performance, and correct abnormal gait patterns. Among their core control architectures, assistive trajectory planning is key to determining system responsiveness and effectiveness under varying l...

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http://ieeexplore.ieee.org/document/11319164

Join the Mailing List

👤 dziura 📅 2025-12-10 📖 1 min 📝 13 words EMBS

Summary: The post Join the Mailing List appeared first on IEEE EMBS.

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https://embc.embs.org/2026/#new_tab

T-MRB Call for Nominations for Associate Editors

👤 Deidre Artis 📅 2026-04-22 📖 1 min 📝 16 words EMBS

> **Summary:** The post T-MRB Call for Nominations for Associate Editors appeared first on IEEE EMBS.

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EMBS MentorLink: Fostering Careers Together

👤 Deidre Artis 📅 2026-08-18 📖 1 min 📝 14 words EMBS

> **Summary:** The post EMBS MentorLink: Fostering Careers Together appeared first on IEEE EMBS.

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https://www.embs.org/featured-news/embs-mentorlink-fostering-careers-together/

2026 Adcom Petition is Open

Deidre Artis | 2026-09-11 | 1 min | 14 words | EMBS

> **Summary:** The post 2026 Adcom Petition is Open appeared first on IEEE EMBS.

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https://www.embs.org/about/embs-officers/2026-embs-administrative-committee-adcom-election-2/#new_tab

Subacute administration of human umbilical cord-derived mesenchymal stem cells is associated with neurobehavioral improvement, myeloid immune modulation, and altered HMGB1 dynamics after experimental ischemic stroke

1 min | 21 words | BRAIN RESEARCH

> **Summary:** Publication date: 15 January 2027 Source: Brain Research, Volume 1895 Author(s): Yuki Takeda, Yoji Kuramoto, Shoichiro Tsuji, Nobutaka Doe, Kenichi Yamahara, Shinichi Yoshimura

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Prenatal cocaine exposure alters synaptic plasticity of hippocampal CA1 pyramidal neurons in postnatal rats

📖 1 min 📝 29 words BRAIN RESEARCH

Summary: Publication date: 15 January 2027 Source: Brain Research, Volume 1895 Author(s): Jian Fu, Cheng-Yan Sun, Qing-Yun Yang, Long Jiang, Yong Tang, Jie Shen, Shun-Nan Ge, Xin Wang, Xue-Lian Wang, Liang Qu

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Enriched environment protects from high-fat diet-induced obesity but not from insulin resistance, glucose intolerance, and metabolic deterioration in the brain

📖 1 min 📝 41 words BRAIN RESEARCH

Summary: Publication date: 1 February 2027 Source: Brain Research, Volume 1896 Author(s): Anne Ladwig, Anna Lena Cremer, Johannes Romer, Valentina Zimmermann, Claus Brandt, Stefan Blaschke, Claudia Orschel, Charlotte Orschel, Marvin Schäfer, Maria A. Rueger, Klarissa H. Stürner, Gereon ...

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The neural circuit implementation of distributional reinforcement learning

Matias, S., Rullan Buxo, C., Xu, S., Campbell, M. G., Lowet, A. S., Lu, X., Wu, Y., Schalek, R., Lichtman, J. W., Watabe-Uchida, M., Drugowitsch, J., Uchida, N.

2026-11-10 1 min 212 words BIORXIV NEUROSCIENCE

Summary: Distributional reinforcement learning (RL) algorithms outperform traditional RL by representing the full probability distribution of future outcomes rather than its mean. Neural signatures consistent with distributional coding have been observed in dopamine neurons and in striatal value representati...

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Open-BEATBox: an open-source autonomous operant chamber for high-throughput data acquisition of complex behaviour

👤 EUVRARD, M., Benzina, N., Duarte-Domingues, D., Mondragon-Gonzalez, S., Huzard, D., Penderia, G., Travert-Jouanneau, Y., Lavielle, O., Lousada, E., Schreiweis, C., LORENZ, A., Pavlov, P., Ajili, Z., Lebegue, L., Tissier, P., BURGUIERE, E.

📅 2026-11-10 📖 1 min 📝 230 words BIORXIV NEUROSCIENCE

Summary: Behavioural neuroscience increasingly relies on rodent models to investigate the neural mechanisms underlying cognition in health and disease. However, conventional operant testing is typically restricted to short daily sessions that require frequent experimenter intervention and often rely on food ...

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Cortical proteomic alterations and exploratory glycopeptide profiling of Parkinson’s disease frontal cortex

📅 2026-11-10 📖 1 min 📝 0 words NATURE NEUROSCIENCE SUBJECTS

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https://www.nature.com/articles/s41598-026-76753-w

Pathways of cross-modal access to the visual cortex in late blindness

📅 2026-11-10 📖 1 min 📝 0 words NATURE NEUROSCIENCE SUBJECTS

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LINC00205 acts as a multivalent scaffold promoting FUSP525L recruitment into Amyotrophic Lateral Sclerosis stress granules

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Common balance tests and their association with falls

📅 2026-11-10 📖 1 min 📝 0 words NATURE NEUROSCIENCE SUBJECTS

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https://www.nature.com/articles/s41598-026-78617-9

The decision to stop “counting” explains numerical underproduction in primates

Laura E. SeidlerLena L. JannaschStephanie WestendorffAndreas Niederahttps://ror.org/03a1kwz48Animal Physiology Unit, Institute of Neurobiology, University of Tübingen, Tübingen 72076, Germany

2026-11-09 1 min 51 words PNAS NEUROSCIENCE

Summary: Proceedings of the National Academy of Sciences, Volume 123, Issue 46, November 2026.
SignificanceProducing a specific number of actions is a fundamental component of counting, yet little is known about how the brain determines when a counting sequence should end. We show that two rhesus macaq...

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https://www.pnas.org/doi/abs/10.1073/pnas.2620933123?af=R

Digital health meets Parkinson’s disease: a bibliometric exploration of wearable sensors research and emerging frontiers (2005–2026)

Lan Xiong

2026-10-08 1 min 228 words FRONTIERS NEUROINFORMATICS

Summary: BackgroundWearable and sensor-based monitoring technologies are increasingly used to support objective, longitudinal assessment in Parkinson’s disease (PD). However, the global development, knowledge structure, and emerging research frontiers of this interdisciplinary field remain incompletely mappe...

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https://www.frontiersin.org/articles/10.3389/fninf.2026.1937585

Epigenetic modification of neurotransmitter pathways during fasting: a systematic review

Majd A. AbuAlrob 2026-11-05 1 min 293 words FRONTIERS HUMAN NEUROSCIENCE

Summary: BackgroundDietary restriction, encompassing intermittent fasting and caloric restriction, represents a prominent non-pharmacological approach to modulating brain aging and neuronal resilience. Emerging evidence links metabolism to long-term neural plasticity through epigenetic mechanisms, yet findin...

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https://www.frontiersin.org/articles/10.3389/fnhum.2026.1969439

Dynamical disorders: synchronization dynamics in complex emotional conditions

S. S. Høgenhaug 2026-11-05 1 min 207 words FRONTIERS HUMAN NEUROSCIENCE

Summary: IntroductionMental health conditions present a particular challenge to categorical nosology. We propose that such conditions are better understood as Dynamical Disorders, defined not only by symptom cluster profiles but by underlying disrupted patterns of coordination across bio-psycho-social scales...

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https://www.frontiersin.org/articles/10.3389/fnhum.2026.1878289

From biomechanical data to robotic execution: design and control of an intelligent Tuina system

Wei Yao 2026-11-02 1 min 181 words FRONTIERS NEUROROBOTICS

Summary: IntroductionTraditional Chinese medicine (TCM) Tuina (massage) relies heavily on practitioners' physical stamina and subjective experience, leading to operational variability that hinders therapeutic standardization and scalable rehabilitation service delivery. To address this limitation, this study...

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https://www.frontiersin.org/articles/10.3389/fnbot.2026.1937295

Path planning algorithm based on pulse-coupled neural network with dual constraints of heading-deviation and node penalty

Xiaoyan Wang 2026-11-05 1 min 321 words FRONTIERS NEUROROBOTICS

Summary: Path planning for mobile robots is a crucial aspect of robot navigation. Traditional search algorithms are limited by heuristic design and search strategies, often struggling to balance search efficiency and path quality in intricate obstacle environments. The Pulse-Coupled Neural Network (PCNN) off...

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Human cortical high-gamma power scales with movement rate in healthy participants and stroke survivors

👤 Fanny Quandt 📅 2025-01-09 📖 1 min 📝 65 words OOSTENVELD ROBERT

Summary: Motor cortical high-gamma oscillations (60-90 Hz) occur at movement onset and are spatially focused over the contralateral primary motor cortex. Although high-gamma oscillations are widely recognized for their significance in human motor control, their precise function on a cortical level remains el...

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NIRS-BIDS: Brain Imaging Data Structure Extended to Near-Infrared Spectroscopy

👤 Luca Pollonini 📅 2025-01-27 📖 1 min 📝 70 words OOSTENVELD ROBERT

Summary: Functional near-infrared spectroscopy (fNIRS) is an increasingly popular neuroimaging technique that measures cortical hemodynamic activity in a non-invasive and portable fashion. Although the fNIRS community has been successful in disseminating open-source processing tools and a standard file forma...

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Pseudonymisation of neuroimages and data protection: Increasing access to data while retaining scientific utility

👤 Lyuba Zehl 📅 2025-06-26 📖 1 min 📝 67 words OOSTENVELD ROBERT

Summary: For a number of years, facial features removal techniques such as 'defacing', 'skull stripping' and 'face masking/blurring', were considered adequate privacy preserving tools to openly share brain images. Scientifically, these measures were already a compromise between data protection requirements a...

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Cycling on the Freeway: The perilous state of open-source neuroscience software

👤 Tim M Tierney 📅 2025-08-13 📖 1 min 📝 74 words OOSTENVELD ROBERT

Summary: Most scientists need software to perform their research (Barker et al., 2020;Carver et al., 2022;Hettrick, 2014;Hettrick et al., 2014;Switters & Osimo, 2019), and neuroscientists are no exception. Whether we work with reaction times, electrophysiological signals, or magnetic resonance imaging data, ...

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Optimal configuration of on-scalp OPMs with fixed channel counts

👤 Robert Oostenveld 📅 2025-08-13 📖 1 min 📝 69 words OOSTENVELD ROBERT

Summary: Recent technological developments have brought optically pumped magnetometers (OPMs) within reach of the larger neuroscientific community. The current state-of-the-art consists of whole-head systems that measure the magnetic field at >100 locations. OPM sensors can be constructed to measure the fiel...

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ARTEM-IS for ERP: Agreed Reporting Template for EEG Methodology-International Standard for Event-Related Potential Experiments

👤 Suzy J Styles 📅 2025-12-08 📖 1 min 📝 73 words OOSTENVELD ROBERT

Summary: The choices we make during the recording, preprocessing and analysis of event-related potentials (ERP) data can affect study outcomes. As such, it is critical that they are transparently reported to allow for reproducibility. Yet, systematic reviews of reporting practices in the field have shown tha...

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Eye-Tracking-BIDS: the Brain Imaging Data Structure extended to gaze position and pupil data

Julia-Katharina Pfarr 2026-02-12 1 min 68 words OOSTENVELD ROBERT

Summary: The Brain Imaging Data Structure (BIDS) is a widely adopted, community-driven standard to organize neuroimaging data and metadata. Although numerous extensions have been developed to incrementally extend coverage to new modalities and data types, an unambiguous, granular specification for eye-tracki...

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Facilitating unusual bodily experiences and out-of-body experiences across wakefulness and sleep: A high-density EEG and neurophenomenology study

Robert Oostenveld 2026-02-12 1 min 67 words OOSTENVELD ROBERT

Summary: Unusual bodily experiences (UBEs) are illusory bodily perceptions that are not coherent with typical wakeful experiences, including flying or floating sensations, body distortions, and out-of-body experiences. This study examines UBEs in a controlled sleep laboratory using meditation and light stimu...

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The Impact of Brain Tumors and Craniotomy Lesions on Scalp EEG

Vitória Piai 2026-02-17 1 min 65 words OOSTENVELD ROBERT

Summary: Electroencephalography (EEG) is widely used in both research and clinical settings, yet its accuracy can be significantly impacted by subject-specific anatomical anomalies such as brain lesions and skull defects. This study investigates the effects of glioma-related brain lesions and craniotomy-indu...

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EMG-BIDS: an extension to the Brain Imaging Data Structure for electromyography

Arnaud Delorme 2026-08-01 1 min 64 words OOSTENVELD ROBERT

Summary: Electromyography (EMG) is fundamental to clinical assessment, rehabilitation, neuromuscular research, and human-machine interfaces. Despite decades of use, no widely adopted standard exists for organizing and sharing EMG data, limiting reusability and large-scale data aggregation. We present EMG-BID...

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Does eHealth literacy matter for health-promoting behaviors among blind and low-vision adults? A mixed methods study

👤 Jessie Chin 📅 2026-11-08 📖 1 min 📝 47 words LOW VISION

Summary: CONCLUSION: eHealth literacy alone does not ensure more health behavior engagement in BLV individuals. To bridge the gap, health systems must prioritize the development of accessible digital health resources, incorporate disability-specific adaptations in health education, and ensure equitable oppor...

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Construction of Highly Viable, Excitable, Aligned Human Induced Pluripotent Stem Cell-Derived Retinal Ganglion Cell Sheets on Poly-ε-caprolactone Nanofibers by Targeting Integrin β1-Mediated Adhesion Pathways

👤 Toru Nakazawa
📅 2026-11-09 📖 1 min 📝 63 words

LOW VISION

Summary: Transplantation of induced pluripotent stem cell-derived retinal ganglion cells (RGCs) is a promising strategy for optic nerve regeneration and vision restoration in glaucoma. However, injections of conventional single-cell suspensions have low engraftment rates, hindering the formation of the compl...

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High-Performance Self-Powered Infrared Photodetector via Semiconducting Graphene Nanoribbon/Germanium Vertical Heterojunctions with Robust Thermal Stability

👤 Aimin Song
📅 2026-11-09 📖 1 min 📝 54 words LOW VISION

Summary: Self-powered photodetectors stand as a cornerstone for next-generation low-power electronics, with promising applications in optical communications, environmental monitoring, night vision, and beyond. However, achieving high performance in these devices, particularly under elevated temperatures, rem...

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Prior-guided whole-slide survival prediction with soft histopathologic priors from vision-language foundation models

👤 Zhihui Feng
📅 2026-11-09 📖 1 min 📝 62 words

LOW VISION

Summary: Whole-slide image (WSI) survival prediction is complicated by heterogeneous tumor morphology, where prognostic evidence is distributed across tumor, stroma, immune infiltration, necrosis, and residual tissue regions and often appears as mixed patch-level tissue evidence. Existing models commonly agg...

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Reduced magnocellular-biased visual responses are associated with working memory and linguistic difficulties in Chinese developmental dyslexia: A hypothesized visual-cognitive cascade

👤 Tsz Wing Leung
📅 2026-11-09 📖 1 min 📝 66 words LOW VISION

Summary: Developmental dyslexia (DD) affects 5-10% of children worldwide. Although phonological theories predominate, they may not fully explain reading difficulties in orthographically dense scripts such as Chinese. We investigated whether Chinese children with DD exhibit differences in early visual cortica...

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During natural vision, semantic novelty modulates fixation-related processing in primate cortex

👤 Lucas C Parra 📅 2026-11-09 📖 1 min 📝 74 words LOW VISION

Summary: We sample visual scenes with short gaze fixations separated by saccades. While low-level integration is known, semantic integration of foveal vision across fixations remains unclear. We hypothesized that the brain responds to changes in semantic information from one fixation to the next, and therefo...

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Health and welfare professionals' global contribution to the International Classification of Functioning, Disability and Health Core Set Vision Loss

👤 Dominique VAN DE Velde 📅 2026-11-10 📖 1 min 📝 64 words LOW VISION

Summary: CONCLUSIONS: A visual impairment exerts a broad and multidimensional impact on functioning, emphasizing the importance of emotional well-being, daily activities, and environmental supports. Regional differences highlight the need for culturally sensitive approaches in assessment and intervention. Th...

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Federated Learning: Envisioning a Pathway to More Accessible and Equitable Cleft Care

👤 Ghulam Qadir Fayyaz 📅 2026-11-10 📖 1 min 📝 66 words LOW VISION

Summary: Cleft lip and palate represents the most common congenital craniofacial anomaly worldwide, with low- and middle-income countries (LMICs) bearing nearly 84% of the global disease burden. Despite the instrumental role of global nongovernmental organizations (NGOs) in expanding access to cleft care acr...

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Dichoptic viewing reveals distinct low- and high-level mechanisms for temporal phase discrimination

👤 Keith A Schneider 📅 2026-11-10 📖 1 min 📝 68 words LOW VISION

Summary: Conscious awareness of temporal visual structure is limited to frequencies below ~10 Hz. Between ~10 and 60 Hz, flicker is visible, but temporal phase cannot be accurately discriminated; above ~60 Hz, stimuli are perceptually fused. Accurately estimating the cognitive limit of phase discrimination r...

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Trends in Near Vision Loss and Age-Sex-SDI Disparities in Belt and Road Countries Over Thirty Years

Xiaopan Li 2026-11-10 1 min 55 words LOW VISION

Summary: CONCLUSIONS: This study represents the first comprehensive analysis of NVL burden and disparities across BRI countries, revealing persistent inequalities by age, sex, and SDI. The findings reflect the diverse socio-economic and health-care contexts of BRI countries and highlight the urgent need for ...

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A single session of high-definition transcranial direct current stimulation does not modulate effects of knee two-point discrimination training or sensorimotor function in healthy adults: Double-blind randomised controlled trial

👤 Paul W Hodges
📅 2026-07-30 📖 1 min 📝 53 words

TACTILE ACUITY

Summary: CONCLUSION: A single 20-minute session of 1 mA HD-tDCS, with or without brief TPD training, does not acutely modify knee somatosensory, sensorimotor, or functional performance in healthy adults. Existing evidence at the hand may not translate to the lower limb. Future work should investigate higher-...

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Variation in glabrous skin physiology across anatomical locations on the human hand

👤 Martina C Meinke 📅 2026-08-13 📖 1 min 📝 76 words TACTILE ACUITY

Summary: Skin physiology varies greatly between anatomical locations, with impact on tactile sensitivity. Research on the variations in morphology and physiology between different anatomical locations of glabrous skin is limited so far. This study investigates the morphology and physiology of glabrous skin o...

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Beyond Hamstring Shortness: Continuous Flexibility Is Associated With Balance and Stability but Not Sensory Measures-A Cross-Sectional Study

👤 Senanur Akıncı 📅 2026-08-14 📖 1 min 📝 66 words TACTILE ACUITY

Summary: CONCLUSIONS: Greater hamstring flexibility was associated with better dynamic balance performance and showed a weaker association with postural stability, whereas no robust associations were observed with sensory measures of postural control. These findings suggest that hamstring flexibility may be ...

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Resting-state EEG microstate dynamics reflect individual differences in tactile angle discriminability

Liu Yang 2026-08-19 1 min 63 words TACTILE ACUITY

Summary: Tactile spatial acuity, the ability to discriminate fine spatial details, is a fundamental sensory capacity modulated by both peripheral and central mechanisms. Here, we investigated whether resting-state EEG microstate dynamics are associated with individual differences in tactile spatial acuity. F...

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A grating orientation task with controlled dome elevation speed and contact force: stimulus conditions and day-to-day reliability

Hideaki Onishi 2026-09-25 1 min 66 words TACTILE ACUITY

Summary: The grating orientation discrimination task (GOT) is typically used to assess tactile spatial acuity. However, performance is influenced by stimulus speed, contact force between the stimulator and the skin, and the method of discrimination. This study aimed to explore suitable stimulus conditions fo...

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Epidermal microstructure and tactile sensitivity in the glabrous skin of hands and feet

Hannes P Saal 2026-09-25 1 min 69 words TACTILE ACUITY

Summary: Glabrous skin on the human palms and soles is specialized for mechanical interaction and is characterized by papillary ridges, an absence of hair follicles, and a thick stratum corneum. Although gross morphological differences, such as the increased thickness of the plantar stratum corneum, are well...

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Neurotrauma During Military Training: Results of the INVestigating traIning assoCiated blasT pAthology (INVICTA) Study

Walter S Carr 2026-10-01 1 min 67 words TACTILE ACUITY

Summary: Concern is growing that subconcussive blast exposure during military exercises such as heavy weapons training (HWT) may adversely affect brain function. HWT involves firing shoulder-mounted weapons that can expose the gunner, assistant gunner, and instructors (range safety officers [RSOs]) to blast ...

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Altered body representation in chronic non-specific low back pain: a testable framework linking body-schema-related disturbance, corticomotor alterations, and altered pain processing

👤 Zhifei Xie
📅 2026-10-24 📖 1 min 📝 66 words

TACTILE ACUITY

Summary: Low back pain is a leading contributor to disability worldwide; in chronic non-specific low back pain (CNSLBP), the processes sustaining pain and disability in the absence of identifiable structural pathology remain incompletely understood. Three partially overlapping research lines have documented ...

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Hyperexcitable primary somatosensory cortex in the MitoPark mouse model of Parkinson's disease

Alan J Emanuel 2026-10-25 1 min 67 words TACTILE ACUITY

Summary: The majority of Parkinson's disease patients suffer from decreased tactile acuity or neuropathic pain, both of which contribute to low quality of life and likely exacerbate motor dysfunction. How Parkinson's disease pathology relates to altered somatosensation remains unclear. One potential contribu...

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The Relationship Between Occlusal Tactile Acuity, Bite Force and Tooth Wear

Birgitta Häggman-Henrikson 2026-11-03 1 min 52 words TACTILE ACUITY

Summary: CONCLUSIONS: Bite force may be a clinically relevant factor associated with accumulated tooth wear, whereas OTA was not independently associated with tooth wear. Longitudinal studies are needed to determine whether bite force can predict tooth wear progression and whether more comprehensive assessme...

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Impact of Braille Booklet, Audio-Tactile Performance and Caregiver-supervised Training on Oral Hygiene Status, Knowledge, Attitude and Practices among Visually Challenged Children- A Randomized Clinical Trial

👤 Priyanga Chandrasekaran
📅 2026-09-22 📖 1 min 📝 52 words

BRAILLE

Summary: CONCLUSION: Both caregiver-supervised training and audio tactile performance using specialized model method of oral health education proved to be more effective than the braille booklet group on plaque control and improvement in the oral hygiene status, oral health knowledge, attitude and practice i...

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Design and functional verification of an automated Perkins Brailler accessory with audio feedback

Ali Alhaj 2026-09-23 1 min 67 words BRAILLE

Summary: CONCLUSIONS: The study did not include learners with visual impairment, Braille teachers, rehabilitation specialists, or other end users, and it did not collect quantitative measures of accuracy across the full character set, writing speed, embossing quality, latency, durability, safety, usability, ...

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Uncertainty-Aware structured decoding for Chinese Braille document transcription

Haoran Sun 2026-09-25 1 min 64 words BRAILLE

Summary: Automatic transcription of braille documents remains challenging in document digitization settings, particularly under real imaging degradations such as uneven embossing, illumination variation, paper texture, skew, blur, and ghosting, which make braille dot presence uncertain and often cause error ...

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Clinical Characteristics and Epidemiology of Isolated Dystonia: Results from a US Claims Analysis

James E Thornton
2026-10-08 1 min 54 words BRAILLE

Summary: CONCLUSIONS: This study defines an isolated dystonia cohort in the US based on real-world evidence and systematic exclusion of secondary causes of dystonia. Estimated prevalence is within range of prior epidemiological studies, and the cohort showed increased rates of comorbid psychiatric disorders....

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Effects of multisensory tactile-auditory integration teaching on systematic anatomy learning in visually impaired medical students

Fanzhen Kong
2026-10-08 1 min 50 words BRAILLE

Summary: CONCLUSION: The tactile-auditory integrated training system significantly improves objective summative knowledge acquisition and subjective learning experiences in visually impaired medical students. This approach provides a scalable, cost-constrained, and replicable pedagogical model for inclusive ...

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Physical and Environmental Accessibility of Dental Services for Persons with Disabilities in Saudi Primary Health Care: A Cross-Sectional Study

👤 Estie Kruger 📅 2026-10-16 📖 1 min 📝 66 words BRAILLE

Summary: Background/Objectives: Persons with disabilities (PWDs) may experience barriers when accessing oral health care, including those related to the physical and sensory environment of the dental-care setting. In this study we assessed the accessibility of primary health care centres (PHCs) providing den...

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Visible Light Driven Diselenide Chemistry for Simultaneous 3D Patterning and Mechanical Reinforcement

👤 Huaping Xu 📅 2026-10-16 📖 1 min 📝 63 words BRAILLE

Summary: Three-dimensional (3D) patterning on polymer surfaces is typically achieved by photolithography or related techniques, which often involve complex procedures, high-energy ultraviolet light, and risk of material damage. Here, we report a visible light-driven strategy that exploits the dual functional...

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An accessible pregnancy test for blind and low-vision users

Vanja Garaj 2026-10-16 1 min 63 words BRAILLE

Summary: CONCLUSIONS: The findings suggest that accessible pregnancy testing must prioritise not only non-visual result reading but also privacy, dignity, reproductive autonomy and independent health management. These results represent the preliminary evidence of acceptability rather than a conclusive test o...

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Genome-Informed Loop-Mediated Isothermal Amplification (LAMP) Assays for Detection and Differentiation of Ralstonia pseudosolanacearum Phylotypes I and III

Jarred Yasuhara-Bell 2026-10-27 1 min 76 words BRAILLE

Summary: The Ralstonia solanacearum species complex (RSSC) causes bacterial wilt in a broad range of economically important hosts. The RSSC has been reclassified into three species, of which R. solanacearum sequevars 1 and 2 in phylotype IIB are designated as Select Agents in the U.S. Many rapid tests for "R...

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Encoding tactile stimuli for Braille recognition with organoids

👤 Benjamin Ward-Cherrier 📅 2026-10-31 📖 1 min 📝 62 words BRAILLE

Summary: This study proposes a transferable encoding strategy that maps tactile sensor data to electrical stimulation patterns, enabling neural organoids to perform an open-loop artificial tactile Braille classification task. Human forebrain organoids cultured on a low-density microelectrode array (MEA) are ...

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The Non-Invasive Brain Stimulation Side Effect Scale (NIBS-SES): development of a brief, multilingual questionnaire for monitoring adverse effects in transcranial magnetic, electrical and ultrasound stimulation

👤 Maximilian Kiebs 📅 2026-11-05 📖 1 min 📝 59 words TDCS TACS TRNS

Summary: CONCLUSIONS: Our findings support the feasibility of repeated session-level assessment with the initial version, from which the final NIBS-SES was derived. This multilingual, open-access instrument comprises 10 within-session and 9 between-session items developed for monitoring across TMS, tES, and ...

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Concurrent prefrontal neurochemical effects of high-definition transcranial direct current stimulation: directional E-field associations with GABA+ and Glx

👤 Jianqi Li 📅 2026-11-05 📖 1 min 📝 58 words TDCS TACS TRNS

Summary: Transcranial direct current stimulation (tDCS) is a promising neuromodulatory technique for treating neuropsychiatric disorders. However, its neurochemical mechanisms remain poorly characterized, and clinical efficacy is limited by substantial inter-individual variability. This study used concurrent...

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Anodal tDCS alters perceived visual contrast in humans

👤 Sang Wook Hong 📅 2026-11-06 📖 1 min 📝 62 words TDCS TACS TRNS

Summary: Cortical gain modulation is thought to alter perceptual experience, most prominently demonstrated by the effects of spatial attention on contrast appearance. Whether externally induced changes in cortical excitability produce comparable perceptual consequences remains unclear. Here, we tested whethe...

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Evaluating OPTION: Benefits and Pitfalls of a Youth Trafficking Diversion Court in Florida

Joan A Reid 2026-11-06 1 min 67 words TDCS TACS TRNS

Summary: Trafficking diversion courts (TDCs) provide an alternative approach to supporting trafficked youth by promoting recovery, preventing revictimization, and diverting youth from the justice system. The current study examines the benefits and challenges of the OPTION (Offering Potentially Trafficked Ind...

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Amelioration of diabetic autonomic neuropathy symptoms following a course of transcranial direct current stimulation

Ahmad Alipour 2026-11-06 1 min 62 words TDCS TACS TRNS

Summary: Cardiovascular autonomic neuropathy (CAN) is a severe and potentially life-threatening complication of diabetes. This randomized, sham-controlled, three-arm trial evaluated the efficacy of anodal transcranial direct current stimulation (tDCS) targeting the left dorsolateral prefrontal cortex for imp...

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From Single Cells to the Circulating Tumor Microenvironment: Label-Free Size-Selection and a Morphological Taxonomy for Liquid Biopsy in Cytopathology

Satoru Kobayashi 2026-11-09 1 min 25 words TDCS TACS TRNS

Summary: CONCLUSION: Size-selection converts a liquid biopsy into a cytology slide, placing interpretation of the circulating tumor ecosystem within the existing competence of the diagnostic laboratory.

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Phase-Dependent Effects of Cerebellar Transcranial Alternating Current Stimulation on Lateral Vestibulospinal Tract Excitability

Tsubasa Mitsutake 2026-11-09 1 min 36 words TDCS TACS TRNS

Summary: CONCLUSIONS: LVST excitability differed according to the phase of the ongoing cerebellar-targeted tACS waveform. These findings provide electrophysiologic evidence of an online phase-dependent effect on vestibulospinal excitability and a physiologic basis for future phase-specific neuromodulation st...

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Effects of Transcranial Direct Current Stimulation on Core Symptoms and Sleep Disturbances in Children With Autism Spectrum Disorder: A Meta-Analysis of Randomized Controlled Trials

👤 Meimei Yang
📅 2026-11-09 📖 1 min 📝 44 words

TDCS TACS TRNS

Summary: CONCLUSION: tDCS, most commonly delivered at 1 mA over the DLPFC, was associated with improvements in some ASD-related behavioral outcomes; preliminary evidence also suggested a potential benefit for sleep disturbances, although this finding was based on only two low-certainty studies and warrants c...

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No effects of single-session concurrent tDCS on explicit motor sequence learning in healthy adults with conventional 2 mA and tangential 6 mA currents

👤 Lucas C Parra
📅 2026-11-09 📖 1 min 📝 45 words

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Summary: CONCLUSION: These findings, together with a survey of the existing literature, suggest that the effects of single-session concurrent tDCS on explicit motor sequence learning are not robust across studies. Our results underscore the necessity of rigorous methodological controls to prevent false disco...

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More than Quaternary Ammonium Compounds: First Evidence of Tertiary Amine Compounds as Novel Contaminants Ubiquitously Present in Indoor Dust

👤 Runzeng Liu 📅 2026-11-10 📖 1 min 📝 65 words TDCS TACS TRNS

Summary: Nitrogen-containing organic compounds have attracted global concern due to their widespread production and potential environmental risks. While the environmental occurrence of quaternary ammonium compounds (QACs) has been widely studied, research on tertiary amine compounds (TACs), another important...

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Prefrontal Activation and Cognitive Heterogeneity in Pediatric Sleep-Disordered Breathing

👤 Amal Isaiah 📅 2026-11-05 📖 1 min 📝 39 words FNIRS

Summary: CONCLUSIONS AND RELEVANCE: In this cross-sectional study, task-related PFC activation was associated with SDB severity and moderated the severity-cognition association. fNIRS may help identify children whose cognitive profiles deviate from conventional severity metrics, thereby supporting individual...

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Mirror Neural System in Hands Motor Imagery with Pictures and Videos Guidance: An EEG-fNIRS Study

👤 Mohamad Sawan 📅 2026-11-05 📖 1 min 📝 64 words FNIRS

Summary: Brain-computer interfaces (BCIs) achieved by motor imagery (MI) have been widely accepted as an effective tool in rehabilitative applications. Functional near-infrared spectroscopy (fNIRS) and electroencephalography (EEG) are two common implementation forms of BCIs, capable of monitoring neuronal po...

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Quality-aware mask-propagation graph attention for fNIRS-based bipolar disorder classification without channel interpolation

👤 Yajing Meng 📅 2026-11-05 📖 1 min 📝 58 words FNIRS

Summary: Bipolar disorder (BD) remains difficult to characterise objectively, motivating a neural marker to complement clinical assessment. Functional near-infrared spectroscopy (fNIRS) offers portable cortical haemodynamic measurement, but channel availability can vary across recordings because of bad chann...

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Brain-Behavior Coupling During Cognitive-Postural Dual-Tasking in Older Adults With Mild Cognitive Impairment: An fNIRS Study

👤 Chuan Guo 📅 2026-11-06 📖 1 min 📝 56 words FNIRS

Summary: CONCLUSIONS: Older adults with MCI exhibit altered brain-behavior coupling during cognitive-postural dual-task conditions involving sensory and postural constraints, characterized by altered relationships between prefrontal-sensorimotor cortical activation and behavioral performance. These findings ...

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Item-Specific Proportion Congruency Uncovers Different Forms of Attentional Control: An event-related fNIRS Study

👤 Murat Perit Çakır 📅 2026-11-07 📖 1 min 📝 70 words FNIRS

Summary: CONCLUSION: While behavioral results mostly support contingency learning, fNIRS data revealed ISPC-related activation in prefrontal channels with sensitivity to BA 46 (lateral, channel 1) and the BA 10/BA 11 overlap zone (medial, channels 10-12), with channel 11 emerging as a convergence point for b...

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Does the emotion regulation of self-directed humor engage the prefrontal cortex? An fNIRS study

Zhonglu Zhang 2026-11-07 1 min 68 words FNIRS

Summary: Previous research has revealed that self-directed humor (self-enhancing and self-defeating) can help regulate negative emotions and that the prefrontal cortex (PFC) plays a critical role in emotion regulation. However, it remains unknown whether emotion regulation of self-directed humor is associate...

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Effects of mini-maps with different reference frames on polarity-free 3D spatial navigation: a behavioural and fNIRS study

Xiaozhou Zhou 2026-11-07 1 min 62 words FNIRS

Summary: Manned spaceflight and deep-sea exploration extend human activities to extreme environments lacking gravitational and visual polarity cues, where spatial disorientation frequently occurs. Mini-map studies largely focus on terrestrial environments, while reference frame design for polarity-free 3D sp...

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Baduanjin improves motor function and cortical activation in children with developmental coordination disorder: a randomized controlled trial

👤 Wei Song 📅 2026-11-09 📖 1 min 📝 40 words FNIRS

Summary: CONCLUSIONS: Baduanjin training improved motor coordination and balance in children with DCD. The observed changes in activation within motor-related cortical regions provide preliminary support for a possible neurophysiological basis of these functional improvements, although the underlying mechani...

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Potential Applications of Brain-Computer Interface Technology in the Rehabilitation of Patients with Anxiety Disorders: A Narrative Review

Hu Lou 2026-11-10 1 min 58 words FNIRS

Summary: Anxiety disorders are among the most prevalent mental disorders worldwide and can cause persistent impairment in emotional regulation, cognition, and social functioning. Pharmacotherapy, psychotherapy, exercise, and non-invasive brain stimulation are widely used, yet their effectiveness may be const...

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Multimodal EEG-fNIRS complexity analysis can reveal the neurovascular coupling change with aging

Yicheng Zhu 2026-11-10 1 min 58 words FNIRS

Summary: CONCLUSION: Aging is associated with task-specific alterations in neural complexity, hemodynamic complexity, and the cross-modal temporal relationship between EEG and fNIRS signals. These findings suggest that age-related changes in NVC-related dynamics are better characterized as dynamic reorganiza...

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Overcoming subject variability in P300 decoding based on quantification by deep learning

👤 Fred R Beyette 📅 2026-11-06 📖 1 min 📝 60 words BRAIN COMPUTER INTERFACE

Summary: Binary P300 decoding is a cornerstone of P300-based brain-computer interfaces, enabling applications such as assistive communication, neurorehabilitation, and cognitive monitoring. However, substantial subject variability in EEG patterns poses a major challenge to the development of generalized P300...

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A ciliated epithelial cell line as a model for respiratory syncytial virus research

👤 Marie-Anne Rameix-Welti 📅 2026-11-06 📖 1 min 📝 74 words BRAIN COMPUTER INTERFACE

Summary: Respiratory syncytial virus (RSV) is the leading cause of severe respiratory infection in infants and an increasingly recognized pathogen in the elderly. Conventional in vitro culture models for RSV rely on poorly differentiated cell lines that lack key airway epithelial features such as polarizatio...

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Neural interface instructs layer 4-appropriate circuit reconstruction with engrafted human brain organoids and improves visually guided behavior

👤 Xiao-Hong Li
📅 2026-11-06 📖 1 min 📝 65 words

BRAIN COMPUTER INTERFACE

Summary: Repairing precise neural microcircuitry after cortical injury remains a pivotal challenge. Human brain organoids (hBOs) are a promising cell source, but disorganized integration limits their therapeutic potential. Here, we demonstrated that targeted neuromodulation, steered by endogenous narrowband ...

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Self-Guided Beta-Responsive Visual Neurofeedback in a Public Exhibition: Feasibility and Usability Randomized Controlled rial

👤 Joseph Ollier
📅 2026-11-07 📖 1 min 📝 57 words

BRAIN COMPUTER INTERFACE

Summary: CONCLUSIONS: This trial evaluated an autonomous consumer-EEG neurofeedback system integrating self-fitting, signal-quality checking, baseline calibration, adaptive feedback, randomization, and self-report without operator support. Many first-time users completed the autonomous workflow with acceptab...

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Auditory rehabilitation in mixed hearing loss due to chronic otitis media sequelae: passive versus active transcutaneous bone conduction implant

👤 Esther Sofía Angulo Martínez
📅 2026-11-08 📖 1 min 📝 47 words
BRAIN COMPUTER INTERFACE

Summary: CONCLUSIONS: The findings of the present study suggest that AT-BCI provides a greater and clinically meaningful auditory gain for HF-PTA in patients with MHL secondary to COM. In ATBCI group a stable audiological performance was achieved as early as the 3-month follow-up and is maintained over time.

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Selective dissociation of coordination bonds enables rapid rigidity switching of hydrogel microfibre bioelectrodes

👤 Lidong Zhang
📅 2026-11-09 📖 1 min 📝 64 words
BRAIN COMPUTER INTERFACE

Summary: Bioelectrodes with tunable rigidity deliver excellent mechanical compliance and can effectively alleviate tissue damage. Herein, we report hydrogel microfibre electrodes (diameter: 50-200 μm) fabricated by cross-linking sodium alginate (SA) microfibres with Ca^(2+)/Al^(3+) dual ions. The resultant S...

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Constructing Neurorights in the Era of Clinical Brain-Computer Interfaces : China's Emerging Governance Pathway

👤 Yuan Lin 📅 2026-11-09 📖 1 min 📝 67 words BRAIN COMPUTER INTERFACE

Summary: In March 2026, China approved the world's first implantable brain-computer interface (BCI) medical device for clinical use outside clinical trials, a milestone that raises urgent questions about how legal and ethical systems should protect the brain and mind. This article examines China's emerging n...

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DicomDoodle: Human-in-the-Loop Annotation of 3D MRI DICOM Images with AI Support and DICOM-SEG Compatibility

👤 Debnath Pal 📅 2026-11-09 📖 1 min 📝 62 words BRAIN COMPUTER INTERFACE

Summary: Accurate medical image annotation requires curation and remains a major bottleneck due to its time-intensive nature and inter-observer variability. To address this, we developed DicomDoodle, an open-source, Python-based tool for human-in-the-loop annotation that integrates a fine-tuned YOLO11 detect...

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Potential Applications of Brain-Computer Interface Technology in the Rehabilitation of Patients with Anxiety Disorders: A Narrative Review

👤 Hu Lou 📅 2026-11-10 📖 1 min 📝 58 words BRAIN COMPUTER INTERFACE

Summary: Anxiety disorders are among the most prevalent mental disorders worldwide and can cause persistent impairment in emotional regulation, cognition, and social functioning. Pharmacotherapy, psychotherapy, exercise, and non-invasive brain stimulation are widely used, yet their effectiveness may be const...

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https://pubmed.ncbi.nlm.nih.gov/43005114/?utm_source=BucketBot&utm_medium=rss&utm_campaign=None&utm_content=1rSUu--tbw4049Wgf_RdKXdtNCvGW0lVBZFpHe7zaN4k4DwoD5&fc=None&ff=20261110142544&v=2.20.1

In Situ Displacement Reaction Enables a Conductive Metallized Hydrogel Interface for Smart Underwater Flexible Electronics

Qigang Wang 2026-11-10 1 min 60 words BRAIN COMPUTER INTERFACE

Summary: Underwater wearable sensing is severely hampered by the intrinsic drawbacks of conventional materials, including water absorption, swelling, and electrical drift during long-term immersion. Here, we develop an anti-swelling hydrogel with a surface-metallized layer, which achieves strong stability of...

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Coping strategies and dreams—The impact of waking-life distress on negative dream emotions.

2026-05-18 1 min 183 words DREAMING

Summary: The relationship between waking life and dreaming seems complex, and various studies show that dreams are influenced by waking-life experiences. However, only a few studies deal with the interactions between coping strategies in waking life and dream emotions. The main goal of this study is to inves...

🔗 **Read full article:**
http://doi.org/10.1037/drm0000346

Consistency of new unpacking syntax in 3.15?

👤 /u/assumptionkrebs1990 📅 2026-11-10 📖 1 min 📝 216 words REDDIT PYTHON

> **Summary:** <!-- SC_OFF --> Not sure if this question belongs here or if the flair is right, but I have a question. You can now (3.15 and up, PEP 448) unpack nested lists and dictionaries with * and ** in comprehension expressions, for example: <pre><code>>>> ranges = [rang...

🔗 **Read full article:**
https://www.reddit.com/r/Python/comments/1xsocqz/consistency_of_new_unpacking_syntax_in_315/

Smart home devices should work without the internet

👤 skittleson 📅 2026-11-10 📖 1 min 📝 13 words HACKER NEWS

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https://docodethatmatters.com/home-automation-overnight-success/

Blender 5.3 released with Gaussian Splat support

👤 amlib 📅 2026-11-10 📖 1 min 📝 13 words HACKER NEWS

> **Summary:** Article URL: https://www.blender.org/download/releases/5-3/ Comments URL: https://news.ycombinator.com/item?id=50442426 Points: 281 # Comments: 37

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https://www.blender.org/download/releases/5-3/

Tailscale's DNS IPv6 spells out "Tailscale"

👤 theanonymousone 📅 2026-11-10 📖 1 min 📝 13 words HACKER NEWS

> **Summary:** Article URL: https://www.reddit.com/r/Tailscale/comments/1xsr24r/tailscales_dns_ipv6_spells_out_tailscale_easter/ Comments URL: <a href="https://news.ycombinator.com/item?id=5...

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.name Grandfathered

👤 p4bl0 📅 2026-11-10 📖 1 min 📝 13 words HACKER NEWS

> **Summary:** Article URL: https://neil.fraser.name/news/2026/11/10/ Comments URL: https://news.ycombinator.com/item?id=50442736 Points: 348 # Comments: 62

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AMD's Zen 6 Family: Feeding the Beast

👤 pixelpoet 📅 2026-11-10 📖 1 min 📝 13 words HACKER NEWS

> **Summary:** Article URL: https://chipsandcheese.com/p/amds-zen-6-family-feeding-the-beast Comments URL: https://news.ycombinator.com/item?id=50443090 Points...

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https://chipsandcheese.com/p/amds-zen-6-family-feeding-the-beast

Val Town is winding down

👤 tosh 📅 2026-11-10 📖 1 min 📝 13 words HACKER NEWS

> **Summary:** Article URL: https://blog.val.town/winding-down Comments URL: https://news.ycombinator.com/item?id=50443427 Points: 108 # Comments: 25

🔗 **Read full article:**
https://blog.val.town/winding-down

Claude Whiteboards, Claude Sheets, and Claude Data

👤 Yorothegoat 📅 2026-11-10 📖 1 min 📝 13 words HACKER NEWS

> **Summary:** Article URL: https://claude.com/resources/articles/sheets-whiteboards-data Comments URL: https://news.ycombinator.com/item?id=50443716 Points: 8</p...

🔗 **Read full article:**
https://claude.com/resources/articles/sheets-whiteboards-data

Show HN: Get your coding agent off ~2019 JavaScript, onto modern web primitives

👤 paulirish 📅 2026-11-10 📖 1 min 📝 364 words HACKER NEWS

> **Summary:** Browsers natively handle a lot of what used to take 50 lines of JS (or a dependency): `CSS.highlights` for text range highlighting without s everywhere, `<dialog>` and showModal() give you backdrops and ideal focus management, `field-sizing` to auto-grow textareas as the user types.. (Somet...

🔗 **Read full article:**
https://github.com/GoogleChrome/modern-web-guidance

Synaptics patent claims stop work on DisplayLink drivers

👤 speckx 📅 2026-11-10 📖 1 min 📝 13 words HACKER NEWS

> **Summary:** Article URL: https://lwn.net/SubscriberLink/1102549/acbc92e00cee3a3f/ Comments URL: https://news.ycombinator.com/item?id=50443824 Points: 62 # Co...

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Starlink Mini 2

👤 robertjfclarke 📅 2026-11-10 📖 1 min 📝 13 words HACKER NEWS

> **Summary:** Article URL: https://starlink.com/mini-2 Comments URL: https://news.ycombinator.com/item?id=50444129 Points: 32 # Comments: 12

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https://starlink.com/mini-2

Automatic computational reproducibility checks with Metacheck

👤 noreply@blogger.com (Daniel Lakens) 📅 2026-10-30 📖 19 min 📝 3854 words

TWENTY PERCENT STATISTICIAN

> **Summary:** When authors share their data and analysis code, the idea is that anyone should be able to run that code on that data and get the results reported in the paper. But sharing data and code in a way that results are reproducible is difficult, and researchers have often not received tr...

🔗 **Read full article:**
http://daniellakens.blogspot.com/2026/10/automatic-computational-reproducibility.html

Registration is now full for the 2024 ERP Boot Camp

👤 Steve Luck 📅 2024-03-16 📖 1 min 📝 106 words ERP BOOT CAMP

> **Summary:** The demand for the 2024 ERP Boot Camp was far beyond our expectations, and we reached our maximum registration of 30 people within one day. We already have a waiting list of over 30 people, so we have closed the registration site. <...

🔗 **Read full article:**
https://erpinfo.org/blog/2024/3/15/registration-full

New Paper: Using Multivariate Pattern Analysis to Increase Effect Sizes for ERP Amplitude Comparisons

👤 Steve Luck 📅 2024-06-10 📖 2 min 📝 521 words ERP BOOT CAMP

> **Summary:** Carrasco, C. D., Bahle, B., Simmons, A. M., & Luck, S. J. (2024). Using multivariate pattern analysis to increase effect sizes for event-related potential analyses. Psychophysiology, 61, e14570. https://doi.org/10.1111/psyp.14570 [<a h...

🔗 **Read full article:**
https://erpinfo.org/blog/2024/6/10/erp-core-decoding-paper

Call for Student/YP Deputy Editor

👤 Deidre Artis 📅 2026-09-29 📖 1 min 📝 14 words EMBS

Summary: The post Call for Student/YP Deputy Editor appeared first on IEEE EMBS.

🔗 **Read full article:**
https://www.embs.org/featured-news/call-for-student-yp-deputy-editor/

Dynamic muscle proprioception impairment in adolescent idiopathic scoliosis revealed by tendon vibration

📖 1 min 📝 29 words NEUROSCIENCE JOURNAL

Summary: Publication date: 9 January 2027 Source: Neuroscience, Volume 620 Author(s): Daniela M Pinzon-Corredor, Christophe Boulay, Alexis Ulian, Guillaume Authier, Nicole Levy, Fabienne Levy, Jean-Luc Jouve, Corentin Petitpas, Sébastien Pesenti, Anne Kavounoudias

🔗 **Read full article:**
https://www.sciencedirect.com/science/article/pii/S0306452226007165?dgcid=rss_sd_all

Altered stability of dynamic functional architecture in the co-occurrence of autism spectrum disorder and attention–deficit/hyperactivity disorder

📖 1 min 📝 32 words NEUROSCIENCE JOURNAL

Summary: Publication date: 9 January 2027 Source: Neuroscience, Volume 620 Author(s): Shuting Zhang, Chao Han, Yuning Si, Jinsha Tan, Xin Pan, Kaixin Li, Mengzhu Wang, Xipeng Yue, Xuan Yu, Wei Wei, Yan Bai, Meiyun Wang

🔗 **Read full article:**
https://www.sciencedirect.com/science/article/pii/S0306452226007049?dgcid=rss_sd_all

Brain-heart decoupling in somatic depression: Desynchronization between salience network neuropsychological connectivity and heart rate variability

📖 1 min 📝 24 words NEUROIMAGE

Summary: Publication date: 1 December 2026 Source: NeuroImage, Volume 343 Author(s): Yi Xia, Junling Sheng, Xiaoqin Wang, Ciqing Bao, Yujue Wang, Xumiao Wang, Zhijian Yao, Qing Lu

🔗 **Read full article:**
https://www.sciencedirect.com/science/article/pii/S1053811926006464?dgcid=rss_sd_all

Do autistic traits modulate the neural processing of logical rule violations in visual and auditory modalities?

📖 1 min 📝 28 words NEUROIMAGE

Summary: Publication date: 1 December 2026 Source: NeuroImage, Volume 343 Author(s): Junyu Wu, Rong Huang, Jingyang Yu, Jiayi Shi, Tianze Wang, Yuting Zhang, Yutong Chen, Jie Li, Shubo Shui, Jing Meng

🔗 **Read full article:**
https://www.sciencedirect.com/science/article/pii/S1053811926006427?dgcid=rss_sd_all

Training interbrain synchrony through dyadic neurofeedback modulates empathy-for-pain

📖 1 min 📝 12 words NEUROIMAGE

Summary: Publication date: 1 December 2026 Source: NeuroImage, Volume 343 Author(s): Mario Francis, Simone Shamay-Tsoory

🔗 **Read full article:**
https://www.sciencedirect.com/science/article/pii/S1053811926006415?dgcid=rss_sd_all

Short-video experience and brain-state dynamics during the processing of rapidly changing video information in youth

📖 1 min 📝 25 words NEUROIMAGE

> **Summary:** Publication date: 1 December 2026 Source: NeuroImage, Volume 343 Author(s): Bijie Tie, Li Geng, Pengcheng Wang, Jason M. Nagata, Jinbo He, Yueyang Xiao, Xueyang Wang, Jiang Qiu

🔗 **Read full article:**
https://www.sciencedirect.com/science/article/pii/S105381192600649X?dgcid=rss_sd_all

Widespread oscillatory suppression in autism persists in the absence of psychotropic medication.

👤 Amorim, A. K. J., Alves, G. M., Pontes, G. C. L., Dias, L., Hondermann, B. M. F., Silva, M. C. C., Costa, L., Ceriatte, C., Pereira, A., Brasil, L. M. B. F., Souza, I. C. N., Machado, C. L. M. V., Lima, B.

📅 2026-11-10 📖 1 min 📝 240 words BIORXIV NEUROSCIENCE

> **Summary:** Purpose: Investigate the neural oscillatory activity associated with autism spectrum disorder (ASD) using electroencephalographic (EEG) recordings in children and adolescents. Compare the corresponding oscillatory power between ASD patients taking psychotropic medication with a drug-naive group. Met...

🔗 **Read full article:**
https://www.biorxiv.org/content/10.64898/2026.11.07.764864v1?rss=1

Accessing Mesoscale Functional Organization from Standard Functional MRI Leveraging AI-Based Super-Resolution

Nasr, S., Li, H. B., Zemlyanker, D., Huber, L., Rosen, M. S., Iglesias, J. E.

2026-11-10 1 min

185 words

BIORXIV NEUROSCIENCE

Summary: Access to the fine-scale functional organization of the human brain is essential for understanding cognition and neurological disorders yet remains beyond the reach of conventional functional MRI (fMRI) because of its limited spatial resolution and contrast-to-noise ratio. To overcome this limitatio...

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https://www.biorxiv.org/content/10.64898/2026.11.07.765069v1?rss=1

Rab10 and Arhgef7 interact to promote axon extension by linking vesicle supply and exocytosis

Olocco, F., Bekker, E., Ravindran, P., Pietralla, T., Puschel, A. W.

2026-11-10 1 min

175 words

BIORXIV NEUROSCIENCE

Summary: The formation of functional neural circuits begins with the extension of a single axon by immature neurons. Axon elongation requires an expansion of the plasma membrane that depends on the polarization of vesicle trafficking and exocytosis. It is not completely understood how the increased transport...

🔗 **Read full article:**
https://www.biorxiv.org/content/10.64898/2026.11.04.764424v1?rss=1

Maternal separation remodels Purkinje cell circuits and impairs cerebellar synaptic plasticity

👤 Lippiello, P., Hoxha, E., Cristiano, C., Pascarella, S., Volpicelli, F., Speranza, L., Cuomo, M., Ferraro, M. G., Stanley, A., Rominto, A. M., Filiz, K. D., Liguori, F., Tropea, D., Di Angelantonio, S., Chiariotti, L., Sulzer, D., Miniaci, M. C.

📅 2026-11-10 📖 1 min 📝 243 words BIORXIV NEUROSCIENCE

Summary: Introduction: Early-life adversity has been associated with cerebellar abnormalities, but the cellular and circuit-level changes involved and their persistence across development remain poorly understood. We examined whether maternal separation (MS) disrupts cerebellar circuit maturation and whether...

🔗 **Read full article:**
https://www.biorxiv.org/content/10.64898/2026.11.08.765140v1?rss=1

Multimodal Reorganization of Gut--Brain Physiology During the Cephalic Phase of Digestion

👤 Kehnemouyi, Y. M., Coleman, T. P.

📅 2026-11-10 📖 1 min 📝 205 words

BIORXIV NEUROSCIENCE

Summary: The vagus nerve plays a central role in gut--brain communication, yet the physiological dynamics associated with cholinergic vagal signaling remain difficult to characterize non-invasively in humans. The cephalic phase response, which prepares the gastrointestinal system for food ingestion through a...

🔗 **Read full article:**
https://www.biorxiv.org/content/10.64898/2026.11.07.765106v1?rss=1

Subcellular Spatial Profiling Unveils Links Between Neuritic Plaques with Selective Tau Vulnerability in Alzheimer's Disease

👤 Li, H., Gross, E. R., Devine, A., Gundavelli, A., Prell, S. A., Woodbury, M. E., Gen, L., Doering, A., Held, A., Hyman, B. T., Serrano-Pozo, A., Zwang, T. J., Bennett, R. E., Das, S.

📅 2026-11-10 📖 1 min 📝 147 words BIORXIV NEUROSCIENCE

Summary: Alzheimer's disease (AD) features widespread extracellular amyloid-{beta} plaques but selective intraneuronal tau accumulation in cortical layers II/III and V, while layer IV remains relatively resistant. Here, we combined subcellular spatial transcriptomics with co-registered A{beta} and pTau immun...

🔗 **Read full article:**
https://www.biorxiv.org/content/10.64898/2026.11.05.764692v1?rss=1

Tendon-specific PIEZO1 activation enhances neural drive to muscle during ballistic movement

Nakamura, K., Kosugi, A., Nakamichi, R., Kurimoto, R., Matsushima, T., Chiba, T., Ko, M., Morita, T., Yoshii, T., Seki, K., Asahara, H.

2026-11-10 1 min 250 words BIORXIV NEUROSCIENCE

Summary: Tendons are generally regarded as passive structures that transmit muscle force to bone, yet they contain the Golgi tendon organ (GTO), whose Ib afferents act on motor neurons either inhibitory or excitatory depending on movement context. Whether a change in tendon mechanical properties alters the n...

Read full article:
https://www.biorxiv.org/content/10.64898/2026.11.07.765061v1?rss=1

Physical activity, fitness, sedentary behaviour, and brain health: a systematic review of Mendelian Randomisation studies

2026-11-10 1 min 0 words NATURE NEUROSCIENCE SUBJECTS

Read full article:
https://www.nature.com/articles/s44437-026-00013-2

Temporal dynamics of premenstrual dysphoric disorder revealed through predictive symptom networks

📅 2026-11-10 📖 1 min 📝 0 words NATURE NEUROSCIENCE SUBJECTS

🔗 **Read full article:**
https://www.nature.com/articles/s41467-026-78590-x

Social status and status loss do not influence alcohol-related behaviors in rats

📅 2026-11-10 📖 1 min 📝 0 words NATURE NEUROSCIENCE SUBJECTS

🔗 **Read full article:**
https://www.nature.com/articles/s41598-026-80056-5

Some people are resistant to dementia — here’s what their genes can teach us

📅 2026-11-10 📖 1 min 📝 0 words NATURE NEUROSCIENCE SUBJECTS

🔗 **Read full article:**
https://www.nature.com/articles/d41586-026-03625-8

The representational geometry of cognitive maps under cognitive control

📅 2026-11-10 📖 1 min 📝 0 words NATURE NEUROSCIENCE SUBJECTS

🔗 **Read full article:**
https://www.nature.com/articles/s41467-026-77621-x

A hybrid neural deep belief networks-gated recurrent unit algorithm with a leakage-aware evaluation protocol for ECG-HRV-based cognitive load classification

📅 2026-11-10 📖 1 min 📝 0 words NATURE NEUROSCIENCE SUBJECTS

🔗 **Read full article:**
https://www.nature.com/articles/s41598-026-80282-x

Long-COVID: Ultrasensitive multiplex protein biomarker profiling reveals subclinical neuroinflammatory and innate immune dysregulation without correlation to symptom severity

📅 2026-11-10 📖 1 min 📝 0 words NATURE NEUROSCIENCE SUBJECTS

🔗 **Read full article:**
https://www.nature.com/articles/s41598-026-77483-9

AutoSNAP enables automated nerve segmentation and morphometry quantification

📅 2026-11-10 📖 1 min 📝 0 words NATURE NEUROSCIENCE SUBJECTS

🔗 **Read full article:**
https://www.nature.com/articles/s41598-026-78196-9

Information formatting in neuronal populations predicts differential functional contributions to behavior

📅 2026-11-10 📖 1 min 📝 0 words NATURE NEUROSCIENCE SUBJECTS

🔗 **Read full article:**
https://www.nature.com/articles/s41593-026-02514-7

Altered developmental profile of the head-direction signal in a rat model of fragile X syndrome

📅 2026-11-10 📖 1 min 📝 0 words NATURE NEUROSCIENCE SUBJECTS

🔗 **Read full article:**
https://www.nature.com/articles/s42003-026-11197-6

Single-neuron encoding of rapidly learned visual information reshapes human perception

Gabriel Kreiman 2026-11-04 1 min 38 words NATURE NEUROSCIENCE

Summary: Nature Neuroscience, Published online: 04 November 2026; doi:10.1038/s41593-026-02496-6 The authors show that rapid learning modulates neuronal activity in the human visual cortex, requiring extra processing time to support recog...

🔗 **Read full article:**
https://www.nature.com/articles/s41593-026-02496-6

The human claustrum tracks uncertainty and prediction error

2026-11-04 1 min 54 words NATURE NEUROSCIENCE

Summary: Nature Neuroscience, Published online: 04 November 2026; doi:10.1038/s41593-026-02471-1 Single-neuron recordings provide direct evidence that the human claustrum tracks uncertainty and prediction errors during adaptive behavior. ...

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https://www.nature.com/articles/s41593-026-02471-1

Information formatting in neuronal populations predicts differential functional contributions to behavior

Marlene R. Cohen
2026-11-10 1 min 32 words NATURE NEUROSCIENCE

Summary: Nature Neuroscience, Published online: 10 November 2026; doi:10.1038/s41593-026-02514-7 Using experiments in monkeys and modeling, the authors demonstrate that the formatting of population information predicts the contribution of...

🔗 **Read full article:**
https://www.nature.com/articles/s41593-026-02514-7

An explainable deep learning framework for differentiating idiopathic normal pressure hydrocephalus from Alzheimer's disease using brain MRI

Sofyan Freihat
2026-11-10 1 min 269 words FRONTIERS NEUROINFORMATICS

Summary: Background and objectivesIdiopathic normal pressure hydrocephalus (NPH) and Alzheimer's disease (AD) may present with overlapping clinical and neuroimaging features, complicating differential diagnosis. This study evaluated an explainable, attention-enhanced deep learning framework for distinguishin...

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https://www.frontiersin.org/articles/10.3389/fninf.2026.1990170

Perceived sound quality and speech recognition are not interchangeable outcome measures in adult cochlear implant users

Andreas Büchner 2026-11-10 1 min 198 words FRONTIERS HUMAN NEUROSCIENCE

Summary: IntroductionThis study examined whether descriptor-based reports of perceived sound quality in adult cochlear implant users are associated with speech-recognition outcomes, preoperative maximum monosyllabic word recognition, and patient- or device-related factors.MethodsIn this retrospective single-...

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Effects of intermittent theta burst stimulation combined with hyperbaric oxygen therapy on cognitive function and cerebral blood flow in patients with post-stroke mild cognitive impairment

Min Tang 2026-11-10 1 min 371 words FRONTIERS HUMAN NEUROSCIENCE

Summary: ObjectiveTo observe the effects of intermittent theta burst stimulation (iTBS) combined with hyperbaric oxygen therapy on cognitive function and cortical hemodynamics in patients with post-stroke mild cognitive impairment using functional near-infrared spectroscopy (fNIRS).MethodsSixty patients with...

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Assessing systematic spatial differences between intraoperative and postoperative Probabilistic Sweet Spots in Deep Brain Stimulation for Essential Tremor

Simone Hemm 2026-11-10 1 min 245 words FRONTIERS HUMAN NEUROSCIENCE

Summary: InroductionDeep Brain Stimulation (DBS) is an established therapy for movement disorders such as Essential Tremor (ET). While electrode displacement across intraoperative and postoperative stages has been studied, Probabilistic Sweet Spot (PSS) analyses rarely account for temporal evolution. Consequ...

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Low-intensity focused ultrasound for neuromodulation: mechanisms, parameters, and emerging clinical applications—a critical review

Yi Yang 2026-11-10 1 min 282 words FRONTIERS NEUROSCIENCE

Summary: BackgroundLow-intensity focused ultrasound (LIFU) is a non-invasive neuromodulation technique that combines millimeter-scale spatial precision with access to cortical and deep targets — including the thalamus, basal ganglia, amygdala, and nucleus accumbens — that otherwise require invasive deep brai...

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Narcolepsy in the postpartum period: current evidence and future directions

Pablo Torterolo 2026-11-10 1 min 214 words FRONTIERS NEUROSCIENCE

Summary: Narcolepsy is a chronic sleep–wake disorder characterized by hypocretin (HCRT) deficiency, most commonly resulting from immune-mediated mechanisms targeting HCRT-producing neurons in the lateral hypothalamus. Although narcolepsy affects men and women at similar rates, increasing evidence suggests th...

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The science of neuronal regeneration: a comprehensive review on the scientific exploration of a new path of regeneration in central nervous system

Sanjay Kumar 2026-11-10 1 min 149 words FRONTIERS NEUROSCIENCE

Summary: The review focuses on neuronal regeneration in the central nervous system and the inhibition of regeneration in the CNS, such as glial scar formation, which is a crucial mechanism of extrinsic inhibition of regeneration. Regeneration-Associated Genes (RAGs), various transcription factors, epigenetic...

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Shared sleep fragmentation in undergraduate roommates: dyadic actigraphy in a dense urban campus setting

Sandra E. Sephton
2026-11-10 1 min 344 words FRONTIERS NEUROSCIENCE

Summary: IntroductionDyadic sleep research has focused on romantic partners, leaving open whether nonromantic people who share a sleep space show related sleep. We examined sleep continuity and roommate concordance among undergraduate roommate dyads on a dense urban campus bordered by an interstate highway, ...

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Visual field performance of the novel DIMS TED spectacle lens compared with DIMS and single-vision spectacle lenses

Natalia Vlasak
2026-11-10 1 min 346 words FRONTIERS NEUROSCIENCE

Summary: PurposeTo evaluate the visual field performance of DIMS TED spectacle lens, an enhanced Defocus Incorporated Multiple Segments (DIMS) spectacle lens design featuring more defocus segments closer to the geometric center, higher defocus power and extended defocus treatment zone, compared with DIMS and...

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The tau–microtubule axis in Alzheimer’s disease: reciprocal mechanisms, biomarker gaps, and therapeutic challenges

👤 Demet Özbabalık Adapınar
📅 2026-11-10 📖 1 min 📝 225 words FRONTIERS NEUROSCIENCE

Summary: Microtubule dysfunction has been implicated in axonal transport failure, synaptic degeneration, and neuronal loss in Alzheimer’s disease, yet its position within the disease process remains unresolved. This critical narrative review evaluates the tau–microtubule axis across physiological, pathologic...

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https://www.frontiersin.org/articles/10.3389/fnins.2026.2004502

Oral citicoline as add-on therapy in open-angle glaucoma: a 12-month prospective, non-randomized comparative cohort study with multimodal microperimetric, structural and OCT angiography assessment

👤 Alessandro Meduri
📅 2026-11-10 📖 1 min 📝 339 words FRONTIERS NEUROSCIENCE

Summary: BackgroundLowering intraocular pressure (IOP) is the only proven treatment for glaucoma, yet many patients progress despite adequate IOP control. Evidence for citicoline, a candidate neuroprotectant, is inconclusive and relies mainly on perimetric and electrophysiological endpoints. We evaluated the...

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Maladaptive state allocation: an orexin-based framework linking sleep disturbance, chronic pain, and substance use disorder

Francisco J. Flores-Ramirez

2026-11-10 1 min 238 words FRONTIERS NEUROSCIENCE

Summary: Sleep disturbance, chronic pain, and substance use disorders are highly prevalent conditions that frequently co-occur and exacerbate one another. Although traditionally studied as distinct clinical entities, growing evidence suggests they share common neurobiological mechanisms. Among these, the ore...

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Source-aware forward modelling and detection limits of acoustoelectric neural recording

Christopher Chare, Rachid Haouari, Klara Volckaert, Agknaton Bottenberg, Sebastian Haesler, Nick Van Helleputte, Xavier Rottenberg, Samer Houri and Peter Peumans

2026-11-10 1 min 298 words JOURNAL NEURAL ENGINEERING

Summary: Objective. Acoustoelectric recording uses focused ultrasound to select the region sampled by scalp or implanted electrodes. We relate its recorded voltage to source structure, terminal conditions and the carrier phase of the focus, and identify configurations that favour neural recording. Approach. ...

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https://iopscience.iop.org/article/10.1088/1741-2552/aebb68

Motor cortex-based intracortical BCI performance is robust to changes in attentional load during dual-tasking

E Canario, C Shearer, M Akcakaya, D Weber, S Chase and J L Collinger

2026-11-10 1 min

291 words

JOURNAL NEURAL ENGINEERING

Summary: Objective. High performance intracortical brain-computer interface (iBCI) control has been demonstrated in research settings, but performance can still vary within and between sessions. One potential source of this variability is the change in attentional load that comes from processing naturally oc...

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https://iopscience.iop.org/article/10.1088/1741-2552/aebb65

Closed-loop neuroprosthetic control through spared neural activity enables proportional foot movements after spinal cord injury

👤 Vlad Cnejevici, Matthias Ponfick, Dietmar Fey, Raul C Sîmpetru and Alessandro Del Vecchio

📅 2026-11-10 📖 1 min 📝 277 words JOURNAL NEURAL ENGINEERING

Summary: Objective. Loss of voluntary foot movement after spinal cord injury (SCI) can significantly limit independent mobility and quality of life. To improve motor output after injury, functional electrical stimulation (FES) is used to deliver stimulation pulses through the skin to affected muscles. While ...

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https://iopscience.iop.org/article/10.1088/1741-2552/aebb64

Golgi: an open-source graphical platform for image-to-recruitment modeling of peripheral nerve stimulation

David Lung, Yuting Jia, Roland Blumer, Lukas Reissig, Lydia M Zopf, Patrick Heimel, Christoph Kraus, Andrea Moro, Matteo Fachino and Max Haberbusch

2026-11-10 1 min 289 words JOURNAL NEURAL ENGINEERING

Summary: Objective. Computational models of peripheral nerve stimulation—coupling finite-element bioelectric fields to biophysical axon models—are essential for designing electrodes and waveforms for neuromodulation, but established open tools are code-first, assume substantial modeling expertise, and often ...

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https://iopscience.iop.org/article/10.1088/1741-2552/aeba17

Recent advances and challenges in autonomous wireless implantable neurologgers for animal studies: neural recording, power supply, and data transmission technologies

👤 Ye Huang, Jiaxu Cao, Xuanqi Wang, Xiaoli You, Ruiyu Bai, Mingyang Li, Yuxiang Hong, Honglong Chang, Haitao Zhao, Minghao Wang, Hao Zhang, Jun Chen and Bowen Ji

📅 2026-11-10 📖 1 min 📝 136 words JOURNAL NEURAL ENGINEERING

Summary: Objective. Wireless implantable neurologgers are emerging as miniature autonomous systems for long-term, untethered neural recording in animal studies. However, their development remains constrained by critical engineering bottlenecks, including neural recording, power supply, and data transmission....

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https://iopscience.iop.org/article/10.1088/1741-2552/aeb97d

Human cortical high-gamma power scales with movement rate in healthy participants and stroke survivors

👤 Fanny Quandt

📅 2025-01-09 📖 1 min 📝 65 words OOSTENVELD ROBERT

Summary: Motor cortical high-gamma oscillations (60-90 Hz) occur at movement onset and are spatially focused over the contralateral primary motor cortex. Although high-gamma oscillations are widely recognized for their significance in human motor control, their precise function on a cortical level remains el...

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NIRS-BIDS: Brain Imaging Data Structure Extended to Near-Infrared Spectroscopy

👤 Luca Pollonini 📅 2025-01-27 📖 1 min 📝 70 words OOSTENVELD ROBERT

> **Summary:** Functional near-infrared spectroscopy (fNIRS) is an increasingly popular neuroimaging technique that measures cortical hemodynamic activity in a non-invasive and portable fashion. Although the fNIRS community has been successful in disseminating open-source processing tools and a standard file forma...

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Pseudonymisation of neuroimages and data protection: Increasing access to data while retaining scientific utility

👤 Lyuba Zehl 📅 2025-06-26 📖 1 min 📝 67 words OOSTENVELD ROBERT

> **Summary:** For a number of years, facial features removal techniques such as 'defacing', 'skull stripping' and 'face masking/blurring', were considered adequate privacy preserving tools to openly share brain images. Scientifically, these measures were already a compromise between data protection requirements a...

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Cycling on the Freeway: The perilous state of open-source neuroscience software

👤 Tim M Tierney 📅 2025-08-13 📖 1 min 📝 74 words OOSTENVELD ROBERT

Summary: Most scientists need software to perform their research (Barker et al., 2020;Carver et al., 2022;Hettrick, 2014;Hettrick et al., 2014;Switters & Osimo, 2019), and neuroscientists are no exception. Whether we work with reaction times, electrophysiological signals, or magnetic resonance imaging data, ...

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Optimal configuration of on-scalp OPMs with fixed channel counts

👤 Robert Oostenveld 📅 2025-08-13 📖 1 min 📝 69 words OOSTENVELD ROBERT

Summary: Recent technological developments have brought optically pumped magnetometers (OPMs) within reach of the larger neuroscientific community. The current state-of-the-art consists of whole-head systems that measure the magnetic field at >100 locations. OPM sensors can be constructed to measure the fiel...

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ARTEM-IS for ERP: Agreed Reporting Template for EEG Methodology-International Standard for Event-Related Potential Experiments

👤 Suzy J Styles 📅 2025-12-08 📖 1 min 📝 73 words OOSTENVELD ROBERT

Summary: The choices we make during the recording, preprocessing and analysis of event-related potentials (ERP) data can affect study outcomes. As such, it is critical that they are transparently reported to allow for reproducibility. Yet, systematic reviews of reporting practices in the field have shown tha...

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Eye-Tracking-BIDS: the Brain Imaging Data Structure extended to gaze position and pupil data

👤 Julia-Katharina Pfarr 📅 2026-02-12 📖 1 min 📝 68 words OOSTENVELD ROBERT

Summary: The Brain Imaging Data Structure (BIDS) is a widely adopted, community-driven standard to organize neuroimaging data and metadata. Although numerous extensions have been developed to incrementally extend coverage to new modalities and data types, an unambiguous, granular specification for eye-tracki...

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Facilitating unusual bodily experiences and out-of-body experiences across wakefulness and sleep: A high-density EEG and neurophenomenology study

👤 Robert Oostenveld 📅 2026-02-12 📖 1 min 📝 67 words OOSTENVELD ROBERT

Summary: Unusual bodily experiences (UBEs) are illusory bodily perceptions that are not coherent with typical wakeful experiences, including flying or floating sensations, body distortions, and out-of-body experiences. This study examines UBEs in a controlled sleep laboratory using meditation and light stimu...

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The Impact of Brain Tumors and Craniotomy Lesions on Scalp EEG

👤 Vitória Piai 📅 2026-02-17 📖 1 min 📝 65 words OOSTENVELD ROBERT

Summary: Electroencephalography (EEG) is widely used in both research and clinical settings, yet its accuracy can be significantly impacted by subject-specific anatomical anomalies such as brain lesions and skull defects. This study investigates the effects of glioma-related brain lesions and craniotomy-indu...

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EMG-BIDS: an extension to the Brain Imaging Data Structure for electromyography

👤 Arnaud Delorme 📅 2026-08-01 📖 1 min 📝 64 words OOSTENVELD ROBERT

Summary: Electromyography (EMG) is fundamental to clinical assessment, rehabilitation, neuromuscular research, and human-machine interfaces. Despite decades of use, no widely adopted standard exists for organizing and sharing EMG data, limiting reusability and large-scale data aggregation. We present EMG-BID...

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Role of smartphone-based digital adherence support for glaucoma medication therapy

👤 David J Mathew 📅 2026-11-07 📖 1 min 📝 57 words LOW VISION

Summary: Sustained adherence to intraocular pressure-lowering medications is essential for preventing progressive visual loss in patients with glaucoma on pharmacotherapy. However, adherence remains suboptimal across patient populations as the result of cognitive, behavioural, physical, and socioeconomic bar...

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Statin Use and Age-related Macular Degeneration: A Systematic Review and Meta-Analysis of Disease Outcomes

Ricardo Noguera Louzada
2026-11-07 1 min 36 words LOW VISION

Summary: CONCLUSION: The evidence was of very low certainty and insufficient to establish a protective or harmful effect of statins on AMD. These findings do not support prescribing statins specifically for AMD or discontinuing otherwise indicated therapy.

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A comparison of diagnostic accuracy between CNN and transformer AI models for periodontal bone loss detection: A systematic review and meta-analysis

Selviana Rizky Pramitha
2026-11-08 1 min 23 words LOW VISION

Summary: CONCLUSION: Current evidence demonstrates greater diagnostic accuracy of CNN-based models for periodontal bone loss detection, while comparative conclusions for transformer-based models remain premature.

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Effectiveness and Safety Outcomes After Switching from Reference Biologics to Biosimilars for Treatment of Non-Infectious Uveitis: A Systematic Review and Meta-Analysis

Rafael N Miranda 2026-11-08 1 min 34 words LOW VISION

Summary: CONCLUSIONS: Currently, the available evidence is of low quality and demonstrates insufficient support for a meaningful difference in terms of effectiveness and safety when switching patients with NIU to biosimilars from their reference biologics.

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Does eHealth literacy matter for health-promoting behaviors among blind and low-vision adults? A mixed methods study

Jessie Chin 2026-11-08 1 min 47 words LOW VISION

Summary: CONCLUSION: eHealth literacy alone does not ensure more health behavior engagement in BLV individuals. To bridge the gap, health systems must prioritize the development of accessible digital health resources, incorporate disability-specific adaptations in health education, and ensure equitable oppor...

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Construction of Highly Viable, Excitable, Aligned Human Induced Pluripotent Stem Cell-Derived Retinal Ganglion Cell Sheets on Poly-ε-caprolactone Nanofibers by Targeting Integrin β1-Mediated Adhesion Pathways

👤 Toru Nakazawa
📅 2026-11-09 📖 1 min 📝 63 words

LOW VISION

Summary: Transplantation of induced pluripotent stem cell-derived retinal ganglion cells (RGCs) is a promising strategy for optic nerve regeneration and vision restoration in glaucoma. However, injections of conventional single-cell suspensions have low engraftment rates, hindering the formation of the compl...

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High-Performance Self-Powered Infrared Photodetector via Semiconducting Graphene Nanoribbon/Germanium Vertical Heterojunctions with Robust Thermal Stability

👤 Aimin Song
📅 2026-11-09 📖 1 min 📝 54 words LOW VISION

Summary: Self-powered photodetectors stand as a cornerstone for next-generation low-power electronics, with promising applications in optical communications, environmental monitoring, night vision, and beyond. However, achieving high performance in these devices, particularly under elevated temperatures, rem...

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Prior-guided whole-slide survival prediction with soft histopathologic priors from vision-language foundation models

👤 Zhihui Feng
📅 2026-11-09 📖 1 min 📝 62 words

LOW VISION

Summary: Whole-slide image (WSI) survival prediction is complicated by heterogeneous tumor morphology, where prognostic evidence is distributed across tumor, stroma, immune infiltration, necrosis, and residual tissue regions and often appears as mixed patch-level tissue evidence. Existing models commonly agg...

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Reduced magnocellular-biased visual responses are associated with working memory and linguistic difficulties in Chinese developmental dyslexia: A hypothesized visual-cognitive cascade

👤 Tsz Wing Leung
📅 2026-11-09 📖 1 min 📝 66 words

LOW VISION

Summary: Developmental dyslexia (DD) affects 5-10% of children worldwide. Although phonological theories predominate, they may not fully explain reading difficulties in orthographically dense scripts such as Chinese. We investigated whether Chinese children with DD exhibit differences in early visual cortica...

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During natural vision, semantic novelty modulates fixation-related processing in primate cortex

👤 Lucas C Parra
📅 2026-11-09 📖 1 min 📝 74 words

LOW VISION

Summary: We sample visual scenes with short gaze fixations separated by saccades. While low-level integration is known, semantic integration of foveal vision across fixations remains unclear. We hypothesized that the brain responds to changes in semantic information from one fixation to the next, and therefo...

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A single session of high-definition transcranial direct current stimulation does not modulate effects of knee two-point discrimination training or sensorimotor function in healthy adults: Double-blind randomised controlled trial

👤 Paul W Hodges
📅 2026-07-30 📖 1 min 📝 53 words

TACTILE ACUITY

Summary: CONCLUSION: A single 20-minute session of 1 mA HD-tDCS, with or without brief TPD training, does not acutely modify knee somatosensory, sensorimotor, or functional performance in healthy adults. Existing evidence at the hand may not translate to the lower limb. Future work should investigate higher-...

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Variation in glabrous skin physiology across anatomical locations on the human hand

Martina C Meinke 2026-08-13 1 min 76 words TACTILE ACUITY

Summary: Skin physiology varies greatly between anatomical locations, with impact on tactile sensitivity. Research on the variations in morphology and physiology between different anatomical locations of glabrous skin is limited so far. This study investigates the morphology and physiology of glabrous skin o...

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Beyond Hamstring Shortness: Continuous Flexibility Is Associated With Balance and Stability but Not Sensory Measures-A Cross-Sectional Study

Senanur Akıncı 2026-08-14 1 min 66 words TACTILE ACUITY

Summary: CONCLUSIONS: Greater hamstring flexibility was associated with better dynamic balance performance and showed a weaker association with postural stability, whereas no robust associations were observed with sensory measures of postural control. These findings suggest that hamstring flexibility may be ...

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Resting-state EEG microstate dynamics reflect individual differences in tactile angle discriminability

Liu Yang 2026-08-19 1 min 63 words TACTILE ACUITY

Summary: Tactile spatial acuity, the ability to discriminate fine spatial details, is a fundamental sensory capacity modulated by both peripheral and central mechanisms. Here, we investigated whether resting-state EEG microstate dynamics are associated with individual differences in tactile spatial acuity. F...

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A grating orientation task with controlled dome elevation speed and contact force: stimulus conditions and day-to-day reliability

Hideaki Onishi 2026-09-25 1 min 66 words TACTILE ACUITY

Summary: The grating orientation discrimination task (GOT) is typically used to assess tactile spatial acuity. However, performance is influenced by stimulus speed, contact force between the stimulator and the skin, and the method of discrimination. This study aimed to explore suitable stimulus conditions fo...

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Epidermal microstructure and tactile sensitivity in the glabrous skin of hands and feet

👤 Hannes P Saal 📅 2026-09-25 📖 1 min 📝 69 words TACTILE ACUITY

Summary: Glabrous skin on the human palms and soles is specialized for mechanical interaction and is characterized by papillary ridges, an absence of hair follicles, and a thick stratum corneum. Although gross morphological differences, such as the increased thickness of the plantar stratum corneum, are well...

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Neurotrauma During Military Training: Results of the INVestigating traIning assoCiated blasT pAthology (INVICTA) Study

👤 Walter S Carr 📅 2026-10-01 📖 1 min 📝 67 words TACTILE ACUITY

Summary: Concern is growing that subconcussive blast exposure during military exercises such as heavy weapons training (HWT) may adversely affect brain function. HWT involves firing shoulder-mounted weapons that can expose the gunner, assistant gunner, and instructors (range safety officers [RSOs]) to blast ...

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Altered body representation in chronic non-specific low back pain: a testable framework linking body-schema-related disturbance, corticomotor alterations, and altered pain processing

👤 Zhifei Xie
📅 2026-10-24 📖 1 min 📝 66 words

TACTILE ACUITY

Summary: Low back pain is a leading contributor to disability worldwide; in chronic non-specific low back pain (CNSLBP), the processes sustaining pain and disability in the absence of identifiable structural pathology remain incompletely understood. Three partially overlapping research lines have documented ...

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Hyperexcitable primary somatosensory cortex in the MitoPark mouse model of Parkinson's disease

Alan J Emanuel 2026-10-25 1 min 67 words TACTILE ACUITY

Summary: The majority of Parkinson's disease patients suffer from decreased tactile acuity or neuropathic pain, both of which contribute to low quality of life and likely exacerbate motor dysfunction. How Parkinson's disease pathology relates to altered somatosensation remains unclear. One potential contribu...

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The Relationship Between Occlusal Tactile Acuity, Bite Force and Tooth Wear

Birgitta Häggman-Henrikson 2026-11-03 1 min 52 words TACTILE ACUITY

Summary: CONCLUSIONS: Bite force may be a clinically relevant factor associated with accumulated tooth wear, whereas OTA was not independently associated with tooth wear. Longitudinal studies are needed to determine whether bite force can predict tooth wear progression and whether more comprehensive assessme...

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Impact of Braille Booklet, Audio-Tactile Performance and Caregiver-supervised Training on Oral Hygiene Status, Knowledge, Attitude and Practices among Visually Challenged Children- A Randomized Clinical Trial

👤 Priyanga Chandrasekaran
📅 2026-09-22 📖 1 min 📝 52 words

BRAILLE

Summary: CONCLUSION: Both caregiver-supervised training and audio tactile performance using specialized model method of oral health education proved to be more effective than the braille booklet group on plaque control and improvement in the oral hygiene status, oral health knowledge, attitude and practice i...

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Design and functional verification of an automated Perkins Brailler accessory with audio feedback

👤 Ali Alhaj 📅 2026-09-23 📖 1 min 📝 67 words BRAILLE

Summary: CONCLUSIONS: The study did not include learners with visual impairment, Braille teachers, rehabilitation specialists, or other end users, and it did not collect quantitative measures of accuracy across the full character set, writing speed, embossing quality, latency, durability, safety, usability, ...

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Uncertainty-Aware structured decoding for Chinese Braille document transcription

👤 Haoran Sun 📅 2026-09-25 📖 1 min 📝 64 words BRAILLE

Summary: Automatic transcription of braille documents remains challenging in document digitization settings, particularly under real imaging degradations such as uneven embossing, illumination variation, paper texture, skew, blur, and ghosting, which make braille dot presence uncertain and often cause error ...

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Clinical Characteristics and Epidemiology of Isolated Dystonia: Results from a US Claims Analysis

James E Thornton 2026-10-08 1 min 54 words BRAILLE

Summary: CONCLUSIONS: This study defines an isolated dystonia cohort in the US based on real-world evidence and systematic exclusion of secondary causes of dystonia. Estimated prevalence is within range of prior epidemiological studies, and the cohort showed increased rates of comorbid psychiatric disorders....

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Effects of multisensory tactile-auditory integration teaching on systematic anatomy learning in visually impaired medical students

Fanzhen Kong 2026-10-08 1 min 50 words BRAILLE

Summary: CONCLUSION: The tactile-auditory integrated training system significantly improves objective summative knowledge acquisition and subjective learning experiences in visually impaired medical students. This approach provides a scalable, cost-constrained, and replicable pedagogical model for inclusive ...

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Physical and Environmental Accessibility of Dental Services for Persons with Disabilities in Saudi Primary Health Care: A Cross-Sectional Study

👤 Estie Kruger 📅 2026-10-16 📖 1 min 📝 66 words BRAILLE

Summary: Background/Objectives: Persons with disabilities (PWDs) may experience barriers when accessing oral health care, including those related to the physical and sensory environment of the dental-care setting. In this study we assessed the accessibility of primary health care centres (PHCs) providing den...

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Visible Light Driven Diselenide Chemistry for Simultaneous 3D Patterning and Mechanical Reinforcement

👤 Huaping Xu 📅 2026-10-16 📖 1 min 📝 63 words BRAILLE

Summary: Three-dimensional (3D) patterning on polymer surfaces is typically achieved by photolithography or related techniques, which often involve complex procedures, high-energy ultraviolet light, and risk of material damage. Here, we report a visible light-driven strategy that exploits the dual functional...

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An accessible pregnancy test for blind and low-vision users

Vanja Garaj 2026-10-16 1 min 63 words BRAILLE

Summary: CONCLUSIONS: The findings suggest that accessible pregnancy testing must prioritise not only non-visual result reading but also privacy, dignity, reproductive autonomy and independent health management. These results represent the preliminary evidence of acceptability rather than a conclusive test o...

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Genome-Informed Loop-Mediated Isothermal Amplification (LAMP) Assays for Detection and Differentiation of Ralstonia pseudosolanacearum Phylotypes I and III

Jarred Yasuhara-Bell 2026-10-27 1 min 76 words BRAILLE

Summary: The Ralstonia solanacearum species complex (RSSC) causes bacterial wilt in a broad range of economically important hosts. The RSSC has been reclassified into three species, of which R. solanacearum sequevars 1 and 2 in phylotype IIB are designated as Select Agents in the U.S. Many rapid tests for "R...

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Encoding tactile stimuli for Braille recognition with organoids

👤 Benjamin Ward-Cherrier 📅 2026-10-31 📖 1 min 📝 62 words BRAILLE

Summary: This study proposes a transferable encoding strategy that maps tactile sensor data to electrical stimulation patterns, enabling neural organoids to perform an open-loop artificial tactile Braille classification task. Human forebrain organoids cultured on a low-density microelectrode array (MEA) are ...

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Feasibility and tolerability of combining 6 Hz transcranial alternating current stimulation (tACS) at 3.2 mA and digital cognitive training in older adults with mild cognitive impairment

👤 Rogério Panizzutti 📅 2026-11-05 📖 1 min 📝 41 words TDCS TACS TRNS

Summary: CONCLUSIONS: Higher-intensity tACS plus DCT was feasible and well tolerated in older adults with MCI. Exploratory findings suggest that active stimulation may improve inhibitory control and facilitate transfer to selected cognitive domains. Confirmation in the ongoing randomized controlled trial is ...

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The Non-Invasive Brain Stimulation Side Effect Scale (NIBS-SES): development of a brief, multilingual questionnaire for monitoring adverse effects in transcranial magnetic, electrical and ultrasound stimulation

👤 Maximilian Kiebs
📅 2026-11-05 📖 1 min 📝 59 words

TDCS TACS TRNS

Summary: CONCLUSIONS: Our findings support the feasibility of repeated session-level assessment with the initial version, from which the final NIBS-SES was derived. This multilingual, open-access instrument comprises 10 within-session and 9 between-session items developed for monitoring across TMS, tES, and ...

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Concurrent prefrontal neurochemical effects of high-definition transcranial direct current stimulation: directional E-field associations with GABA+ and Glx

👤 Jianqi Li 📅 2026-11-05 📖 1 min 📝 58 words TDCS TACS TRNS

Summary: Transcranial direct current stimulation (tDCS) is a promising neuromodulatory technique for treating neuropsychiatric disorders. However, its neurochemical mechanisms remain poorly characterized, and clinical efficacy is limited by substantial inter-individual variability. This study used concurrent...

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Anodal tDCS alters perceived visual contrast in humans

👤 Sang Wook Hong 📅 2026-11-06 📖 1 min 📝 62 words TDCS TACS TRNS

Summary: Cortical gain modulation is thought to alter perceptual experience, most prominently demonstrated by the effects of spatial attention on contrast appearance. Whether externally induced changes in cortical excitability produce comparable perceptual consequences remains unclear. Here, we tested whethe...

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Evaluating OPTION: Benefits and Pitfalls of a Youth Trafficking Diversion Court in Florida

👤 Joan A Reid 📅 2026-11-06 📖 1 min 📝 67 words TDCS TACS TRNS

Summary: Trafficking diversion courts (TDCs) provide an alternative approach to supporting trafficked youth by promoting recovery, preventing revictimization, and diverting youth from the justice system. The current study examines the benefits and challenges of the OPTION (Offering Potentially Trafficked Ind...

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Amelioration of diabetic autonomic neuropathy symptoms following a course of transcranial direct current stimulation

👤 Ahmad Alipour 📅 2026-11-06 📖 1 min 📝 62 words TDCS TACS TRNS

Summary: Cardiovascular autonomic neuropathy (CAN) is a severe and potentially life-threatening complication of diabetes. This randomized, sham-controlled, three-arm trial evaluated the efficacy of anodal transcranial direct current stimulation (tDCS) targeting the left dorsolateral prefrontal cortex for imp...

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From Single Cells to the Circulating Tumor Microenvironment: Label-Free Size-Selection and a Morphological Taxonomy for Liquid Biopsy in Cytopathology

Satoru Kobayashi 2026-11-09 1 min 25 words TDCS TACS TRNS

Summary: CONCLUSION: Size-selection converts a liquid biopsy into a cytology slide, placing interpretation of the circulating tumor ecosystem within the existing competence of the diagnostic laboratory.

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Phase-Dependent Effects of Cerebellar Transcranial Alternating Current Stimulation on Lateral Vestibulospinal Tract Excitability

Tsubasa Mitsutake 2026-11-09 1 min 36 words TDCS TACS TRNS

Summary: CONCLUSIONS: LVST excitability differed according to the phase of the ongoing cerebellar-targeted tACS waveform. These findings provide electrophysiologic evidence of an online phase-dependent effect on vestibulospinal excitability and a physiologic basis for future phase-specific neuromodulation st...

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Effects of Transcranial Direct Current Stimulation on Core Symptoms and Sleep Disturbances in Children With Autism Spectrum Disorder: A Meta-Analysis of Randomized Controlled Trials

👤 Meimei Yang
📅 2026-11-09 📖 1 min 📝 44 words

TDCS TACS TRNS

Summary: CONCLUSION: tDCS, most commonly delivered at 1 mA over the DLPFC, was associated with improvements in some ASD-related behavioral outcomes; preliminary evidence also suggested a potential benefit for sleep disturbances, although this finding was based on only two low-certainty studies and warrants c...

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No effects of single-session concurrent tDCS on explicit motor sequence learning in healthy adults with conventional 2 mA and tangential 6 mA currents

👤 Lucas C Parra
📅 2026-11-09 📖 1 min 📝 45 words

TDCS TACS TRNS

Summary: CONCLUSION: These findings, together with a survey of the existing literature, suggest that the effects of single-session concurrent tDCS on explicit motor sequence learning are not robust across studies. Our results underscore the necessity of rigorous methodological controls to prevent false disco...

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Penetrating needling from Qiuxu (GB40) to Zhaohai (KI6) versus separate needling for post-stroke lower-limb motor dysfunction: an assessor-blinded randomized controlled trial protocol

Chunxiao Wan
2026-11-05 1 min 63 words FNIRS

Summary: Post-stroke lower-limb motor dysfunction limits mobility, balance, and independence and remains a major target of neurorehabilitation. Acupuncture is widely used as an adjunctive intervention after stroke, but direct randomized evidence comparing penetrating needling with conventional separate needl...

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Advancing Repetitive Transcranial Magnetic Stimulation for Depression: Novel Protocols, Precision Targeting, and Future Directions

👤 Youguo Tan
📅 2026-11-05 📖 1 min 📝 57 words FNIRS

Summary: Major depressive disorder (MDD) remains a significant global health challenge, with substantial limitations inherent in current pharmacotherapeutic and psychotherapeutic approaches. Repetitive transcranial magnetic stimulation (rTMS) has emerged as a safe, effective, and non-invasive neuromodulation...

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Transforming neurophysiological data into measurable stress metric using machine learning-based stress quantification framework for smart factory workers

👤 Azfar Khalid 📅 2026-11-05 📖 1 min 📝 67 words FNIRS

Summary: INTRODUCTION: Mental stress is a common issue in demanding occupational setups, such as smart industrial settings, particularly when working with robots, being one of the primary reasons for decreased performance and productivity. Estimating and evaluating stress are critical for worker safety, perf...

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Prefrontal Activation and Cognitive Heterogeneity in Pediatric Sleep-Disordered Breathing

👤 Amal Isaiah 📅 2026-11-05 📖 1 min 📝 39 words FNIRS

Summary: CONCLUSIONS AND RELEVANCE: In this cross-sectional study, task-related PFC activation was associated with SDB severity and moderated the severity-cognition association. fNIRS may help identify children whose cognitive profiles deviate from conventional severity metrics, thereby supporting individual...

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Mirror Neural System in Hands Motor Imagery with Pictures and Videos Guidance: An EEG-fNIRS Study

👤 Mohamad Sawan 📅 2026-11-05 📖 1 min 📝 64 words FNIRS

Summary: Brain-computer interfaces (BCIs) achieved by motor imagery (MI) have been widely accepted as an effective tool in rehabilitative applications. Functional near-infrared spectroscopy (fNIRS) and electroencephalography (EEG) are two common implementation forms of BCIs, capable of monitoring neuronal po...

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Quality-aware mask-propagation graph attention for fNIRS-based bipolar disorder classification without channel interpolation

👤 Yajing Meng 📅 2026-11-05 📖 1 min 📝 58 words FNIRS

Summary: Bipolar disorder (BD) remains difficult to characterise objectively, motivating a neural marker to complement clinical assessment. Functional near-infrared spectroscopy (fNIRS) offers portable cortical haemodynamic measurement, but channel availability can vary across recordings because of bad chann...

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Brain-Behavior Coupling During Cognitive-Postural Dual-Tasking in Older Adults With Mild Cognitive Impairment: An fNIRS Study

👤 Chuan Guo 📅 2026-11-06 📖 1 min 📝 56 words FNIRS

Summary: CONCLUSIONS: Older adults with MCI exhibit altered brain-behavior coupling during cognitive-postural dual-task conditions involving sensory and postural constraints, characterized by altered relationships between prefrontal-sensorimotor cortical activation and behavioral performance. These findings ...

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Item-Specific Proportion Congruency Uncovers Different Forms of Attentional Control: An event-related fNIRS Study

👤 Murat Perit Çakır 📅 2026-11-07 📖 1 min 📝 70 words FNIRS

Summary: CONCLUSION: While behavioral results mostly support contingency learning, fNIRS data revealed ISPC-related activation in prefrontal channels with sensitivity to BA 46 (lateral, channel 1) and the BA 10/BA 11 overlap zone (medial, channels 10-12), with channel 11 emerging as a convergence point for b...

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Does the emotion regulation of self-directed humor engage the prefrontal cortex? An fNIRS study

Zhonglu Zhang 2026-11-07 1 min 68 words FNIRS

Summary: Previous research has revealed that self-directed humor (self-enhancing and self-defeating) can help regulate negative emotions and that the prefrontal cortex (PFC) plays a critical role in emotion regulation. However, it remains unknown whether emotion regulation of self-directed humor is associate...

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Effects of mini-maps with different reference frames on polarity-free 3D spatial navigation: a behavioural and fNIRS study

Xiaozhou Zhou 2026-11-07 1 min 62 words FNIRS

Summary: Manned spaceflight and deep-sea exploration extend human activities to extreme environments lacking gravitational and visual polarity cues, where spatial disorientation frequently occurs. Mini-map studies largely focus on terrestrial environments, while reference frame design for polarity-free 3D sp...

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Mirror Neural System in Hands Motor Imagery with Pictures and Videos Guidance: An EEG-fNIRS Study

Mohamad Sawan 2026-11-05 1 min 64 words BRAIN COMPUTER INTERFACE

Summary: Brain-computer interfaces (BCIs) achieved by motor imagery (MI) have been widely accepted as an effective tool in rehabilitative applications. Functional near-infrared spectroscopy (fNIRS) and electroencephalography (EEG) are two common implementation forms of BCIs, capable of monitoring neuronal po...

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Changes to task relevance in redundant motor environments distinguish structure learning from adaptation

Amy L Orsborn 2026-11-06 1 min 68 words BRAIN COMPUTER INTERFACE

Summary: Motor actions often exceed the dimensionality of task feedback, which creates control redundancy that requires learners to identify which movement dimensions influence task outcomes. The computations that shape learning in redundant tasks, especially when task-relevant spaces change, remain unclear....

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CEP-MI: a novel cue-execution dual-stage motor imagery paradigm with PGCANet decoding architecture

Caili Gong 2026-11-06 1 min 34 words BRAIN COMPUTER INTERFACE

Summary: CONCLUSION: Targeting the above-mentioned research gaps, the CEP-MI paradigm provides valuable insights into the imagery-execution transition mechanism. PGCANet enhances MI decoding accuracy via multi-branch collaboration, laying a foundation for MI-BCI's clinical application in neurorehabilitation.

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Overcoming subject variability in P300 decoding based on quantification by deep learning

Fred R Beyette 2026-11-06 1 min 60 words BRAIN COMPUTER INTERFACE

Summary: Binary P300 decoding is a cornerstone of P300-based brain-computer interfaces, enabling applications such as assistive communication, neurorehabilitation, and cognitive monitoring. However, substantial subject variability in EEG patterns poses a major challenge to the development of generalized P300...

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A ciliated epithelial cell line as a model for respiratory syncytial virus research

👤 Marie-Anne Rameix-Welti 📅 2026-11-06 📖 1 min 📝 74 words BRAIN COMPUTER INTERFACE

Summary: Respiratory syncytial virus (RSV) is the leading cause of severe respiratory infection in infants and an increasingly recognized pathogen in the elderly. Conventional in vitro culture models for RSV rely on poorly differentiated cell lines that lack key airway epithelial features such as polarizatio...

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Neural interface instructs layer 4-appropriate circuit reconstruction with engrafted human brain organoids and improves visually guided behavior

👤 Xiao-Hong Li 📅 2026-11-06 📖 1 min 📝 65 words BRAIN COMPUTER INTERFACE

Summary: Repairing precise neural microcircuitry after cortical injury remains a pivotal challenge. Human brain organoids (hBOs) are a promising cell source, but disorganized integration limits their therapeutic potential. Here, we demonstrated that targeted neuromodulation, steered by endogenous narrowband ...

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Self-Guided Beta-Responsive Visual Neurofeedback in a Public Exhibition: Feasibility and Usability Randomized Controlled rial

👤 Joseph Ollier
📅 2026-11-07 📖 1 min 📝 57 words

BRAIN COMPUTER INTERFACE

Summary: CONCLUSIONS: This trial evaluated an autonomous consumer-EEG neurofeedback system integrating self-fitting, signal-quality checking, baseline calibration, adaptive feedback, randomization, and self-report without operator support. Many first-time users completed the autonomous workflow with acceptab...

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Auditory rehabilitation in mixed hearing loss due to chronic otitis media sequelae: passive versus active transcutaneous bone conduction implant

Esther Sofía Angulo Martínez 2026-11-08 1 min 47 words BRAIN COMPUTER INTERFACE

Summary: CONCLUSIONS: The findings of the present study suggest that AT-BCI provides a greater and clinically meaningful auditory gain for HF-PTA in patients with MHL secondary to COM. In ATBCI group a stable audiological performance was achieved as early as the 3-month follow-up and is maintained over time.

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Selective dissociation of coordination bonds enables rapid rigidity switching of hydrogel microfibre bioelectrodes

Lidong Zhang 2026-11-09 1 min 64 words BRAIN COMPUTER INTERFACE

Summary: Bioelectrodes with tunable rigidity deliver excellent mechanical compliance and can effectively alleviate tissue damage. Herein, we report hydrogel microfibre electrodes (diameter: 50-200 μm) fabricated by cross-linking sodium alginate (SA) microfibres with Ca^(2+)/Al^(3+) dual ions. The resultant S...

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Constructing Neurorights in the Era of Clinical Brain-Computer Interfaces : China's Emerging Governance Pathway

Yuan Lin 2026-11-09 1 min 67 words BRAIN COMPUTER INTERFACE

Summary: In March 2026, China approved the world's first implantable brain-computer interface (BCI) medical device for clinical use outside clinical trials, a milestone that raises urgent questions about how legal and ethical systems should protect the brain and mind. This article examines China's emerging n...

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Implicit learning of melodic structure: A role for pitch?

2024-01-22 1 min 180 words PSYCHOMUSICOLOGY

Summary: Growing evidence suggests that pitch influences musical processing, with melodic processing being enhanced in higher pitch ranges (e.g., Fujioka et al., 2005) and rhythmic processing being enhanced in lower pitches, and these effects may have a basis in elementary properties of the auditory system (...

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To those who feel unloved, misunderstood, or afraid

📅 2026-09-22 📖 3 min 📝 687 words FMHY

> **Summary:** I understand the world can feel devoid of love, but I swear, there is a spark within humanity that has the capability to care deeply. To love unconditionally. It doesn't always shine through us all, but it exists, it's real, it's tangible, and no one is exempt from it. We often see it shin...

🔗 **Read full article:**
https://fmhy.net/posts/love

Comparing two Python game development libraries: Pygame vs Arcade

👤 /u/Python_Data_Science 📅 2026-11-10 📖 1 min 📝 86 words REDDIT PYTHON

> **Summary:** <!-- SC_OFF --> I built the exact same simple mini-game using both Pygame and Arcade to see how they compare for beginners. My takeaway is that while Pygame's loop structure is great for teaching absolute programming fundamentals, Arcade is vastly superior for actual game developme...

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https://www.reddit.com/r/Python/comments/1xsho6k/comparing_two_python_game_development_libraries/

Your TV can see through walls

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Amazon saves millions by replacing Sonnet 4.6 with Strands Decider

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ClickHouse Acquires Xata

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Fireworks AI security breach impacts Notion and their customers

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M John Harrison becomes first science fiction author to win the Booker prize

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Fossil CO2 emissions decrease by 0.4% in 2026

👤 giuliomagnifico 📅 2026-11-10 📖 1 min 📝 13 words HACKER NEWS

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Queens Public library under ransomware attack e-borrowing suspended

👤 rustyhancock 📅 2026-11-10 📖 1 min 📝 13 words HACKER NEWS

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AI usage generates a fragile codebase

👤 plun9 📅 2026-11-10 📖 1 min 📝 13 words HACKER NEWS

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Meet the Sad Wives of AI

👤 rzk 📅 2026-11-10 📖 1 min 📝 13 words HACKER NEWS

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What Constant Surveillance Is Doing to Children

👤 FinnLobsien 📅 2026-11-10 📖 1 min 📝 13 words HACKER NEWS

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Advancing Embodied Intelligence in Robotic-Assisted Endovascular Procedures: A Systematic Review of AI Solutions

📅 2025-12-16 📖 1 min 📝 191 words REVIEWS BIOMEDICAL ENGINEERING

Summary: Endovascular procedures have revolutionized vascular disease treatment, yet their manual execution is challenged by the demands for high precision, operator fatigue, and radiation exposure. Robotic systems have emerged as transformative solutions to mitigate these inherent limitations. A crucial mom...

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http://ieeexplore.ieee.org/document/11301748

Charitable giving with earned and windfall money: behavioral and ERP responses to donation-risk information

📖 1 min 📝 17 words BRAIN RESEARCH

Summary: Publication date: 15 January 2027 Source: Brain Research, Volume 1895 Author(s): Jing Wang, Xiaoya Li, Siqi Jian, Peng Li

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Transcutaneous auricular vagus nerve stimulation attenuates post-stroke dysphagia by modulating microglial polarization via the endothelial VWF/PI3K-Akt axis

📖 1 min 📝 17 words BRAIN RESEARCH

> **Summary:** Publication date: 15 January 2027 Source: Brain Research, Volume 1895 Author(s): Fan Yu, Yi Feng, Yun Miao, Guangxu Xu

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TDP-43 dysfunction destabilizes glial homeostatic identity in ALS/FTD

📖 1 min 📝 14 words BRAIN RESEARCH

> **Summary:** Publication date: 15 January 2027 Source: Brain Research, Volume 1895 Author(s): Anna E Dale, Rita Sattler

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A caudate-centered network for alexithymia derived from patients with brain tumors

👤 McMahon, J. T., Healey, M. R., Sanchez-Gama, Y., Campanella, F., Skrap, M., Ferguson, M. A.

📅 2026-11-09 📖 1 min 📝 191 words BIORXIV NEUROSCIENCE

Summary: Alexithymia is a trait characterized by impaired emotional awareness. Previous accounts examining the neurobiological underpinnings of this trait have centered on a variety of structures, and alexithymia-causing lesions occur in disparate locations. We use lesion network mapping to derive a neural c...

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Linking Covariates to State Durations in Dynamic Brain Networks via Hidden Semi-Markov Models

👤 Shappell, H., Lindquist, M.

📅 2026-11-09 📖 1 min 📝 212 words BIORXIV NEUROSCIENCE

Summary: Hidden semi-Markov models (HSMMs) provide a flexible framework for studying dynamic functional connectivity because they represent brain activity as transitions between latent network states with explicit dwell time distributions. Although dwell times are often associated with behavioral and demogra...

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Electrophysiological Dynamics of Default Network Suppression

👤 Sun, W., Libowitz, M. R., Cowan, R. L., Tobel, T., Du, J., Feldman, D. A., Davis, T. S., King, J. B., Rahimpour, S., Smith, E. H., Buckner, R. L., Shofty, B.

📅 2026-11-09 📖 1 min 📝 206 words BIORXIV NEUROSCIENCE

Summary: Distributed brain networks compete with one another in the service of information processing. One dramatic example is antagonism between networks that support internally directed cognition, often collectively referred to as the default network, and networks that orient to external events and drive a...

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Sensory experience-induced plasticity of somatostatin interneurons occurs through two-pore potassium (K2P) channels

Bohannon, B. M., Alonso, N. L., Walker, M. G., Wang, M., Kaimaktchiev, K., Yang, C., Feldman, D.

2026-11-09 1 min 150 words BIORXIV NEUROSCIENCE

Summary: GABAergic interneurons (INs) are powerful modulators of plasticity and learning in cortical networks, but plasticity of INs themselves is not well understood. We studied experience-dependent plasticity in somatostatin (SST) INs, which critically regulate dendritic integration, sensory tuning, and pl...

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BVLab-analyzer: a unified visual toolbox for terabyte-scale whole-brain cell detection, neuron reconstruction, and vascular network extraction

Tingwei Quan

2026-10-07 1 min 139 words FRONTIERS NEUROINFORMATICS

Summary: BVLab-Analyzer is a standalone graphical software tool for end-to-end analysis of terabyte-scale whole-brain microscopy images. Operating through a zero-code interface, it handles data import (TIFF, BioImageVision, OME-Zarr), deep learning-based semantic segmentation, 3D skeleton reconstruction, and...

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Natural language processing for suicidal behavior detection in short Spanish clinical notes from first-contact interviews

👤 Ansel Y. Rodríguez-González
📅 2026-10-07 📖 1 min 📝 267 words

FRONTIERS NEUROINFORMATICS

Summary: IntroductionSuicide is a multifactorial phenomenon influenced by social, cultural, biological, and psychological factors, whose early detection remains challenging. According to the World Health Organization, suicide is the third leading cause of death among individuals aged 15–29 years, representin...

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Transformers for fMRI analysis: from spatiotemporal modeling to brain foundation models

👤 Bocheng Wang
📅 2026-10-07 📖 1 min 📝 217 words

FRONTIERS NEUROINFORMATICS

Summary: Functional magnetic resonance imaging (fMRI) serves as a pivotal tool for investigating the functional organization and dynamic connectivity of the human brain, indirectly reflecting neural activity through blood oxygen level-dependent (BOLD) signals. However, the high-dimensional spatiotemporal str...

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Multi-receptor whole-brain modeling reveals affinity-driven chemoarchitectural mechanism of virtual atomoxetine action

👤 Gustavo Deco 📅 2026-11-04 📖 1 min 📝 219 words

FRONTIERS COMPUTATIONAL NEUROSCIENCE

Summary: Linking molecular mechanisms with large-scale brain dynamics remains a fundamental challenge in computational psychiatry. Receptor-informed whole-brain models have provided mechanistic insights into pharmacological interventions, yet they have largely relied on single-receptor topographies that cann...

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https://www.frontiersin.org/articles/10.3389/fncom.2026.1854866

Editorial: Exploring the role of criticality in neural systems

👤 Tiago L. Ribeiro 📅 2026-11-04 📖 1 min 📝 0 words

FRONTIERS COMPUTATIONAL NEUROSCIENCE

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Music with monaural beats for anxiety reduction and mood enhancement in individuals with moderate trait-anxiety

Gianluca Micchi 2026-11-04 1 min 279 words FRONTIERS HUMAN NEUROSCIENCE

Summary: Monaural beats embedded in music are a popular anxiety-relief intervention, yet implementation parameters and efficacy remain poorly understood. Across two studies of online participants with moderate trait anxiety, we tested whether theta-frequency (4–7 Hz) monaural beats in music produce anxiolyti...

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The association between dietary protein intake and mental disorders in older adults: a scoping review

Aliza Haslinda Hamirudin 2026-11-04 1 min 280 words

FRONTIERS HUMAN NEUROSCIENCE

Summary: BackgroundMental disorders are increasingly prevalent among older adults, significantly affecting their quality of life and overall wellbeing. Recent evidence suggests that dietary factors, particularly protein intake, may play a role in mental health outcomes. However, the association between dieta...

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https://www.frontiersin.org/articles/10.3389/fnhum.2026.1949231

Editorial: Women in speech and language 2025

Anastasia M. Raymer 2026-11-04 1 min 0 words FRONTIERS HUMAN NEUROSCIENCE

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https://www.frontiersin.org/articles/10.3389/fnhum.2026.2009604

Brain-inspired predictive visual attention for spatiotemporal semantic understanding of complex film and television videos

Peng Han 2026-10-22 1 min 275 words FRONTIERS NEUROROBOTICS

Summary: Due to the unique characteristics of film and television, such as montage editing and narrative structure jumps, semantically salient targets frequently experience temporal feature drift, making it difficult for existing models to maintain stable semantic understanding output when keyframes are lost...

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MS-3D-CNN with quantum-inspired chaotic hybrid adaptive wolf-particle swarm optimization for tactical pattern classification in basketball from player tracking data

Haoyan Liu 2026-10-30 1 min 229 words FRONTIERS NEUROROBOTICS

Summary: IntroductionAI sport analysis methods have been considerably improved by developments in knowledge of tactically information and its codification. Nevertheless, methods for capturing multi-levels spatio-temporal information for complex team-sport behaviors are still limited. In opposed to motion rec...

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MA-Pose and DMMS-GCN: skeleton-based martial-arts category recognition with edge-device deployment

Shilin Guo 2026-11-04 1 min 238 words FRONTIERS NEUROROBOTICS

Summary: With the development of visual sensor technology and human pose estimation algorithms, skeleton-based action recognition has received extensive attention in sports analysis and training systems. The recognition of martial-arts actions is difficult because different styles can share similar poses whi...

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https://www.frontiersin.org/articles/10.3389/fnbot.2026.1967140

Deep reinforcement learning–driven model-based adaptive PID strategy for six-wheeled rocker–bogie mobile platform

Hongxue Feng 2026-11-04 1 min 202 words FRONTIERS NEUROROBOTICS

Summary: IntroductionSix-wheel rocker–bogie rovers encounter nonlinear dynamics, variable traction, and wheel slip that can degrade the performance of fixed-gain proportional–integral–derivative (PID) controllers. This study develops a hybrid deep reinforcement learning–PID framework to improve steering and ...

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Reliability-aware multimodal sensor fusion for human motion monitoring with safety-constrained neurorobotic feedback in sports training

Jiaoyu Wang 2026-11-04 1 min 247 words FRONTIERS NEUROROBOTICS

Summary: IntroductionEmbodied intelligent systems require reliable perception and safety-constrained feedback mechanisms for human motion understanding under heterogeneous sensing conditions. In this study, volleyball training is considered as a representative dynamic human-motion scenario rather than the so...

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https://www.frontiersin.org/articles/10.3389/fnbot.2026.1973789

The butyrate-m6A-BDNF axis: a hypothesis of gut-brain signal transduction in PSD

👤 Dajun Gu 📅 2026-11-04 📖 1 min 📝 151 words FRONTIERS NEUROSCIENCE

Summary: A reduction in brain-derived neurotrophic factor (BDNF) protein levels has been reported in post-stroke depression (PSD). However, whether transcript abundance and translation efficiency change in parallel or become uncoupled in PSD remains to be determined. In addition to classical transcriptional ...

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https://www.frontiersin.org/articles/10.3389/fnins.2026.1869522

TEMPO-taVNS: a multimodal platform optimizing thermal, electrical, and mechanical auricular stimulation

👤 Matthew P. Ward 📅 2026-11-04 📖 1 min 📝 176 words FRONTIERS NEUROSCIENCE

Summary: Transcutaneous auricular vagus nerve stimulation (taVNS) is typically delivered electrically, while alternative stimulation modalities and their potential for personalization remain underexplored. We present TEMPO-taVNS (Thermal, Electrical, and Mechanical Parameter Optimization for taVNS), a human-...

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Baseline cognition accounts for most predictive information on cognitive recovery during rehabilitation after carbon monoxide poisoning

Peihui Liu 2026-11-04 1 min 271 words FRONTIERS NEUROSCIENCE

Summary: BackgroundCognitive change during rehabilitation after carbon monoxide (CO) poisoning may differ from prognosis at acute presentation. We examined whether information recorded in a rehabilitation pathway added predictive information beyond baseline cognition.MethodsThis single-center retrospective c...

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A purinergic receptor dominance model for microglial state transitions: balancing perturbation and restraint during neuroinflammation

Minke Tang 2026-11-04 1 min 183 words FRONTIERS NEUROSCIENCE

Summary: Microglia show substantial functional heterogeneity during neuroinflammatory responses, yet the upstream mechanisms through which they interpret extracellular perturbation-associated signals and transition between functional states remain incompletely understood. Purinergic signaling is well positio...

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Retraction: Hypothermia impairs glymphatic drainage in traumatic brain injury as assessed by dynamic contrast-enhanced MRI with intrathecal contrast

📅 2026-11-04 📖 1 min 📝 0 words FRONTIERS NEUROSCIENCE

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https://www.frontiersin.org/articles/10.3389/fnins.2026.2046766

Delineating neural contributions to electroencephalogram-based speech decoding

👤 Motoshige Sato, Eri Hatakeyama, Masakazu Inoue, Yasuo Kabe, Sensho Nobe, Akito Yoshida, Atsushi Yamamoto, Yuya Kita, Mayumi Shimizu, Kenichi Tomeoka, Michael X Cohen and Shuntaro Sasai

📅 2026-10-30 📖 1 min 📝 249 words JOURNAL NEURAL ENGINEERING

Summary: Objective. Speech brain–computer interfaces (BCIs) have emerged as a pivotal technology in facilitating communication for individuals with speech impairments. Utilizing electroencephalography (EEG) for noninvasive speech BCIs offers an accessible and affordable solution, potentially benefiting a bro...

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Diff-ADN: a diffusion-guided artifact denoising network with deterministic residual refinement for EEG

Mudasir Jamil, Muhammad Zulkifal Aziz, Binwen Huang and Xiaojun Yu

2026-10-30 1 min

241 words

JOURNAL NEURAL ENGINEERING

Summary: Objective. Physiological artifacts degrade electroencephalographic (EEG) recordings and can affect downstream brain–computer interface (BCI) analysis. This study develops a compact single-channel framework for ocular, muscular, cardiac, and mixed-artifact removal. Approach. The Diffusion-Guided Arti...

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NeuroTest: a benchtop testbed for evaluating sensing-capable electrophysiology and neurostimulation systems

Morrow JK, H Cho, M Benjaber, Denison TJ, Widge AS and Herron JA

2026-11-04 1 min

247 words

JOURNAL NEURAL ENGINEERING

Summary: Objective. The development of increasingly complex implantable neuromodulation devices requires extensive benchtop validation and testing prior to in vivo implementation. Our goal is to enable this in vitro testing by creating a benchtop system that is accessible, easy-to-use, and functions reliably...

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Towards semantic brain-computer interfacing: the graded inventory of semantic triggers (GIST) paradigm

👤 Joshua James Podmore, Moein Radman, Riccardo Poli, Silke Paulmann and Ian Daly

📅 2026-11-04

📖 1 min 📝 299 words

JOURNAL NEURAL ENGINEERING

Summary: Objective. Semantic brain-computer interfaces (sBCIs) aim to decode conceptual information directly from neural activity, enabling more efficient and natural communication for individuals with severe motor impairments. However, existing approaches are often confounded by stimulus-driven perceptual p...

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Human cortical high-gamma power scales with movement rate in healthy participants and stroke survivors

👤 Fanny Quandt

📅 2025-01-09 📖 1 min 📝 65 words

OOSTENVELD ROBERT

Summary: Motor cortical high-gamma oscillations (60-90 Hz) occur at movement onset and are spatially focused over the contralateral primary motor cortex. Although high-gamma oscillations are widely recognized for their significance in human motor control, their precise function on a cortical level remains el...

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NIRS-BIDS: Brain Imaging Data Structure Extended to Near-Infrared Spectroscopy

👤 Luca Pollonini 📅 2025-01-27 📖 1 min 📝 70 words OOSTENVELD ROBERT

> **Summary:** Functional near-infrared spectroscopy (fNIRS) is an increasingly popular neuroimaging technique that measures cortical hemodynamic activity in a non-invasive and portable fashion. Although the fNIRS community has been successful in disseminating open-source processing tools and a standard file forma...

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Pseudonymisation of neuroimages and data protection: Increasing access to data while retaining scientific utility

👤 Lyuba Zehl 📅 2025-06-26 📖 1 min 📝 67 words OOSTENVELD ROBERT

> **Summary:** For a number of years, facial features removal techniques such as 'defacing', 'skull stripping' and 'face masking/blurring', were considered adequate privacy preserving tools to openly share brain images. Scientifically, these measures were already a compromise between data protection requirements a...

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Cycling on the Freeway: The perilous state of open-source neuroscience software

👤 Tim M Tierney 📅 2025-08-13 📖 1 min 📝 74 words OOSTENVELD ROBERT

Summary: Most scientists need software to perform their research (Barker et al., 2020;Carver et al., 2022;Hettrick, 2014;Hettrick et al., 2014;Switters & Osimo, 2019), and neuroscientists are no exception. Whether we work with reaction times, electrophysiological signals, or magnetic resonance imaging data, ...

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Optimal configuration of on-scalp OPMs with fixed channel counts

👤 Robert Oostenveld 📅 2025-08-13 📖 1 min 📝 69 words OOSTENVELD ROBERT

Summary: Recent technological developments have brought optically pumped magnetometers (OPMs) within reach of the larger neuroscientific community. The current state-of-the-art consists of whole-head systems that measure the magnetic field at >100 locations. OPM sensors can be constructed to measure the fiel...

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ARTEM-IS for ERP: Agreed Reporting Template for EEG Methodology-International Standard for Event-Related Potential Experiments

👤 Suzy J Styles 📅 2025-12-08 📖 1 min 📝 73 words OOSTENVELD ROBERT

Summary: The choices we make during the recording, preprocessing and analysis of event-related potentials (ERP) data can affect study outcomes. As such, it is critical that they are transparently reported to allow for reproducibility. Yet, systematic reviews of reporting practices in the field have shown tha...

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Eye-Tracking-BIDS: the Brain Imaging Data Structure extended to gaze position and pupil data

👤 Julia-Katharina Pfarr 📅 2026-02-12 📖 1 min 📝 68 words OOSTENVELD ROBERT

Summary: The Brain Imaging Data Structure (BIDS) is a widely adopted, community-driven standard to organize neuroimaging data and metadata. Although numerous extensions have been developed to incrementally extend coverage to new modalities and data types, an unambiguous, granular specification for eye-tracki...

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Facilitating unusual bodily experiences and out-of-body experiences across wakefulness and sleep: A high-density EEG and neurophenomenology study

👤 Robert Oostenveld 📅 2026-02-12 📖 1 min 📝 67 words OOSTENVELD ROBERT

Summary: Unusual bodily experiences (UBEs) are illusory bodily perceptions that are not coherent with typical wakeful experiences, including flying or floating sensations, body distortions, and out-of-body experiences. This study examines UBEs in a controlled sleep laboratory using meditation and light stimu...

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The Impact of Brain Tumors and Craniotomy Lesions on Scalp EEG

👤 Vitória Piai 📅 2026-02-17 📖 1 min 📝 65 words OOSTENVELD ROBERT

Summary: Electroencephalography (EEG) is widely used in both research and clinical settings, yet its accuracy can be significantly impacted by subject-specific anatomical anomalies such as brain lesions and skull defects. This study investigates the effects of glioma-related brain lesions and craniotomy-indu...

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EMG-BIDS: an extension to the Brain Imaging Data Structure for electromyography

👤 Arnaud Delorme 📅 2026-08-01 📖 1 min 📝 64 words OOSTENVELD ROBERT

Summary: Electromyography (EMG) is fundamental to clinical assessment, rehabilitation, neuromuscular research, and human-machine interfaces. Despite decades of use, no widely adopted standard exists for organizing and sharing EMG data, limiting reusability and large-scale data aggregation. We present EMG-BID...

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Efficacy and safety of repeated low-level red light therapy in myopia control: an updated and comprehensive systematic review and meta-analysis

👤 Junguo Duan 📅 2026-11-07 📖 1 min 📝 72 words LOW VISION

Summary: CONCLUSIONS: This meta analysis suggests that, in the short term (within 12 months of follow up), RLRL may be associated with slower axial length elongation, less myopic refractive progression, and increased choroidal thickness in children with myopia. However, these findings should be interpreted w...

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Insecticide resistance profile of the Anopheles gambiae s.l. mosquitoes in Nigeria: a 5-year malaria vector surveillance report

Awolola Samson 2026-11-07 1 min 60 words LOW VISION

Summary: CONCLUSION: The predominance of Anopheles coluzzii in tested collections, coupled with pervasive pyrethroid resistance and widespread oxidase-based metabolic resistant mechanisms, poses significant challenges to malaria vector control efforts in Nigeria. These findings underscore the urgent need to ...

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Association between physical fitness performance and myopia risk (AL/CR) in primary school children aged 6 to 12 years in Shenzhen, China

Hong-Wei Deng 2026-11-07 1 min 24 words LOW VISION

Summary: CONCLUSIONS: Rope-skipping performance was inversely associated with AL/CR ≥3, suggesting a potential protective role against myopia. Further studies are needed to confirm this association.

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Role of smartphone-based digital adherence support for glaucoma medication therapy

David J Mathew 2026-11-07 1 min 57 words LOW VISION

Summary: Sustained adherence to intraocular pressure-lowering medications is essential for preventing progressive visual loss in patients with glaucoma on pharmacotherapy. However, adherence remains suboptimal across patient populations as the result of cognitive, behavioural, physical, and socioeconomic bar...

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Statin Use and Age-related Macular Degeneration: A Systematic Review and Meta-Analysis of Disease Outcomes

Ricardo Noguera Louzada 2026-11-07 1 min 36 words LOW VISION

Summary: CONCLUSION: The evidence was of very low certainty and insufficient to establish a protective or harmful effect of statins on AMD. These findings do not support prescribing statins specifically for AMD or discontinuing otherwise indicated therapy.

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A comparison of diagnostic accuracy between CNN and transformer AI models for periodontal bone loss detection: A systematic review and meta-analysis

Selviana Rizky Pramitha 2026-11-08 1 min 23 words LOW VISION

Summary: CONCLUSION: Current evidence demonstrates greater diagnostic accuracy of CNN-based models for periodontal bone loss detection, while comparative conclusions for transformer-based models remain premature.

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Effectiveness and Safety Outcomes After Switching from Reference Biologics to Biosimilars for Treatment of Non-Infectious Uveitis: A Systematic Review and Meta-Analysis

Rafael N Miranda 2026-11-08 1 min 34 words LOW VISION

Summary: CONCLUSIONS: Currently, the available evidence is of low quality and demonstrates insufficient support for a meaningful difference in terms of effectiveness and safety when switching patients with NIU to biosimilars from their reference biologics.

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Does eHealth literacy matter for health-promoting behaviors among blind and low-vision adults? A mixed methods study

👤 Jessie Chin 📅 2026-11-08 📖 1 min 📝 47 words LOW VISION

Summary: CONCLUSION: eHealth literacy alone does not ensure more health behavior engagement in BLV individuals. To bridge the gap, health systems must prioritize the development of accessible digital health resources, incorporate disability-specific adaptations in health education, and ensure equitable oppor...

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Construction of Highly Viable, Excitable, Aligned Human Induced Pluripotent Stem Cell-Derived Retinal Ganglion Cell Sheets on Poly-ε-caprolactone Nanofibers by Targeting Integrin β1-Mediated Adhesion Pathways

👤 Toru Nakazawa
📅 2026-11-09 📖 1 min 📝 63 words

LOW VISION

Summary: Transplantation of induced pluripotent stem cell-derived retinal ganglion cells (RGCs) is a promising strategy for optic nerve regeneration and vision restoration in glaucoma. However, injections of conventional single-cell suspensions have low engraftment rates, hindering the formation of the compl...

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High-Performance Self-Powered Infrared Photodetector via Semiconducting Graphene Nanoribbon/Germanium Vertical Heterojunctions with Robust Thermal Stability

👤 Aimin Song
📅 2026-11-09 📖 1 min 📝 54 words LOW VISION

Summary: Self-powered photodetectors stand as a cornerstone for next-generation low-power electronics, with promising applications in optical communications, environmental monitoring, night vision, and beyond. However, achieving high performance in these devices, particularly under elevated temperatures, rem...

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A single session of high-definition transcranial direct current stimulation does not modulate effects of knee two-point discrimination training or sensorimotor function in healthy adults: Double-blind randomised controlled trial

👤 Paul W Hodges
📅 2026-07-30 📖 1 min 📝 53 words

TACTILE ACUITY

Summary: CONCLUSION: A single 20-minute session of 1 mA HD-tDCS, with or without brief TPD training, does not acutely modify knee somatosensory, sensorimotor, or functional performance in healthy adults. Existing evidence at the hand may not translate to the lower limb. Future work should investigate higher-...

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Variation in glabrous skin physiology across anatomical locations on the human hand

Martina C Meinke 2026-08-13 1 min 76 words TACTILE ACUITY

Summary: Skin physiology varies greatly between anatomical locations, with impact on tactile sensitivity. Research on the variations in morphology and physiology between different anatomical locations of glabrous skin is limited so far. This study investigates the morphology and physiology of glabrous skin o...

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Beyond Hamstring Shortness: Continuous Flexibility Is Associated With Balance and Stability but Not Sensory Measures-A Cross-Sectional Study

Senanur Akıncı 2026-08-14 1 min 66 words TACTILE ACUITY

Summary: CONCLUSIONS: Greater hamstring flexibility was associated with better dynamic balance performance and showed a weaker association with postural stability, whereas no robust associations were observed with sensory measures of postural control. These findings suggest that hamstring flexibility may be ...

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Resting-state EEG microstate dynamics reflect individual differences in tactile angle discriminability

Liu Yang 2026-08-19 1 min 63 words TACTILE ACUITY

Summary: Tactile spatial acuity, the ability to discriminate fine spatial details, is a fundamental sensory capacity modulated by both peripheral and central mechanisms. Here, we investigated whether resting-state EEG microstate dynamics are associated with individual differences in tactile spatial acuity. F...

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A grating orientation task with controlled dome elevation speed and contact force: stimulus conditions and day-to-day reliability

Hideaki Onishi 2026-09-25 1 min 66 words TACTILE ACUITY

Summary: The grating orientation discrimination task (GOT) is typically used to assess tactile spatial acuity. However, performance is influenced by stimulus speed, contact force between the stimulator and the skin, and the method of discrimination. This study aimed to explore suitable stimulus conditions fo...

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Epidermal microstructure and tactile sensitivity in the glabrous skin of hands and feet

👤 Hannes P Saal 📅 2026-09-25 📖 1 min 📝 69 words TACTILE ACUITY

Summary: Glabrous skin on the human palms and soles is specialized for mechanical interaction and is characterized by papillary ridges, an absence of hair follicles, and a thick stratum corneum. Although gross morphological differences, such as the increased thickness of the plantar stratum corneum, are well...

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Neurotrauma During Military Training: Results of the INVestigating traIning assoCiated blasT pAthology (INVICTA) Study

👤 Walter S Carr 📅 2026-10-01 📖 1 min 📝 67 words TACTILE ACUITY

Summary: Concern is growing that subconcussive blast exposure during military exercises such as heavy weapons training (HWT) may adversely affect brain function. HWT involves firing shoulder-mounted weapons that can expose the gunner, assistant gunner, and instructors (range safety officers [RSOs]) to blast ...

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Altered body representation in chronic non-specific low back pain: a testable framework linking body-schema-related disturbance, corticomotor alterations, and altered pain processing

👤 Zhifei Xie
📅 2026-10-24 📖 1 min 📝 66 words

TACTILE ACUITY

Summary: Low back pain is a leading contributor to disability worldwide; in chronic non-specific low back pain (CNSLBP), the processes sustaining pain and disability in the absence of identifiable structural pathology remain incompletely understood. Three partially overlapping research lines have documented ...

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Hyperexcitable primary somatosensory cortex in the MitoPark mouse model of Parkinson's disease

Alan J Emanuel 2026-10-25 1 min 67 words TACTILE ACUITY

Summary: The majority of Parkinson's disease patients suffer from decreased tactile acuity or neuropathic pain, both of which contribute to low quality of life and likely exacerbate motor dysfunction. How Parkinson's disease pathology relates to altered somatosensation remains unclear. One potential contribu...

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The Relationship Between Occlusal Tactile Acuity, Bite Force and Tooth Wear

Birgitta Häggman-Henrikson 2026-11-03 1 min 52 words TACTILE ACUITY

Summary: CONCLUSIONS: Bite force may be a clinically relevant factor associated with accumulated tooth wear, whereas OTA was not independently associated with tooth wear. Longitudinal studies are needed to determine whether bite force can predict tooth wear progression and whether more comprehensive assessme...

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Impact of Braille Booklet, Audio-Tactile Performance and Caregiver-supervised Training on Oral Hygiene Status, Knowledge, Attitude and Practices among Visually Challenged Children- A Randomized Clinical Trial

👤 Priyanga Chandrasekaran
📅 2026-09-22 📖 1 min 📝 52 words

BRAILLE

Summary: CONCLUSION: Both caregiver-supervised training and audio tactile performance using specialized model method of oral health education proved to be more effective than the braille booklet group on plaque control and improvement in the oral hygiene status, oral health knowledge, attitude and practice i...

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Design and functional verification of an automated Perkins Brailler accessory with audio feedback

👤 Ali Alhaj 📅 2026-09-23 📖 1 min 📝 67 words BRAILLE

Summary: CONCLUSIONS: The study did not include learners with visual impairment, Braille teachers, rehabilitation specialists, or other end users, and it did not collect quantitative measures of accuracy across the full character set, writing speed, embossing quality, latency, durability, safety, usability, ...

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Uncertainty-Aware structured decoding for Chinese Braille document transcription

👤 Haoran Sun 📅 2026-09-25 📖 1 min 📝 64 words BRAILLE

Summary: Automatic transcription of braille documents remains challenging in document digitization settings, particularly under real imaging degradations such as uneven embossing, illumination variation, paper texture, skew, blur, and ghosting, which make braille dot presence uncertain and often cause error ...

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Clinical Characteristics and Epidemiology of Isolated Dystonia: Results from a US Claims Analysis

James E Thornton 2026-10-08 1 min 54 words BRAILLE

Summary: CONCLUSIONS: This study defines an isolated dystonia cohort in the US based on real-world evidence and systematic exclusion of secondary causes of dystonia. Estimated prevalence is within range of prior epidemiological studies, and the cohort showed increased rates of comorbid psychiatric disorders....

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Effects of multisensory tactile-auditory integration teaching on systematic anatomy learning in visually impaired medical students

Fanzhen Kong 2026-10-08 1 min 50 words BRAILLE

Summary: CONCLUSION: The tactile-auditory integrated training system significantly improves objective summative knowledge acquisition and subjective learning experiences in visually impaired medical students. This approach provides a scalable, cost-constrained, and replicable pedagogical model for inclusive ...

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Physical and Environmental Accessibility of Dental Services for Persons with Disabilities in Saudi Primary Health Care: A Cross-Sectional Study

👤 Estie Kruger 📅 2026-10-16 📖 1 min 📝 66 words BRAILLE

Summary: Background/Objectives: Persons with disabilities (PWDs) may experience barriers when accessing oral health care, including those related to the physical and sensory environment of the dental-care setting. In this study we assessed the accessibility of primary health care centres (PHCs) providing den...

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Visible Light Driven Diselenide Chemistry for Simultaneous 3D Patterning and Mechanical Reinforcement

👤 Huaping Xu 📅 2026-10-16 📖 1 min 📝 63 words BRAILLE

Summary: Three-dimensional (3D) patterning on polymer surfaces is typically achieved by photolithography or related techniques, which often involve complex procedures, high-energy ultraviolet light, and risk of material damage. Here, we report a visible light-driven strategy that exploits the dual functional...

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An accessible pregnancy test for blind and low-vision users

Vanja Garaj 2026-10-16 1 min 63 words BRAILLE

Summary: CONCLUSIONS: The findings suggest that accessible pregnancy testing must prioritise not only non-visual result reading but also privacy, dignity, reproductive autonomy and independent health management. These results represent the preliminary evidence of acceptability rather than a conclusive test o...

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Genome-Informed Loop-Mediated Isothermal Amplification (LAMP) Assays for Detection and Differentiation of Ralstonia pseudosolanacearum Phylotypes I and III

Jarred Yasuhara-Bell 2026-10-27 1 min 76 words BRAILLE

Summary: The Ralstonia solanacearum species complex (RSSC) causes bacterial wilt in a broad range of economically important hosts. The RSSC has been reclassified into three species, of which R. solanacearum sequevars 1 and 2 in phylotype IIB are designated as Select Agents in the U.S. Many rapid tests for "R...

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Encoding tactile stimuli for Braille recognition with organoids

👤 Benjamin Ward-Cherrier 📅 2026-10-31 📖 1 min 📝 62 words BRAILLE

Summary: This study proposes a transferable encoding strategy that maps tactile sensor data to electrical stimulation patterns, enabling neural organoids to perform an open-loop artificial tactile Braille classification task. Human forebrain organoids cultured on a low-density microelectrode array (MEA) are ...

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Correction: Transcranial direct current stimulation to improve cognitive function in patients with vascular cognitive impairment: a literature review

👤 Ling Guan 📅 2026-11-05 📖 1 min 📝 6 words TDCS TACS TRNS

Summary: [This corrects the article DOI: 10.3389/fnagi.2026.1829594.].

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Feasibility and tolerability of combining 6 Hz transcranial alternating current stimulation (tACS) at 3.2 mA and digital cognitive training in older adults with mild cognitive impairment

👤 Rogério Panizzutti
📅 2026-11-05 📖 1 min 📝 41 words

TDCS TACS TRNS

Summary: CONCLUSIONS: Higher-intensity tACS plus DCT was feasible and well tolerated in older adults with MCI. Exploratory findings suggest that active stimulation may improve inhibitory control and facilitate transfer to selected cognitive domains. Confirmation in the ongoing randomized controlled trial is ...

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The Non-Invasive Brain Stimulation Side Effect Scale (NIBS-SES): development of a brief, multilingual questionnaire for monitoring adverse effects in transcranial magnetic, electrical and ultrasound stimulation

👤 Maximilian Kiebs
📅 2026-11-05 📖 1 min 📝 59 words

TDCS TACS TRNS

Summary: CONCLUSIONS: Our findings support the feasibility of repeated session-level assessment with the initial version, from which the final NIBS-SES was derived. This multilingual, open-access instrument comprises 10 within-session and 9 between-session items developed for monitoring across TMS, tES, and ...

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Concurrent prefrontal neurochemical effects of high-definition transcranial direct current stimulation: directional E-field associations with GABA+ and Glx

👤 Jianqi Li 📅 2026-11-05 📖 1 min 📝 58 words TDCS TACS TRNS

Summary: Transcranial direct current stimulation (tDCS) is a promising neuromodulatory technique for treating neuropsychiatric disorders. However, its neurochemical mechanisms remain poorly characterized, and clinical efficacy is limited by substantial inter-individual variability. This study used concurrent...

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Anodal tDCS alters perceived visual contrast in humans

👤 Sang Wook Hong 📅 2026-11-06 📖 1 min 📝 62 words TDCS TACS TRNS

Summary: Cortical gain modulation is thought to alter perceptual experience, most prominently demonstrated by the effects of spatial attention on contrast appearance. Whether externally induced changes in cortical excitability produce comparable perceptual consequences remains unclear. Here, we tested whethe...

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Evaluating OPTION: Benefits and Pitfalls of a Youth Trafficking Diversion Court in Florida

👤 Joan A Reid 📅 2026-11-06 📖 1 min 📝 67 words TDCS TACS TRNS

Summary: Trafficking diversion courts (TDCs) provide an alternative approach to supporting trafficked youth by promoting recovery, preventing revictimization, and diverting youth from the justice system. The current study examines the benefits and challenges of the OPTION (Offering Potentially Trafficked Ind...

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Amelioration of diabetic autonomic neuropathy symptoms following a course of transcranial direct current stimulation

👤 Ahmad Alipour 📅 2026-11-06 📖 1 min 📝 62 words TDCS TACS TRNS

Summary: Cardiovascular autonomic neuropathy (CAN) is a severe and potentially life-threatening complication of diabetes. This randomized, sham-controlled, three-arm trial evaluated the efficacy of anodal transcranial direct current stimulation (tDCS) targeting the left dorsolateral prefrontal cortex for imp...

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From Single Cells to the Circulating Tumor Microenvironment: Label-Free Size-Selection and a Morphological Taxonomy for Liquid Biopsy in Cytopathology

Satoru Kobayashi 2026-11-09 1 min 25 words TDCS TACS TRNS

Summary: CONCLUSION: Size-selection converts a liquid biopsy into a cytology slide, placing interpretation of the circulating tumor ecosystem within the existing competence of the diagnostic laboratory.

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Phase-Dependent Effects of Cerebellar Transcranial Alternating Current Stimulation on Lateral Vestibulospinal Tract Excitability

Tsubasa Mitsutake 2026-11-09 1 min 36 words TDCS TACS TRNS

Summary: CONCLUSIONS: LVST excitability differed according to the phase of the ongoing cerebellar-targeted tACS waveform. These findings provide electrophysiologic evidence of an online phase-dependent effect on vestibulospinal excitability and a physiologic basis for future phase-specific neuromodulation st...

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Effects of Transcranial Direct Current Stimulation on Core Symptoms and Sleep Disturbances in Children With Autism Spectrum Disorder: A Meta-Analysis of Randomized Controlled Trials

👤 Meimei Yang
📅 2026-11-09 📖 1 min 📝 44 words

TDCS TACS TRNS

Summary: CONCLUSION: tDCS, most commonly delivered at 1 mA over the DLPFC, was associated with improvements in some ASD-related behavioral outcomes; preliminary evidence also suggested a potential benefit for sleep disturbances, although this finding was based on only two low-certainty studies and warrants c...

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Penetrating needling from Qiuxu (GB40) to Zhaohai (KI6) versus separate needling for post-stroke lower-limb motor dysfunction: an assessor-blinded randomized controlled trial protocol

Chunxiao Wan
2026-11-05 1 min 63 words FNIRS

Summary: Post-stroke lower-limb motor dysfunction limits mobility, balance, and independence and remains a major target of neurorehabilitation. Acupuncture is widely used as an adjunctive intervention after stroke, but direct randomized evidence comparing penetrating needling with conventional separate needl...

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Advancing Repetitive Transcranial Magnetic Stimulation for Depression: Novel Protocols, Precision Targeting, and Future Directions

👤 Youguo Tan 📅 2026-11-05 📖 1 min 📝 57 words FNIRS

Summary: Major depressive disorder (MDD) remains a significant global health challenge, with substantial limitations inherent in current pharmacotherapeutic and psychotherapeutic approaches. Repetitive transcranial magnetic stimulation (rTMS) has emerged as a safe, effective, and non-invasive neuromodulation...

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Transforming neurophysiological data into measurable stress metric using machine learning-based stress quantification framework for smart factory workers

👤 Azfar Khalid 📅 2026-11-05 📖 1 min 📝 67 words FNIRS

Summary: INTRODUCTION: Mental stress is a common issue in demanding occupational setups, such as smart industrial settings, particularly when working with robots, being one of the primary reasons for decreased performance and productivity. Estimating and evaluating stress are critical for worker safety, perf...

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Prefrontal Activation and Cognitive Heterogeneity in Pediatric Sleep-Disordered Breathing

👤 Amal Isaiah 📅 2026-11-05 📖 1 min 📝 39 words FNIRS

Summary: CONCLUSIONS AND RELEVANCE: In this cross-sectional study, task-related PFC activation was associated with SDB severity and moderated the severity-cognition association. fNIRS may help identify children whose cognitive profiles deviate from conventional severity metrics, thereby supporting individual...

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Mirror Neural System in Hands Motor Imagery with Pictures and Videos Guidance: An EEG-fNIRS Study

👤 Mohamad Sawan
📅 2026-11-05 📖 1 min 📝 64 words FNIRS

Summary: Brain-computer interfaces (BCIs) achieved by motor imagery (MI) have been widely accepted as an effective tool in rehabilitative applications. Functional near-infrared spectroscopy (fNIRS) and electroencephalography (EEG) are two common implementation forms of BCIs, capable of monitoring neuronal po...

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